

Worked examples for Lecture 12: Finite elasticity

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These examples accompany Lecture 12. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Example 1: Simple shear. A body with referential density $\rho(\mathbf{x})$ undergoes the motion

$$\varphi(\mathbf{x}, t) = (x_1 + \gamma t x_2, x_2, x_3),$$

where γ is a constant shear rate, and the reference body is the position of the body at $t = 0$. (a) Compute the referential velocity, the deformation gradient and the Jacobian, and show that the motion preserves volume. (b) Find the inverse motion $\varphi^{-1}(\cdot, t)$, and hence the spatial density and the spatial velocity field, i.e. the velocity of whichever particle occupies the spatial point $\mathbf{y}$ at time t . Compare these with their referential counterparts. (c) Compute $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, show that $\delta \mathbf{x} \cdot \mathbf{C} \delta \mathbf{x}$ is the squared length of the image of a small material line element $\delta \mathbf{x}$, and find the eigenvalues and eigenvectors of $\mathbf{C}$. The square roots of the eigenvalues are the *principal stretches*.

Solution. (a) Differentiating the motion with respect to time at fixed $\mathbf{x}$, and with respect to $\mathbf{x}$ at fixed t , we obtain

$$\mathbf{v}(\mathbf{x}, t) = (\gamma x_2, 0, 0), \quad \mathbf{F}(\mathbf{x}, t) = \begin{pmatrix} 1 & \gamma t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (1)$$

The deformation gradient is upper triangular with unit diagonal, and so

$$J = \det \mathbf{F} = 1 \quad (2)$$

for all $\mathbf{x}$ and t . Applying the change of variables formula with $f = 1$ to any sub-body U , we find that the volume of U_t equals that of U , so the motion is *isochoric*. Mass conservation, $\rho(\mathbf{x}) = J \varrho[\varphi(\mathbf{x}, t), t]$, then gives $\varrho[\varphi(\mathbf{x}, t), t] = \rho(\mathbf{x})$: each particle retains its initial density.

(b) Setting $\mathbf{y} = \varphi(\mathbf{x}, t)$ and solving for $\mathbf{x}$ gives the inverse motion

$$\varphi^{-1}(\mathbf{y}, t) = (y_1 - \gamma t y_2, y_2, y_3), \quad (3)$$

which is well-defined for all t , as it must be since $J \neq 0$. The spatial density is therefore

$$\varrho(\mathbf{y}, t) = \rho(y_1 - \gamma t y_2, y_2, y_3). \quad (4)$$

Although $J = 1$, the spatial density at a fixed point $\mathbf{y}$ is in general time-dependent. For example, if $\rho(\mathbf{x}) = \rho_0(1 + \alpha x_1)$, then $\rho(\mathbf{y}, t) = \rho_0[1 + \alpha(y_1 - \gamma t y_2)]$, because particles of differing density are carried past the point $\mathbf{y}$ by the shear. Only for uniform ρ is ρ constant in time. For the velocity, substituting eq. (3) into eq. (1) gives the spatial velocity field

$$\mathbf{v}[\varphi^{-1}(\mathbf{y}, t), t] = (\gamma y_2, 0, 0). \quad (5)$$

For this particular motion the spatial and referential velocity fields have the same functional form, because the velocity depends only on x_2 , and $x_2 = y_2$ is unchanged by the motion. Their meanings are nevertheless different: eq. (1) gives the velocity of the particle labelled $\mathbf{x}$, while eq. (5) gives the velocity observed at the fixed spatial point $\mathbf{y}$. Example 4 provides a case in which the two fields differ in form.

(c) From eq. (1), and writing $a = \gamma t$ for brevity,

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} = \begin{pmatrix} 1 & a & 0 \\ a & 1 + a^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (6)$$

By the Taylor expansion of the motion, a small material line element $\delta \mathbf{x}$ at $\mathbf{x}$ is mapped to $\mathbf{F} \delta \mathbf{x}$, whose squared length is

$$\|\mathbf{F} \delta \mathbf{x}\|^2 = \delta \mathbf{x} \cdot \mathbf{F}^T \mathbf{F} \delta \mathbf{x} = \delta \mathbf{x} \cdot \mathbf{C} \delta \mathbf{x}. \quad (7)$$

Hence $\mathbf{C}$ is symmetric and positive-definite, and a unit line element along an eigenvector of $\mathbf{C}$ with eigenvalue λ has its length multiplied by $\sqrt{\lambda}$. The x_3 -direction is an eigenvector with eigenvalue 1. In the (x_1, x_2) -plane the characteristic equation is

$$\lambda^2 - (2 + a^2)\lambda + 1 = 0, \quad (8)$$

with roots

$$\lambda_{\pm} = 1 + \frac{1}{2}a^2 \pm a\sqrt{1 + \frac{1}{4}a^2}. \quad (9)$$

The product of the roots is $\lambda_+ \lambda_- = 1 = \det \mathbf{C} = J^2$, as it must be for an isochoric motion. The principal stretches take the simple form

$$\sqrt{\lambda_{\pm}} = \sqrt{1 + \frac{1}{4}a^2} \pm \frac{1}{2}a, \quad (10)$$

as may be verified by squaring the right-hand side. The eigenvector equation $(1 - \lambda)u_1 + au_2 = 0$ gives $\mathbf{u} \propto (1, (\lambda - 1)/a, 0)$, and using eqs. (9) and (10) we have $(\lambda_{\pm} - 1)/a = \frac{1}{2}a \pm \sqrt{1 + \frac{1}{4}a^2} = \pm \sqrt{\lambda_{\pm}}$, so that

$$\mathbf{u}_{\pm} \propto (1, \pm \sqrt{\lambda_{\pm}}, 0). \quad (11)$$

These are orthogonal, since $\mathbf{u}_+ \cdot \mathbf{u}_- = 1 - \sqrt{\lambda_+ \lambda_-} = 0$. For small a we have $\lambda_{\pm} = 1 \pm a + O(a^2)$, principal stretches $1 \pm \frac{1}{2}a$, and eigenvectors tending to $(1, \pm 1, 0)/\sqrt{2}$: to first order, simple shear is an extension along one diagonal of the (x_1, x_2) -plane and an equal contraction along the other, which is the familiar statement of small-strain theory. For large a the largest stretch is approximately a , and its direction $(1, \sqrt{\lambda_+}, 0)$ tends towards the x_2 -axis, since material lines initially along x_2 are drawn out most strongly by the shear.

Example 2: Polar decomposition by hand. Consider the plane deformation gradient

$$\mathbf{F} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix},$$

the third direction being trivial ($F_{33} = 1, F_{13} = F_{23} = F_{31} = F_{32} = 0$). The polar decomposition theorem of Lecture 13 states that $\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{V}\mathbf{R}$ with $\mathbf{R}$ a rotation matrix and $\mathbf{U}, \mathbf{V}$ symmetric and positive-definite. Compute $\mathbf{C} = \mathbf{F}^T\mathbf{F}$ and its eigenvalues and eigenvectors; construct $\mathbf{U} = \sqrt{\mathbf{C}}$ from the eigen-decomposition; show that $\mathbf{R} = \mathbf{F}\mathbf{U}^{-1}$ is a rotation and find its angle; compute $\mathbf{V} = \mathbf{R}\mathbf{U}^T$; and describe the image of the unit circle.

Solution. Direct multiplication gives

$$\mathbf{C} = \mathbf{F}^T\mathbf{F} = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}, \quad (12)$$

with $\text{tr } \mathbf{C} = 6$ and $\det \mathbf{C} = 4 = (\det \mathbf{F})^2$. The characteristic equation $\lambda^2 - 6\lambda + 4 = 0$ has roots

$$\lambda_{\pm} = 3 \pm \sqrt{5}, \quad (13)$$

and the eigenvector equation $(4 - \lambda)u_1 + 2u_2 = 0$ gives $\mathbf{u} \propto (2, \lambda - 4)$, so that

$$\mathbf{u}_{\pm} \propto (2, -1 \pm \sqrt{5}), \quad \|\mathbf{u}_{\pm}\|^2 = 10 \mp 2\sqrt{5}. \quad (14)$$

These are orthogonal, as $4 - (\sqrt{5} - 1)(\sqrt{5} + 1) = 0$. The eigenvector $\mathbf{u}_+$ makes an angle $\alpha = \arctan[(\sqrt{5} - 1)/2] \approx 31.7^\circ$ with the x_1 -axis. The square roots of the eigenvalues have the closed forms

$$\sqrt{\lambda_{\pm}} = \frac{\sqrt{5} \pm 1}{\sqrt{2}}, \quad (15)$$

since $(\sqrt{5} \pm 1)^2/2 = (6 \pm 2\sqrt{5})/2 = 3 \pm \sqrt{5}$; these are the principal stretches, with product $2 = \det \mathbf{F}$.

Writing $\mathbf{q}_{\pm} = \mathbf{u}_{\pm}/\|\mathbf{u}_{\pm}\|$, the square root of $\mathbf{C}$ is

$$\mathbf{U} = \sqrt{\lambda_+} \mathbf{q}_+ \mathbf{q}_+^T + \sqrt{\lambda_-} \mathbf{q}_- \mathbf{q}_-^T. \quad (16)$$

Using $1/(10 - 2\sqrt{5}) = (5 + \sqrt{5})/40$, the first projector is

$$\mathbf{q}_+ \mathbf{q}_+^T = \frac{5 + \sqrt{5}}{40} \begin{pmatrix} 4 & 2\sqrt{5} - 2 \\ 2\sqrt{5} - 2 & 6 - 2\sqrt{5} \end{pmatrix} = \begin{pmatrix} \frac{5+\sqrt{5}}{10} & \frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{5-\sqrt{5}}{10} \end{pmatrix}, \quad (17)$$

and $\mathbf{q}_- \mathbf{q}_-^T = \mathbf{1} - \mathbf{q}_+ \mathbf{q}_+^T$. Substituting into eq. (16) with eq. (15), the entries of $\mathbf{U}$ are

$$\begin{aligned} U_{11} &= \frac{(\sqrt{5} + 1)(5 + \sqrt{5}) + (\sqrt{5} - 1)(5 - \sqrt{5})}{10\sqrt{2}} = \frac{12\sqrt{5}}{10\sqrt{2}} = \frac{6}{\sqrt{10}}, \\ U_{22} &= \frac{(\sqrt{5} + 1)(5 - \sqrt{5}) + (\sqrt{5} - 1)(5 + \sqrt{5})}{10\sqrt{2}} = \frac{8\sqrt{5}}{10\sqrt{2}} = \frac{4}{\sqrt{10}}, \\ U_{12} &= \frac{\sqrt{5}}{5} (\sqrt{\lambda_+} - \sqrt{\lambda_-}) = \frac{2}{\sqrt{10}}. \end{aligned} \quad (18)$$

Hence

$$\mathbf{U} = \frac{1}{\sqrt{10}} \begin{pmatrix} 6 & 2 \\ 2 & 4 \end{pmatrix} = \sqrt{\frac{2}{5}} \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \quad (19)$$

and we check that $\mathbf{U}^2 = \frac{2}{5} \begin{pmatrix} 10 & 5 \\ 5 & 5 \end{pmatrix} = \mathbf{C}$, $\text{tr } \mathbf{U} = \sqrt{10} = \sqrt{\lambda_+} + \sqrt{\lambda_-}$ and $\det \mathbf{U} = 2 = \det \mathbf{F}$. (In two dimensions there is a shortcut: the Cayley–Hamilton theorem gives $\mathbf{U}^2 - (\text{tr } \mathbf{U})\mathbf{U} + (\det \mathbf{U})\mathbf{1} = \mathbf{0}$, so $\mathbf{U} = (\mathbf{C} + \sqrt{\det \mathbf{C}}\mathbf{1})/\text{tr } \mathbf{U}$ with $(\text{tr } \mathbf{U})^2 = \text{tr } \mathbf{C} + 2\sqrt{\det \mathbf{C}} = 10$, which reproduces eq. (19) immediately.)

Inverting the 2×2 matrix,

$$\mathbf{U}^{-1} = \frac{1}{\det \mathbf{U}} \begin{pmatrix} U_{22} & -U_{12} \\ -U_{12} & U_{11} \end{pmatrix} = \frac{1}{\sqrt{10}} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}, \quad (20)$$

and therefore

$$\mathbf{R} = \mathbf{F}\mathbf{U}^{-1} = \frac{1}{\sqrt{10}} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 & 1 \\ -1 & 3 \end{pmatrix}. \quad (21)$$

The columns of $\mathbf{R}$ are orthonormal, so $\mathbf{R}^T \mathbf{R} = \mathbf{1}$, and $\det \mathbf{R} = (9 + 1)/10 = 1$; hence $\mathbf{R}$ is a rotation. Comparing with $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ we read off $\cos \theta = 3/\sqrt{10}$ and $\sin \theta = -1/\sqrt{10}$, so that

$$\tan \theta = -\frac{1}{3}, \quad \theta \approx -18.4^\circ, \quad (22)$$

a clockwise rotation. The left stretch is most easily found from $\mathbf{V} = \mathbf{R}\mathbf{U}\mathbf{R}^T = \mathbf{F}\mathbf{R}^T$:

$$\mathbf{V} = \frac{1}{\sqrt{10}} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix} = \frac{1}{\sqrt{10}} \begin{pmatrix} 7 & 1 \\ 1 & 3 \end{pmatrix}, \quad (23)$$

which is symmetric with $\text{tr } \mathbf{V} = \sqrt{10}$ and $\det \mathbf{V} = 2$, so it has the same eigenvalues as $\mathbf{U}$, and $\mathbf{V}^2 = \begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix} = \mathbf{F}\mathbf{F}^T$. As a final check,

$$\mathbf{V}\mathbf{R} = \frac{1}{10} \begin{pmatrix} 7 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ -1 & 3 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 20 & 10 \\ 0 & 10 \end{pmatrix} = \mathbf{F}, \quad (24)$$

and similarly $\mathbf{R}\mathbf{U} = \mathbf{F}$. The eigenvectors of $\mathbf{V}$ are those of $\mathbf{F}\mathbf{F}^T$, namely $\mathbf{w}_\pm \propto (1, -2 \pm \sqrt{5})$, and one verifies that $\mathbf{R}\mathbf{u}_+ \propto \mathbf{w}_+$: the eigenvectors of $\mathbf{V}$ are the rotated eigenvectors of $\mathbf{U}$. Indeed $\mathbf{w}_+$ makes an angle $\beta = \arctan(\sqrt{5} - 2) \approx 13.3^\circ = \alpha + \theta$ with the x_1 -axis.

Geometrically (Fig. 1), $\mathbf{F}$ maps the unit circle to an ellipse with semi-axes $\sqrt{\lambda_\pm} \approx 2.29$ and 0.87 along $\mathbf{w}_\pm$, whose area $\pi\sqrt{\lambda_+\lambda_-} = 2\pi$ is J times that of the circle. The decomposition $\mathbf{F} = \mathbf{R}\mathbf{U}$ realises this as a stretch along the fixed directions $\mathbf{q}_\pm$ followed by the rotation, while $\mathbf{F} = \mathbf{V}\mathbf{R}$ rotates first and then stretches along the co-rotated directions $\mathbf{w}_\pm = \mathbf{R}\mathbf{q}_\pm$. Note that this matrix is the deformation gradient of the simple shear of Example 1 with $\gamma t = 1$, composed with a stretch by a factor of 2 along x_1 ; neither ingredient is a rotation, yet the deformation contains a rotation of 18.4° that only the polar decomposition makes visible.

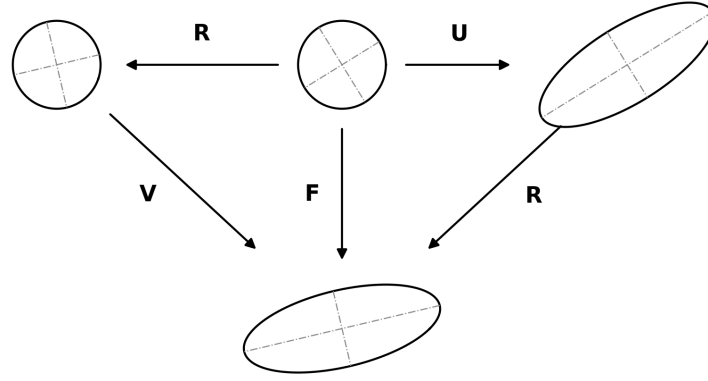


Fig. 1: The polar decomposition of the deformation gradient of Example 2, drawn in the manner of the corresponding figure in Lecture 13. The unit circle (centre of the upper row) is stretched by $\mathbf{U}$ along its eigenvectors (dash-dotted lines, at 31.7° and 121.7°) into an ellipse with semi-axes 2.29 and 0.87, and rotated by $\mathbf{R}$ through -18.4° . Either order of operations produces the same final ellipse, whose axes lie along the eigenvectors of $\mathbf{V}$.

Example 3: Rigid body motions. Let $\varphi(\mathbf{x}, t) = \mathbf{a}(t) + \mathbf{Q}(t)\mathbf{x}$, with $\mathbf{a}(t)$ a translation and $\mathbf{Q}(t)$ a rotation matrix. (a) Show that $\mathbf{F} = \mathbf{Q}$, $J = 1$ and $\mathbf{C} = \mathbf{1}$. (b) Show that, if the origin of the reference body is placed at its centre of mass, the kinetic energy is $\frac{1}{2}M\|\dot{\mathbf{a}}\|^2$ plus a rotational term, and identify the latter. (c) Show that any strain energy function satisfying material frame indifference has $W(\mathbf{x}, \mathbf{Q}) = W(\mathbf{x}, \mathbf{1})$, so that the potential energy is unchanged by the motion. (d) Determine which of $W_1 = \frac{1}{2}\mu \operatorname{tr}(\mathbf{F}^T \mathbf{F} - \mathbf{1})$, $W_2 = \frac{1}{2}\mu \operatorname{tr}(\mathbf{F} - \mathbf{1})$ and $W_3 = \frac{1}{2}\mu \|\mathbf{F} - \mathbf{1}\|^2$, where $\|\mathbf{A}\|^2 = A_{ij}A_{ij}$, are frame indifferent, giving an explicit rotation $\mathbf{Q}$ for those that are not.

Solution. (a) Since $\varphi_i = a_i(t) + Q_{ij}(t)x_j$, we have $F_{ij} = \partial\varphi_i/\partial x_j = Q_{ij}$. A rotation matrix satisfies $\mathbf{Q}^T \mathbf{Q} = \mathbf{1}$ and $\det \mathbf{Q} = 1$, so $J = 1$ and $\mathbf{C} = \mathbf{Q}^T \mathbf{Q} = \mathbf{1}$. By eq. (7), no material line element changes its length, which is what we mean by a rigid motion.

(b) The referential velocity is $\mathbf{v} = \dot{\mathbf{a}} + \dot{\mathbf{Q}}\mathbf{x}$. Differentiating $\mathbf{Q}^T \mathbf{Q} = \mathbf{1}$ gives $\dot{\mathbf{Q}}^T \mathbf{Q} + \mathbf{Q}^T \dot{\mathbf{Q}} = \mathbf{0}$, so the matrix $\boldsymbol{\Omega} = \mathbf{Q}^T \dot{\mathbf{Q}}$ is antisymmetric and can be written $\Omega_{ij} = -\epsilon_{ijk}\omega_k$ for some vector $\boldsymbol{\omega}(t)$, the angular velocity referred to the reference body. Then $\boldsymbol{\Omega}\mathbf{x} = \boldsymbol{\omega} \times \mathbf{x}$ and $\dot{\mathbf{Q}}\mathbf{x} = \mathbf{Q}(\boldsymbol{\omega} \times \mathbf{x})$, so that

$$\|\mathbf{v}\|^2 = \|\dot{\mathbf{a}}\|^2 + 2\dot{\mathbf{a}} \cdot \mathbf{Q}(\boldsymbol{\omega} \times \mathbf{x}) + \|\boldsymbol{\omega} \times \mathbf{x}\|^2, \quad (25)$$

where we used that $\mathbf{Q}$ preserves lengths. Integrating against ρ over M , the middle term contains $\int_M \rho \mathbf{x} d^3\mathbf{x}$, which vanishes when the origin is at the centre of mass. Using $\|\boldsymbol{\omega} \times \mathbf{x}\|^2 = \|\boldsymbol{\omega}\|^2 \|\mathbf{x}\|^2 - (\boldsymbol{\omega} \cdot \mathbf{x})^2$, the kinetic energy becomes

$$\mathcal{T} = \frac{1}{2}M\|\dot{\mathbf{a}}\|^2 + \frac{1}{2}\omega_i I_{ij} \omega_j, \quad I_{ij} = \int_M \rho(\mathbf{x}) (\|\mathbf{x}\|^2 \delta_{ij} - x_i x_j) d^3\mathbf{x}, \quad (26)$$

where $M = \int_M \rho d^3\mathbf{x}$ is the total mass and I_{ij} is the inertia tensor of the reference body, which is constant in time. This is the familiar decomposition of the kinetic energy of

a rigid body into translational and rotational parts, here obtained directly from the referential formula of the lecture.

(c) Material frame indifference requires $W(\mathbf{x}, \mathbf{QF}) = W(\mathbf{x}, \mathbf{F})$ for all deformation gradients and all rotations. Setting $\mathbf{F} = \mathbf{1}$ gives $W(\mathbf{x}, \mathbf{Q}) = W(\mathbf{x}, \mathbf{1})$, and hence for the rigid motion

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \mathbf{Q}(t)] d^3\mathbf{x} = \int_M W(\mathbf{x}, \mathbf{1}) d^3\mathbf{x}, \quad (27)$$

which is independent of time. The Lagrangian therefore reduces, up to a constant, to the kinetic energy of eq. (26), and Hamilton's principle would return the equations of motion of a free rigid body.

(d) For W_1 we have $W_1(\mathbf{QF}) = \frac{1}{2}\mu \operatorname{tr}(\mathbf{F}^T \mathbf{Q}^T \mathbf{QF} - \mathbf{1}) = W_1(\mathbf{F})$ for every rotation, so W_1 is frame indifferent; it depends on $\mathbf{F}$ only through $\mathbf{C}$. For the other two it suffices to exhibit a single counterexample. Take the rotation by $\pi/2$ about the x_3 -axis,

$$\mathbf{Q} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \operatorname{tr} \mathbf{Q} = 1, \quad (28)$$

and compare $\mathbf{F} = \mathbf{1}$ with $\mathbf{F} = \mathbf{Q}$. We find $W_2(\mathbf{1}) = 0$ but $W_2(\mathbf{Q}) = \frac{1}{2}\mu(1 - 3) = -\mu$, while $W_3(\mathbf{1}) = 0$ but

$$W_3(\mathbf{Q}) = \frac{1}{2}\mu(Q_{ij} - \delta_{ij})(Q_{ij} - \delta_{ij}) = \frac{1}{2}\mu(Q_{ij}Q_{ij} - 2Q_{ii} + 3) = \mu(3 - \operatorname{tr} \mathbf{Q}) = 2\mu. \quad (29)$$

Neither W_2 nor W_3 is frame indifferent: both assign a change of potential energy to a rigid rotation of the undeformed body. More generally, for a rotation through an angle ψ about any axis $\operatorname{tr} \mathbf{Q} = 1 + 2\cos \psi$, so $W_3(\mathbf{Q}) = 2\mu(1 - \cos \psi)$. The function W_3 is what one obtains by naively treating the full displacement gradient, rather than its symmetric part, as a measure of strain, and this calculation shows why the linearised theory of the next lecture must be built from the symmetric strain tensor. Note finally that frame indifference is necessary but not sufficient for a physically sensible strain energy: W_1 passes the test, yet $\partial W_1 / \partial F_{ij} = \mu F_{ij}$ does not vanish at $\mathbf{F} = \mathbf{1}$, so a body with this strain energy is not stress-free in its reference configuration.

Example 4: Uniform stretch of a cube. Let the reference body M be the unit cube $[0, 1]^3$ with uniform referential density ρ , and consider the motion $\varphi(\mathbf{x}, t) = (\lambda_1 x_1, \lambda_2 x_2, \lambda_3 x_3)$ with $\lambda_i > 0$. (a) Compute the mass, the volume of M_t and the spatial density. (b) Now let $\lambda_i = \lambda_i(t)$. Compute the kinetic energy from the referential formula of the lecture, and again as an integral over M_t of $\frac{1}{2}\varrho \|\mathbf{v}\|^2$ with the spatial velocity field, and check that the two agree.

Solution. (a) The deformation gradient is $\mathbf{F} = \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$ with $J = \lambda_1 \lambda_2 \lambda_3 > 0$. The instantaneous body is the box $M_t = [0, \lambda_1] \times [0, \lambda_2] \times [0, \lambda_3]$, whose volume is $\lambda_1 \lambda_2 \lambda_3 = J$, in agreement with the change of variables formula $\int_{M_t} d^3\mathbf{y} = \int_M J d^3\mathbf{x}$. The mass is $\int_M \rho d^3\mathbf{x} = \rho$, and mass conservation gives the spatial density

$$\varrho(\mathbf{y}, t) = \frac{\rho}{J} = \frac{\rho}{\lambda_1 \lambda_2 \lambda_3}, \quad (30)$$

which is uniform, and satisfies $\int_{M_t} \varrho d^3\mathbf{y} = (\rho/J)J = \rho$ as it must.

(b) The referential velocity is $v_i = \dot{\lambda}_i x_i$ (no sum on i), so the kinetic energy is

$$\mathcal{T} = \frac{1}{2}\rho \int_M \sum_{i=1}^3 \dot{\lambda}_i^2 x_i^2 d^3\mathbf{x} = \frac{1}{2}\rho \sum_{i=1}^3 \dot{\lambda}_i^2 \int_0^1 x_i^2 dx_i = \frac{\rho}{6} (\dot{\lambda}_1^2 + \dot{\lambda}_2^2 + \dot{\lambda}_3^2). \quad (31)$$

For the spatial calculation, the particle at $\mathbf{y} \in M_t$ is $x_i = y_i / \lambda_i$, so the spatial velocity field is

$$v_i[\varphi^{-1}(\mathbf{y}, t), t] = \frac{\dot{\lambda}_i}{\lambda_i} y_i, \quad (32)$$

which, in contrast to Example 1, differs in form from the referential field. Using eq. (30) and $\int_{M_t} y_1^2 d^3\mathbf{y} = \frac{1}{3}\lambda_1^3\lambda_2\lambda_3 = \frac{1}{3}J\lambda_1^2$ (and similarly for y_2, y_3),

$$\frac{1}{2} \int_{M_t} \varrho \|\mathbf{v}\|^2 d^3\mathbf{y} = \frac{\rho}{2J} \sum_{i=1}^3 \frac{\dot{\lambda}_i^2}{\lambda_i^2} \cdot \frac{J\lambda_i^2}{3} = \frac{\rho}{6} \sum_{i=1}^3 \dot{\lambda}_i^2, \quad (33)$$

in agreement with eq. (31). The two calculations are related by the change of variables $\mathbf{y} = \varphi(\mathbf{x}, t)$, under which $\varrho d^3\mathbf{y} = \rho d^3\mathbf{x}$; the referential form is the more convenient because its domain of integration does not move.

Example 5: Hamilton's principle for an elastic rod. A thin rod of unit cross-sectional area occupies $x \in [0, L]$ in its reference configuration and has uniform referential density ρ . Its motion is a scalar function $\varphi(x, t)$, with $v = \partial\varphi/\partial t$ and $F = \partial\varphi/\partial x$, and its strain energy function is $W(F) = \frac{1}{2}E(F - 1)^2$ with $E > 0$ a constant. Starting from the action with Lagrangian density $\frac{1}{2}\rho v^2 - W(F)$, derive the equation of motion and the natural boundary conditions at $x = 0$ and $x = L$. Identify the first Piola–Kirchhoff stress. Then linearise about the reference configuration by writing $\varphi = x + su$ and obtain the wave equation and the wave speed.

Solution. The action is

$$\mathcal{S}(\varphi) = \int_0^T \int_0^L \left\{ \frac{1}{2}\rho \left(\frac{\partial\varphi}{\partial t} \right)^2 - \frac{1}{2}E \left(\frac{\partial\varphi}{\partial x} - 1 \right)^2 \right\} dx dt. \quad (34)$$

For a variation $\delta\varphi$ vanishing at $t = 0$ and $t = T$, the first variation is

$$\delta\mathcal{S} = \int_0^T \int_0^L \left\{ \rho \frac{\partial\varphi}{\partial t} \frac{\partial\delta\varphi}{\partial t} - T \frac{\partial\delta\varphi}{\partial x} \right\} dx dt, \quad T = \frac{dW}{dF} = E \left(\frac{\partial\varphi}{\partial x} - 1 \right). \quad (35)$$

Integrating the first term by parts in time, the boundary contribution vanishes by the fixed end-point condition, while the second term is integrated by parts in x :

$$\delta\mathcal{S} = \int_0^T \int_0^L \left\{ -\rho \frac{\partial^2\varphi}{\partial t^2} + \frac{\partial T}{\partial x} \right\} \delta\varphi dx dt - \int_0^T [T \delta\varphi]_{x=0}^{x=L} dt. \quad (36)$$

For $\delta\mathcal{S}$ to vanish for all admissible variations, the volume and boundary terms must vanish separately. Choosing $\delta\varphi$ supported in the interior gives the Euler–Lagrange equation

$$\rho \frac{\partial^2\varphi}{\partial t^2} = \frac{\partial T}{\partial x} = E \frac{\partial^2\varphi}{\partial x^2}, \quad (37)$$

and then choosing $\delta\varphi$ with arbitrary values at the ends gives the natural boundary conditions $T(0, t) = T(L, t) = 0$, i.e.

$$\frac{\partial\varphi}{\partial x} = 1 \quad \text{at} \quad x = 0 \quad \text{and} \quad x = L. \quad (38)$$

Comparing with the lecture, $T = \partial W / \partial F$ is the one-dimensional first Piola–Kirchhoff stress, eq. (37) is $\rho \partial v / \partial t = \partial T / \partial x$, and eq. (38) is the traction-free condition $T\hat{n} = 0$ with $\hat{n} = \pm 1$ the outward normal at the two ends. Physically, an element with $F > 1$ is stretched and carries the tension $E(F - 1)$, an element with $F < 1$ is compressed, and the free ends of the rod carry no force. Had one end been held fixed instead, say $\varphi(0, t) = 0$, the admissible variations would satisfy $\delta\varphi(0, t) = 0$, no boundary term would arise there, and the condition at that end would be the prescribed (essential) one rather than a natural one.

Linearising, set $\varphi(x, t) = x + su(x, t)$ with s small. Then $F = 1 + s \partial u / \partial x$, $T = sE \partial u / \partial x$, and eq. (37) becomes, after dividing by s ,

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad c = \sqrt{\frac{E}{\rho}}, \quad (39)$$

with $\partial u / \partial x = 0$ at the free ends. The normal modes are $u = \cos(n\pi x / L) \cos(\omega_n t)$ with $\omega_n = n\pi c / L$. Because W is quadratic in F the exact equation (37) was already linear, so here the linearisation is exact. This cannot happen in three dimensions: a strain energy that is quadratic in $\mathbf{F}$ and frame indifferent must be a linear function of $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and then $\partial W / \partial F_{ij}$ is linear in $\mathbf{F}$ and non-zero at $\mathbf{F} = \mathbf{1}$ (as for W_1 in Example 3), so the reference configuration is not stress-free. The linearisation carried out in Lecture 13 is therefore a genuine approximation.

Example 6: Traction as force per unit reference area. Consider again the uniform stretch of Example 4, and suppose that the strain energy function is $W(\mathbf{F}) = \frac{1}{2}\kappa(J - 1)^2$ with $\kappa > 0$ constant and $J = \det \mathbf{F}$. (a) Show that $\partial J / \partial F_{ij} = J F_{ji}^{-1}$, and hence compute the first Piola–Kirchhoff stress. (b) Compute the traction $\mathbf{t} = \mathbf{T}\hat{\mathbf{n}}$ on the face $x_1 = 1$ of the reference cube, the total force acting on the image of this face, and the corresponding force per unit *current* area (the Cauchy traction). Show that the two tractions differ by the factor $\lambda_2 \lambda_3$.

Solution. (a) We first note that W is frame indifferent, since $\det(\mathbf{QF}) = \det \mathbf{Q} \det \mathbf{F} = J$. To differentiate the determinant, let $\delta\mathbf{F}$ be a small perturbation and write $\mathbf{F} + \delta\mathbf{F} = \mathbf{F}(\mathbf{1} + \mathbf{F}^{-1}\delta\mathbf{F})$. For any matrix $\mathbf{A}$, expanding the determinant shows that $\det(\mathbf{1} + \mathbf{A}) = 1 + \text{tr} \mathbf{A} + O(\|\mathbf{A}\|^2)$, because every term in the expansion other than the product of the diagonal entries is at least quadratic in the entries of $\mathbf{A}$. Hence

$$\det(\mathbf{F} + \delta\mathbf{F}) = J \left[1 + \text{tr}(\mathbf{F}^{-1}\delta\mathbf{F}) \right] + O(\|\delta\mathbf{F}\|^2) = J + J F_{ji}^{-1} \delta F_{ij} + O(\|\delta\mathbf{F}\|^2), \quad (40)$$

and comparing with $\delta J = (\partial J / \partial F_{ij}) \delta F_{ij}$ we obtain

$$\frac{\partial J}{\partial F_{ij}} = J F_{ji}^{-1}, \quad (41)$$

which is Jacobi's formula; note the transposed indices on the right-hand side. The chain rule then gives the first Piola–Kirchhoff stress

$$T_{ij} = \frac{\partial W}{\partial F_{ij}} = \kappa(J-1) \frac{\partial J}{\partial F_{ij}} = \kappa(J-1) J F_{ji}^{-1}, \quad \text{i.e.} \quad \mathbf{T} = \kappa(J-1) J \mathbf{F}^{-T}. \quad (42)$$

For the uniform stretch $\mathbf{F} = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, $J = \lambda_1 \lambda_2 \lambda_3$ and $\mathbf{F}^{-T}$ is diagonal with entries λ_i^{-1} , so that

$$\mathbf{T} = \kappa(J-1) \text{diag}(\lambda_2 \lambda_3, \lambda_3 \lambda_1, \lambda_1 \lambda_2). \quad (43)$$

(b) On the face $x_1 = 1$ of the reference cube the outward unit normal is $\hat{\mathbf{n}} = (1, 0, 0)$, and from eq. (43) the traction is

$$\mathbf{t} = \mathbf{T} \hat{\mathbf{n}} = \kappa(J-1) \lambda_2 \lambda_3 (1, 0, 0). \quad (44)$$

This is the force per unit *reference* area exerted on the body across the surface. The face has unit area in the reference body, so the total force on its image is $\kappa(J-1) \lambda_2 \lambda_3$ in the x_1 -direction. That image is the face $y_1 = \lambda_1$ of the box M_t , which has current area $\lambda_2 \lambda_3$, so the force per unit current area is

$$\mathbf{t}^{\text{Cauchy}} = \frac{\kappa(J-1) \lambda_2 \lambda_3}{\lambda_2 \lambda_3} (1, 0, 0) = \kappa(J-1) (1, 0, 0), \quad (45)$$

and the two tractions differ by the factor $\lambda_2 \lambda_3 = J / \lambda_1$, the ratio of current to reference area of the face. For $J > 1$ the traction is directed outwards: an outward pull must be applied to hold the body in a dilated state, and conversely a compressed body ($J < 1$) pushes back on whatever confines it.

The same argument applied to the other faces shows that the Cauchy traction on every face is $\kappa(J-1) \hat{\mathbf{n}}_t$, with $\hat{\mathbf{n}}_t$ the current outward normal, so the Cauchy stress is the isotropic tensor $\kappa(J-1) \mathbf{1}$: a body with this strain energy behaves as a compressible fluid with pressure $-\kappa(J-1)$, and for small volume changes $J-1 \approx \nabla \cdot \mathbf{u}$ we recover the linear relation between pressure and dilatation with bulk modulus κ . The general relation between the two stress tensors is $\boldsymbol{\sigma} = J^{-1} \mathbf{T} \mathbf{F}^T$, which indeed reduces to $\kappa(J-1) \mathbf{1}$ for eq. (42); the Piola–Kirchhoff and Cauchy stresses coincide only when $\mathbf{F} = \mathbf{1}$, which is why their distinction can be ignored in linearised theories about a stress-free reference state.

Worked examples for Lecture 13: Constitutive theory

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These examples accompany Lecture 13. They are intended as practice in the concrete calculations that underlie the lecture – differentiating strain energy functions, linearising about a stress-free equilibrium, and handling the symmetries of the elastic tensor – and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory. Throughout we write $H_{ij} = \partial u_i / \partial x_j$ for the displacement gradient of a linearised motion and $e_{ij} = \frac{1}{2}(H_{ij} + H_{ji})$ for the linearised strain tensor.

Example 1: The St Venant–Kirchhoff material. The simplest strain energy function compatible with material frame indifference and isotropy is that of the St Venant–Kirchhoff material,

$$U(\mathbf{C}) = \frac{1}{2}\lambda (\text{tr } \mathbf{E})^2 + \mu \text{tr}(\mathbf{E}^2), \quad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}), \quad (1)$$

where λ and μ are constants and $\mathbf{E}$ is the finite strain tensor. (a) Compute the second Piola–Kirchhoff stress $\mathbf{S} = 2\partial U / \partial \mathbf{C}$ and the first Piola–Kirchhoff stress $\mathbf{T} = \mathbf{F}\mathbf{S}$, and verify that $\mathbf{F}^{-1}\mathbf{T}$ is symmetric. (b) Linearise about $\mathbf{F} = \mathbf{I}$ by writing $\mathbf{F} = \mathbf{I} + s\mathbf{H}$ and show that $T_{ij} = s [\lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})] H_{kl} + O(s^2)$, so that the elastic tensor is the isotropic one quoted in the lecture. (c) Check the result by expanding $W(\mathbf{F}) = U(\mathbf{F}^T \mathbf{F})$ directly to second order in s .

Solution. (a) The chain-rule argument of the lecture treats the nine components C_{kl} as independent variables, so we must regard eq. (1) as defining U on all 3×3 matrices, with $\text{tr}(\mathbf{E}^2) = E_{kl}E_{lk}$. This extension is invariant under $\mathbf{C} \rightarrow \mathbf{C}^T$, which is what the lecture’s derivation of $\mathbf{T} = 2\mathbf{F}\partial U / \partial \mathbf{C}$ requires. From $\partial E_{kl} / \partial C_{ij} = \frac{1}{2}\delta_{ki}\delta_{lj}$ we have

$$\frac{\partial \text{tr } \mathbf{E}}{\partial C_{ij}} = \frac{1}{2}\delta_{ij}, \quad \frac{\partial}{\partial C_{ij}} (E_{kl}E_{lk}) = 2E_{lk} \frac{\partial E_{kl}}{\partial C_{ij}} = E_{ji}, \quad (2)$$

and hence, for symmetric $\mathbf{C}$,

$$S_{ij} = 2 \frac{\partial U}{\partial C_{ij}} = \lambda (\text{tr } \mathbf{E}) \delta_{ij} + 2\mu E_{ij}, \quad \mathbf{T} = \mathbf{F}\mathbf{S} = \mathbf{F} [\lambda (\text{tr } \mathbf{E}) \mathbf{I} + 2\mu \mathbf{E}]. \quad (3)$$

The stress–strain relation between $\mathbf{S}$ and $\mathbf{E}$ therefore has exactly the form of Hooke’s law, but here it holds for finite deformations. Since $\mathbf{F}^{-1}\mathbf{T} = \mathbf{S}$ and $\mathbf{E}$ is symmetric, $\mathbf{F}^{-1}\mathbf{T}$ is symmetric, as material frame indifference requires. Note that $\mathbf{T}$ itself is not

symmetric: for the simple shear $\mathbf{F} = \mathbf{I} + g \hat{\mathbf{e}}_1 \hat{\mathbf{e}}_2^T$ a short calculation from eq. (3) gives $T_{21} = \mu g$ but $T_{12} = \mu g + g^3(\frac{1}{2}\lambda + \mu)$.

(b) With $\mathbf{F} = \mathbf{I} + s\mathbf{H}$,

$$\mathbf{C} = \mathbf{I} + s(\mathbf{H} + \mathbf{H}^T) + s^2\mathbf{H}^T\mathbf{H}, \quad \mathbf{E} = s\mathbf{e} + \frac{1}{2}s^2\mathbf{H}^T\mathbf{H}, \quad (4)$$

so the finite strain tensor reduces to the linearised strain at first order. Substituting into eq. (3) and discarding $O(s^2)$ terms, including those generated by the factor $\mathbf{F} = \mathbf{I} + s\mathbf{H}$ multiplying $\mathbf{S}$,

$$T_{ij} = s [\lambda e_{kk} \delta_{ij} + 2\mu e_{ij}] + O(s^2) = s [\lambda \delta_{ij} \delta_{kl} + \mu(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})] H_{kl} + O(s^2), \quad (5)$$

where in the second step we used $2e_{ij} = H_{ij} + H_{ji} = (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) H_{kl}$. Because $\mathbf{H}$ is an arbitrary matrix, not necessarily symmetric, the coefficient of H_{kl} is uniquely determined, and comparison with Hooke's law $T_{ij} = s A_{ijkl} \partial u_k / \partial x_l$ identifies

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (6)$$

which is the isotropic elastic tensor of the lecture, with λ and μ the Lamé parameters.

(c) Since $W(\mathbf{I}) = 0$ and $\mathbf{T}(\mathbf{I}) = \mathbf{0}$, Taylor's theorem gives

$$W(\mathbf{I} + s\mathbf{H}) = \frac{1}{2}s^2 A_{ijkl} H_{ij} H_{kl} + O(s^3), \quad A_{ijkl} = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{I}). \quad (7)$$

On the other hand, from eq. (4), $(\text{tr } \mathbf{E})^2 = s^2(e_{kk})^2 + O(s^3)$ and $\text{tr}(\mathbf{E}^2) = s^2 e_{ij} e_{ij} + O(s^3)$, so that

$$W(\mathbf{I} + s\mathbf{H}) = s^2 \left[\frac{1}{2} \lambda (e_{kk})^2 + \mu e_{ij} e_{ij} \right] + O(s^3). \quad (8)$$

Using $e_{ij} e_{ij} = \frac{1}{2}(H_{ij} H_{ij} + H_{ij} H_{ji})$ one checks that the bracket equals $\frac{1}{2} A_{ijkl} H_{ij} H_{kl}$ with A_{ijkl} as in eq. (6), as required. Writing $e_{ij} = e'_{ij} + \frac{1}{3} e_{kk} \delta_{ij}$ with e'_{ij} traceless, the quadratic form becomes

$$\frac{1}{2} A_{ijkl} H_{ij} H_{kl} = \frac{1}{2} \kappa (e_{kk})^2 + \mu e'_{ij} e'_{ij}, \quad \kappa = \lambda + \frac{2}{3} \mu, \quad (9)$$

so the requirement that the stress-free configuration be a minimum of the elastic energy is $\kappa > 0$ and $\mu > 0$: the bulk modulus of the lecture controls the energy of volume changes, and the shear modulus that of shape changes.

Example 2: An elastic fluid and the Cauchy stress. An elastic fluid has strain energy function $W(\mathbf{F}) = V(J)$ with $J = \det \mathbf{F}$. (a) Using the cofactor expansion of the determinant, show that $\partial J / \partial F_{ij} = J (\mathbf{F}^{-1})_{ji}$, i.e. $\partial J / \partial \mathbf{F} = J \mathbf{F}^{-T}$. (b) Deduce that $\mathbf{T} = V'(J) J \mathbf{F}^{-T}$ and check that $\mathbf{F}^{-1} \mathbf{T}$ is symmetric. (c) Let dS be a surface element of the reference body with unit normal $\hat{\mathbf{n}}$, and dS_t , $\hat{\mathbf{n}}_t$ its image under the motion. The Cauchy stress $\boldsymbol{\sigma}$ is defined so that the force on the element is $\boldsymbol{\sigma} \hat{\mathbf{n}}_t dS_t = \mathbf{T} \hat{\mathbf{n}} dS$. Show that $\hat{\mathbf{n}}_t dS_t = J \mathbf{F}^{-T} \hat{\mathbf{n}} dS$, and hence that $\boldsymbol{\sigma} = J^{-1} \mathbf{T} \mathbf{F}^T$. Deduce that the Cauchy stress in the fluid is $-p \mathbf{I}$ with $p = -V'(J)$. (d) Linearise about a stress-free equilibrium, $V'(1) = 0$, and show that $A_{ijkl} = \kappa \delta_{ij} \delta_{kl}$ with $\kappa = V''(1)$.

Solution. (a) Expanding the determinant along the i th row,

$$J = \sum_{j=1}^3 F_{ij} \text{cof}_{ij}(\mathbf{F}) \quad (\text{no sum on } i), \quad (10)$$

where the cofactor $\text{cof}_{ij}(\mathbf{F})$ is $(-1)^{i+j}$ times the determinant of the 2×2 matrix obtained by deleting row i and column j . Since no element of row i appears in any $\text{cof}_{ij}(\mathbf{F})$, differentiating eq. (10) with respect to F_{ij} gives $\partial J / \partial F_{ij} = \text{cof}_{ij}(\mathbf{F})$. Cramer's rule states that $\mathbf{F}^{-1} = J^{-1} \text{cof}(\mathbf{F})^T$, so $\text{cof}_{ij}(\mathbf{F}) = J(\mathbf{F}^{-1})_{ji}$ and

$$\frac{\partial J}{\partial F_{ij}} = J(\mathbf{F}^{-1})_{ji} = J(\mathbf{F}^{-T})_{ij}. \quad (11)$$

At $\mathbf{F} = \mathbf{I}$ this reads $\partial J / \partial F_{ij} = \delta_{ij}$, i.e. $\det(\mathbf{I} + s\mathbf{H}) = 1 + s \text{tr } \mathbf{H} + O(s^2)$, which we use in part (d).

(b) By the chain rule and eq. (11),

$$T_{ij} = \frac{\partial W}{\partial F_{ij}} = V'(J) \frac{\partial J}{\partial F_{ij}} = V'(J) J(\mathbf{F}^{-T})_{ij}, \quad (12)$$

and $\mathbf{F}^{-1}\mathbf{T} = V'(J) J \mathbf{F}^{-1}\mathbf{F}^{-T} = V'(J) J \mathbf{C}^{-1}$ is symmetric; equivalently the second Piola–Kirchhoff stress is $\mathbf{S} = V'(J) J \mathbf{C}^{-1}$. The first Piola–Kirchhoff stress $\mathbf{T}$ is proportional to $\mathbf{F}^{-T}$ and is symmetric only when $\mathbf{F}$ is.

(c) A surface element at $\mathbf{x}$ spanned by the small vectors $d\mathbf{a}$ and $d\mathbf{b}$ has $\hat{\mathbf{n}} dS = d\mathbf{a} \times d\mathbf{b}$, and under the motion the spanning vectors become $\mathbf{F} d\mathbf{a}$ and $\mathbf{F} d\mathbf{b}$, so $\hat{\mathbf{n}}_t dS_t = (\mathbf{F} d\mathbf{a}) \times (\mathbf{F} d\mathbf{b})$. For any vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ the scalar triple product is a determinant, so

$$[(\mathbf{F}\mathbf{a}) \times (\mathbf{F}\mathbf{b})] \cdot (\mathbf{F}\mathbf{c}) = \det(\mathbf{F}[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]) = J(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}, \quad (13)$$

where $[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$ is the matrix with the given columns. The left-hand side equals $\mathbf{F}^T[(\mathbf{F}\mathbf{a}) \times (\mathbf{F}\mathbf{b})] \cdot \mathbf{c}$, and as $\mathbf{c}$ is arbitrary,

$$(\mathbf{F}\mathbf{a}) \times (\mathbf{F}\mathbf{b}) = J\mathbf{F}^{-T}(\mathbf{a} \times \mathbf{b}), \quad \text{hence} \quad \hat{\mathbf{n}}_t dS_t = J\mathbf{F}^{-T}\hat{\mathbf{n}} dS, \quad (14)$$

which is Nanson's formula; the matrix $J\mathbf{F}^{-T}$ is the cofactor matrix of part (a), which is no coincidence, as cofactors are signed areas. Substituting into the definition of the Cauchy stress, $\boldsymbol{\sigma} J\mathbf{F}^{-T}\hat{\mathbf{n}} = \mathbf{T}\hat{\mathbf{n}}$ for every $\hat{\mathbf{n}}$, and therefore

$$\boldsymbol{\sigma} = J^{-1}\mathbf{T}\mathbf{F}^T = J^{-1}\mathbf{F}(\mathbf{F}^{-1}\mathbf{T})\mathbf{F}^T = J^{-1}\mathbf{F}\mathbf{S}\mathbf{F}^T. \quad (15)$$

The second form shows that $\boldsymbol{\sigma}$ is symmetric precisely when $\mathbf{F}^{-1}\mathbf{T}$ is, so the symmetry condition of the lecture is the referential expression of the familiar symmetry of the Cauchy stress. For the fluid, eq. (12) gives

$$\boldsymbol{\sigma} = J^{-1} V'(J) J \mathbf{F}^{-T} \mathbf{F}^T = V'(J) \mathbf{I} = -p \mathbf{I}, \quad p = -V'(J). \quad (16)$$

The identification of p as a pressure is thermodynamically sensible: a reference volume dV has energy $V(J) dV$ and current volume $dv = J dV$, so $p = -\partial(V dV) / \partial(dv) =$

$-V'(J)$ is minus the derivative of the energy with respect to volume, and compressing the fluid ($J < 1$, $V' < 0$) produces a positive pressure. The first Piola–Kirchhoff stress $\mathbf{T} = -pJ\mathbf{F}^{-T}$ is the same isotropic pressure referred, through Nanson’s formula, to reference rather than current area, and this is why it is not itself isotropic.

(d) With $\mathbf{F} = \mathbf{I} + s\mathbf{H}$ we have $J = 1 + s \operatorname{tr} \mathbf{H} + O(s^2)$, so $V'(J) = V'(1) + V''(1)(J - 1) + O(s^2) = s \kappa \operatorname{tr} \mathbf{H} + O(s^2)$ with $\kappa = V''(1)$, while $J\mathbf{F}^{-T} = \mathbf{I} + O(s)$. Eq. (12) then gives

$$T_{ij} = s \kappa \delta_{ij} H_{kk} + O(s^2) = s \kappa \delta_{ij} \delta_{kl} H_{kl} + O(s^2), \quad \text{so} \quad A_{ijkl} = \kappa \delta_{ij} \delta_{kl}, \quad (17)$$

which is the isotropic tensor of Example 1 with $\mu = 0$, in which case $\lambda + \frac{2}{3}\mu$ reduces to λ , and the lecture’s κ is this remaining modulus. The name is appropriate: $p = -V'(J) = -\kappa(J - 1) + O((J - 1)^2)$, so $\kappa = -dp/dJ$ at $J = 1$ is the bulk modulus in the usual sense. Had the equilibrium not been stress-free, say $V'(1) = -p^0$, the expansion $J\mathbf{F}^{-T} = \mathbf{I} + s(\operatorname{tr} \mathbf{H} \mathbf{I} - \mathbf{H}^T) + O(s^2)$ would have contributed the additional first-order term $-p^0(\delta_{ij}\delta_{kl} - \delta_{il}\delta_{jk})H_{kl}$ to T_{ij} , and the resulting coefficient of H_{kl} does not possess the symmetry $A_{ijkl} = A_{jikl}$. This is a first glimpse of why the pre-stressed theory needed later in the course is more complicated.

Example 3: The material symmetry group. Fix a point $\mathbf{x}$ and let $\mathcal{G}$ be the set of rotation matrices $\mathbf{Q}$ such that $W(\mathbf{FQ}) = W(\mathbf{F})$ for every deformation gradient $\mathbf{F}$. (a) Show directly from this definition that $\mathcal{G}$ is a group under matrix multiplication. (b) Show that if $U(\mathbf{C}) = f(\operatorname{tr} \mathbf{C}, \operatorname{tr} \mathbf{C}^2, \det \mathbf{C})$ then every rotation lies in $\mathcal{G}$, and explain why every isotropic strain energy function is of this form. (c) Let $\hat{\nu}$ be a unit vector and $U(\mathbf{C}) = f(\hat{\nu} \cdot \mathbf{C}\hat{\nu}, \hat{\nu} \cdot \mathbf{C}^2\hat{\nu}, \operatorname{tr} \mathbf{C}, \operatorname{tr} \mathbf{C}^2, \det \mathbf{C})$. Show that $\mathbf{Q} \in \mathcal{G}$ whenever $\mathbf{Q}\hat{\nu} = \pm\hat{\nu}$, and conversely that if $\mathbf{Q} \in \mathcal{G}$ for every f of this form then $\mathbf{Q}\hat{\nu} = \pm\hat{\nu}$. (d) Show that if $\mathbf{Q} \in \mathcal{G}$ then the elastic tensor satisfies $A_{ijkl} = Q_{im}Q_{jn}Q_{kp}Q_{lq}A_{mnpq}$.

Solution. (a) The identity lies in $\mathcal{G}$ trivially, and associativity is inherited from matrix multiplication. If $\mathbf{Q}_1, \mathbf{Q}_2 \in \mathcal{G}$ then, applying the defining property twice,

$$W(\mathbf{FQ}_1\mathbf{Q}_2) = W((\mathbf{FQ}_1)\mathbf{Q}_2) = W(\mathbf{FQ}_1) = W(\mathbf{F}), \quad (18)$$

so $\mathbf{Q}_1\mathbf{Q}_2 \in \mathcal{G}$; here we used that $\mathbf{FQ}_1$ is itself an admissible deformation gradient, having positive determinant. If $\mathbf{Q} \in \mathcal{G}$ then $\mathbf{Q}^{-1} = \mathbf{Q}^T$ is a rotation and, applying the defining property to the deformation gradient $\mathbf{FQ}^{-1}$,

$$W(\mathbf{FQ}^{-1}) = W((\mathbf{FQ}^{-1})\mathbf{Q}) = W(\mathbf{F}), \quad (19)$$

so $\mathbf{Q}^{-1} \in \mathcal{G}$. Hence $\mathcal{G}$ is a subgroup of the rotation group.

(b) Under $\mathbf{F} \rightarrow \mathbf{FQ}$ the right Cauchy–Green tensor transforms as $\mathbf{C} \rightarrow \mathbf{Q}^T\mathbf{CQ}$, and by the cyclic property of the trace and the multiplicativity of the determinant

$$\operatorname{tr}(\mathbf{Q}^T\mathbf{CQ}) = \operatorname{tr} \mathbf{C}, \quad \operatorname{tr}[(\mathbf{Q}^T\mathbf{CQ})^2] = \operatorname{tr}(\mathbf{Q}^T\mathbf{C}^2\mathbf{Q}) = \operatorname{tr} \mathbf{C}^2, \quad \det(\mathbf{Q}^T\mathbf{CQ}) = \det \mathbf{C}. \quad (20)$$

Thus $U(\mathbf{Q}^T\mathbf{CQ}) = U(\mathbf{C})$ for every rotation, which is the isotropy condition of the lecture. Conversely, the lecture shows that an isotropic U depends on $\mathbf{C}$ only through its eigenvalues c_1, c_2, c_3 . These are the roots of the characteristic polynomial

$$\begin{aligned} \det(\mathbf{C} - c\mathbf{I}) &= -c^3 + I_1c^2 - I_2c + I_3, \\ I_1 &= \operatorname{tr} \mathbf{C}, \quad I_2 = \frac{1}{2}[(\operatorname{tr} \mathbf{C})^2 - \operatorname{tr} \mathbf{C}^2], \quad I_3 = \det \mathbf{C}, \end{aligned} \quad (21)$$

so the eigenvalues are functions of $(\text{tr } \mathbf{C}, \text{tr } \mathbf{C}^2, \det \mathbf{C})$ and every isotropic strain energy function can be written in the stated form. The St Venant–Kirchhoff energy of Example 1 is an instance, with $\text{tr } \mathbf{E} = \frac{1}{2}(I_1 - 3)$ and $\text{tr}(\mathbf{E}^2) = \frac{1}{4}(\text{tr } \mathbf{C}^2 - 2I_1 + 3)$.

(c) Since $(\mathbf{Q}^T \mathbf{C} \mathbf{Q})^2 = \mathbf{Q}^T \mathbf{C}^2 \mathbf{Q}$, the two new arguments transform as

$$\hat{\nu} \cdot (\mathbf{Q}^T \mathbf{C} \mathbf{Q}) \hat{\nu} = (\mathbf{Q} \hat{\nu}) \cdot \mathbf{C} (\mathbf{Q} \hat{\nu}), \quad \hat{\nu} \cdot (\mathbf{Q}^T \mathbf{C} \mathbf{Q})^2 \hat{\nu} = (\mathbf{Q} \hat{\nu}) \cdot \mathbf{C}^2 (\mathbf{Q} \hat{\nu}), \quad (22)$$

and both are quadratic in $\mathbf{Q} \hat{\nu}$, so they are unchanged when $\mathbf{Q} \hat{\nu} = \pm \hat{\nu}$; together with eq. (20) this shows that such $\mathbf{Q}$ lie in $\mathcal{G}$. For the converse it suffices to consider f depending only on its first argument, say $U(\mathbf{C}) = \hat{\nu} \cdot \mathbf{C} \hat{\nu}$, and to choose the particular deformations $\mathbf{C} = \mathbf{I} + \tau \hat{\nu} \hat{\nu}^T$, which are symmetric and positive-definite for $\tau > -1$. Then $\hat{\nu} \cdot \mathbf{C} \hat{\nu} = 1 + \tau$ while $(\mathbf{Q} \hat{\nu}) \cdot \mathbf{C} (\mathbf{Q} \hat{\nu}) = 1 + \tau(\hat{\nu} \cdot \mathbf{Q} \hat{\nu})^2$, and equality for all τ forces $(\hat{\nu} \cdot \mathbf{Q} \hat{\nu})^2 = 1$. As both vectors are unit vectors, the Cauchy–Schwarz inequality then gives $\mathbf{Q} \hat{\nu} = \pm \hat{\nu}$. The material symmetry group of a strain energy function of this form is therefore the set of rotations preserving the axis $\pm \hat{\nu}$: all rotations about $\hat{\nu}$, which is the defining property of transverse isotropy in the lecture, together with the rotations through π about axes perpendicular to $\hat{\nu}$, which reverse $\hat{\nu}$. That the latter are automatically symmetries reflects the fact that $\hat{\nu}$ enters only quadratically, and correspondingly the transversely isotropic elastic tensor of the lecture is unchanged by $\hat{\nu} \rightarrow -\hat{\nu}$.

(d) Material symmetry and material frame indifference together give, for $\mathbf{Q} \in \mathcal{G}$,

$$W(\mathbf{Q}^T \mathbf{F} \mathbf{Q}) = W(\mathbf{Q}^T \mathbf{F}) = W(\mathbf{F}), \quad (23)$$

where the first equality is material symmetry applied to the deformation gradient $\mathbf{Q}^T \mathbf{F}$ and the second is frame indifference with the rotation $\mathbf{Q}^T$. Writing $G_{mn} = Q_{am} F_{ab} Q_{bn}$ for the components of $\mathbf{Q}^T \mathbf{F} \mathbf{Q}$ we have $\partial G_{mn} / \partial F_{ij} = Q_{im} Q_{jn}$, and differentiating eq. (23) twice by the chain rule,

$$\frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{F}) = Q_{im} Q_{jn} Q_{kp} Q_{lq} \frac{\partial^2 W}{\partial G_{mn} \partial G_{pq}}(\mathbf{Q}^T \mathbf{F} \mathbf{Q}). \quad (24)$$

Setting $\mathbf{F} = \mathbf{I}$, for which $\mathbf{Q}^T \mathbf{F} \mathbf{Q} = \mathbf{I}$ as well, gives $A_{ijkl} = Q_{im} Q_{jn} Q_{kp} Q_{lq} A_{mnpq}$. The elastic tensor is thus invariant under the symmetry group, and this is the form in which material symmetry is used in linearised elasticity; the point of the argument is that the material symmetry condition alone relates the second derivatives at $\mathbf{I}$ to those at $\mathbf{Q}$, and frame indifference is needed to bring them back to $\mathbf{I}$. Numerically, the transversely isotropic tensor of the lecture with $\hat{\nu} = \hat{\mathbf{e}}_3$ is invariant under rotations about $\hat{\mathbf{e}}_3$ and under a rotation through π about $\hat{\mathbf{e}}_1$, but not under a generic rotation unless $\gamma = \xi = \zeta = 0$.

Example 4: Counting components and Voigt notation. Define the Voigt map from symmetric index pairs to a single index,

$$(11, 22, 33, 23, 13, 12) \longrightarrow (1, 2, 3, 4, 5, 6). \quad (25)$$

(a) Show that the symmetries $A_{ijkl} = A_{jikl} = A_{ijlk} = A_{klij}$ are exactly what is needed for the elastic tensor to be represented by a symmetric 6×6 matrix C_{IJ} , and hence that it has 21 independent components. (b) Write the isotropic elastic tensor as a 6×6 matrix. (c) For the transversely isotropic tensor of the lecture with $\hat{\nu} = \hat{\mathbf{e}}_3$, compute the 6×6 matrix and identify the five independent constants

$A_{1111}, A_{3333}, A_{1133}, A_{2323}$ and A_{1212} in terms of $\lambda, \mu, \gamma, \xi, \zeta$.

Solution. (a) The symmetry $A_{ijkl} = A_{jikl}$ says that A_{ijkl} depends on (i, j) only through the unordered pair $\{i, j\}$, and $A_{ijkl} = A_{ijlk}$ says the same of (k, l) . There are six unordered pairs, and eq. (25) labels them by $I, J \in \{1, \dots, 6\}$, so

$$C_{IJ} = A_{ijkl}, \quad I \leftrightarrow \{i, j\}, \quad J \leftrightarrow \{k, l\}, \quad (26)$$

is well defined and every one of the 81 components of A_{ijkl} can be read off from it. The hyperelastic symmetry $A_{ijkl} = A_{klij}$ is then precisely the statement $C_{IJ} = C_{JI}$. A symmetric 6×6 matrix has 6 diagonal and 15 off-diagonal entries, giving the 21 independent elastic constants quoted in the lecture. For comparison, the hyperelastic symmetry alone makes the 9×9 matrix indexed by ordered pairs (ij) and (kl) symmetric, which gives the 45 constants mentioned in the lecture before the remaining symmetries are imposed. When Hooke's law $T_{ij} = A_{ijkl}e_{kl}$ is written in this notation as $T_I = \sum_J C_{IJ}\tilde{e}_J$ one must take $\tilde{e}_J = e_{kl}$ for $J \leq 3$ but $\tilde{e}_J = 2e_{kl}$ for $J \geq 4$, because both (k, l) and (l, k) contribute to the sum over the ordered pair; these are the so-called engineering shear strains.

(b) For $A_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ the first term contributes λ to C_{IJ} whenever $I, J \leq 3$, and the second contributes 2μ on the diagonal for $I \leq 3$ and μ on the diagonal for $I \geq 4$, so

$$\mathbf{C}^{\text{iso}} = \begin{pmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{pmatrix}. \quad (27)$$

(c) With $\hat{v}_i = \delta_{i3}$ every factor of $\hat{v}$ in the lecture's expression forces the corresponding index to equal 3, and the remaining Kronecker deltas must pair equal indices. Consider first the components with all indices in $\{1, 2\}$: only the λ and μ terms survive, giving $A_{1111} = A_{2222} = \lambda + 2\mu$, $A_{1122} = \lambda$ and $A_{1212} = \mu$, exactly as in the isotropic case. For A_{3333} every term contributes: the γ term gives 8γ , the two ξ terms give $4\xi + 4\xi$, and each of the four ζ terms gives $-\zeta$, so

$$A_{3333} = \lambda + 2\mu + 8\gamma + 8\xi - 4\zeta. \quad (28)$$

For A_{1133} the μ, γ and ζ terms all require a delta between the index 1 and the index 3 and vanish, while the second ξ term $\delta_{ij}\hat{v}_k\hat{v}_l$ contributes 4ξ , so $A_{1133} = A_{2233} = \lambda + 4\xi$. For A_{2323} the μ term $\delta_{ik}\delta_{jl}$ gives μ , the ξ terms vanish since $\delta_{23} = 0$, and of the ζ terms only $\hat{v}_j\hat{v}_l\delta_{ik}$ survives, giving $A_{2323} = A_{1313} = \mu - \zeta$. Every remaining entry vanishes: in any non-vanishing term each index is either set equal to 3 by a factor of $\hat{v}$ or paired with an equal index by a Kronecker delta, so each of the values 1, 2 and 3 must occur an even number of times among (i, j, k, l) , and the components computed above, together with those related to them by the symmetries of part (a), are the only ones with this property. Collecting the results,

$$\mathbf{C}^{\text{TI}} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{13} & 0 & 0 & 0 \\ C_{13} & C_{13} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{66} \end{pmatrix}, \quad (29)$$

with

$$\begin{aligned} C_{11} &= A_{1111} = \lambda + 2\mu, & C_{33} &= A_{3333} = \lambda + 2\mu + 8\gamma + 8\zeta - 4\zeta, \\ C_{13} &= A_{1133} = \lambda + 4\zeta, & C_{44} &= A_{2323} = \mu - \zeta, \\ C_{66} &= A_{1212} = \mu, & C_{12} &= A_{1122} = \lambda = C_{11} - 2C_{66}. \end{aligned} \quad (30)$$

The matrix contains six distinct entries, but C_{12} is fixed by C_{11} and C_{66} , so there are five independent constants, and the map $(\lambda, \mu, \gamma, \zeta, \zeta) \mapsto (C_{11}, C_{33}, C_{13}, C_{44}, C_{66})$ is invertible: $\mu = C_{66}$, $\lambda = C_{11} - 2C_{66}$, $\zeta = \frac{1}{4}(C_{13} - \lambda)$, $\zeta = \mu - C_{44}$ and $\gamma = \frac{1}{8}(C_{33} - \lambda - 2\mu - 8\zeta + 4\zeta)$. Isotropy is recovered when $\gamma = \zeta = \zeta = 0$. For the sample values $\lambda = 2$, $\mu = 1$, $\gamma = 0.3$, $\zeta = 0.2$ and $\zeta = 0.1$ the script builds all 81 components, confirms the three symmetries of part (a), and gives $C_{11} = 4$, $C_{33} = 7.6$, $C_{13} = 2.8$, $C_{44} = 0.9$, $C_{66} = 1$ and $C_{12} = 2$, in agreement with eq. (30). The relation $C_{66} = \frac{1}{2}(C_{11} - C_{12})$ is what makes the material isotropic within the plane perpendicular to $\hat{\nu}$; the five constants are often written $A = C_{11}$, $C = C_{33}$, $F = C_{13}$, $L = C_{44}$ and $N = C_{66}$ in seismology, and this is the form in which the anisotropy of the uppermost mantle is parameterised in earth models such as PREM.

Example 5: From the elastic tensor to the Navier equation. Insert the isotropic elastic tensor into the linearised equation of motion $\rho \partial^2 u_i / \partial t^2 = \partial_j (A_{ijkl} \partial_l u_k)$. (a) For constant λ and μ obtain $\rho \ddot{\mathbf{u}} = (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u}$. (b) Show that in a heterogeneous isotropic body additional terms appear, and write them in terms of $\nabla \lambda$, $\nabla \mu$ and the linearised strain e_{ij} .

Solution. We first form the stress. Contracting the isotropic tensor with the displacement gradient,

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} = \lambda \delta_{ij} \frac{\partial u_k}{\partial x_k} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) = \lambda \delta_{ij} e_{kk} + 2\mu e_{ij}, \quad (31)$$

which is Hooke's law for an isotropic body, with only the symmetric part of the displacement gradient entering, as the lecture notes. Taking the divergence, and allowing λ and μ to depend on position,

$$\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial u_k}{\partial x_k} \right) + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right]. \quad (32)$$

(a) If λ and μ are constant they pass through the derivatives, and using the equality of mixed partial derivatives in the last term,

$$\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = \lambda \frac{\partial^2 u_k}{\partial x_i \partial x_k} + \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j} + \mu \frac{\partial^2 u_k}{\partial x_i \partial x_k} = (\lambda + \mu) \frac{\partial}{\partial x_i} (\nabla \cdot \mathbf{u}) + \mu \nabla^2 u_i, \quad (33)$$

so the linearised equation of motion becomes

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u}, \quad (34)$$

which is the form quoted in the lecture. Using the identity $\nabla^2 \mathbf{u} = \nabla(\nabla \cdot \mathbf{u}) - \nabla \times (\nabla \times \mathbf{u})$ it can also be written as $\rho \ddot{\mathbf{u}} = (\lambda + 2\mu) \nabla(\nabla \cdot \mathbf{u}) - \mu \nabla \times (\nabla \times \mathbf{u})$, which separates

the dilatational and rotational parts of the motion; the combinations $(\lambda + 2\mu)/\rho$ and μ/ρ will reappear as the squared P- and S-wave speeds.

(b) In general the product rule applied to eq. (32) produces, in addition to the terms of eq. (33) evaluated with the local values of λ and μ , the terms in which the derivative falls on a modulus,

$$\frac{\partial \lambda}{\partial x_i} \frac{\partial u_k}{\partial x_k} + \frac{\partial \mu}{\partial x_j} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) = (\nabla \cdot \mathbf{u}) \frac{\partial \lambda}{\partial x_i} + 2e_{ij} \frac{\partial \mu}{\partial x_j}, \quad (35)$$

so that the equation of motion in a heterogeneous isotropic body reads

$$\begin{aligned} \rho \frac{\partial^2 u_i}{\partial t^2} &= (\lambda + \mu) \frac{\partial}{\partial x_i} (\nabla \cdot \mathbf{u}) + \mu \nabla^2 u_i \\ &\quad + (\nabla \cdot \mathbf{u}) \frac{\partial \lambda}{\partial x_i} + 2e_{ij} \frac{\partial \mu}{\partial x_j}. \end{aligned} \quad (36)$$

In vector notation the extra terms are $(\nabla \cdot \mathbf{u}) \nabla \lambda + (\nabla \mathbf{u} + \nabla \mathbf{u}^T) \nabla \mu$. They are of first order in the gradients of the moduli and first order in the derivatives of $\mathbf{u}$, whereas the retained terms involve second derivatives of $\mathbf{u}$. For waves whose wavelength is short compared with the scale over which λ and μ vary they are therefore small, which is the starting point of ray theory later in the course; but they cannot be dropped near sharp gradients, and across a discontinuity the divergence form $\partial_j (A_{ijkl} \partial_l u_k)$ must be retained, since it is this form that yields the continuity of traction as a boundary condition. The script confirms eq. (36) for a quadratic displacement field and linearly varying moduli, for which central differences of the stress are exact.

Example 6: Uniform dilatation and the bulk modulus. Consider the uniform dilatation $\mathbf{F} = (1 + \epsilon) \mathbf{I}$ of the St Venant–Kirchhoff material of Example 1. (a) Evaluate $\mathbf{E}$, $\mathbf{S}$ and $\mathbf{T}$ exactly, and expand $\mathbf{T}$ to second order in ϵ ; show that the first-order term is $\mathbf{T} = 3\kappa \epsilon \mathbf{I}$ with $\kappa = \lambda + \frac{2}{3}\mu$. (b) Compute the Cauchy stress $\boldsymbol{\sigma} = J^{-1} \mathbf{T} \mathbf{F}^T$ and the pressure p to second order in ϵ , and show that $p = -\kappa(J - 1)$ to first order. Give the exact dependence of p on J .

Solution. (a) With $\mathbf{F} = (1 + \epsilon) \mathbf{I}$ we have $\mathbf{C} = (1 + \epsilon)^2 \mathbf{I}$ and $J = (1 + \epsilon)^3$, so

$$\mathbf{E} = \frac{1}{2} \left[(1 + \epsilon)^2 - 1 \right] \mathbf{I} = \left(\epsilon + \frac{1}{2} \epsilon^2 \right) \mathbf{I}, \quad \text{tr } \mathbf{E} = 3 \left(\epsilon + \frac{1}{2} \epsilon^2 \right). \quad (37)$$

From eq. (3) the second Piola–Kirchhoff stress is $\mathbf{S} = (3\lambda + 2\mu)(\epsilon + \frac{1}{2}\epsilon^2) \mathbf{I}$, and since $3\lambda + 2\mu = 3\kappa$,

$$\mathbf{T} = \mathbf{F} \mathbf{S} = 3\kappa (1 + \epsilon) \left(\epsilon + \frac{1}{2} \epsilon^2 \right) \mathbf{I} = 3\kappa \left(\epsilon + \frac{3}{2} \epsilon^2 + \frac{1}{2} \epsilon^3 \right) \mathbf{I}, \quad (38)$$

which is exact. To first order $\mathbf{T} = 3\kappa \epsilon \mathbf{I}$, in agreement with Hooke’s law for the displacement gradient $H_{kl} = \epsilon \delta_{kl}$, for which the isotropic tensor of eq. (6) gives $A_{ijkl} H_{kl} = (3\lambda + 2\mu) \epsilon \delta_{ij}$. The traction on any reference surface element is $3\kappa \epsilon \hat{\mathbf{n}}$ per unit reference area, so an outward tension is needed to hold the body in its expanded state, as expected.

(b) Since $\mathbf{F}^T = (1 + \epsilon)\mathbf{I}$ and $J = (1 + \epsilon)^3$, eq. (15) gives

$$\boldsymbol{\sigma} = J^{-1}\mathbf{T}\mathbf{F}^T = 3\kappa \frac{\epsilon + \frac{1}{2}\epsilon^2}{1 + \epsilon} \mathbf{I} = 3\kappa \left(\epsilon - \frac{1}{2}\epsilon^2 \right) \mathbf{I} + O(\epsilon^3), \quad (39)$$

and the pressure is

$$p = -\frac{1}{3}\text{tr } \boldsymbol{\sigma} = -3\kappa \left(\epsilon - \frac{1}{2}\epsilon^2 \right) + O(\epsilon^3). \quad (40)$$

At first order $p = -3\kappa\epsilon$, and as $J - 1 = 3\epsilon + O(\epsilon^2)$ this is $p = -\kappa(J - 1)$: the constant κ defined in the lecture through λ and μ is indeed the bulk modulus, $\kappa = -J \, dp/dJ$ evaluated at $J = 1$, exactly as for the elastic fluid of Example 2. The exact result follows on noting that $(\epsilon + \frac{1}{2}\epsilon^2)/(1 + \epsilon) = \frac{1}{2} [(1 + \epsilon) - (1 + \epsilon)^{-1}]$ and $1 + \epsilon = J^{1/3}$, whence

$$p = -\frac{3\kappa}{2} \left(J^{1/3} - J^{-1/3} \right). \quad (41)$$

For $\epsilon = 0.01$ and $\lambda = 2$, $\mu = 1.3$ (so that $\kappa = 2.8667$) the script gives $p = -0.085574$, compared with $-3\kappa\epsilon = -0.086$ at first order and $-3\kappa(\epsilon - \frac{1}{2}\epsilon^2) = -0.085570$ at second order.

Two remarks. First, the first Piola–Kirchhoff and Cauchy stresses agree at first order in ϵ , as the lecture asserts for linearised motions about a stress-free equilibrium, but their second-order terms have coefficients $+\frac{3}{2}$ and $-\frac{1}{2}$ respectively; the difference is entirely the geometric factor $J^{-1}\mathbf{F}^T$ converting force per unit reference area to force per unit current area, and it is invisible in the linearised theory precisely because the equilibrium stress vanishes. Had the reference configuration carried a stress $\mathbf{T}^0 \neq \mathbf{0}$, this factor would contribute at first order, as in the remark at the end of Example 2. Second, the strain energy for this deformation is $W = \frac{9}{8}\kappa (J^{2/3} - 1)^2$, which tends to the finite value $\frac{9}{8}\kappa$ as $J \rightarrow 0$, so that crushing the body to zero volume costs a finite energy. This is a well-known deficiency of the St Venant–Kirchhoff model, which is intended only for small strains, and it is one reason why the general framework of the lecture is formulated for an arbitrary function $U(\mathbf{C})$ rather than for this particular choice.

Worked examples for Lecture 14: Seismic sources

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These examples accompany Lecture 14. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Example 1: The stress glut of a fault in an isotropic body, and the moment tensor. An idealised fault lies in the (x_1, x_2) -plane of an isotropic body with Lamé parameters λ and μ , and the displacement discontinuity vector $\mathbf{w}$ is tangential to the fault, so that $w_3 = 0$.

- (a) Starting from the stress glut $\bar{S}_{ij} = A_{ijk}w_k\delta(x_3)$ obtained in the lecture, show that

$$\bar{S}_{ij} = \mu (w_i\delta_{j3} + w_j\delta_{i3}) \delta(x_3),$$

and comment on the absence of λ .

- (b) The **moment tensor** of a source is defined as the volume integral of the stress glut,

$$M_{ij}(t) = \int_M \bar{S}_{ij}(\mathbf{x}, t) d^3\mathbf{x}.$$

Suppose that for $t > 0$ the slip $\mathbf{w} = D \hat{\mathbf{e}}_1$ is uniform over a patch Σ of the fault having area $\mathcal{A}$, that μ is constant on Σ , and that Σ is small compared with the wavelengths of interest. Explain why the stress glut can then be replaced by $M_{ij}\delta(\mathbf{x})$, determine M_{ij} , and find its eigenvalues and eigenvectors. Express $\mathbf{M}$ in terms of its eigenvectors.

Solution. (a) For an isotropic material the elastic tensor takes the form

$$A_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \quad (1)$$

and so, setting $l = 3$ and contracting with w_k ,

$$A_{ijk}w_k = \lambda\delta_{ij}\delta_{k3}w_k + \mu(\delta_{ik}\delta_{j3} + \delta_{i3}\delta_{jk})w_k = \lambda w_3\delta_{ij} + \mu(w_i\delta_{j3} + w_j\delta_{i3}). \quad (2)$$

The first term is proportional to w_3 , which vanishes because the displacement discontinuity is tangential to the fault, and hence

$$\bar{S}_{ij} = \mu(w_i\delta_{j3} + w_j\delta_{i3})\delta(x_3). \quad (3)$$

Written out, the only non-zero components are

$$\bar{S}_{13} = \bar{S}_{31} = \mu w_1\delta(x_3), \quad \bar{S}_{23} = \bar{S}_{32} = \mu w_2\delta(x_3), \quad (4)$$

so that the stress glut is symmetric, as it must be within the linearised theory, and also traceless, $\bar{S}_{ii} = 2\mu w_3 \delta(x_3) = 0$. The same calculation for a general fault surface with unit normal $\hat{\mathbf{n}}$ gives $\bar{S}_{ij} = \mu(w_i \hat{n}_j + w_j \hat{n}_i) \delta_\Sigma$ whenever $\mathbf{w} \cdot \hat{\mathbf{n}} = 0$. The Lamé parameter λ enters only through the term $\lambda w_3 \delta_{ij}$, which is to say only if the fault opens or the two sides interpenetrate. For a shear dislocation the source is therefore completely insensitive to λ ; the radiated waves depend on λ only through their propagation, not through their generation.

(b) The stress glut enters the linearised equations of motion through $-\partial \bar{S}_{ij} / \partial x_j$, and it acts on the displacement field only through integrals against smooth functions. For any smooth ϕ , using the defining property of $\delta(x_3)$ to reduce the volume integral to one over the fault,

$$\int_M \bar{S}_{ij} \phi \, d^3 \mathbf{x} = \int_\Sigma \mu (w_i \delta_{j3} + w_j \delta_{i3}) \phi \, dS \approx \phi(\mathbf{0}) \int_\Sigma \mu (w_i \delta_{j3} + w_j \delta_{i3}) \, dS = \phi(\mathbf{0}) M_{ij}, \quad (5)$$

provided that ϕ varies little over Σ , which is the case for wavefields whose wavelengths are large compared with the patch. Since $\int_M M_{ij} \delta(\mathbf{x}) \phi \, d^3 \mathbf{x} = M_{ij} \phi(\mathbf{0})$, the replacement $\bar{S}_{ij} \rightarrow M_{ij} \delta(\mathbf{x})$ preserves precisely this zeroth moment of the glut, and this is the **point-source approximation**. Evaluating the integral for uniform slip $\mathbf{w} = D \hat{\mathbf{e}}_1$ and constant μ gives, for $t > 0$,

$$M_{ij} = \mu \mathcal{A} D (\delta_{i1} \delta_{j3} + \delta_{i3} \delta_{j1}), \quad \text{i.e.} \quad \mathbf{M} = M_0 \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad M_0 = \mu \mathcal{A} D, \quad (6)$$

where M_0 is the **scalar seismic moment**, with units of N m. A moment tensor of this form is called a **double couple**, for reasons made clear in the next example.

To find the eigenvalues we expand $\det(\mathbf{M} - \sigma \mathbf{I})$ along its second row, which contains the single non-zero entry $-\sigma$, to obtain

$$\det(\mathbf{M} - \sigma \mathbf{I}) = -\sigma (\sigma^2 - M_0^2), \quad (7)$$

so the eigenvalues are $M_0, 0$ and $-M_0$. The corresponding unit eigenvectors are found by inspection, since $\mathbf{M}$ simply exchanges the first and third components of a vector (scaled by M_0) and annihilates the second,

$$\hat{\mathbf{t}} = \frac{1}{\sqrt{2}} (\hat{\mathbf{e}}_1 + \hat{\mathbf{e}}_3), \quad \hat{\mathbf{b}} = \hat{\mathbf{e}}_2, \quad \hat{\mathbf{p}} = \frac{1}{\sqrt{2}} (\hat{\mathbf{e}}_1 - \hat{\mathbf{e}}_3), \quad (8)$$

with $\mathbf{M} \hat{\mathbf{t}} = M_0 \hat{\mathbf{t}}$, $\mathbf{M} \hat{\mathbf{b}} = \mathbf{0}$ and $\mathbf{M} \hat{\mathbf{p}} = -M_0 \hat{\mathbf{p}}$. The vector $\hat{\mathbf{t}}$ lies at 45° to the fault plane in the direction of slip and is known as the **tension** or T-axis; $\hat{\mathbf{p}}$, also at 45° to the fault, is the **pressure** or P-axis; and $\hat{\mathbf{b}}$, which lies in the fault plane perpendicular to the slip, is the **null** or B-axis. Because $\mathbf{M}$ is symmetric, it has the spectral decomposition

$$\mathbf{M} = M_0 (\hat{\mathbf{t}} \hat{\mathbf{t}}^T - \hat{\mathbf{p}} \hat{\mathbf{p}}^T), \quad (9)$$

which can be checked directly since $\hat{\mathbf{t}} \hat{\mathbf{t}}^T - \hat{\mathbf{p}} \hat{\mathbf{p}}^T = \hat{\mathbf{e}}_1 \hat{\mathbf{e}}_3^T + \hat{\mathbf{e}}_3 \hat{\mathbf{e}}_1^T$. Eq. (9) says that a double couple is equivalent to a pair of equal and opposite linear dipoles, a tensile one along $\hat{\mathbf{t}}$ and a compressive one along $\hat{\mathbf{p}}$, at 45° to the fault.

Two features of the result are worth noting. First, the trace of $\mathbf{M}$, which is the sum of its eigenvalues, vanishes; a double couple involves no change in volume. Second, $\mathbf{M}$ in eq. (6) is unchanged if $\hat{\mathbf{e}}_1$ and $\hat{\mathbf{e}}_3$ are interchanged, i.e. if the slip direction and the fault normal swap roles. Slip $D\hat{\mathbf{e}}_3$ on a fault with normal $\hat{\mathbf{e}}_1$ produces an identical moment tensor. This second plane is known as the **auxiliary plane**, and the moment tensor alone cannot distinguish it from the true fault plane; both planes contain the null axis and bisect the P- and T-axes.

Example 2: The equivalent body force for a point source. A point source in the interior of a body has stress glut $\bar{S}_{ij}(\mathbf{x}, t) = M_{ij}(t)\delta(\mathbf{x})$, where the moment tensor M_{ij} is symmetric.

- (a) Show that the linearised equations of motion and boundary conditions are those of the same body subject to a body force $f_i = -M_{ij}\partial\delta(\mathbf{x})/\partial x_j$ and no applied surface traction.
- (b) Show that this force exerts no net force and no net torque on the body,

$$\int_M f_i d^3\mathbf{x} = 0, \quad \int_M \epsilon_{ijk} x_j f_k d^3\mathbf{x} = 0,$$

and identify which property of M_{ij} is responsible for each.

- (c) For the double couple of Example 1, interpret $\mathbf{f}$ as the limit of two force couples, and verify that their torques cancel.

Solution. (a) The linearised equations of motion obtained in the lecture are

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j}, \quad (10)$$

together with the boundary condition $A_{ijkl}(\partial u_k/\partial x_l)\hat{n}_j = \bar{S}_{ij}\hat{n}_j$ on ∂M . Substituting $\bar{S}_{ij} = M_{ij}(t)\delta(\mathbf{x})$, and noting that M_{ij} does not depend on position, the right hand side of eq. (10) becomes

$$f_i(\mathbf{x}, t) = -M_{ij}(t) \frac{\partial \delta}{\partial x_j}(\mathbf{x}), \quad (11)$$

while $\delta(\mathbf{x})$ vanishes identically on ∂M because the source lies in the interior, so the boundary condition reduces to the traction-free condition $A_{ijkl}(\partial u_k/\partial x_l)\hat{n}_j = 0$. The problem is therefore identical to that for the same elastic body driven by the body force $\mathbf{f}$, which is why the lecture refers to the right hand side of eq. (10) as an effective body force. The derivative of the delta function is understood in the sense of distributions: for any smooth ϕ ,

$$\int_M \phi \frac{\partial \delta}{\partial x_j} d^3\mathbf{x} = - \int_M \frac{\partial \phi}{\partial x_j} \delta d^3\mathbf{x} = - \frac{\partial \phi}{\partial x_j}(\mathbf{0}), \quad (12)$$

where the surface term arising from integration by parts vanishes because δ is supported only at the origin.

(b) Applying eq. (12) with $\phi = 1$ gives the net force

$$\int_M f_i d^3\mathbf{x} = -M_{ij} \int_M \frac{\partial \delta}{\partial x_j} d^3\mathbf{x} = M_{ij} \frac{\partial(1)}{\partial x_j} = 0, \quad (13)$$

and this holds for any M_{ij} , symmetric or not; it is a consequence of the source entering only as a divergence. For the torque about the origin we apply eq. (12) with $\phi = x_j$,

$$\int_M \epsilon_{ijk} x_j f_k d^3\mathbf{x} = -\epsilon_{ijk} M_{kl} \int_M x_j \frac{\partial \delta}{\partial x_l} d^3\mathbf{x} = \epsilon_{ijk} M_{kl} \left. \frac{\partial x_j}{\partial x_l} \right|_{\mathbf{x}=0} = \epsilon_{ijk} M_{kl} \delta_{jl} = \epsilon_{ijk} M_{kj}. \quad (14)$$

This is a contraction of the symmetric tensor M_{kj} with the alternating symbol, which is antisymmetric in j and k , and so it vanishes. Since the net force is zero, the torque about any other point is the same, and hence also zero. Conversely, if M_{ij} had an antisymmetric part, eq. (14) shows that the source would exert a net torque equal to minus twice the axial vector of that part. The symmetry of the stress glut, which in the lecture was a consequence of the invariance of the strain energy under superimposed rigid rotations, is thus exactly the condition that the source does not alter the angular momentum of the body, while the divergence form guarantees that it does not alter the linear momentum. An earthquake is an internal process, and cannot do either.

(c) For the double couple $M_{13} = M_{31} = M_0$ of Example 1, eq. (11) gives

$$f_1 = -M_0 \frac{\partial \delta}{\partial x_3}, \quad f_2 = 0, \quad f_3 = -M_0 \frac{\partial \delta}{\partial x_1}. \quad (15)$$

A derivative of the delta function is the limit of a difference of two shifted delta functions,

$$\frac{\partial \delta}{\partial x_3}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{1}{h} \left[\delta\left(\mathbf{x} + \frac{h}{2} \hat{\mathbf{e}}_3\right) - \delta\left(\mathbf{x} - \frac{h}{2} \hat{\mathbf{e}}_3\right) \right], \quad (16)$$

as follows by integrating both sides against a smooth ϕ and Taylor expanding. Hence

$$f_1 = \lim_{h \rightarrow 0} \frac{M_0}{h} \left[\delta\left(\mathbf{x} - \frac{h}{2} \hat{\mathbf{e}}_3\right) - \delta\left(\mathbf{x} + \frac{h}{2} \hat{\mathbf{e}}_3\right) \right], \quad (17)$$

which is a point force of magnitude $F = M_0/h$ directed along $+\hat{\mathbf{e}}_1$ at $\mathbf{x} = +\frac{h}{2}\hat{\mathbf{e}}_3$, together with an equal and opposite force at $\mathbf{x} = -\frac{h}{2}\hat{\mathbf{e}}_3$. This is a **force couple** with arm h and moment $Fh = M_0$, which remains finite as $h \rightarrow 0$; the forces act along the slip direction and are separated along the fault normal. The side $x_3 > 0$ is pushed in the $+\hat{\mathbf{e}}_1$ direction and the side $x_3 < 0$ in the $-\hat{\mathbf{e}}_1$ direction, consistent with the displacement jumping by $+D \hat{\mathbf{e}}_1$ on crossing the fault from $x_3 < 0$ to $x_3 > 0$. Its torque is

$$\frac{h}{2} \hat{\mathbf{e}}_3 \times F \hat{\mathbf{e}}_1 + \left(-\frac{h}{2} \hat{\mathbf{e}}_3\right) \times (-F \hat{\mathbf{e}}_1) = Fh \hat{\mathbf{e}}_3 \times \hat{\mathbf{e}}_1 = M_0 \hat{\mathbf{e}}_2. \quad (18)$$

In exactly the same way f_3 is a couple of forces $\pm F \hat{\mathbf{e}}_3$ at $\pm \frac{h}{2} \hat{\mathbf{e}}_1$, with forces along the fault normal separated along the slip direction, and torque

$$Fh \hat{\mathbf{e}}_1 \times \hat{\mathbf{e}}_3 = -M_0 \hat{\mathbf{e}}_2. \quad (19)$$

The two torques cancel, as they must by part (b). The first couple is the one we might have written down intuitively from the picture of slip on the fault; the second, which corresponds to slip on the auxiliary plane, is forced upon us by the requirement that the source exerts no net torque, and it is the pair together that constitutes the **double couple**. A single couple would set the body spinning.

Example 3: An explosive source generates only P-waves. An explosion at the origin of a homogeneous isotropic whole space, with Lamé parameters λ and μ and density ρ , is modelled by the isotropic stress glut

$$\bar{S}_{ij}(\mathbf{x}, t) = M_0 \delta_{ij} \delta(\mathbf{x}) H(t).$$

Find the equivalent body force, write down the resulting equation of motion for $\mathbf{u}$, and by taking its curl show that the displacement field is irrotational for all time. Deduce that only P-waves are generated.

Solution. The equivalent body force is

$$f_i = -\frac{\partial}{\partial x_j} [M_0 \delta_{ij} \delta(\mathbf{x}) H(t)] = -M_0 H(t) \frac{\partial \delta}{\partial x_i}(\mathbf{x}), \quad \text{i.e.} \quad \mathbf{f} = -M_0 H(t) \nabla \delta(\mathbf{x}), \quad (20)$$

which is the gradient of a scalar. For a homogeneous isotropic body we substitute eq. (1) into the elastic operator and use the constancy of λ and μ ,

$$\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = \lambda \frac{\partial^2 u_k}{\partial x_i \partial x_k} + \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j} + \mu \frac{\partial^2 u_j}{\partial x_j \partial x_i} = (\lambda + \mu) \frac{\partial}{\partial x_i} (\nabla \cdot \mathbf{u}) + \mu \nabla^2 u_i, \quad (21)$$

so that the equation of motion is the one given in Lecture 13 for a homogeneous isotropic body, now with a body force,

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} - (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) - \mu \nabla^2 \mathbf{u} = -M_0 H(t) \nabla \delta(\mathbf{x}), \quad (22)$$

to be solved with $\mathbf{u} = \partial \mathbf{u} / \partial t = \mathbf{0}$ at $t = 0$. Now take the curl of eq. (22) and write $\boldsymbol{\omega} = \nabla \times \mathbf{u}$. The curl of a gradient vanishes, which disposes of both the $(\lambda + \mu)$ term and the body force, while the curl commutes with ∇^2 and with $\partial^2 / \partial t^2$ because the coefficients are constant. We are left with

$$\rho \frac{\partial^2 \boldsymbol{\omega}}{\partial t^2} - \mu \nabla^2 \boldsymbol{\omega} = \mathbf{0}, \quad \boldsymbol{\omega}(\mathbf{x}, 0) = \mathbf{0}, \quad \frac{\partial \boldsymbol{\omega}}{\partial t}(\mathbf{x}, 0) = \mathbf{0}, \quad (23)$$

the initial conditions following by taking the curl of those for $\mathbf{u}$, which hold at every point of the body. Each Cartesian component of $\boldsymbol{\omega}$ thus satisfies the homogeneous scalar wave equation with speed $\beta = \sqrt{\mu / \rho}$ and vanishing initial data, and the only such solution is zero. To see this, consider the energy

$$\mathcal{E}(t) = \frac{1}{2} \int \left(\rho \left\| \frac{\partial \boldsymbol{\omega}}{\partial t} \right\|^2 + \mu \frac{\partial \omega_i}{\partial x_j} \frac{\partial \omega_i}{\partial x_j} \right) d^3 \mathbf{x}, \quad (24)$$

whose time derivative, after an integration by parts, is $\int \dot{\omega}_i (\rho \ddot{\omega}_i - \mu \nabla^2 \omega_i) d^3 \mathbf{x} = 0$; the surface term at infinity vanishes since, at any finite time, the disturbance has only travelled a finite distance from the source. Thus $\mathcal{E}(t) = \mathcal{E}(0) = 0$, and as $\mathcal{E}$ is a sum of non-negative terms both $\partial \boldsymbol{\omega} / \partial t$ and the gradient of $\boldsymbol{\omega}$ vanish; $\boldsymbol{\omega}$ is therefore constant in space and time, and equal to its initial value $\mathbf{0}$. The displacement is **irrotational** at all times.

In a whole space an irrotational field can be written as a gradient, $\mathbf{u} = \nabla\phi$. Substituting into eq. (22), and using $\nabla^2\nabla\phi = \nabla\nabla^2\phi$ and $\nabla \cdot \nabla\phi = \nabla^2\phi$, every term becomes a gradient,

$$\nabla \left[\rho \frac{\partial^2 \phi}{\partial t^2} - (\lambda + 2\mu) \nabla^2 \phi + M_0 H(t) \delta(\mathbf{x}) \right] = \mathbf{0}. \quad (25)$$

The bracket is therefore a function of t alone, which can be absorbed into ϕ without changing $\mathbf{u}$, and so

$$\rho \frac{\partial^2 \phi}{\partial t^2} - (\lambda + 2\mu) \nabla^2 \phi = -M_0 H(t) \delta(\mathbf{x}). \quad (26)$$

This is the scalar wave equation with wave speed $\alpha = \sqrt{(\lambda + 2\mu)/\rho}$, the P-wave speed of Lecture 15: the explosion radiates a spherically symmetric P-wave and nothing else. The isotropic moment tensor $M_0 \delta_{ij}$ is invariant under all rotations, so there is no fault plane and no preferred direction in the radiation. This should be contrasted with the double couple of Example 1, whose moment tensor is traceless. In practice explosions do produce S-waves, because the Earth is neither homogeneous nor unbounded and P-waves are partly converted to S-waves at the free surface and at internal discontinuities, but these are secondary. The size of the isotropic part of the moment tensor, essentially zero for an earthquake, is one of the discriminants used to distinguish nuclear tests from earthquakes.

Example 4: Finite rupture and the moment rate function. A rectangular fault occupies the region $0 \leq x_1 \leq L$, $0 \leq x_2 \leq W_f$ of the plane $x_3 = 0$ in a body with constant shear modulus μ . Rupture begins at $x_1 = 0$ at time $t = 0$ and propagates in the x_1 -direction at constant speed v_r , with the slip at each point taking place instantaneously as the rupture front passes,

$$w_1(x_1, x_2, t) = D H\left(t - \frac{x_1}{v_r}\right), \quad w_2 = w_3 = 0.$$

Find the time-dependent scalar moment $M_0(t)$ and the **moment rate** $\dot{M}_0(t)$, and show that the latter is a boxcar of duration L/v_r . Estimate the rupture duration and the total moment for $L = 100$ km, $v_r = 3$ km s⁻¹, $W_f = 20$ km, $D = 2$ m and $\mu = 3 \times 10^{10}$ Pa.

Solution. From eq. (4) the stress glut has the single independent component $\bar{S}_{13} = \bar{S}_{31} = \mu w_1 \delta(x_3)$, and so by the definition of the moment tensor in Example 1 the only non-zero components are

$$M_{13}(t) = M_{31}(t) = \mu \int_{\Sigma} w_1 dS \equiv M_0(t), \quad (27)$$

which defines the time-dependent scalar moment. The moment tensor has the double-couple form of Example 1 at every instant, with the whole time dependence carried by $M_0(t)$. Substituting the assumed slip, and performing the trivial integral over x_2 ,

$$M_0(t) = \mu W_f D \int_0^L H\left(t - \frac{x_1}{v_r}\right) dx_1. \quad (28)$$

The step function equals one where $x_1 < v_r t$ and zero elsewhere, so the integral is the length of the ruptured portion of the fault at time t , namely

$$M_0(t) = \begin{cases} 0, & t < 0, \\ \mu W_f D v_r t, & 0 \leq t \leq T_r, \\ \mu W_f L D, & t > T_r, \end{cases} \quad T_r = \frac{L}{v_r}. \quad (29)$$

The moment grows linearly while the rupture front crosses the fault and then stays at its final value $\mu \mathcal{A} D$, with $\mathcal{A} = L W_f$, in agreement with Example 1. Differentiating under the integral sign, and using $H' = \delta$,

$$\dot{M}_0(t) = \mu W_f \int_0^L \frac{\partial w_1}{\partial t} dx_1 = \mu W_f D \int_0^L \delta\left(t - \frac{x_1}{v_r}\right) dx_1 = \mu W_f D v_r \int_0^{T_r} \delta(t - \tau) d\tau, \quad (30)$$

where we substituted $\tau = x_1/v_r$. The final integral is one if $0 < t < T_r$ and zero otherwise, and so

$$\dot{M}_0(t) = \mu W_f D v_r [H(t) - H(t - T_r)], \quad (31)$$

a boxcar of height $\mu W_f D v_r$ and duration T_r , in agreement with the derivative of eq. (29). Its area is $\mu W_f D v_r T_r = \mu \mathcal{A} D$, the total moment, as it must be.

For the given values the rupture duration is

$$T_r = \frac{100 \text{ km}}{3 \text{ km s}^{-1}} \approx 33 \text{ s}, \quad (32)$$

and the total moment is

$$M_0 = \mu \mathcal{A} D = (3 \times 10^{10} \text{ Pa}) (10^5 \text{ m}) (2 \times 10^4 \text{ m}) (2 \text{ m}) = 1.2 \times 10^{20} \text{ N m}, \quad (33)$$

with a moment rate of $3.6 \times 10^{18} \text{ N m s}^{-1}$ during the rupture. For orientation, the moment magnitude is defined by $M_w = \frac{2}{3} [\log_{10}(M_0/\text{N m}) - 9.1]$, which here gives $M_w \approx 7.3$, a large earthquake and a plausible size for a rupture of these dimensions. The moment rate function, often called the source time function, is what a distant seismometer effectively records: the far-field displacement radiated by a point source is proportional to $\dot{M}_0$, so the pulse from this event would be roughly 33 s long. Two refinements are easily incorporated. If the slip at each point rises over a finite rise time τ_r rather than instantaneously, $\dot{M}_0$ becomes the convolution of the boxcar with a second boxcar of width τ_r , i.e. a trapezoid of total duration $T_r + \tau_r$. And for an observer in the direction of rupture propagation the pulse is compressed, as successive parts of the fault radiate from progressively closer points, an effect known as directivity. Note finally that the point-source approximation of Example 1 requires wavelengths large compared with L , which for this event means periods well in excess of T_r .

Example 5: Well- and ill-posedness of polynomial interpolation. Given distinct points $x_1, \dots, x_n$ and values $y_1, \dots, y_n$, we seek a polynomial $p(x) = \sum_{k=0}^m c_k x^k$ with $p(x_i) = y_i$. Write the problem as a linear system $\mathbf{V}\mathbf{c} = \mathbf{y}$ for the coefficient vector $\mathbf{c}$, and then:

- (a) for $n = 3$ and $m = 2$, show explicitly that a solution exists, is unique, and depends continuously on $\mathbf{y}$;
- (b) for $n = 3$ and $m = 1$, derive the condition that the data must satisfy for a solution to exist;
- (c) for $n = 2$ and $m = 2$, exhibit the one-parameter family of solutions;
- (d) for equally spaced points on $[0, 1]$ with $m = n - 1$, compute the condition number of $\mathbf{V}$ for $n = 5, 10, 15$ and 20 , and comment on what continuous dependence is worth in practice.

Solution. The interpolation conditions are linear in the unknown coefficients,

$$\sum_{k=0}^m x_i^k c_k = y_i, \quad i = 1, \dots, n, \quad \text{i.e.} \quad \mathbf{V}\mathbf{c} = \mathbf{y}, \quad V_{ik} = x_i^k, \quad (34)$$

where $\mathbf{V}$ is the $n \times (m + 1)$ **Vandermonde matrix**. In the language of the lecture the inputs are $\mathbf{y}$ and the outputs are $\mathbf{c}$. A solution exists if and only if $\mathbf{y}$ lies in the column space of $\mathbf{V}$, it is unique if and only if the null space of $\mathbf{V}$ is trivial, and, whenever the solution is unique, it depends continuously on $\mathbf{y}$ because it is then given by a linear map between finite-dimensional spaces, and every such map is continuous.

(a) With $n = 3$ and $m = 2$ the matrix is square, and subtracting the first row from the other two,

$$\det \mathbf{V} = \det \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} = \det \begin{pmatrix} x_2 - x_1 & x_2^2 - x_1^2 \\ x_3 - x_1 & x_3^2 - x_1^2 \end{pmatrix} = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2), \quad (35)$$

where the last step factors $x_j^2 - x_1^2 = (x_j - x_1)(x_j + x_1)$ from each row. This is non-zero for distinct points, so $\mathbf{V}$ is invertible and the problem has the unique solution $\mathbf{c} = \mathbf{V}^{-1}\mathbf{y}$ for every $\mathbf{y}$. The solution can also be written down directly in Lagrange form,

$$p(x) = \sum_{i=1}^3 y_i \ell_i(x), \quad \ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}, \quad (36)$$

since $\ell_i(x_j) = \delta_{ij}$; this establishes existence, and uniqueness follows because the difference of two solutions is a quadratic with three distinct roots, hence identically zero. Continuous dependence is quantified by

$$\|\delta \mathbf{c}\| \leq \|\mathbf{V}^{-1}\| \|\delta \mathbf{y}\|, \quad \frac{\|\delta \mathbf{c}\|}{\|\mathbf{c}\|} \leq \text{cond}(\mathbf{V}) \frac{\|\delta \mathbf{y}\|}{\|\mathbf{y}\|}, \quad \text{cond}(\mathbf{V}) = \|\mathbf{V}\| \|\mathbf{V}^{-1}\|, \quad (37)$$

where $\|\cdot\|$ is the matrix norm induced by the Euclidean vector norm and $\text{cond}(\mathbf{V})$ is the **condition number**, which for this norm is the ratio of the largest to the smallest

singular value of $\mathbf{V}$. The second inequality follows from the first together with $\|\mathbf{y}\| \leq \|\mathbf{V}\| \|\mathbf{c}\|$. The problem is well-posed.

(b) With $n = 3$ and $m = 1$ we have three equations $c_0 + c_1 x_i = y_i$ for two unknowns. The two columns of $\mathbf{V}$, namely $(1, 1, 1)^T$ and $(x_1, x_2, x_3)^T$, are linearly independent, so the column space is a plane in $\mathbb{R}^3$ and $\mathbf{y}$ must lie in it. The plane is the orthogonal complement of any non-zero vector $\mathbf{a}$ with $\mathbf{a}^T \mathbf{V} = \mathbf{0}$, and

$$\mathbf{a} = (x_2 - x_3, x_3 - x_1, x_1 - x_2)^T \quad (38)$$

does the job, since its components sum to zero and $(x_2 - x_3)x_1 + (x_3 - x_1)x_2 + (x_1 - x_2)x_3 = 0$. Existence therefore requires

$$(x_2 - x_3)y_1 + (x_3 - x_1)y_2 + (x_1 - x_2)y_3 = 0, \quad (39)$$

and this condition is also sufficient. Rearranging, it reads $(y_3 - y_1)(x_2 - x_1) = (y_2 - y_1)(x_3 - x_1)$, which states that the slope from the first point to the third equals the slope from the first to the second: the three points (x_i, y_i) must be collinear. Equivalently, the determinant of the matrix with rows $(1, x_i, y_i)$ vanishes. For equally spaced points $x_i = x_1 + (i - 1)h$ the condition is $y_1 - 2y_2 + y_3 = 0$, the vanishing of the second difference. When it holds, the solution is $c_1 = (y_2 - y_1)/(x_2 - x_1)$ and $c_0 = y_1 - c_1 x_1$. Data chosen at random satisfy eq. (39) with probability zero, so for noisy measurements the problem as posed has no solution, and one is led, as noted in the lecture, to relax what is meant by a solution, for instance to the $\mathbf{c}$ that minimises $\|\mathbf{V}\mathbf{c} - \mathbf{y}\|$.

(c) With $n = 2$ and $m = 2$ there are two equations for three unknowns. A vector $\mathbf{c}$ lies in the null space if and only if the corresponding quadratic vanishes at x_1 and x_2 , that is, if and only if $p(x) = \alpha(x - x_1)(x - x_2)$ for some α , so the null space is spanned by

$$\mathbf{c}_0 = (x_1 x_2, -(x_1 + x_2), 1)^T, \quad (40)$$

as may be checked directly: $x_1 x_2 - (x_1 + x_2)x_i + x_i^2 = (x_i - x_1)(x_i - x_2) = 0$. A particular solution is the straight line through the two points, and the general solution is

$$p(x) = y_1 + \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) + \alpha(x - x_1)(x - x_2), \quad (41)$$

or in terms of coefficients, with $s = (y_2 - y_1)/(x_2 - x_1)$,

$$c_2 = \alpha, \quad c_1 = s - \alpha(x_1 + x_2), \quad c_0 = y_1 - s x_1 + \alpha x_1 x_2, \quad (42)$$

for arbitrary α . Every quadratic through the two data points fits them equally well, and the data provide no information whatever about α . This is the underdetermined case of the lecture, and it is the situation in which almost all geophysical inverse problems find themselves. To select one member of the family we must add information not contained in the data, for example by choosing the α that minimises $\|\mathbf{c}\|$, and later lectures will have much to say about how such choices are made and justified.

(d) For $m = n - 1$ the matrix is square and, by the argument of part (a) generalised to n points, invertible, so the problem is well-posed for every n . Taking equally spaced points $x_i = (i - 1)/(n - 1)$ on $[0, 1]$ and computing the condition number in

the Euclidean norm numerically,¹ we find

$$\begin{array}{c|cccc} n & 5 & 10 & 15 & 20 \\ \hline \text{cond}(\mathbf{V}) & 6.9 \times 10^2 & 1.5 \times 10^7 & 4.0 \times 10^{11} & 1.1 \times 10^{16} \end{array} . \quad (43)$$

The growth is exponential in n . By eq. (37) the condition number bounds the factor by which relative errors in the data can be amplified in the coefficients, and the bound is attained for suitably chosen perturbations. For $n = 10$, data accurate to one part in 10^7 can produce coefficients with errors of order one hundred per cent; for $n = 20$, data accurate to sixteen significant figures, i.e. to the full precision of double arithmetic, can produce coefficients with no correct digits at all. The problem is well-posed in the strict sense of the lecture, since the solution does depend continuously on the data, but the constant in that continuity is so large that the solution is practically useless. Once measurement errors and finite precision are admitted, the distinction between an ill-posed problem and a badly conditioned one all but disappears, and both call for the same kind of remedy. It is worth adding that the ill-conditioning here lies in the choice of unknowns: the monomials x^k on $[0, 1]$ become nearly parallel as k grows, so the columns of $\mathbf{V}$ are nearly linearly dependent. The values of the interpolating polynomial are far better determined than its monomial coefficients, and a wiser parameterisation, for instance in terms of Chebyshev polynomials, greatly reduces the condition number. Choosing the model parameters well is often the first step in taming an inverse problem.

¹ For $n = 20$ the condition number is comparable with the reciprocal of double-precision machine epsilon, and a floating-point computation of it is itself uncertain at the level of a few per cent; a routine such as `numpy.linalg.cond` returns 1.2×10^{16} . The value quoted was obtained by inverting $\mathbf{V}$ in exact rational arithmetic before taking norms.

Worked examples for Lecture 15: Plane wave propagation

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These examples accompany Lecture 15. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Throughout, ρ is the referential density, A_{ijkl} the elastic tensor, $\Gamma(\hat{\mathbf{p}})$ the Christoffel matrix for a unit propagation direction $\hat{\mathbf{p}}$, and the phase speeds c and polarisation vectors $\mathbf{a}$ satisfy the Christoffel equation $\Gamma(\hat{\mathbf{p}})\mathbf{a} = c^2\mathbf{a}$. Numerical results were checked with the scripts in the python directory.

Example 1: P- and S-waves with numbers. At a depth of 100 km the Preliminary Reference Earth Model (PREM) has, to three significant figures, $\rho = 3.38 \times 10^3 \text{ kg m}^{-3}$, bulk modulus $\kappa = 1.30 \times 10^{11} \text{ Pa}$ and shear modulus $\mu = 6.80 \times 10^{10} \text{ Pa}$. Compute the Lamé parameter λ , the P- and S-wave speeds α and β , the ratio α/β , and Poisson's ratio $\sigma = \lambda/[2(\lambda + \mu)]$. Show that α/β depends on the elastic moduli only through σ , and check that $\alpha/\beta = \sqrt{3}$ for a Poisson solid, for which $\lambda = \mu$.

Solution. We recall from the lecture that the bulk modulus of an isotropic solid is $\kappa = \lambda + \frac{2}{3}\mu$, and hence

$$\lambda = \kappa - \frac{2}{3}\mu = 1.30 \times 10^{11} - 0.453 \times 10^{11} = 8.47 \times 10^{10} \text{ Pa.} \quad (1)$$

The P- and S-wave speeds follow from the solution of the Christoffel equation for an isotropic medium. It is convenient to write the P-wave modulus as $\lambda + 2\mu = \kappa + \frac{4}{3}\mu = 2.21 \times 10^{11} \text{ Pa}$, so that

$$\begin{aligned} \alpha &= \sqrt{\frac{\lambda + 2\mu}{\rho}} = \sqrt{\frac{2.207 \times 10^{11}}{3.38 \times 10^3}} \text{ m s}^{-1} = 8.08 \text{ km s}^{-1}, \\ \beta &= \sqrt{\frac{\mu}{\rho}} = \sqrt{\frac{6.80 \times 10^{10}}{3.38 \times 10^3}} \text{ m s}^{-1} = 4.49 \text{ km s}^{-1}. \end{aligned} \quad (2)$$

Note that with all quantities in SI units the speeds come out in m s^{-1} ; a common slip is to mix GPa with kg m^{-3} and obtain speeds that are wrong by a factor of $\sqrt{10^9}$. The ratio of the speeds and Poisson's ratio are

$$\frac{\alpha}{\beta} = \frac{8.080}{4.485} = 1.80, \quad \sigma = \frac{\lambda}{2(\lambda + \mu)} = \frac{8.47}{2 \times 15.27} = 0.277. \quad (3)$$

To relate the two, we note that the definition of σ can be inverted to give

$$\frac{\lambda}{\mu} = \frac{2\sigma}{1-2\sigma}, \quad (4)$$

and hence

$$\frac{\alpha^2}{\beta^2} = \frac{\lambda + 2\mu}{\mu} = 2 + \frac{\lambda}{\mu} = \frac{2(1-\sigma)}{1-2\sigma}. \quad (5)$$

The ratio of the wave speeds is, therefore, a function of Poisson's ratio alone, and conversely a measurement of α/β determines σ without knowledge of the density. With $\sigma = 0.277$ eq. (5) gives $\alpha/\beta = 1.80$, in agreement with eq. (3). For a Poisson solid $\lambda = \mu$ gives $\sigma = 1/4$, and then $\alpha^2/\beta^2 = 2 \times \frac{3}{4}/\frac{1}{2} = 3$, so that $\alpha/\beta = \sqrt{3} = 1.73$.

The values in PREM are close to, but slightly above, those of a Poisson solid; a value of α/β between 1.7 and 1.8 is typical of the crust and mantle, which is why the Poisson solid is so often used as a rough model in seismology. It is worth remembering that in practice the logic runs the other way from this example: seismology measures α and β , and the moduli in PREM are inferred from them together with the density.

Example 2: The Christoffel matrix in a cubic crystal. A crystal with cubic symmetry has an elastic tensor of the form

$$A_{ijkl} = c_{12} \delta_{ij} \delta_{kl} + c_{44} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + (c_{11} - c_{12} - 2c_{44}) \sum_{n=1}^3 \delta_{in} \delta_{jn} \delta_{kn} \delta_{ln}, \quad (6)$$

where c_{11} , c_{12} and c_{44} are the three independent elastic constants in Voigt's notation and the co-ordinate axes are aligned with the crystal axes. Take the round values $c_{11} = 320$ GPa, $c_{12} = 70$ GPa, $c_{44} = 80$ GPa and $\rho = 3300$ kg m⁻³, which are of the magnitude found for olivine (though olivine itself is orthorhombic rather than cubic). Obtain the Christoffel matrix for a general unit vector $\hat{\mathbf{p}}$, and find the phase speeds and polarisation vectors for propagation along the [100], [110] and [111] directions, i.e. for $\hat{\mathbf{p}}$ proportional to (1,0,0), (1,1,0) and (1,1,1). For what relation between the constants is the material isotropic?

Solution. Writing $d = c_{11} - c_{12} - 2c_{44}$ for the coefficient of the last term in eq. (6), and contracting the elastic tensor with $\hat{p}_j \hat{p}_l$ term by term, we find

$$\rho \Gamma_{ik}(\hat{\mathbf{p}}) = A_{ijkl} \hat{p}_j \hat{p}_l = c_{12} \hat{p}_i \hat{p}_k + c_{44} (\delta_{ik} \hat{p}_j \hat{p}_j + \hat{p}_i \hat{p}_k) + d \sum_{n=1}^3 \delta_{in} \delta_{kn} \hat{p}_n^2, \quad (7)$$

where in the last term the sum over j and l has collapsed onto $j = l = n$. Since $\hat{p}_j \hat{p}_j = 1$, this can be written in matrix form as

$$\rho \Gamma(\hat{\mathbf{p}}) = (c_{12} + c_{44}) \hat{\mathbf{p}} \hat{\mathbf{p}}^T + c_{44} \mathbf{I} + d \text{diag}(\hat{p}_1^2, \hat{p}_2^2, \hat{p}_3^2). \quad (8)$$

The first two terms are exactly the isotropic Christoffel matrix of the lecture with $\lambda = c_{12}$ and $\mu = c_{44}$, and the third term, which is not invariant under rotations because it singles out the co-ordinate axes, carries all of the anisotropy. The material is therefore isotropic if and only if $d = 0$, i.e. $c_{11} - c_{12} = 2c_{44}$, in which case $c_{11} = \lambda + 2\mu$ as it

should. For our numbers $d = 320 - 70 - 160 = 90$ GPa, and the ratio $2c_{44}/(c_{11} - c_{12}) = 0.64$, which would be unity for an isotropic material, indicates a strongly anisotropic crystal.

Propagation along [100]. With $\hat{\mathbf{p}} = (1, 0, 0)$ the matrix $\hat{\mathbf{p}}\hat{\mathbf{p}}^T$ and the diagonal term are both non-zero only in the $(1, 1)$ entry, so that

$$\rho \Gamma = \text{diag}(c_{12} + 2c_{44} + d, c_{44}, c_{44}) = \text{diag}(c_{11}, c_{44}, c_{44}). \quad (9)$$

The eigenvectors are the co-ordinate axes: a longitudinal wave polarised along $\hat{\mathbf{e}}_1 = \hat{\mathbf{p}}$ with $\rho c^2 = c_{11}$, and a two-fold degenerate transverse wave with $\rho c^2 = c_{44}$ whose polarisation can be any vector in the $(\hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3)$ -plane. Numerically $c = \sqrt{320 \times 10^9 / 3300} = 9.85 \text{ km s}^{-1}$ and $c = \sqrt{80 \times 10^9 / 3300} = 4.92 \text{ km s}^{-1}$.

Propagation along [110]. With $\hat{\mathbf{p}} = (1, 1, 0)/\sqrt{2}$ we have $\hat{p}_1^2 = \hat{p}_2^2 = \frac{1}{2}$, $\hat{p}_3 = 0$, and eq. (8) gives

$$\rho \Gamma = \frac{c_{12} + c_{44}}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + c_{44} \mathbf{I} + \frac{d}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (10)$$

Rather than expanding the characteristic polynomial we guess the eigenvectors from the structure of the matrix, as was done for the isotropic case in the lecture. The vector $(1, 1, 0)/\sqrt{2}$ is an eigenvector of the first matrix with eigenvalue $c_{12} + c_{44}$ and of the third with eigenvalue $d/2$; the vector $(1, -1, 0)/\sqrt{2}$ is annihilated by the first matrix and is an eigenvector of the third with eigenvalue $d/2$; and $(0, 0, 1)$ is annihilated by both. Hence

$$\begin{aligned} \mathbf{a} = \frac{1}{\sqrt{2}}(1, 1, 0) : \quad \rho c^2 &= c_{12} + 2c_{44} + \frac{1}{2}d = \frac{1}{2}(c_{11} + c_{12} + 2c_{44}) = 275 \text{ GPa}, \\ \mathbf{a} = \frac{1}{\sqrt{2}}(1, -1, 0) : \quad \rho c^2 &= c_{44} + \frac{1}{2}d = \frac{1}{2}(c_{11} - c_{12}) = 125 \text{ GPa}, \\ \mathbf{a} = (0, 0, 1) : \quad \rho c^2 &= c_{44} = 80 \text{ GPa}, \end{aligned} \quad (11)$$

giving phase speeds of 9.13, 6.15 and 4.92 km s⁻¹ respectively. The first wave is longitudinal and the other two transverse, but the two transverse waves now have different speeds: this is the shear wave splitting discussed at the end of the lecture, and here it is far from slight, the two S-wave speeds differing by some 25%. The polarisation of the faster S-wave lies in the $(1, 2)$ -plane, that of the slower one along the third crystal axis.

Propagation along [111]. With $\hat{\mathbf{p}} = (1, 1, 1)/\sqrt{3}$ all three $\hat{p}_n^2$ equal $\frac{1}{3}$, so that the anisotropic term is proportional to the identity and

$$\rho \Gamma = (c_{12} + c_{44}) \hat{\mathbf{p}}\hat{\mathbf{p}}^T + \left(c_{44} + \frac{1}{3}d\right) \mathbf{I}. \quad (12)$$

This has the same form as the isotropic Christoffel matrix, and so can be solved in the same way: $\hat{\mathbf{p}}$ is an eigenvector, and so is any vector orthogonal to it,

$$\begin{aligned} \mathbf{a} = \hat{\mathbf{p}} : \quad \rho c^2 &= c_{12} + 2c_{44} + \frac{1}{3}d = \frac{1}{3}(c_{11} + 2c_{12} + 4c_{44}) = 260 \text{ GPa}, \\ \mathbf{a} \perp \hat{\mathbf{p}} : \quad \rho c^2 &= c_{44} + \frac{1}{3}d = \frac{1}{3}(c_{11} - c_{12} + c_{44}) = 110 \text{ GPa}, \end{aligned} \quad (13)$$

with phase speeds 8.88 km s⁻¹ and (two-fold degenerate) 5.77 km s⁻¹. All of these eigenvalues were confirmed with a numerical eigen-decomposition of eq. (8).

Two features are worth noting. First, along the high-symmetry directions [100], [110] and [111] the waves are purely longitudinal or purely transverse; in a general direction the eigenvectors of eq. (8) are neither parallel nor perpendicular to $\hat{\mathbf{p}}$, which is why one speaks of quasi P- and quasi S-waves. Second, the S-wave degeneracy is not split along [111] even though the crystal is anisotropic: the [111] direction is a three-fold symmetry axis of the cube, and a symmetric 2×2 matrix acting in the plane transverse to such an axis that is invariant under rotations by 120° must be a multiple of the identity. Degeneracies in the slowness surface are thus often dictated by symmetry.

Example 3: The SH sheet of the slowness surface in a transversely isotropic medium. A transversely isotropic medium with symmetry axis $\hat{\nu}$ has elastic tensor

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{kj}) + 8\gamma \hat{\nu}_i \hat{\nu}_j \hat{\nu}_k \hat{\nu}_l + 4\zeta (\hat{\nu}_i \hat{\nu}_j \delta_{kl} + \delta_{ij} \hat{\nu}_k \hat{\nu}_l) - \zeta (\hat{\nu}_i \hat{\nu}_k \delta_{jl} + \hat{\nu}_j \hat{\nu}_k \delta_{il} + \hat{\nu}_j \hat{\nu}_l \delta_{ik} + \hat{\nu}_i \hat{\nu}_l \delta_{jk}), \quad (14)$$

as in Lecture 13 and the first problem set. Take $\hat{\nu} = \hat{\mathbf{e}}_3$ and $\hat{\mathbf{p}} = (\sin \theta, 0, \cos \theta)$. Show that $\hat{\mathbf{e}}_2$ is an exact eigenvector of $\Gamma(\hat{\mathbf{p}})$ for every θ , and find the corresponding phase speed. Show that the associated sheet of the slowness surface is an ellipsoid of revolution, and compare the exact phase speed with the prediction of first-order degenerate perturbation theory. Illustrate the result for the sample values $\rho = 3300 \text{ kg m}^{-3}$, $\lambda = 80$, $\mu = 70$, $\gamma = 2$, $\zeta = 1$ and $\xi = 8 \text{ GPa}$.

Solution. We first form the Christoffel matrix for a general direction. Contracting each term of eq. (14) with $\hat{p}_j \hat{p}_l$, and writing $\hat{\nu} \cdot \hat{\mathbf{p}} = \cos \theta$, we obtain

$$\rho \Gamma_{ik}(\hat{\mathbf{p}}) = (\lambda + \mu) \hat{p}_i \hat{p}_k + \mu \delta_{ik} + 8\gamma \cos^2 \theta \hat{\nu}_i \hat{\nu}_k + 4\zeta \cos \theta (\hat{\nu}_i \hat{p}_k + \hat{p}_i \hat{\nu}_k) - \zeta [\hat{\nu}_i \hat{\nu}_k + \cos \theta (\hat{\nu}_i \hat{p}_k + \hat{p}_i \hat{\nu}_k) + \cos^2 \theta \delta_{ik}]. \quad (15)$$

Now let this matrix act on $\hat{\mathbf{e}}_2$. Every term in eq. (15) other than the two proportional to δ_{ik} contains a factor of either $\hat{p}_k$ or $\hat{\nu}_k$, and both $\hat{p}_k \hat{e}_{2k} = 0$ and $\hat{\nu}_k \hat{e}_{2k} = 0$ because $\hat{\mathbf{e}}_2$ is orthogonal to the plane containing the propagation direction and the symmetry axis. Hence

$$\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{e}}_2 = \frac{\mu - \zeta \cos^2 \theta}{\rho} \hat{\mathbf{e}}_2, \quad (16)$$

so that $\hat{\mathbf{e}}_2$ is an exact eigenvector for all θ , with squared phase speed

$$c_{\text{SH}}^2 = \frac{\mu - \zeta \cos^2 \theta}{\rho} = \frac{\mu \sin^2 \theta + (\mu - \zeta) \cos^2 \theta}{\rho}. \quad (17)$$

This wave is transverse and polarised horizontally (if we think of $\hat{\nu}$ as vertical), and so is called the SH wave. Its speed varies from $\sqrt{\mu/\rho}$ for horizontal propagation to $\sqrt{(\mu - \zeta)/\rho}$ along the axis, and a positive squared speed along the axis requires $\mu > \zeta$. Because eigenvectors of a symmetric matrix with distinct eigenvalues are orthogonal, the remaining two polarisation vectors lie in the $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_3)$ -plane spanned by $\hat{\mathbf{p}}$ and $\hat{\nu}$; they are the quasi P- and quasi SV-waves, whose speeds require the solution of a genuine 2×2 eigenvalue problem and are not in general given by such a simple formula. Nothing in the argument depended on $\hat{\mathbf{p}}$ lying in the $(1, 3)$ -plane: by the rotational

symmetry about $\hat{\nu}$, for any $\hat{\mathbf{p}}$ making an angle θ with the axis the vector $\hat{\nu} \times \hat{\mathbf{p}} / \sin \theta$ is an exact eigenvector with the eigenvalue in eq. (17).

The slowness sheet. The slowness vector of the SH wave is $\mathbf{p} = \hat{\mathbf{p}} / c_{\text{SH}}$, so that $\|\mathbf{p}\|^2 = \rho / (\mu - \zeta \cos^2 \theta)$ and $p_3 = \|\mathbf{p}\| \cos \theta$. Multiplying out,

$$\mu \|\mathbf{p}\|^2 - \zeta p_3^2 = \rho \iff \mu(p_1^2 + p_2^2) + (\mu - \zeta)p_3^2 = \rho, \quad (18)$$

which is an ellipsoid of revolution about the symmetry axis with equatorial semi-axis $\sqrt{\rho/\mu}$ and polar semi-axis $\sqrt{\rho/(\mu - \zeta)}$. For $\zeta > 0$ the sheet is prolate, the slowness being largest, and the speed smallest, along the axis. That one sheet of the slowness surface of a transversely isotropic medium is exactly an ellipsoid is a well-known special property; the quasi-P and quasi-SV sheets are quartic surfaces.

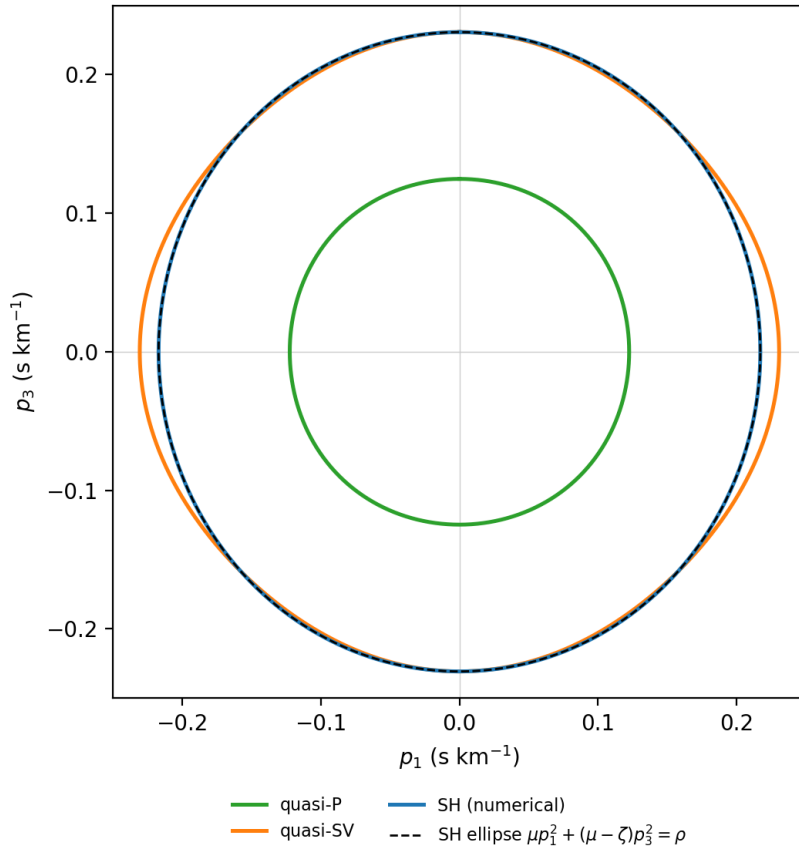


Fig. 1: Section through the slowness surface in the (p_1, p_3) -plane for the sample transversely isotropic medium of Example 3, with the symmetry axis vertical. The three sheets were obtained by numerical diagonalisation of the Christoffel matrix and are labelled by their polarisation; the dashed curve is the ellipse of eq. (18), on which the numerical SH sheet lies to machine precision. The SH and quasi-SV sheets touch on the symmetry axis, where the two S-waves are degenerate, and cross at oblique angles.

Comparison with perturbation theory. Treating the terms in γ , $\tilde{\zeta}$ and ζ as the perturbation $\delta\Gamma$, with the isotropic part having $\beta^2 = \mu/\rho$, the degenerate perturbation theory of the lecture asks us to diagonalise $\hat{\mathbf{t}}_a \cdot \delta\Gamma(\hat{\mathbf{p}})\hat{\mathbf{t}}_b$ on the plane orthogonal to $\hat{\mathbf{p}}$. Taking $\hat{\mathbf{t}}_2 =$

$\hat{\mathbf{e}}_2$ and $\hat{\mathbf{t}}_1 = \hat{\mathbf{e}}_2 \times \hat{\mathbf{p}} = (\cos \theta, 0, -\sin \theta)$, eq. (16) shows that $\delta\Gamma\hat{\mathbf{e}}_2 = -(\zeta \cos^2 \theta / \rho)\hat{\mathbf{e}}_2$, and hence

$$\hat{\mathbf{t}}_1 \cdot \delta\Gamma(\hat{\mathbf{p}})\hat{\mathbf{t}}_2 = 0, \quad \hat{\mathbf{t}}_2 \cdot \delta\Gamma(\hat{\mathbf{p}})\hat{\mathbf{t}}_2 = -\frac{\zeta \cos^2 \theta}{\rho}. \quad (19)$$

The 2×2 matrix is diagonal in this basis, so $\hat{\mathbf{e}}_2$ is one of the zeroth-order polarisations (as it must be, being an exact eigenvector), and the first-order speed perturbation of the SH wave is

$$\delta\beta_{\text{SH}} = \frac{1}{2\beta} \hat{\mathbf{e}}_2 \cdot \delta\Gamma(\hat{\mathbf{p}})\hat{\mathbf{e}}_2 = -\frac{\zeta \cos^2 \theta}{2\rho\beta} = -\frac{\zeta}{2\mu}\beta \cos^2 \theta. \quad (20)$$

The exact result, eq. (17), can be written

$$c_{\text{SH}} = \beta \sqrt{1 - \frac{\zeta}{\mu} \cos^2 \theta} = \beta \left[1 - \frac{\zeta}{2\mu} \cos^2 \theta - \frac{\zeta^2}{8\mu^2} \cos^4 \theta - \dots \right], \quad (21)$$

which agrees with $\beta + \delta\beta_{\text{SH}}$ to first order in ζ/μ , the error being of second order as it should be. The perturbation theory gives the eigenvalue c^2 exactly here; the only approximation is the linearisation of the square root.

For the sample values the SH speed is 4.61 km s^{-1} horizontally and 4.33 km s^{-1} along the axis, and with $\zeta/\mu = 0.114$ the first-order formula is in error by at most 8 m s^{-1} , or 0.2%. Fig. 1 shows the section of the slowness surface through the (p_1, p_3) -plane. Along the axis ($\theta = 0$) the SH and quasi-SV sheets touch, since eq. (15) shows that $\Gamma(\hat{\nu})$ acts as $(\mu - \zeta)/\rho$ times the identity on the plane orthogonal to $\hat{\nu}$: the two S-waves are degenerate along a symmetry axis, exactly as along $[111]$ in the cubic crystal.

Example 4: First-order perturbation theory checked against exact results. Write the elastic tensor of the cubic crystal of Example 2 as $A_{ijkl} = c_{12}\delta_{ij}\delta_{kl} + c_{44}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + s\delta A_{ijkl}$ with

$$\delta A_{ijkl} = d \sum_{n=1}^3 \delta_{in}\delta_{jn}\delta_{kn}\delta_{ln}, \quad d = c_{11} - c_{12} - 2c_{44}, \quad (22)$$

so that $s = 0$ is an isotropic medium with $\lambda = c_{12}$, $\mu = c_{44}$, and $s = 1$ is the crystal itself. (a) Obtain $\delta\Gamma(\hat{\mathbf{p}})$ and the first-order quasi-P speed perturbation $\delta\alpha$ for a general direction. (b) For $\hat{\mathbf{p}}$ along $[110]$ compare $\alpha + \delta\alpha$ with the exact speed found in Example 2, and account for the discrepancy. (c) Solve the 2×2 degenerate problem for the quasi S-waves along $[110]$ and show that it reproduces the exact splitting between $\rho c^2 = \frac{1}{2}(c_{11} - c_{12})$ and $\rho c^2 = c_{44}$. (d) For the general direction $\hat{\mathbf{p}} = (2, 3, 6)/7$ compare the three first-order eigenvalues with the exact ones, computed numerically, for $s = 1$, $\frac{1}{2}$ and $\frac{1}{4}$, and hence exhibit the order of the error.

Solution. (a) The unperturbed medium has $\alpha^2 = (c_{12} + 2c_{44})/\rho$ and $\beta^2 = c_{44}/\rho$, which for the numbers of Example 2 give $\alpha = 8.35 \text{ km s}^{-1}$ and $\beta = 4.92 \text{ km s}^{-1}$. From eq. (8), or directly from the definition $\delta\Gamma_{ik} = \rho^{-1}\delta A_{ijkl}\hat{p}_j\hat{p}_l$,

$$\rho \delta\Gamma(\hat{\mathbf{p}}) = d \text{diag}(\hat{p}_1^2, \hat{p}_2^2, \hat{p}_3^2). \quad (23)$$

The non-degenerate first-order formula of the lecture then gives, for the quasi P-wave,

$$\delta\alpha = \frac{1}{2\alpha} \hat{\mathbf{p}} \cdot \delta\mathbf{\Gamma}(\hat{\mathbf{p}}) \hat{\mathbf{p}} = \frac{d}{2\rho\alpha} (\hat{p}_1^4 + \hat{p}_2^4 + \hat{p}_3^4). \quad (24)$$

Since $\sum_n \hat{p}_n^4$ ranges from $\frac{1}{3}$ (along [111]) to 1 (along a cube axis), the quasi-P speed varies by an amount of order $d/(3\rho\alpha)$ with direction; for $d > 0$ the fastest direction is a cube axis and the slowest is a body diagonal, in agreement with the exact results of Example 2.

(b) Along [110] we have $\hat{p}_1^2 = \hat{p}_2^2 = \frac{1}{2}$ and $\hat{p}_3 = 0$, so that $\sum_n \hat{p}_n^4 = \frac{1}{2}$ and

$$\delta\alpha = \frac{d}{4\rho\alpha} = \frac{90 \times 10^9}{4 \times 3300 \times 8348} \text{ m s}^{-1} = 0.817 \text{ km s}^{-1}, \quad \alpha + s \delta\alpha|_{s=1} = 9.17 \text{ km s}^{-1}. \quad (25)$$

The exact speed from Example 2 is $\sqrt{275 \times 10^9 / 3300} = 9.13 \text{ km s}^{-1}$, so the first-order result is too large by 37 m s^{-1} , or 0.4%. To see where this comes from, note that along [110] the matrix $\rho \delta\mathbf{\Gamma} = \frac{1}{2}d \text{ diag}(1, 1, 0)$ maps $\hat{\mathbf{p}}$ to $\frac{1}{2}d \hat{\mathbf{p}}$, so $\hat{\mathbf{p}}$ is an exact eigenvector of the full Christoffel matrix, with the exact eigenvalue

$$c^2 = \alpha^2 + s \frac{d}{2\rho} = \alpha^2 + 2\alpha s \delta\alpha. \quad (26)$$

First-order perturbation theory is therefore exact for the eigenvalue c^2 ; the entire discrepancy arises from the expansion of the square root,

$$c = \alpha \sqrt{1 + \frac{2s \delta\alpha}{\alpha}} = \alpha + s \delta\alpha - \frac{s^2 \delta\alpha^2}{2\alpha} + \dots, \quad (27)$$

and the neglected second-order term, $-\delta\alpha^2/(2\alpha) = -d^2/(32\rho^2\alpha^3) = -40 \text{ m s}^{-1}$ at $s = 1$, accounts for the difference to within the size of the third-order term. The expansion parameter here is $2\delta\alpha/\alpha = d/(2\rho\alpha^2) = 90/460 = 0.20$, so an error of order its square is to be expected.

(c) For the quasi S-waves along [110] we take the orthonormal transverse vectors $\hat{\mathbf{t}}_1 = (0, 0, 1)$ and $\hat{\mathbf{t}}_2 = (1, -1, 0)/\sqrt{2}$. With $\rho \delta\mathbf{\Gamma} = \frac{1}{2}d \text{ diag}(1, 1, 0)$ we find $\hat{\mathbf{t}}_1 \cdot \delta\mathbf{\Gamma} \hat{\mathbf{t}}_1 = 0$, since $\hat{\mathbf{t}}_1$ has no component in the (1,2)-plane, $\hat{\mathbf{t}}_2 \cdot \delta\mathbf{\Gamma} \hat{\mathbf{t}}_2 = d/(2\rho)$, and $\hat{\mathbf{t}}_1 \cdot \delta\mathbf{\Gamma} \hat{\mathbf{t}}_2 = 0$. The degenerate eigenvalue problem of the lecture is thus already diagonal,

$$\begin{pmatrix} 0 & 0 \\ 0 & d/(2\rho) \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 2\beta \delta\beta \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad (28)$$

with solutions $\delta\beta = 0$ for polarisation $\hat{\mathbf{t}}_1 = (0, 0, 1)$ and $\delta\beta = d/(4\rho\beta)$ for polarisation $\hat{\mathbf{t}}_2 = (1, -1, 0)/\sqrt{2}$. These are precisely the exact polarisations found in Example 2, and the corresponding eigenvalues are

$$c^2 = \beta^2 = \frac{c_{44}}{\rho}, \quad c^2 = \beta^2 + 2\beta \delta\beta = \frac{c_{44} + \frac{1}{2}d}{\rho} = \frac{c_{11} - c_{12}}{2\rho}, \quad (29)$$

which are the exact values. The reason is the same as in part (b): because $\delta\mathbf{\Gamma}$ is diagonal in the basis $\{\hat{\mathbf{p}}, \hat{\mathbf{t}}_1, \hat{\mathbf{t}}_2\}$, these three vectors are exact eigenvectors of $\mathbf{\Gamma} + s \delta\mathbf{\Gamma}$ for every s , and the eigenvalues are exactly linear in s . The first-order theory correctly predicts

that the S-wave degeneracy is split, and which polarisation is the faster. For the speeds themselves, however, the linearised result $\beta + \delta\beta = 4.92 + 1.38 = 6.31 \text{ km s}^{-1}$ compares poorly with the exact $\sqrt{125 \times 10^9 / 3300} = 6.15 \text{ km s}^{-1}$, the error of 2.5% again coming from the square root; the expansion parameter is now $d/(2\rho\beta^2) = 45/80 = 0.56$, and the anisotropy of this crystal is simply not slight as far as its S-waves are concerned.

(d) In a general direction none of the zeroth-order vectors is an exact eigenvector and the first-order eigenvalues are genuinely approximate. For $\hat{\mathbf{p}} = (2, 3, 6)/7$ we have $\sum_n \hat{p}_n^4 = 1393/2401 = 0.580$, so the first-order quasi-P eigenvalue is

$$\rho c^2 = c_{12} + 2c_{44} + s d \sum_n \hat{p}_n^4 = (230 + 52.22 s) \text{ GPa}. \quad (30)$$

For the quasi S-waves we choose $\hat{\mathbf{t}}_1 \propto \hat{\mathbf{p}} \times \hat{\mathbf{e}}_3$ and $\hat{\mathbf{t}}_2 = \hat{\mathbf{p}} \times \hat{\mathbf{t}}_1$, and evaluate the 2×2 matrix numerically from eq. (23),

$$\rho (\hat{\mathbf{t}}_a \cdot \delta \Gamma \hat{\mathbf{t}}_b) = \begin{pmatrix} 10.17 & -3.63 \\ -3.63 & 27.61 \end{pmatrix} \text{ GPa}, \quad (31)$$

whose eigenvalues are 9.45 and 28.34 GPa (their sum, 37.78 GPa, equals $d(1 - \sum_n \hat{p}_n^4)$, as it must since the trace of $\rho \delta \Gamma$ is d). The first-order quasi-S eigenvalues are therefore $\rho c^2 = (80 + 9.45 s)$ and $(80 + 28.34 s)$ GPa. Comparing with the exact eigenvalues of $\Gamma + s \delta \Gamma$ obtained numerically, all in GPa, we find:

s	quasi S ₁			quasi S ₂			quasi P		
	first order	exact	error	first order	exact	error	first order	exact	error
1	89.45	89.18	+0.27	108.34	105.58	+2.76	282.22	285.25	-3.03
1/2	84.72	84.65	+0.07	94.17	93.41	+0.76	256.11	256.93	-0.83
1/4	82.36	82.34	+0.02	87.08	86.89	+0.20	243.05	243.27	-0.22

Halving s divides each error by a factor close to four (between 3.8 and 3.9, the shortfall being due to third-order terms), confirming that the first-order eigenvalues are in error at order s^2 . Two further features of the table are instructive. The three errors in each row sum to zero, because the trace of $\Gamma + s \delta \Gamma$ is exactly linear in s and is reproduced by first-order theory. And the quasi-P eigenvalue is underestimated while both quasi-S eigenvalues are overestimated: this is the familiar “level repulsion” of second-order perturbation theory, the second-order shift in c^2 of the quasi P-wave being

$$s^2 \sum_{a=1}^2 \frac{[\hat{\mathbf{t}}_a \cdot \delta \Gamma(\hat{\mathbf{p}}) \hat{\mathbf{p}}]^2}{\alpha^2 - \beta^2} = 3.60 s^2 \text{ GPa} / \rho, \quad (32)$$

which is positive because the P-wave has the largest eigenvalue, and which reproduces the quasi-P errors in the table for the smaller values of s (at $s = \frac{1}{4}$ it gives 0.225 GPa against the observed 0.22 GPa). Translating to speeds at $s = 1$, the lecture formulae give $\alpha + \delta\alpha = 9.30$, $\beta + \delta\beta_1 = 5.21$ and $\beta + \delta\beta_2 = 5.80 \text{ km s}^{-1}$, against exact values of 9.30, 5.20 and 5.66 km s^{-1} . The remarkable accuracy of the quasi-P speed here is partly fortuitous: the neglected second-order shift in c^2 (positive) and the neglected term in the square-root expansion (negative) happen almost to cancel in this direction, whereas for the faster quasi S-wave both errors have the same sign and add.

The moral is that first-order perturbation theory captures the qualitative picture – the directional dependence of the quasi-P speed, the splitting of the S-waves and

their polarisations – exactly as the lecture describes, while its quantitative accuracy is governed by the ratio of the anisotropic perturbation to the relevant isotropic modulus. For the weak anisotropy of a few per cent typical of the Earth's mantle this ratio is small and the theory is excellent; for a single olivine-like crystal it is not.

Example 5: Energy flux of a plane wave. On the first problem set it is shown that, in the absence of a stress glut, the energy density $E = \frac{1}{2}\rho \partial_t u_i \partial_t u_i + \frac{1}{2}A_{ijkl} \partial_j u_i \partial_l u_k$ satisfies $\partial_t E + \partial_j s_j = 0$, where the elastic Poynting vector is $s_j = -A_{ijkl} \partial_l u_k \partial_t u_i$. Consider the plane wave $u_i(\mathbf{x}, t) = f(t - p_j x_j) a_i$ with $\|\mathbf{a}\| = 1$ satisfying the Christoffel equation. Show that the energy flux is along the vector $A_{ijkl} a_i a_k p_l$, and that the **energy velocity** $\mathbf{v} = \mathbf{s}/E$ satisfies $\mathbf{v} \cdot \mathbf{p} = 1$. Show that for a P-wave in an isotropic medium the time-averaged flux is $\langle \mathbf{s} \rangle = \rho \alpha \langle \dot{u}^2 \rangle \hat{\mathbf{p}}$, with the corresponding result for an S-wave, and find the energy velocity of the SH wave of Example 3.

Solution. For the plane wave the derivatives were computed in the lecture,

$$\frac{\partial u_k}{\partial x_l} = -f'(t - p_m x_m) a_k p_l, \quad \frac{\partial u_i}{\partial t} = f'(t - p_m x_m) a_i, \quad (33)$$

and so, on substituting into the definitions,

$$s_j = f'^2 A_{ijkl} a_i a_k p_l, \quad E = \frac{1}{2} f'^2 (\rho a_i a_i + A_{ijkl} a_i p_j a_k p_l) = \frac{1}{2} f'^2 \rho (1 + \mathbf{a} \cdot \mathbf{\Gamma}(\mathbf{p}) \mathbf{a}) = \rho f'^2, \quad (34)$$

where in the last step we used the Christoffel equation $\mathbf{\Gamma}(\mathbf{p}) \mathbf{a} = \mathbf{a}$ in its unnormalised form (recall $\mathbf{\Gamma}(\mathbf{p}) = c^{-2} \mathbf{\Gamma}(\hat{\mathbf{p}})$) together with $\|\mathbf{a}\| = 1$. Thus the kinetic and strain energies of a plane wave are equal at every instant, and the energy flux at every instant is along the fixed vector $A_{ijkl} a_i a_k p_l$; the same is therefore true of the time average, which for a periodic f is $\langle s_j \rangle = \langle f'^2 \rangle A_{ijkl} a_i a_k p_l$. The energy velocity is independent of f ,

$$v_j = \frac{s_j}{E} = \frac{1}{\rho} A_{ijkl} a_i a_k p_l, \quad \mathbf{v} \cdot \mathbf{p} = \frac{1}{\rho} A_{ijkl} a_i p_j a_k p_l = \mathbf{a} \cdot \mathbf{\Gamma}(\mathbf{p}) \mathbf{a} = 1. \quad (35)$$

Since $\mathbf{p} = \hat{\mathbf{p}}/c$, this says that $\mathbf{v} \cdot \hat{\mathbf{p}} = c$: the component of the energy velocity along the propagation direction is always the phase speed, but in an anisotropic medium $\mathbf{v}$ need not be parallel to $\hat{\mathbf{p}}$, in which case $\|\mathbf{v}\| > c$ and energy travels obliquely to the wavefronts. This energy velocity is what is called the group velocity in the study of dispersive or anisotropic waves.

Isotropic P- and S-waves. Inserting the isotropic elastic tensor,

$$A_{ijkl} a_i a_k p_l = \lambda a_j (\mathbf{a} \cdot \mathbf{p}) + \mu (\mathbf{a} \cdot \mathbf{a}) p_j + \mu a_j (\mathbf{a} \cdot \mathbf{p}). \quad (36)$$

For a P-wave $\mathbf{a} = \hat{\mathbf{p}}$ and $\mathbf{p} = \hat{\mathbf{p}}/\alpha$, so this becomes $(\lambda + 2\mu)\hat{\mathbf{p}}/\alpha = \rho \alpha \hat{\mathbf{p}}$, while for an S-wave $\mathbf{a} \cdot \mathbf{p} = 0$ and $\mathbf{p} = \hat{\mathbf{p}}/\beta$ give $\mu \hat{\mathbf{p}}/\beta = \rho \beta \hat{\mathbf{p}}$. Since the particle velocity is $\dot{u}_i = f' a_i$ with $\|\mathbf{a}\| = 1$, we have $\dot{u}^2 = f'^2$, and hence

$$\langle \mathbf{s} \rangle = \rho \alpha \langle \dot{u}^2 \rangle \hat{\mathbf{p}} \quad (\text{P-wave}), \quad \langle \mathbf{s} \rangle = \rho \beta \langle \dot{u}^2 \rangle \hat{\mathbf{p}} \quad (\text{S-wave}). \quad (37)$$

In an isotropic medium, then, energy is carried along the propagation direction at the phase speed, with a flux equal to the wave speed times the energy density $\rho \langle \dot{u}^2 \rangle$, and

the product $\rho\alpha$ or $\rho\beta$ that appears is the acoustic impedance familiar from the theory of reflection and transmission.

The SH wave in a transversely isotropic medium. With $\mathbf{a} = \hat{\mathbf{e}}_2$ and $\hat{\mathbf{v}} = \hat{\mathbf{e}}_3$ we need $A_{2j2l}p_l$. From eq. (14), and using $\hat{v}_2 = 0$, the only surviving terms are $A_{2j2l} = \lambda\delta_{2j}\delta_{2l} + \mu(\delta_{jl} + \delta_{2j}\delta_{2l}) - \zeta\hat{v}_j\hat{v}_l$, and since $p_2 = 0$,

$$\mathbf{v} = \frac{1}{\rho} (\mu\mathbf{p} - \zeta(\hat{\mathbf{v}} \cdot \mathbf{p})\hat{\mathbf{v}}) = \frac{1}{\rho c_{\text{SH}}} (\mu \sin \theta, 0, (\mu - \zeta) \cos \theta). \quad (38)$$

This is parallel to $\hat{\mathbf{p}} = (\sin \theta, 0, \cos \theta)$ only for $\theta = 0$ or $\pi/2$; in between, the energy velocity is deflected away from the symmetry axis (for $\zeta > 0$), by an angle whose maximum for the sample values of Example 3 is about 3.5° , at $\theta \approx 43^\circ$. One can check directly from eq. (17) that

$$\mathbf{v} \cdot \hat{\mathbf{p}} = \frac{\mu \sin^2 \theta + (\mu - \zeta) \cos^2 \theta}{\rho c_{\text{SH}}} = c_{\text{SH}}, \quad (39)$$

as required by eq. (35). Finally, the gradient with respect to $\mathbf{p}$ of the function $\mu(p_1^2 + p_2^2) + (\mu - \zeta)p_3^2 - \rho$ defining the SH slowness sheet in eq. (18) is $2(\mu p_1, \mu p_2, (\mu - \zeta)p_3)$, which is parallel to $\mathbf{v}$: the energy velocity is normal to the slowness surface. This is in fact a general property of plane waves in anisotropic media, and it is the reason that the shape of the slowness surface, rather than the phase speed alone, governs the geometry of wave propagation.

Worked examples for Lecture 16: Ray theory

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These examples accompany Lecture 16. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Example 1: The isotropic eikonal equation and a linear velocity gradient. (a) Show that in an isotropic medium with P-wave speed $\alpha(\mathbf{x})$ the eikonal equation for the P-wave sheet of the local slowness surface reduces to $\|\nabla T\| = 1/\alpha$, and that the corresponding ray Hamiltonian is $H(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\alpha(\mathbf{x})^2\|\mathbf{p}\|^2$. (b) Consider a two-dimensional model with horizontal coordinate x , depth z measured positively downwards, and P-wave speed $\alpha(z) = \alpha_0 + gz$ with $\alpha_0 > 0$ and $g > 0$. A ray leaves the origin with take-off angle θ_0 measured from the downward vertical. Solve Hamilton's equations for this ray. Show that the horizontal slowness $p_x = \sin \theta_0 / \alpha_0$ is conserved, that the ray is a circular arc of radius $R = 1/(gp_x)$ centred at the depth $z = -\alpha_0/g$, and that the travel time along the ray is

$$T = \frac{1}{g} \ln \left[\frac{\tan(\theta/2)}{\tan(\theta_0/2)} \right],$$

where θ is the angle the ray makes with the downward vertical at the point in question.

(c) Find the horizontal distance X at which the ray returns to the surface $z = 0$ and the travel time $T(X)$, and evaluate both for $\alpha_0 = 5 \text{ km s}^{-1}$, $g = 0.05 \text{ s}^{-1}$ and $\theta_0 = 30^\circ$.

Solution. (a) In an isotropic medium the elastic tensor is $A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$, and so the local Christoffel operator defined in the lecture takes the form

$$\Gamma_{ik}(\mathbf{x}, \mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l = \alpha^2 p_i p_k + \beta^2 (\|\mathbf{p}\|^2 \delta_{ik} - p_i p_k), \quad (1)$$

with $\alpha^2 = (\lambda + 2\mu)/\rho$ and $\beta^2 = \mu/\rho$, exactly as for plane waves in Lecture 15 but with ρ , λ and μ now functions of $\mathbf{x}$. The two terms are, respectively, $\|\mathbf{p}\|^2$ times the projections parallel and orthogonal to $\hat{\mathbf{p}}$, so the eigenvalues of Γ are $\alpha^2\|\mathbf{p}\|^2$, with eigenvector $\hat{\mathbf{p}}$, and $\beta^2\|\mathbf{p}\|^2$, with the two-dimensional eigenspace orthogonal to $\hat{\mathbf{p}}$. The eikonal equation $\det(\Gamma - \mathbf{I}) = 0$ therefore factorises as

$$(\alpha^2\|\mathbf{p}\|^2 - 1) (\beta^2\|\mathbf{p}\|^2 - 1)^2 = 0, \quad (2)$$

which is the lecture's general factorised form with the local phase speeds $c_1 = \alpha$ and $c_2 = c_3 = \beta$ independent of the direction $\hat{\mathbf{p}}$. Selecting the P-wave sheet and recalling

that $\mathbf{p} = \nabla T$, the reduced eikonal equation is

$$\alpha(\mathbf{x})^2 \|\nabla T(\mathbf{x})\|^2 = 1, \quad \text{i.e.} \quad \|\nabla T\| = \frac{1}{\alpha}, \quad (3)$$

and the ray Hamiltonian $H = \frac{1}{2} \|\mathbf{p}\|^2 c_1^2$ of the lecture becomes

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2} \alpha(\mathbf{x})^2 \|\mathbf{p}\|^2, \quad (4)$$

with the eikonal equation being $H[\mathbf{x}, \nabla T(\mathbf{x})] = \frac{1}{2}$. The polarisation $\hat{\mathbf{a}}^0 = \hat{\mathbf{p}}$ is, as for plane waves, along the local direction of propagation.

(b) We work in the (x, z) -plane, with $\mathbf{x} = (x, z)$ and $\mathbf{p} = (p_x, p_z)$, and the sign conventions stated in the question: z increases downwards, so that α increases with depth, and a ray travelling straight down has $\theta = 0$. The initial point in phase space is $\mathbf{x}^0 = (0, 0)$ together with

$$\mathbf{p}^0 = \frac{1}{\alpha_0} (\sin \theta_0, \cos \theta_0), \quad (5)$$

which has $\|\mathbf{p}^0\| = 1/\alpha_0$ and so satisfies $H = \frac{1}{2}$ as required. From eq. (4), $\partial H / \partial p_i = \alpha^2 p_i$ and $\partial H / \partial x_i = \alpha \|\mathbf{p}\|^2 \partial \alpha / \partial x_i$, and since α depends only on z with $d\alpha/dz = g$, Hamilton's canonical equations read

$$\frac{dx}{d\sigma} = \alpha^2 p_x, \quad \frac{dz}{d\sigma} = \alpha^2 p_z, \quad \frac{dp_x}{d\sigma} = 0, \quad \frac{dp_z}{d\sigma} = -\|\mathbf{p}\|^2 \alpha g = -\frac{g}{\alpha}, \quad (6)$$

where in the last equation we have used the conservation of the Hamiltonian, $\alpha^2 \|\mathbf{p}\|^2 = 1$, along the solution. The third equation shows at once that the horizontal slowness is conserved,

$$p_x = \frac{\sin \theta_0}{\alpha_0} \equiv p, \quad (7)$$

which is the conserved quantity associated with the translational symmetry of the model in x ; it is often called the ray parameter. Because $\|\mathbf{p}\| = 1/\alpha$ everywhere along the ray, we can introduce an angle $\theta(\sigma)$ through

$$p_x = \frac{\sin \theta}{\alpha}, \quad p_z = \frac{\cos \theta}{\alpha}, \quad (8)$$

with $\theta(0) = \theta_0$. The first two equations in eq. (6) then give $d\mathbf{x}/d\sigma = \alpha(\sin \theta, \cos \theta)$, so θ is precisely the angle between the ray and the downward vertical, and $ds/d\sigma = \alpha$ for the arc length s along the ray. Combining eqs. (7) and (8) we obtain Snell's law for this model,

$$\frac{\sin \theta}{\alpha(z)} = p, \quad (9)$$

so that θ increases with depth: the ray turns away from the vertical as it enters faster material. Differentiating eq. (9) with respect to σ and using $dz/d\sigma = \alpha \cos \theta$ gives $\cos \theta d\theta/d\sigma = pg\alpha \cos \theta$, and hence

$$\frac{d\theta}{d\sigma} = pg\alpha = g \sin \theta. \quad (10)$$

It is now simplest to use θ rather than σ as the parameter along the ray. Dividing $dx/d\sigma = \alpha \sin \theta$ and $dz/d\sigma = \alpha \cos \theta$ by eq. (10), and writing $R = 1/(gp)$, we find

$$\frac{dx}{d\theta} = \frac{\alpha}{g} = R \sin \theta, \quad \frac{dz}{d\theta} = \frac{\alpha \cos \theta}{g \sin \theta} = R \cos \theta, \quad (11)$$

where we have used $\alpha = \sin \theta / p$ from eq. (9). Integrating from θ_0 , where the ray is at the origin,

$$x = R(\cos \theta_0 - \cos \theta), \quad z = R(\sin \theta - \sin \theta_0), \quad (12)$$

and therefore

$$(x - R \cos \theta_0)^2 + (z + R \sin \theta_0)^2 = R^2. \quad (13)$$

The ray is a circle of radius

$$R = \frac{1}{gp} = \frac{\alpha_0}{g \sin \theta_0}, \quad (14)$$

whose centre lies at horizontal position $R \cos \theta_0$ and at the depth $-R \sin \theta_0 = -\alpha_0/g$. This depth is the same for all take-off angles: it is the height above the surface at which the linear velocity profile would extrapolate to zero. Note also that eq. (11) gives $ds/d\theta = R$, so the curvature $d\theta/ds = 1/R$ is indeed constant along the ray.

For the travel time we use the lecture's result that the generating parameter is the travel time, $T[\mathbf{x}(\sigma)] = \sigma$, together with eq. (10):

$$T = \int_0^\sigma d\sigma' = \int_{\theta_0}^\theta \frac{d\theta'}{g \sin \theta'} = \frac{1}{g} \ln \left[\frac{\tan(\theta/2)}{\tan(\theta_0/2)} \right], \quad (15)$$

where we have used $\frac{d}{d\theta} \ln \tan(\theta/2) = 1/\sin \theta$. The same result follows from $dT = ds/\alpha = R d\theta/\alpha$ and eq. (9).

(c) The ray is horizontal, $\theta = \pi/2$, at its turning point, which from eq. (12) lies at the depth

$$z_t = R(1 - \sin \theta_0) = \frac{\alpha_0}{g} \left(\frac{1}{\sin \theta_0} - 1 \right), \quad (16)$$

and by the symmetry of the circle it returns to $z = 0$ when $\theta = \pi - \theta_0$, at the distance

$$X = 2R \cos \theta_0 = \frac{2\alpha_0}{g} \cot \theta_0. \quad (17)$$

Setting $\theta = \pi - \theta_0$ in eq. (15) and using $\tan[(\pi - \theta_0)/2] = \cot(\theta_0/2)$,

$$T(X) = \frac{2}{g} \ln \cot \frac{\theta_0}{2} = \frac{2}{g} \operatorname{arcsinh} \left(\frac{gX}{2\alpha_0} \right), \quad (18)$$

where the second form follows from $\cot(\theta_0/2) = \cot \theta_0 + \sqrt{1 + \cot^2 \theta_0}$ and $\operatorname{arcsinh} u = \ln(u + \sqrt{1 + u^2})$, with $u = \cot \theta_0 = gX/(2\alpha_0)$ from eq. (17). Eq. (18) is the travel-time curve for this model. For small X it reduces to $T \approx X/\alpha_0$, while for large X the travel time grows only logarithmically because the ray spends most of its time in fast material at depth. For the given numbers, $p = 0.1 \text{ s km}^{-1}$, $R = 200 \text{ km}$, the centre of the circle lies at $(173.2, -100) \text{ km}$, the turning depth is $z_t = 100 \text{ km}$, the ray returns to the surface at $X = 346.4 \text{ km}$, and $T(X) = 40 \ln \cot 15^\circ = 52.68 \text{ s}$. This is considerably less than the 69.28 s that would be taken along the straight line at speed α_0 . Integrating

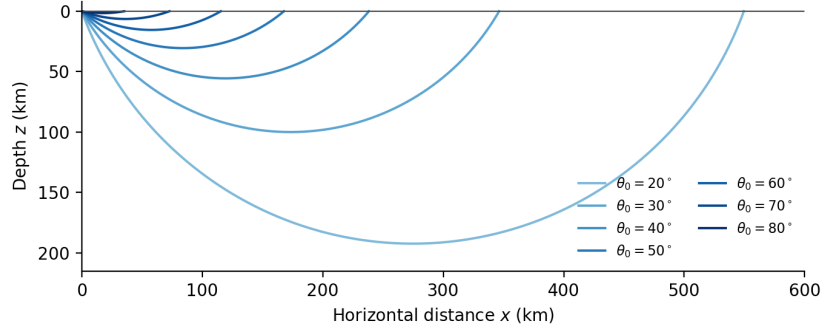


Fig. 1: Fan of rays leaving the origin in the model $\alpha(z) = \alpha_0 + gz$ with $\alpha_0 = 5 \text{ km s}^{-1}$ and $g = 0.05 \text{ s}^{-1}$, for take-off angles between 20° and 80° . Each ray is a circular arc centred at the depth $-\alpha_0/g = -100 \text{ km}$.

Hamilton's equations in eq. (6) numerically with `scipy.integrate.solve_ivp` reproduces the circle to 10^{-7} km and the travel time to 10^{-8} s . Fig. 1 shows the resulting fan of rays.

This example is the simplest illustration of how a velocity that increases with depth returns rays to the surface, which is the mechanism that makes body-wave seismology possible; the spherical analogue is developed in Lecture 17.

Example 2: From Hamilton's equations to the ray equation. For the isotropic P-wave Hamiltonian $H(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\alpha(\mathbf{x})^2\|\mathbf{p}\|^2$ write down Hamilton's equations. Show that the arc length s along a ray satisfies $ds/d\sigma = \alpha$, and hence that rays satisfy the ray equation

$$\frac{d}{ds} \left(\frac{1}{\alpha} \frac{d\mathbf{x}}{ds} \right) = \nabla \left(\frac{1}{\alpha} \right).$$

Deduce that rays are straight lines when α is constant, and show that in general a ray has curvature $\kappa = \|\hat{\mathbf{t}} \times \nabla \ln \alpha\|$, where $\hat{\mathbf{t}}$ is the unit tangent, and bends towards regions of lower wave speed. Check the result against Example 1.

Solution. The partial derivatives of the Hamiltonian are $\partial H / \partial p_i = \alpha^2 p_i$ and $\partial H / \partial x_i = \alpha \|\mathbf{p}\|^2 \partial \alpha / \partial x_i$, and so Hamilton's equations are

$$\frac{d\mathbf{x}}{d\sigma} = \alpha^2 \mathbf{p}, \quad \frac{d\mathbf{p}}{d\sigma} = -\|\mathbf{p}\|^2 \alpha \nabla \alpha. \quad (19)$$

Along a solution the Hamiltonian is conserved and equal to $\frac{1}{2}$, so $\|\mathbf{p}\| = 1/\alpha$, and the equations simplify to

$$\frac{d\mathbf{x}}{d\sigma} = \alpha \hat{\mathbf{p}}, \quad \frac{d\mathbf{p}}{d\sigma} = -\frac{\nabla \alpha}{\alpha} = -\nabla \ln \alpha. \quad (20)$$

The first of these shows that the ray is everywhere tangent to the local slowness vector, and that

$$\frac{ds}{d\sigma} = \left\| \frac{d\mathbf{x}}{d\sigma} \right\| = \alpha, \quad (21)$$

which is just the statement $dT = ds/\alpha$ that the travel time accumulates at the local slowness. The unit tangent is therefore

$$\hat{\mathbf{t}} = \frac{d\mathbf{x}}{ds} = \frac{1}{\alpha} \frac{d\mathbf{x}}{d\sigma} = \hat{\mathbf{p}}, \quad \text{so that} \quad \mathbf{p} = \frac{1}{\alpha} \frac{d\mathbf{x}}{ds}. \quad (22)$$

Differentiating this expression with respect to s and using eqs. (20) and (21),

$$\frac{d}{ds} \left(\frac{1}{\alpha} \frac{d\mathbf{x}}{ds} \right) = \frac{d\mathbf{p}}{ds} = \frac{1}{\alpha} \frac{d\mathbf{p}}{d\sigma} = -\frac{\nabla\alpha}{\alpha^2} = \nabla \left(\frac{1}{\alpha} \right), \quad (23)$$

which is the ray equation. It is a second-order equation for the ray path alone, with the slowness vector eliminated; it is also the Euler–Lagrange equation for Fermat’s principle of stationary travel time $\int ds/\alpha$, though we do not need that here. If α is constant the right hand side vanishes and $d^2\mathbf{x}/ds^2 = \mathbf{0}$, so rays are straight lines, as they must be if the theory is to reproduce plane waves in a homogeneous body.

To find the curvature we expand the left hand side of eq. (23), using the chain rule in the form $d(1/\alpha)/ds = \hat{\mathbf{t}} \cdot \nabla(1/\alpha)$, which gives

$$\frac{1}{\alpha} \frac{d\hat{\mathbf{t}}}{ds} + \hat{\mathbf{t}} \left[\hat{\mathbf{t}} \cdot \nabla \left(\frac{1}{\alpha} \right) \right] = \nabla \left(\frac{1}{\alpha} \right). \quad (24)$$

Multiplying by α and using $\alpha \nabla(1/\alpha) = -\nabla \ln \alpha$,

$$\frac{d\hat{\mathbf{t}}}{ds} = - \left[\nabla \ln \alpha - \hat{\mathbf{t}}(\hat{\mathbf{t}} \cdot \nabla \ln \alpha) \right] = -(\nabla \ln \alpha)_{\perp}, \quad (25)$$

where $(\nabla \ln \alpha)_{\perp}$ denotes the component of $\nabla \ln \alpha$ orthogonal to the ray. By the Frenet formula $d\hat{\mathbf{t}}/ds = \kappa \hat{\mathbf{n}}$, with $\kappa \geq 0$ the curvature and $\hat{\mathbf{n}}$ the unit principal normal, and since for any vector $\|\hat{\mathbf{t}} \times \mathbf{v}\| = \|\mathbf{v}_{\perp}\|$,

$$\kappa = \|\hat{\mathbf{t}} \times \nabla \ln \alpha\| = -\hat{\mathbf{n}} \cdot \nabla \ln \alpha, \quad \hat{\mathbf{n}} = -\frac{(\nabla \ln \alpha)_{\perp}}{\|(\nabla \ln \alpha)_{\perp}\|}. \quad (26)$$

The principal normal, which is the direction in which the ray is turning, is anti-parallel to the transverse component of $\nabla \ln \alpha$: rays bend towards regions of low wave speed, and do so more sharply the larger the fractional transverse velocity gradient. A ray travelling directly along $\nabla \alpha$ has zero curvature, which is why a vertical ray in Example 1 is straight.

In Example 1, $\nabla \ln \alpha = (g/\alpha)\hat{\mathbf{z}}$ and $\hat{\mathbf{t}} = (\sin \theta, \cos \theta)$, so eq. (26) gives

$$\kappa = \frac{g \sin \theta}{\alpha} = gp = \frac{1}{R}, \quad (27)$$

in agreement with the constant radius found there, and $\hat{\mathbf{n}} = (\cos \theta, -\sin \theta)$ points upwards, towards the slower material. The script `verify_examples5.py` also integrates eq. (19) through a model containing a localised low-velocity anomaly and confirms that the numerically measured curvature of the ray agrees with eq. (26) to better than 0.1%. The bending of rays towards low velocities is what produces the focusing and caustic seen in the lecture’s figure of ray tracing through a low-velocity region.

Example 3: The transport equation in one dimension. A thin elastic rod has density $\rho(x)$ and Young's modulus $E(x)$, both smooth, so that its longitudinal displacement satisfies

$$\rho \frac{\partial^2 u}{\partial t^2} - \frac{\partial}{\partial x} \left(E \frac{\partial u}{\partial x} \right) = 0.$$

(a) Insert the one-dimensional ray series $u(x, \omega) = \sum_{n=0}^{\infty} a^n(x) (i\omega)^{-n} e^{-i\omega T(x)}$ into the frequency-domain equation of motion and, following the lecture's derivation, obtain the eikonal equation and the equations satisfied by a^0 , a^1 and the general a^n .

(b) Solve the eikonal equation, and show that the first transport equation gives $a^0 \propto (\rho c)^{-1/2}$ with $c = \sqrt{E/\rho}$.

(c) Interpret this result in terms of energy flux, and explain what goes wrong if the impedance ρc has a jump discontinuity.

Solution. (a) In the frequency domain, with the convention $\partial_t \rightarrow i\omega$ used in the lecture, the equation of motion becomes

$$(i\omega)^2 \rho u - \frac{d}{dx} \left(E \frac{du}{dx} \right) = 0, \quad (28)$$

where u now denotes the Fourier transform. The one-dimensional slowness is $T' = dT/dx$, which plays the role of $p_i = \partial T / \partial x_i$ in the lecture. Differentiating the ray series term by term,

$$\frac{du}{dx} = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} (a^{n'} - i\omega T' a^n) e^{-i\omega T}, \quad (29)$$

$$\frac{d}{dx} \left(E \frac{du}{dx} \right) = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left[(E a^{n'})' - i\omega (E T' a^n)' - i\omega E T' a^{n''} + (i\omega)^2 E T'^2 a^n \right] e^{-i\omega T}, \quad (30)$$

where a prime on a^n or on a bracket denotes d/dx , and the two terms proportional to $i\omega$ in eq. (30) come respectively from differentiating the amplitude factor $E T' a^n$ and from differentiating the exponential in the term $E a^{n'} e^{-i\omega T}$. These are the one-dimensional counterparts of the lecture's expressions for $\partial u_k / \partial x_l$ and for the divergence of the stress. Substituting into eq. (28) and multiplying by -1 ,

$$\sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left[(E a^{n'})' - i\omega (E T' a^n)' - i\omega E T' a^{n''} + (i\omega)^2 (E T'^2 - \rho) a^n \right] e^{-i\omega T} = 0. \quad (31)$$

This must hold for all ω , so the coefficient of each power of $i\omega$ vanishes. The highest power, $(i\omega)^2$, arises only from the $n = 0$ term and gives

$$(E T'^2 - \rho) a^0 = 0, \quad (32)$$

the coefficient of $(i\omega)^1$ collects the last term with $n = 1$ and the middle terms with $n = 0$,

$$(E T'^2 - \rho) a^1 = (E T' a^0)' + E T' a^{0'}, \quad (33)$$

and for $(i\omega)^{2-n}$ with $n \geq 2$ three successive terms of the series contribute,

$$(ET'^2 - \rho) a^n = (ET' a^{n-1})' + ET' a^{n-1'} - (E a^{n-2'})'. \quad (34)$$

These are exactly the lecture's three equations with $A_{ijkl} \rightarrow E$, $p_i \rightarrow T'$ and $\delta_{ik} \rightarrow 1$. For a non-trivial a^0 , eq. (32) requires the eikonal equation

$$ET'^2 = \rho, \quad \text{i.e.} \quad T' = \pm \frac{1}{c}, \quad c = \sqrt{\frac{E}{\rho}}. \quad (35)$$

In one dimension the local Christoffel operator is the scalar $\Gamma = ET'^2/\rho = c^2 T'^2$, the ray Hamiltonian is $H = \frac{1}{2} c^2 p^2$, and the local slowness surface consists of the two points $p = \pm 1/c$; there is no polarisation to determine. The important structural difference from three dimensions now appears: once eq. (35) holds, the left hand sides of eqs. (33) and (34) vanish identically, and so every one of these equations is a constraint on the previous amplitude. Thus

$$2ET' a^{0'} + (ET')' a^0 = 0, \quad (36)$$

which is the first transport equation, and for $n \geq 1$

$$2ET' a^{n'} + (ET')' a^n = (E a^{n-1'})', \quad (37)$$

so each higher amplitude satisfies the same first-order equation, forced by the second derivative of its predecessor.

(b) Taking the right-going wave, $T' = 1/c$, the travel time is $T(x) = \int_0^x dx'/c(x')$ and $ET' = E/c = \sqrt{\rho E} = \rho c$. Eq. (36) then reads

$$\frac{a^{0'}}{a^0} = -\frac{1}{2} \frac{(\rho c)'}{\rho c}, \quad \text{so that} \quad a^0(x) = \frac{C}{\sqrt{\rho(x)c(x)}}, \quad (38)$$

with C a constant. Equivalently, multiplying eq. (36) by a^0 gives the conservation law

$$\frac{d}{dx} [ET' (a^0)^2] = \frac{d}{dx} [\rho c (a^0)^2] = 0. \quad (39)$$

The zeroth-order ray solution is therefore

$$u(x, \omega) \approx \frac{C}{\sqrt{\rho(x)c(x)}} \exp \left[-i\omega \int_0^x \frac{dx'}{c(x')} \right], \quad (40)$$

which you may recognise as the WKB approximation.

(c) For the time-harmonic wave $u = a \cos \omega(t - T)$ the energy flux along the rod is $-E \partial_x u \partial_t u$. To leading order in ω we have $\partial_x u \approx (a\omega/c) \sin \omega(t - T)$ and $\partial_t u = -a\omega \sin \omega(t - T)$, and so the flux averaged over a period is

$$\langle F \rangle = \frac{1}{2} \frac{E}{c} \omega^2 a^2 = \frac{1}{2} \rho c \omega^2 a^2. \quad (41)$$

Eq. (39) is therefore the statement that the energy flux carried by the wave is independent of position: in a smoothly varying rod nothing is reflected at leading order,

and the amplitude adjusts so that the same power is transmitted through every cross-section. The quantity $Z = \rho c = \sqrt{\rho E}$ is the acoustic impedance, and it is only through Z that the material affects the amplitude; a rod in which c varies but ρc is constant has a^0 constant.

If Z jumps from Z_1 to Z_2 at some point, eq. (38) would predict that the transmitted amplitude is $\sqrt{Z_1/Z_2}$ times the incident amplitude, with a^0 discontinuous and $a^{0'}$ containing a delta function, so that the assumption of smooth amplitude functions underlying the ray series fails. The exact matching conditions (continuity of u and of $E \partial_x u$) instead give a transmitted amplitude $2Z_1/(Z_1 + Z_2)$ together with a reflected wave of relative amplitude $(Z_1 - Z_2)/(Z_1 + Z_2)$ whose phase $T' = -1/c$ belongs to the other point of the local slowness surface. A single ray series carries only one phase function and so cannot represent this reflected wave; a discontinuity has to be handled by matching separate ray series across it, as remarked at the end of the lecture. Writing $Z_2 = Z_1(1 + \delta)$, both $\sqrt{Z_1/Z_2}$ and $2Z_1/(Z_1 + Z_2)$ equal $1 - \delta/2 + O(\delta^2)$, so the ray-theoretic amplitude is correct precisely to the order at which reflection can be neglected. As a numerical check, `verify_examples5.py` integrates eq. (28) exactly through a smooth rod and compares with eq. (40): the relative error is 0.8%, 0.4% and 0.2% at $\omega = 20, 40$ and 80 (in units where the rod has length 10 and $c \approx 3$), halving each time the frequency doubles, as expected for a neglected term of order $1/\omega$.

Example 4: Method of characteristics for a point source. A homogeneous and isotropic medium in two dimensions has constant P-wave speed α and density ρ . A source at the origin is modelled by the initial data $T = 0$ at $\mathbf{x} = \mathbf{0}$, so that the initial surface S^0 in phase space consists of the points $(\mathbf{x}^0, \mathbf{p}^0) = (\mathbf{0}, \hat{\mathbf{p}}/\alpha)$ with $\hat{\mathbf{p}} = (\cos \phi, \sin \phi)$ for all $\phi \in [0, 2\pi)$. (a) Integrate Hamilton's equations from this initial data to show that $T(\mathbf{x}) = \|\mathbf{x}\|/\alpha$, verify that this satisfies the eikonal equation, and check the symmetry $\partial p_i / \partial x_j = \partial p_j / \partial x_i$ of the resulting slowness field. (b) Show that for a homogeneous isotropic medium the $i\omega$ equation of the ray series, projected onto $\hat{\mathbf{p}}$, gives the transport equation $\nabla \cdot (A_0^2 \nabla T) = 0$ for the scalar P-wave amplitude A_0 defined by $\mathbf{a}^0 = A_0 \hat{\mathbf{p}}$. Solve it for the point source and comment on the behaviour at the origin.

Solution. (a) With α constant, eq. (19) reduces to $d\mathbf{x}/d\sigma = \alpha^2 \mathbf{p}$ and $d\mathbf{p}/d\sigma = \mathbf{0}$, so from the initial point labelled by ϕ we obtain

$$\mathbf{p}(\sigma) = \frac{\hat{\mathbf{p}}}{\alpha}, \quad \mathbf{x}(\sigma) = \alpha\sigma \hat{\mathbf{p}} = \alpha\sigma(\cos \phi, \sin \phi). \quad (42)$$

The rays are straight lines radiating from the origin, in accordance with Example 2. Given any $\mathbf{x} \neq \mathbf{0}$, there is exactly one ray through it, namely that with ϕ equal to the polar angle of $\mathbf{x}$, and the point is reached when $\sigma = \|\mathbf{x}\|/\alpha$. Since the generating parameter is the travel time,

$$T(\mathbf{x}) = \frac{\|\mathbf{x}\|}{\alpha} = \frac{r}{\alpha}, \quad (43)$$

whose wavefronts $T = \text{constant}$ are circles centred on the source. Directly, $\nabla T = \hat{\mathbf{x}}/\alpha$ and so $\|\nabla T\| = 1/\alpha$, which is the isotropic eikonal equation of Example 1. The

slowness field constructed by the method of characteristics is $p_i(\mathbf{x}) = x_i/(\alpha r)$, with

$$\frac{\partial p_i}{\partial x_j} = \frac{1}{\alpha r} (\delta_{ij} - \hat{x}_i \hat{x}_j), \quad (44)$$

which is symmetric in i and j as the lecture's general argument requires, so that $\mathbf{p}$ is indeed a gradient. Note that the initial surface S^0 here is a circle of slowness vectors sitting above a single point, so it is not the graph of a vector field over the initial data; the level surface $H = \frac{1}{2}$ swept out by the characteristics, $\{(\mathbf{x}, \hat{\mathbf{x}}/\alpha)\}$, becomes the graph of ∇T only away from the origin.

(b) For a P-wave $\hat{\mathbf{a}}^0 = \hat{\mathbf{p}}$, so we write

$$\mathbf{a}^0 = A_0 \hat{\mathbf{p}} = \varphi \mathbf{p}, \quad \varphi = \alpha A_0, \quad (45)$$

using $\hat{\mathbf{p}} = \alpha \mathbf{p}$ on the P-wave sheet. The lecture's equation at order $i\omega$ is

$$[A_{ijkl} p_j p_l - \rho \delta_{ik}] a_k^1 = \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^0) + A_{ijkl} p_j \frac{\partial a_k^0}{\partial x_l} \equiv F_i, \quad (46)$$

and since the matrix on the left is $\rho(\Gamma - \mathbf{I})$, which is symmetric and annihilates $\hat{\mathbf{p}}$, contracting with $\hat{p}_i$ removes the unknown $\mathbf{a}^1$ entirely and leaves the solvability condition $\hat{p}_i F_i = 0$. To evaluate it we use two facts about the slowness field: $\partial_j p_i = \partial_i p_j$, and, because $\|\mathbf{p}\| = 1/\alpha$ is constant in a homogeneous medium, $p_i \partial_j p_i = \frac{1}{2} \partial_j \|\mathbf{p}\|^2 = 0$, which together imply $p_j \partial_j p_i = p_j \partial_i p_j = 0$ as well. With $A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$ constant, the first term of F_i contracted with $\hat{p}_i$ is

$$\hat{p}_i A_{ijkl} \partial_j (\varphi p_l p_k) = \lambda \hat{p}_j \partial_j (\varphi \|\mathbf{p}\|^2) + 2\mu \hat{p}_k \partial_l (\varphi p_l p_k) = \frac{\lambda}{\alpha^2} \hat{\mathbf{p}} \cdot \nabla \varphi + \frac{2\mu}{\alpha} \nabla \cdot (\varphi \mathbf{p}), \quad (47)$$

where the two μ terms are equal by the symmetry of $p_l p_k$, and where we have expanded $\hat{p}_k \partial_l (\varphi p_l p_k) = \hat{p}_k p_k \partial_l (\varphi p_l) + \varphi p_l \hat{p}_k \partial_l p_k$ and used $\hat{p}_k p_k = 1/\alpha$ and $p_l \partial_l p_k = 0$. The second term gives

$$\begin{aligned} \hat{p}_i A_{ijkl} p_j \partial_l (\varphi p_k) &= \lambda \hat{p}_j p_j \partial_k (\varphi p_k) + \mu \hat{p}_k p_j \partial_j (\varphi p_k) + \mu \hat{p}_l p_j \partial_l (\varphi p_j) \\ &= \frac{\lambda}{\alpha} \nabla \cdot (\varphi \mathbf{p}) + \frac{2\mu}{\alpha^2} \hat{\mathbf{p}} \cdot \nabla \varphi, \end{aligned} \quad (48)$$

since $\hat{p}_k p_j \partial_j (\varphi p_k) = \hat{p}_k p_k p_j \partial_j \varphi = \alpha^{-2} \hat{\mathbf{p}} \cdot \nabla \varphi$ and $\hat{p}_l p_j \partial_l (\varphi p_j) = \|\mathbf{p}\|^2 \hat{p}_l \partial_l \varphi = \alpha^{-2} \hat{\mathbf{p}} \cdot \nabla \varphi$. Adding eqs. (47) and (48), and using $\hat{\mathbf{p}} = \alpha \mathbf{p}$,

$$\hat{p}_i F_i = \frac{\lambda + 2\mu}{\alpha} [\mathbf{p} \cdot \nabla \varphi + \nabla \cdot (\varphi \mathbf{p})] = \frac{\lambda + 2\mu}{\alpha} [2\mathbf{p} \cdot \nabla \varphi + \varphi \nabla \cdot \mathbf{p}] = \frac{\lambda + 2\mu}{\alpha \varphi} \nabla \cdot (\varphi^2 \mathbf{p}). \quad (49)$$

Setting this to zero, and recalling that $\mathbf{p} = \nabla T$ and that $\varphi = \alpha A_0$ with α constant, we arrive at the transport equation

$$\nabla \cdot (A_0^2 \nabla T) = 0. \quad (50)$$

This is the two- and three-dimensional counterpart of eq. (39). In a heterogeneous isotropic medium the same calculation, with rather more bookkeeping, gives the transport equation in the form

$$\nabla \cdot (\rho \alpha^2 A_0^2 \nabla T) = 0, \quad (51)$$

of which eq. (39) is the one-dimensional case with $\rho c^2 = E$.

For the point source $\nabla T = \hat{\mathbf{x}}/\alpha$, and in plane polar coordinates eq. (50) becomes

$$\frac{1}{r} \frac{d}{dr} (r A_0^2) = 0, \quad \text{so that} \quad A_0(r) = \frac{C}{\sqrt{r}}, \quad (52)$$

while in three dimensions the same argument with spherical wavefronts gives $A_0 \propto 1/r$. Physically, $\rho \alpha \omega^2 A_0^2$ times the length $2\pi r$ of the wavefront is the power crossing it, and this must be the same for every r . The amplitude is infinite at $r = 0$, where all of the rays intersect: the source point is a caustic, in fact the most degenerate kind, and ray theory can describe the field radiated by the source but not the source region itself. In the script `verify_examples5.py` the full right hand side F_i of eq. (46) is evaluated by finite differences for the isotropic tensor with $\mathbf{a}^0 = r^{-1/2} \hat{\mathbf{p}}$, and $\hat{p}_i F_i$ is found to vanish to rounding error, whereas it does not for $\mathbf{a}^0 = r^{-1} \hat{\mathbf{p}}$.

Example 5: Where is the approximation? For a plane wave entering a region in which the material parameters vary smoothly over a length scale L , estimate the size of the neglected $n = 1$ term of the ray series relative to the $n = 0$ term. Express the result in terms of the frequency ω and the time $\tau = L/\alpha$ taken by the wave to cross the heterogeneity, and in terms of the wavelength λ . Evaluate it for $L = 1000 \text{ km}$, $\alpha = 10 \text{ km s}^{-1}$ and a period of 20 s, and again for $L = 50 \text{ km}$.

Solution. The ratio of successive terms in the ray series is $|\mathbf{a}^1|/(\omega|\mathbf{a}^0|)$, so we need the size of $\mathbf{a}^1$, which is fixed by the order $i\omega$ equation, eq. (46). On the left, $A_{ijkl} p_j p_l$ and $\rho \delta_{ik}$ are each of order ρ , since $A_{ijkl} \sim \rho \alpha^2$ and $\|\mathbf{p}\| = 1/\alpha$. On the right, each term contains one spatial derivative acting on a product of A_{ijkl} , $\mathbf{p}$ and $\mathbf{a}^0$, all of which vary over the length scale L of the heterogeneity (the travel time and amplitude inherit their length scale from the medium), and so

$$\rho |\mathbf{a}^1| \sim \frac{\rho \alpha^2}{\alpha} \frac{|\mathbf{a}^0|}{L}, \quad \text{i.e.} \quad |\mathbf{a}^1| \sim \frac{\alpha}{L} |\mathbf{a}^0|. \quad (53)$$

The matrix on the left of eq. (46) is singular, so strictly this fixes only the components of $\mathbf{a}^1$ orthogonal to $\hat{\mathbf{a}}^0$; the remaining component is fixed by the solvability condition of the $n = 2$ equation, which has the same structure and gives the same estimate. The one-dimensional case makes this concrete: eq. (37) with $n = 1$ reads $2\rho c a^{1'} + (\rho c)' a^1 = (E a^{0'})'$, whose right hand side is of order $E a^0/L^2$, so that over a distance L we obtain $a^1 \sim E a^0/(\rho c L) = c a^0/L$. Hence the relative size of the first correction is

$$\frac{|\mathbf{a}^1|}{\omega |\mathbf{a}^0|} \sim \frac{\alpha}{\omega L} = \frac{1}{\omega \tau} = \frac{\lambda}{2\pi L}, \quad (54)$$

where we have used $\omega = 2\pi\alpha/\lambda$. The ray series is thus an expansion in the number of wavelengths, over 2π , that fit inside the heterogeneity, or equivalently in the number of radians of phase the wave accumulates while crossing it; this is the quantitative form of the condition $L/\lambda \gg 1$ under which the lecture introduced ray theory. For the first set of numbers, $\lambda = 200 \text{ km}$, $\tau = 100 \text{ s}$, $\omega = 0.314 \text{ s}^{-1}$ and $\omega\tau = 31.4$, so the neglected term is about 3% of the leading one and ray theory should work well; heterogeneity on this scale is typical of the lower mantle. For $L = 50 \text{ km}$, which is

more representative of crustal structure, the ratio is 0.64 and the expansion is useless at this period: the wave is scattered rather than refracted, and one is in the regime $L/\lambda \approx 1$ where the Born approximation, not ray theory, is the appropriate tool. Note finally that this estimate is local. Even when it is small, the errors accumulate along a ray, and it says nothing about the smooth, possibly large, error term discussed in the lecture, nor about caustics, where $\mathbf{a}^0$ itself fails.

Worked examples for Lecture 17: The spherical Earth

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These examples accompany Lecture 17. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Throughout, b denotes the radius of the earth model, $\alpha(r)$ the P-wave speed, and $q = r \sin \theta / \alpha$ the ray parameter conserved along each ray. Numerical results were checked with the scripts in the python directory.

Example 1: The homogeneous sphere. Consider a spherically symmetric earth model of radius b in which the P-wave speed α is constant. For a ray leaving the surface with ray parameter q , evaluate the lecture's integrals for the epicentral angle Δ and the travel time T in closed form. Check the results against the elementary geometry of a chord, and verify directly that $dT/d\Delta = q$.

Solution. When α is constant the turning-point condition $q = r_t / \alpha(r_t)$ gives $r_t = \alpha q$, and the integral for the epicentral angle becomes

$$\Delta = \int_{\alpha q}^b \frac{2\alpha q}{r \sqrt{r^2 - \alpha^2 q^2}} dr. \quad (1)$$

Writing $a = \alpha q$ for the moment, a direct differentiation shows that

$$\frac{d}{dr} \arccos\left(\frac{a}{r}\right) = \frac{a}{r^2} \frac{1}{\sqrt{1 - a^2/r^2}} = \frac{a}{r \sqrt{r^2 - a^2}}, \quad (2)$$

so the integrand in eq. (1) is an exact derivative and

$$\Delta = 2 \left[\arccos\left(\frac{\alpha q}{r}\right) \right]_{\alpha q}^b = 2 \arccos\left(\frac{\alpha q}{b}\right), \quad (3)$$

the contribution from the lower limit vanishing because $\arccos 1 = 0$. In the same way, because $d\sqrt{r^2 - a^2}/dr = r/\sqrt{r^2 - a^2}$, the travel-time integral gives

$$T = \int_{\alpha q}^b \frac{2r}{\alpha \sqrt{r^2 - \alpha^2 q^2}} dr = \frac{2}{\alpha} \sqrt{b^2 - \alpha^2 q^2}. \quad (4)$$

We can check these results against elementary geometry. When α is constant the second of Hamilton's equations gives $d\mathbf{p}/d\sigma = \mathbf{0}$, so the rays are straight lines, and a ray between two surface points separated by the epicentral angle Δ is a chord of length $2b \sin(\Delta/2)$. Its travel time is therefore

$$T = \frac{2b}{\alpha} \sin \frac{\Delta}{2}. \quad (5)$$

From eq. (3) we have $\alpha q/b = \cos(\Delta/2)$, and hence $\sqrt{b^2 - \alpha^2 q^2} = b \sin(\Delta/2)$, which shows that eq. (4) agrees with eq. (5). The take-off angle is also consistent with the geometry: at the surface $q = b \sin \theta_0 / \alpha$, so $\sin \theta_0 = \cos(\Delta/2)$ and the chord makes an angle $\pi/2 - \Delta/2$ with the inward radius, exactly the base angle of the isosceles triangle formed by the chord and the two radii.

Finally, differentiating eq. (5) with respect to Δ and using $\cos(\Delta/2) = \alpha q/b$,

$$\frac{dT}{d\Delta} = \frac{b}{\alpha} \cos \frac{\Delta}{2} = q. \quad (6)$$

Note that Δ decreases as q increases, since from eq. (3) $d\Delta/dq = -2\alpha/\sqrt{b^2 - \alpha^2 q^2} < 0$: rays that leave the surface closer to the horizontal (larger q) travel less far. Consequently $d^2T/d\Delta^2 = dq/d\Delta < 0$, and the travel-time curve is concave, as are the observed curves shown in the lecture. The result $dT/d\Delta = q$ is not special to the homogeneous sphere, as the next example shows.

Example 2: The Benndorf relation. Let $\alpha(r)$ be such that $r/\alpha(r)$ increases monotonically with r , so that every ray leaving the surface with $0 < q < b/\alpha(b)$ has a unique turning point $r_t(q)$ determined by $q = r_t/\alpha(r_t)$. Starting from the lecture's integrals for $T(q)$ and $\Delta(q)$, show that

$$\frac{dT}{dq} = q \frac{d\Delta}{dq},$$

and hence that $dT/d\Delta = q$ along the travel-time curve. Explain how this allows the ray parameter, and hence the P-wave speed at the turning point, to be read off an observed travel-time curve.

Solution. Both T and Δ depend on q through the integrand and through the lower limit $r_t(q)$, and the integrands are singular at r_t . It is easier to work with the combination $\tau(q) = T - q\Delta$. Subtracting q times the Δ integral from the T integral and placing the two terms over a common denominator,

$$\tau(q) = T - q\Delta = \int_{r_t}^b \frac{2(r^2 - \alpha^2 q^2)}{\alpha r \sqrt{r^2 - \alpha^2 q^2}} dr = \int_{r_t}^b \frac{2}{\alpha r} \sqrt{r^2 - \alpha^2 q^2} dr. \quad (7)$$

This integrand is continuous and vanishes at the lower limit, because $r_t = \alpha(r_t)q$. Differentiating with respect to q using Leibniz's rule, the term arising from the motion of the lower limit is proportional to the integrand evaluated at r_t and so is zero, while differentiating under the integral sign gives

$$\frac{d\tau}{dq} = \int_{r_t}^b \frac{2}{\alpha r} \frac{-\alpha^2 q}{\sqrt{r^2 - \alpha^2 q^2}} dr = - \int_{r_t}^b \frac{2\alpha q}{r \sqrt{r^2 - \alpha^2 q^2}} dr = -\Delta(q). \quad (8)$$

The differentiated integrand is singular at r_t , but the singularity is of inverse square-root type and is integrable, and the interchange of differentiation and integration is legitimate. On the other hand, differentiating the definition $\tau = T - q\Delta$ directly,

$$\frac{d\tau}{dq} = \frac{dT}{dq} - \Delta - q \frac{d\Delta}{dq}, \quad (9)$$

and comparison with eq. (8) gives

$$\frac{dT}{dq} = q \frac{d\Delta}{dq}. \quad (10)$$

Wherever $d\Delta/dq \neq 0$ we can regard T as a function of Δ along the travel-time curve, and dividing eq. (10) by $d\Delta/dq$ yields

$$\frac{dT}{d\Delta} = q, \quad (11)$$

which is known as the Benndorf relation. Example 1 verified it for the homogeneous sphere.

The relation has a simple physical meaning. At the surface $q = b \sin \theta_0 / \alpha(b)$, so q/b is the horizontal component of the slowness vector of the emerging wavefront; equivalently the wavefront sweeps along the surface with apparent speed $b d\Delta/dT = b/q$. Its importance for the determination of Earth structure is the following. Suppose that we have measured the travel-time curve $T(\Delta)$ of a phase such as P. Its slope at a distance Δ gives the ray parameter $q(\Delta)$ of the ray emerging there, without any knowledge of the velocity model. The turning-point condition then tells us that

$$\frac{r_t}{\alpha(r_t)} = q(\Delta) \quad (12)$$

for the deepest point of that ray. A single number cannot fix both r_t and $\alpha(r_t)$, but the missing information is contained in the way q varies with Δ . Indeed, the integral in eq. (8) is of Abel type, and it can be inverted to give the turning radius r_1 of the ray with parameter q_1 arriving at Δ_1 as

$$\ln \frac{b}{r_1} = \frac{1}{\pi} \int_0^{\Delta_1} \cosh^{-1} \left[\frac{q(\Delta)}{q_1} \right] d\Delta, \quad (13)$$

after which $\alpha(r_1) = r_1/q_1$. This is the Herglotz–Wiechert formula, and it was the basis of the earliest determinations of the Earth's radial velocity structure. Its derivation requires Δ to be a monotonic function of q , which fails when r/α is not monotonic, for instance across a low-velocity zone, and this is one reason why the ad hoc methods mentioned in the lecture were needed in practice. For the homogeneous sphere the right-hand side of eq. (13) can be evaluated numerically using $q(\Delta) = (b/\alpha) \cos(\Delta/2)$, and it reproduces $\ln(b/r_1) = -\ln \cos(\Delta_1/2)$, as required by $r_1 = \alpha q_1$.

Example 3: The flat-earth transformation. Given a spherically symmetric model $\alpha(r)$, define a depth coordinate z and a transformed wave speed $\alpha_f(z)$ by

$$z = b \ln \frac{b}{r}, \quad \alpha_f(z) = \frac{b}{r} \alpha(r),$$

and set $q_f = q/b$. Show that the spherical integrals for T and Δ become the integrals for a horizontally stratified half-space obtained in Problem 7 of the first problem set, with q_f as the ray parameter and horizontal distance $X = b\Delta$. Use the transformation to find $\Delta(q)$ and $T(q)$ in closed form for the power-law model

$$\alpha(r) = \alpha_0 \left(\frac{b}{r} \right)^\nu, \quad \nu > -1,$$

and comment on the cases $\nu = 0$, $\nu = 1$ and $\nu \rightarrow -1$.

Solution. From $z = b \ln(b/r)$ we have $r = b e^{-z/b}$, so that $dr/r = -dz/b$, and the range $r_t \leq r \leq b$ maps to $0 \leq z \leq z_t$ with $z_t = b \ln(b/r_t)$, the orientation of the interval being reversed. The definitions of α_f and q_f have been arranged so that

$$\frac{\alpha(r)q}{r} = \frac{b}{r} \alpha(r) \frac{q}{b} = \alpha_f(z) q_f, \quad (14)$$

which is just the statement that $\sin \theta$ takes the same value at corresponding points of the two models. Dividing the numerator and denominator of the lecture's Δ integrand by r and using eq. (14),

$$\Delta = \int_{r_t}^b \frac{2\alpha q/r}{\sqrt{1 - \alpha^2 q^2/r^2}} \frac{dr}{r} = \frac{1}{b} \int_0^{z_t} \frac{2q_f \alpha_f}{\sqrt{1 - q_f^2 \alpha_f^2}} dz = \frac{X}{b}, \quad (15)$$

where X is precisely the horizontal distance integral of Problem 7. For the travel time we use $1/\alpha = (b/r)/\alpha_f$ together with $(b/r) dr = -dz$ to obtain

$$T = \int_{r_t}^b \frac{2}{\alpha \sqrt{1 - \alpha^2 q^2/r^2}} dr = \int_0^{z_t} \frac{2}{\alpha_f \sqrt{1 - q_f^2 \alpha_f^2}} dz, \quad (16)$$

which is the flat-earth travel time. The turning-point conditions also correspond, because $q = r_t/\alpha(r_t)$ is equivalent to $q_f \alpha_f(z_t) = 1$. The transformation is exact for the kinematics of rays in any spherically symmetric model, and it allows methods developed for horizontally stratified media to be applied to the Earth.

For the power-law model the transformed wave speed is

$$\alpha_f(z) = \frac{b}{r} \alpha_0 \left(\frac{b}{r} \right)^\nu = \alpha_0 e^{(1+\nu)z/b} = \alpha_0 e^{z/h}, \quad h = \frac{b}{1+\nu}, \quad (17)$$

so that every power law becomes an exponential flat-earth model, with scale height $h > 0$ precisely when $\nu > -1$. In the flat-earth integrals we substitute $u = q_f \alpha_f(z) = q_f \alpha_0 e^{z/h}$, for which $dz = h du/u$, and u runs from $u_0 = q_f \alpha_0 = \alpha_0 q/b$ at the surface to 1 at the turning depth. Then

$$X = 2h \int_{u_0}^1 \frac{du}{\sqrt{1 - u^2}} = 2h \arccos u_0, \quad (18)$$

and, using $1/\alpha_f = q_f/u$ together with $d(-\sqrt{1 - u^2}/u)/du = 1/(u^2 \sqrt{1 - u^2})$,

$$T = 2h q_f \int_{u_0}^1 \frac{du}{u^2 \sqrt{1 - u^2}} = 2h q_f \left[-\frac{\sqrt{1 - u^2}}{u} \right]_{u_0}^1 = \frac{2h}{\alpha_0} \sqrt{1 - u_0^2}. \quad (19)$$

Transforming back with $\Delta = X/b$, $h = b/(1 + \nu)$ and $u_0 = \alpha_0 q/b$, we arrive at

$$\Delta(q) = \frac{2}{1 + \nu} \arccos \left(\frac{\alpha_0 q}{b} \right), \quad T(q) = \frac{2}{(1 + \nu) \alpha_0} \sqrt{b^2 - \alpha_0^2 q^2}, \quad (20)$$

with turning radius $r_t = b e^{-z_t/b} = b(\alpha_0 q/b)^{1/(1+\nu)}$, which agrees with the direct solution of $q = r_t/\alpha(r_t)$. Eliminating q , the travel-time curve is

$$T(\Delta) = \frac{2b}{(1+\nu)\alpha_0} \sin \frac{(1+\nu)\Delta}{2}, \quad (21)$$

and differentiating gives $dT/d\Delta = (b/\alpha_0) \cos[(1+\nu)\Delta/2] = q$, in agreement with the Benndorf relation. These closed forms were checked against numerical quadrature of the original spherical integrals; for example with $b = 6371$ km, $\alpha_0 = 8$ km s⁻¹, $\nu = 1$ and $q = 300$ s rad⁻¹ we find $\Delta = 67.87^\circ$, $T = 737.7$ s and $r_t = 3910$ km.

For $\nu = 0$ we recover the results of Example 1. For $\nu = 1$, in which $\alpha \propto 1/r$, a ray with a given take-off angle subtends exactly half the epicentral angle, and takes exactly half the time, of the corresponding ray in the homogeneous sphere with $\alpha = \alpha_0$: the increase of wave speed with depth bends the ray back to the surface more quickly. Since $\alpha \rightarrow \infty$ as $r \rightarrow 0$, this model is only sensible for a shell such as the mantle, and eq. (20) should be used for rays whose turning radius lies within that shell; with the numbers above the surface speed of 8 km s⁻¹ increases to 14.6 km s⁻¹ at $r = 3480$ km. As $\nu \rightarrow -1$ the scale height $h \rightarrow \infty$, and for $\nu = -1$ the flat-earth model is homogeneous, so that the transformed rays are straight lines that never turn. In the sphere $\alpha \propto r$, the quantity $\sin \theta = \alpha q/r$ is constant along each ray, and the rays are logarithmic spirals that wind into the centre. More generally, a ray turns only where r/α increases outwards, that is where $d\alpha/dr < \alpha/r$; a wave speed that increases with depth faster than this critical rate, as happens for $\nu < -1$, produces rays that never return to the surface.

Example 4: Hypocentre location by linearised least squares. Consider a homogeneous half-space with known P-wave speed α , and use coordinates in which x_3 measures depth below the free surface. Seismometers at surface positions $\mathbf{x}_i$ ($i = 1, \dots, N_s$) record the first arrival times t_i from an event with unknown hypocentre $\mathbf{x}_h$ and origin time t_h . (a) Show that

$$\frac{\partial T(\mathbf{x}_i, \mathbf{x}_h)}{\partial \mathbf{x}_h} = -\frac{\mathbf{x}_i - \mathbf{x}_h}{\alpha \|\mathbf{x}_i - \mathbf{x}_h\|},$$

and interpret the result. (b) Set up the Gauss–Newton iteration for minimising the least-squares misfit $J(\mathbf{x}_h, t_h)$ of the lecture. (c) Show that if all the seismometers lie at the same horizontal distance from the epicentre, the normal matrix is singular, and identify the combination of parameters that cannot be determined. (d) Illustrate the iteration numerically with $\alpha = 6$ km s⁻¹ and five stations.

Solution. (a) In a homogeneous medium rays are straight lines, so $T(\mathbf{x}_i, \mathbf{x}_h) = \|\mathbf{x}_i - \mathbf{x}_h\|/\alpha$. Using $\partial \|\mathbf{y}\|/\partial \mathbf{y} = \mathbf{y}/\|\mathbf{y}\|$ with $\mathbf{y} = \mathbf{x}_i - \mathbf{x}_h$, and the chain rule with $\partial \mathbf{y}/\partial \mathbf{x}_h = -\mathbf{I}$,

$$\frac{\partial T(\mathbf{x}_i, \mathbf{x}_h)}{\partial \mathbf{x}_h} = -\frac{1}{\alpha} \hat{\mathbf{n}}_i, \quad \hat{\mathbf{n}}_i = \frac{\mathbf{x}_i - \mathbf{x}_h}{\|\mathbf{x}_i - \mathbf{x}_h\|}. \quad (22)$$

Here $\hat{\mathbf{n}}_i/\alpha$ is the slowness vector $\mathbf{p}$ of the ray at the source, so the derivative is minus the slowness vector at the source. This is a general property of ray theory: the travel time increases at the rate $\|\mathbf{p}\| = 1/\alpha$ when the source is moved away from the receiver

along the ray, and does not change to first order when it is moved perpendicular to the ray. Moving the source has the same effect as moving the receiver in the opposite direction, which is why the sign is negative. The derivative with respect to the origin time is simply $\partial(t_h + T)/\partial t_h = 1$.

(b) Collect the unknowns into $\mathbf{m} = (\mathbf{x}_h, t_h)$ and define the weighted residuals $r_i(\mathbf{m}) = [t_h + T(\mathbf{x}_i, \mathbf{x}_h) - t_i]/\sigma_i$, so that $J = \sum_i r_i^2$. Given a current estimate $\mathbf{m}$, the residuals for a nearby model are linearised as

$$r_i(\mathbf{m} + \delta\mathbf{m}) \approx r_i(\mathbf{m}) + \sum_{j=1}^4 G_{ij} \delta m_j, \quad G_{ij} = \frac{1}{\sigma_i} \frac{\partial(t_h + T_i)}{\partial m_j}, \quad (23)$$

where by eq. (22) the i th row of the $N_s \times 4$ matrix $\mathbf{G}$ is $\sigma_i^{-1}(-\hat{\mathbf{n}}_i^T/\alpha, 1)$. Minimising the resulting quadratic approximation to J with respect to $\delta\mathbf{m}$ gives the normal equations

$$\mathbf{G}^T \mathbf{G} \delta\mathbf{m} = -\mathbf{G}^T \mathbf{r}, \quad (24)$$

after which $\mathbf{m}$ is replaced by $\mathbf{m} + \delta\mathbf{m}$ and the process is repeated until $\delta\mathbf{m}$ is negligible. Each iteration requires the travel times and their derivatives for the current hypocentre, which in a realistic earth model are obtained by ray tracing. At least four stations in general position are needed for the 4×4 normal matrix $\mathbf{G}^T \mathbf{G}$ to be invertible.

(c) Let the hypocentre lie at depth h beneath the origin of horizontal coordinates, and let the i th station lie at horizontal distance d_i , so that $D_i = \|\mathbf{x}_i - \mathbf{x}_h\| = \sqrt{d_i^2 + h^2}$. Because the stations are on the surface, the depth component of $\hat{\mathbf{n}}_i$ is $-h/D_i$, and the third and fourth entries of the i th row of $\mathbf{G}$ are

$$G_{i3} = \frac{1}{\sigma_i} \frac{h}{\alpha D_i}, \quad G_{i4} = \frac{1}{\sigma_i}. \quad (25)$$

If all the stations are at the same horizontal distance d , then $D_i = D$ for every i and the third column of $\mathbf{G}$ is $h/(\alpha D)$ times the fourth. The vector $\mathbf{v} = (0, 0, 1, -h/(\alpha D))^T$ then satisfies $\mathbf{G}\mathbf{v} = \mathbf{0}$, so the normal matrix has a null vector and eq. (24) cannot be solved uniquely. Physically, deepening the source by δh lengthens every ray by the same amount and delays every arrival by $h \delta h/(\alpha D)$, which can be undone exactly by making the origin time earlier by the same amount. In this configuration the trade-off is in fact exact rather than merely linear, since all the predicted arrival times depend on h and t_h only through the single combination $t_h + \sqrt{d^2 + h^2}/\alpha$. The depth of an earthquake is resolved only by stations at a range of epicentral distances, for which the take-off angles differ, and in practice depth is the most poorly determined of the four hypocentral parameters.

(d) We take stations at the horizontal positions (30, 0), (0, 40), (−25, 15), (−10, −35) and (20, −20) km, a true hypocentre at (3, −2, 12) km with $t_h = 0$, and equal uncertainties $\sigma_i = 0.1$ s. The synthetic arrival times are 4.936, 7.297, 5.814, 6.241 and 4.586 s, with no noise added so that the exact answer is known. Starting from $\mathbf{x}_h = (0, 0, 5)$ km and $t_h = 1$ s, the Gauss–Newton iteration gives

iteration	$x_{h,1}$ (km)	$x_{h,2}$ (km)	$x_{h,3}$ (km)	t_h (s)
0	0.000	0.000	5.000	1.0000
1	2.795	−1.868	13.377	0.0966
2	3.004	−2.001	12.034	0.0017
3	3.000	−2.000	12.000	0.0000

with the misfit falling from $J = 370$ to 18 , 6.7×10^{-3} and 3.4×10^{-9} . The convergence is quadratic once the iterate is close to the solution, as expected for a Gauss–Newton method applied to data that can be fitted exactly. If instead the same five stations are placed at azimuths $0^\circ, 72^\circ, \dots, 288^\circ$ on a circle of radius 30 km about the epicentre of a source at 12 km depth, the eigenvalues of $\mathbf{G}^T \mathbf{G}$ are found to be 0, 5.99, 5.99 and 502 (in units of s^{-2} with distances in kilometres), the null vector being proportional to $(0, 0, 1, -0.0619)$ in agreement with $h/(\alpha D) = 12/(6 \times 32.31) = 0.0619$ from eq. (25).

Example 5: The P-wave shadow zone. Consider a two-layer earth model of radius $b = 6371$ km consisting of a homogeneous mantle with $\alpha_m = 13 \text{ km s}^{-1}$ above a homogeneous core of radius $c = 3480$ km with $\alpha_c = 8 \text{ km s}^{-1}$. For a source at the surface: (a) find the epicentral angle Δ_1 of the mantle ray that grazes the core; (b) derive $\Delta(q)$ and $T(q)$ for rays refracted through the core, using the fact that the ray parameter is continuous across the core–mantle boundary (CMB); (c) find the smallest angular distance at which core-refracted P-waves arrive, and hence the extent of the P-wave shadow zone. Compare with the observed shadow zone, which extends from about 103° to 142° .

Solution. (a) In the homogeneous mantle rays are straight lines, and by Example 1 a ray with parameter q would turn at radius $\alpha_m q$. It remains in the mantle if $\alpha_m q > c$, and it grazes the core when $q = q_1 = c/\alpha_m = 267.7 \text{ s rad}^{-1}$. Its epicentral angle follows from eq. (3),

$$\Delta_1 = 2 \arccos \left(\frac{\alpha_m q_1}{b} \right) = 2 \arccos \frac{c}{b} = 113.8^\circ, \quad (26)$$

and its travel time from eq. (4) is $T_1 = 2\sqrt{b^2 - c^2}/\alpha_m = 821 \text{ s}$. Mantle P-waves therefore arrive at all distances $0 < \Delta < 113.8^\circ$, and at no greater distance, as the grazing ray is the one that leaves the surface most steeply, with $\sin \theta_0 = c/b$, that is at 33.1° from the vertical, without entering the core.

(b) A ray with $q < q_1$ reaches the CMB. The down-going mantle leg from $r = b$ to $r = c$ subtends an angle obtained from the antiderivative in eq. (2),

$$\Delta_m(q) = \int_c^b \frac{\alpha_m q}{r \sqrt{r^2 - \alpha_m^2 q^2}} dr = \arccos \left(\frac{\alpha_m q}{b} \right) - \arccos \left(\frac{\alpha_m q}{c} \right). \quad (27)$$

At the CMB the ray is refracted. Since $q = r \sin \theta / \alpha$ and $r = c$ on both sides of the boundary, continuity of q is the statement that $\sin \theta_c / \alpha_c = \sin \theta_m / \alpha_m$, which is Snell's law; it expresses the fact that the wavefront, and hence the phase along the boundary, is continuous across it. Because $\alpha_c < \alpha_m$ the ray is bent towards the radial direction on entering the core, and there is no critical angle: every mantle ray that reaches the CMB is transmitted. Inside the homogeneous core the ray is again a straight line with the same q , and its turning radius $\alpha_c q$ is automatically less than c . By eq. (3) with b replaced by c , the core leg subtends $2 \arccos(\alpha_c q / c)$, and adding the symmetric up-going mantle leg,

$$\Delta(q) = 2 \arccos \left(\frac{\alpha_m q}{b} \right) - 2 \arccos \left(\frac{\alpha_m q}{c} \right) + 2 \arccos \left(\frac{\alpha_c q}{c} \right), \quad 0 \leq q < q_1. \quad (28)$$

The travel time is assembled in the same way from eq. (4),

$$T(q) = \frac{2}{\alpha_m} \left[\sqrt{b^2 - \alpha_m^2 q^2} - \sqrt{c^2 - \alpha_m^2 q^2} \right] + \frac{2}{\alpha_c} \sqrt{c^2 - \alpha_c^2 q^2}, \quad (29)$$

and one may check, either directly or numerically, that $dT/dq = q d\Delta/dq$ continues to hold, so that the Benndorf relation survives the presence of the discontinuity.

(c) At the two ends of the range of q we have

$$\lim_{q \rightarrow q_1} \Delta(q) = \Delta_1 + 2 \arccos \frac{\alpha_c}{\alpha_m} = 113.8^\circ + 104.0^\circ = 217.8^\circ, \quad \Delta(0) = 180^\circ, \quad (30)$$

the first because a ray entering the core at grazing incidence is refracted to an angle θ_c from the radial direction given by $\sin \theta_c = \alpha_c/\alpha_m$, that is 38.0° , and then crosses the core along a long chord, the second because the vertical ray passes through the centre. As q decreases from q_1 the second term in eq. (28) grows rapidly, with $d\Delta/dq \rightarrow +\infty$ at q_1 , and Δ falls steeply before rising again towards 180° . Minimising eq. (28) numerically, the stationary point is at $q_B = 157.6 \text{ s rad}^{-1}$ where $\Delta_B = 172.1^\circ$. This is a caustic, at which neighbouring rays cross and ray theory predicts an infinite amplitude, and it is the analogue for this model of the caustic responsible for the strong PKP arrivals observed near 144° in the Earth.

The stationary point is not, however, the nearest point to the source at which core-refracted energy arrives. An epicentral angle $\Delta > 180^\circ$ means that the ray has passed beyond the antipode, and the station lies at the angular distance $360^\circ - \Delta$ from the source on the far side. The relevant quantity is therefore $\delta(q) = \min[\Delta(q), 360^\circ - \Delta(q)]$, and a numerical search over $0 \leq q < q_1$ shows that its infimum is attained in the grazing limit,

$$\delta_{\min} = 360^\circ - \lim_{q \rightarrow q_1} \Delta(q) = 360^\circ - 113.8^\circ - 104.0^\circ = 142.2^\circ. \quad (31)$$

Core-refracted P-waves thus arrive at angular distances between 142.2° and 180° ; between 172.1° and 180° three distinct core rays arrive at each distance, one from each monotonic branch of $\Delta(q)$, while between 142.2° and 172.1° there is a single arrival, from the branch with q close to q_1 . Combining this with part (a), no geometrical P-wave arrives in the interval

$$113.8^\circ < \delta < 142.2^\circ, \quad (32)$$

which is the P-wave shadow zone of this model. Figure 1 shows a fan of rays, the gap in surface coverage being clearly visible.

The far edge of the shadow zone agrees remarkably well with the observed value of 142° , although this is partly fortuitous: in the real Earth it is set by the PKP caustic rather than by the grazing ray, and its position depends on the details of the core velocity structure. The near edge is predicted at 113.8° rather than the observed 103° because the wave speed of the real mantle increases with depth. The mantle legs of the grazing ray are then curved back towards the surface and each subtends less than $\arccos(c/b) = 56.9^\circ$, and the whole of Oldham's shadow zone moves closer to the source. Within the shadow, seismograms are not silent: the core reflection PcP, waves diffracted around the CMB, and S-waves and surface waves are all recorded there, but the sharp, direct P-wave that dominates the first arrival at shorter distances is absent, and it was this absence that led Oldham to infer the existence of the core.

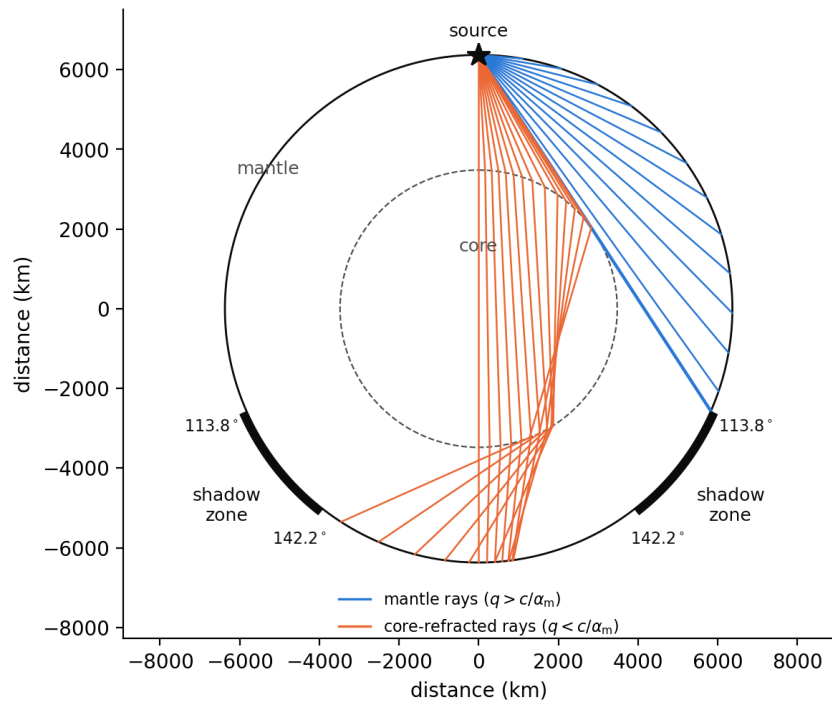


Fig. 1: Ray paths for a surface source in the two-layer model of Example 5. Mantle rays (blue) are straight lines that emerge at distances up to 113.8° ; rays that enter the core (orange) are refracted towards the radial direction at the CMB and emerge no closer to the source than 142.2° . The thick arcs mark the shadow zone on either side of the source.

Worked examples for Lecture 18: Delay time tomography

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These examples accompany Lecture 18. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Throughout we use a two-block toy model. The reference P-wave velocity is a constant α , and the model consists of two adjacent blocks, each of length ℓ , in which the velocity is perturbed by constants $\delta\alpha_1$ and $\delta\alpha_2$. Rays are straight lines that pass along the blocks, and so the basis functions of the lecture are the indicator functions φ_j of the blocks. The elements of the matrix $\mathbf{A}$ are then

$$A_{ij} = - \int_{\text{ray}_i} \frac{\varphi_j}{\alpha^2} ds = - \frac{\ell_{ij}}{\alpha^2}, \quad (1)$$

where ℓ_{ij} is the length of the i th ray within the j th block. The factor $-1/\alpha^2$ is common to every element, and it is tidier to take as model parameters the perturbations to the slowness $1/\alpha$ in each block,

$$\delta u_j = \delta \left(\frac{1}{\alpha} \right)_j = - \frac{\delta\alpha_j}{\alpha^2}, \quad (2)$$

so that $\mathbf{m} = (\delta u_1, \delta u_2)^T$ and $A_{ij} = \ell_{ij}$ is simply a path length. The two descriptions differ only by a rescaling of the columns of $\mathbf{A}$, and the velocity perturbations can be read off at the end from $\delta\alpha_j/\alpha = -\alpha \delta u_j$. For numerical values we take $\ell = 100$ km and $\alpha = 8$ km s⁻¹, so that the unperturbed travel time along one block is 12.5 s.

Example 1: Toy tomography with two blocks. Consider the following three sets of straight rays through the two-block model.

- (a) A single ray that passes along both blocks, with delay time d_1 .
- (b) The ray of (a) together with a second ray that passes along block 1 only, with delay time d_2 .
- (c) The rays of (b) together with a third ray that passes along block 2 only, with delay time d_3 .

In each case write down $\mathbf{A}$ and the equation $\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{e}$. Find the null space of $\mathbf{A}$ in case (a), and solve the problem exactly in case (b). In case (c) the data

$d_1 = 1.2 \text{ s}$, $d_2 = 0.5 \text{ s}$ and $d_3 = 0.4 \text{ s}$ are inconsistent; find the least squares solution, the predicted data and the residuals, taking the errors to have equal variances.

Solution. In case (a) there is one datum and two unknowns, and by eq. (1) with slowness parameters

$$\mathbf{A} = \begin{pmatrix} \ell & \ell \end{pmatrix}, \quad d_1 = \ell \delta u_1 + \ell \delta u_2 + e_1. \quad (3)$$

The null space consists of the vectors $\mathbf{m}_0$ with $\mathbf{A}\mathbf{m}_0 = \mathbf{0}$, i.e. $\delta u_1 + \delta u_2 = 0$, and so is the line spanned by $(1, -1)^T$. The single ray measures the sum of the two slowness perturbations and says nothing whatever about their difference: adding any multiple of $(1, -1)^T$ to a solution leaves the datum unchanged, and the problem has no unique solution even for error-free data. Note that $\mathbf{A}^T \mathbf{A} = \ell^2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has eigenvectors $(1, 1)^T$ and $(1, -1)^T$ with eigenvalues $2\ell^2$ and 0; the zero eigenvalue is the null space again.

In case (b) the second ray has length ℓ in block 1 and zero in block 2, so

$$\mathbf{A} = \begin{pmatrix} \ell & \ell \\ \ell & 0 \end{pmatrix}, \quad \det \mathbf{A} = -\ell^2 \neq 0, \quad \mathbf{A}^{-1} = \frac{1}{\ell} \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}. \quad (4)$$

The null space is now trivial and, neglecting errors, the unique solution is

$$\delta u_1 = \frac{d_2}{\ell}, \quad \delta u_2 = \frac{d_1 - d_2}{\ell}. \quad (5)$$

With $d_1 = 1.2 \text{ s}$ and $d_2 = 0.5 \text{ s}$ this gives $\delta u_1 = 5 \times 10^{-3} \text{ s km}^{-1}$ and $\delta u_2 = 7 \times 10^{-3} \text{ s km}^{-1}$, or $\delta \alpha_1 / \alpha = -4.0 \%$ and $\delta \alpha_2 / \alpha = -5.6 \%$: both blocks are slow, as they must be for positive delays. Because $\mathbf{A}$ is square and invertible the data are fit exactly whatever their errors, and the errors pass straight into the model, $\delta \mathbf{m} = \mathbf{A}^{-1} \mathbf{e}$. For uncorrelated errors of variance σ_d^2 the model covariance is

$$\sigma_d^2 \mathbf{A}^{-1} \mathbf{A}^{-T} = \frac{\sigma_d^2}{\ell^2} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad (6)$$

so block 2, which is seen only through the difference $d_1 - d_2$, has twice the variance of block 1, and the two estimates are anticorrelated.

In case (c) we have three data and two unknowns,

$$\mathbf{A} = \begin{pmatrix} \ell & \ell \\ \ell & 0 \\ 0 & \ell \end{pmatrix}, \quad \mathbf{d} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}. \quad (7)$$

Exact solutions exist only if $d_1 = d_2 + d_3$, and for the given data $d_2 + d_3 = 0.9 \text{ s} \neq d_1$. With equal variances the covariance matrix is $\mathbf{C} = \sigma_d^2 \mathbf{I}$, the factor σ_d^{-2} cancels from the least squares equations of the lecture, and we must solve $\mathbf{A}^T \mathbf{A} \mathbf{m} = \mathbf{A}^T \mathbf{d}$. Here

$$\mathbf{A}^T \mathbf{A} = \ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad (\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{3\ell^2} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{A}^T \mathbf{d} = \ell \begin{pmatrix} d_1 + d_2 \\ d_1 + d_3 \end{pmatrix}, \quad (8)$$

and hence

$$\mathbf{m} = \frac{1}{3\ell} \begin{pmatrix} d_1 + 2d_2 - d_3 \\ d_1 - d_2 + 2d_3 \end{pmatrix} = \begin{pmatrix} 6 \times 10^{-3} \\ 5 \times 10^{-3} \end{pmatrix} \text{ s km}^{-1}, \quad (9)$$

which corresponds to $\delta\alpha_1/\alpha = -4.8\%$ and $\delta\alpha_2/\alpha = -4.0\%$. The predicted data are $\mathbf{A}\mathbf{m} = (1.1, 0.6, 0.5)^T$ s, and the residuals are

$$\mathbf{A}\mathbf{m} - \mathbf{d} = -\frac{d_1 - d_2 - d_3}{3} \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} -0.1 \\ 0.1 \\ 0.1 \end{pmatrix} \text{ s.} \quad (10)$$

The residual vector is orthogonal to both columns of $\mathbf{A}$, as it must be at a least squares minimum, and is proportional to $(1, -1, -1)^T$: the only combination of the data that the model cannot reproduce is the inconsistency $d_1 - d_2 - d_3$, and least squares distributes it equally between the three measurements. The example shows the three regimes met in practice: an underdetermined problem with a null space, an exactly determined one in which noise is amplified, and an overdetermined one whose data can be fit only in the least squares sense.

Example 2: Regularised least squares in the toy problem. Return to case (a) of Example 1, with a single datum d , and measure delay times in units of their standard deviation so that $\mathbf{C} = \mathbf{I}$. Taking $\mathbf{B} = \mathbf{I}$, compute the regularised least squares solution

$$\mathbf{m}(\lambda) = (\mathbf{A}^T \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^T d \quad (11)$$

explicitly, and discuss its limits as $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$. Find the data misfit $\|\mathbf{A}\mathbf{m} - d\|$ and the model norm $\|\mathbf{m}\|$ as functions of λ , and hence the trade-off curve between them. Evaluate the results for $d = 1.2$ s and $\lambda/\ell^2 = 0, 0.5, 2$ and 8 .

Solution. With $\mathbf{A} = (\ell \ \ell)$ we have

$$\mathbf{A}^T \mathbf{A} + \lambda \mathbf{I} = \begin{pmatrix} \ell^2 + \lambda & \ell^2 \\ \ell^2 & \ell^2 + \lambda \end{pmatrix}, \quad \det(\mathbf{A}^T \mathbf{A} + \lambda \mathbf{I}) = (\ell^2 + \lambda)^2 - \ell^4 = \lambda(2\ell^2 + \lambda), \quad (12)$$

which is positive for every $\lambda > 0$, in agreement with the general argument in the lecture that the regularised normal matrix is positive definite. Inverting the two-by-two matrix,

$$(\mathbf{A}^T \mathbf{A} + \lambda \mathbf{I})^{-1} = \frac{1}{\lambda(2\ell^2 + \lambda)} \begin{pmatrix} \ell^2 + \lambda & -\ell^2 \\ -\ell^2 & \ell^2 + \lambda \end{pmatrix}, \quad (13)$$

and since $\mathbf{A}^T d = \ell d (1, 1)^T$ we find

$$\mathbf{m}(\lambda) = \frac{\ell d}{\lambda(2\ell^2 + \lambda)} \begin{pmatrix} \lambda \\ \lambda \end{pmatrix} = \frac{\ell d}{2\ell^2 + \lambda} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (14)$$

The same result follows more quickly from the eigenvectors noted in Example 1. The matrix $\mathbf{A}^T \mathbf{A} + \lambda \mathbf{I}$ has eigenvalue $2\ell^2 + \lambda$ on $(1, 1)^T$ and λ on $(1, -1)^T$; the right-hand side $\mathbf{A}^T d$ lies entirely along $(1, 1)^T$, and so is simply divided by $2\ell^2 + \lambda$. In particular the regularised solution never acquires a component along the null-space direction $(1, -1)^T$: with $\mathbf{B} = \mathbf{I}$ and $\mathbf{m}_0 = \mathbf{0}$, damping selects the solution that puts nothing into the null space.

As $\lambda \rightarrow 0$, $\mathbf{m} \rightarrow (d/2\ell)(1, 1)^T$. This fits the datum exactly, since $\ell(d/2\ell + d/2\ell) = d$, and among all exact solutions $(d/2\ell)(1, 1)^T + a(1, -1)^T$ it has the smallest norm, $\|\mathbf{m}\|^2 = d^2/2\ell^2 + 2a^2$ being minimised at $a = 0$. Thus the small damping limit is

the minimum-norm solution, which attributes the delay equally to the two blocks. As $\lambda \rightarrow \infty$, $\mathbf{m} \rightarrow \mathbf{0}$: the regularisation term dominates and the model is pulled entirely to the reference model, whatever the data say.

For the misfit and norm we have $\mathbf{A}\mathbf{m} = 2\ell^2 d / (2\ell^2 + \lambda)$, and so

$$\|\mathbf{A}\mathbf{m} - d\| = \frac{\lambda d}{2\ell^2 + \lambda}, \quad \|\mathbf{m}\| = \frac{\sqrt{2} \ell d}{2\ell^2 + \lambda}. \quad (15)$$

The misfit rises monotonically from 0 to d as λ increases, while the norm falls monotonically from $d/\sqrt{2}\ell$ to 0. Eliminating λ between the two expressions,

$$\|\mathbf{A}\mathbf{m} - d\| = d - \sqrt{2} \ell \|\mathbf{m}\|, \quad (16)$$

so the trade-off curve in the (norm, misfit) plane is a straight line joining the minimum-norm solution $(d/\sqrt{2}\ell, 0)$ to the reference model $(0, d)$. For $d = 1.2$ s and $\ell = 100$ km the numbers are as follows.

λ/ℓ^2	$\delta u_1 = \delta u_2$ (s km ⁻¹)	$\ \mathbf{A}\mathbf{m} - d\ $ (s)	$\ \mathbf{m}\ $ (s km ⁻¹)
0	6.0×10^{-3}	0	8.49×10^{-3}
0.5	4.8×10^{-3}	0.24	6.79×10^{-3}
2	3.0×10^{-3}	0.60	4.24×10^{-3}
8	1.2×10^{-3}	0.96	1.70×10^{-3}

The value $\lambda = 2\ell^2$, equal to the single non-zero eigenvalue of $\mathbf{A}^T \mathbf{A}$, halves the amplitude of the solution and misfits the datum by half its value. The trade-off curve here is a straight line because $\mathbf{A}$ has rank one; the characteristic bend, or “corner”, of trade-off curves in real problems arises only when the singular values of $\mathbf{A}$ span a range, different components of the model then being switched off at different values of λ .

Example 3: Fermat’s principle checked exactly. A medium of uniform P-wave velocity α contains a slab $0 < x < h$ in which the velocity is $\alpha(1 + \epsilon)$. A source at $(-a, -b)$ and a receiver at $(h + a, b)$ lie on opposite sides of the slab, so that the unperturbed ray is a straight line crossing the slab obliquely at the angle θ to its normal, where $\tan \theta = 2b/(2a + h)$. Show from Snell’s law that the exact travel time is given by

$$T(\epsilon) = \frac{2a}{\alpha \cos \theta_1} + \frac{h}{\alpha(1 + \epsilon) \cos \theta_2},$$

$$\sin \theta_2 = (1 + \epsilon) \sin \theta_1, \quad 2a \tan \theta_1 + h \tan \theta_2 = 2b. \quad (17)$$

Show that the linearised delay time of the lecture is $\delta T = -\epsilon \ell_{\text{in}}/\alpha$, with $\ell_{\text{in}} = h/\cos \theta$ the length of the unperturbed ray inside the slab, that the first-order term of $T(\epsilon)$ agrees with this, and that the contribution of ray bending to the delay time is $O(\epsilon^2)$ and negative. Confirm this numerically for $\alpha = 8$ km s⁻¹, $h = 100$ km, $a = 200$ km, $b = 250$ km and $\epsilon = 0.01, 0.03$ and 0.1 .

Solution. Within each homogeneous region the perturbed ray is straight, and at the two plane interfaces it refracts according to Snell’s law, the horizontal slowness $p = \sin \theta_1/\alpha = \sin \theta_2/\alpha(1 + \epsilon)$ being conserved. Let θ_1 be the angle to the slab normal of the

two outer segments and θ_2 that of the segment inside the slab. The outer segments each have length $a/\cos\theta_1$ and the inner segment $h/\cos\theta_2$, which gives the travel time in eq. (17); the transverse distances covered, $a\tan\theta_1$ twice and $h\tan\theta_2$ once, must sum to $2b$, which is the final condition. For given ϵ these two conditions determine θ_1 through a single transcendental equation, which is readily solved numerically. At $\epsilon = 0$ we have $\theta_1 = \theta_2 = \theta$ and the ray is the straight line with $T_0 = (2a + h)/\alpha \cos\theta$.

The velocity perturbation is $\delta v = \epsilon\alpha$ inside the slab and zero outside, and the linearised delay time of the lecture is the integral along the *unperturbed* ray,

$$\delta T = - \int_{\text{ray}} \frac{\delta v}{v^2} ds = - \frac{\epsilon\alpha}{\alpha^2} \ell_{\text{in}} = - \frac{\epsilon \ell_{\text{in}}}{\alpha}, \quad \ell_{\text{in}} = \frac{h}{\cos\theta}. \quad (18)$$

To compare with the exact result it is clearest to make the Fermat argument explicit. Any three-segment path is fixed by its entry point $(0, y_1)$ and exit point (h, y_2) , and has travel time

$$T(\epsilon; \mathbf{y}) = T_{\text{out}}(\mathbf{y}) + \frac{T_{\text{in}}(\mathbf{y})}{1 + \epsilon},$$

$$T_{\text{out}} = \frac{\sqrt{a^2 + (y_1 + b)^2} + \sqrt{a^2 + (b - y_2)^2}}{\alpha}, \quad T_{\text{in}} = \frac{\sqrt{h^2 + (y_2 - y_1)^2}}{\alpha}, \quad (19)$$

with $\mathbf{y} = (y_1, y_2)^T$. Snell's law is precisely the condition $\nabla_{\mathbf{y}} T(\epsilon; \mathbf{y}) = \mathbf{0}$, and the true ray, $\mathbf{y}(\epsilon)$, is the minimiser of $T(\epsilon; \cdot)$; in particular the unperturbed ray $\mathbf{y}_0$ satisfies $\nabla T_{\text{out}}(\mathbf{y}_0) = -\nabla T_{\text{in}}(\mathbf{y}_0) \equiv -\mathbf{g}$. Writing $\mathbf{y}(\epsilon) = \mathbf{y}_0 + \delta\mathbf{y}$ and expanding,

$$T(\epsilon) = T(\epsilon; \mathbf{y}_0) + \nabla_{\mathbf{y}} T(\epsilon; \mathbf{y}_0) \cdot \delta\mathbf{y} + \frac{1}{2} \delta\mathbf{y}^T \mathbf{H} \delta\mathbf{y} + \dots, \quad (20)$$

where $\mathbf{H}$ is the Hessian of $T_0(\mathbf{y}) = T(0; \mathbf{y})$ at $\mathbf{y}_0$. The first term is the travel time through the perturbed slab along the unperturbed path,

$$T(\epsilon; \mathbf{y}_0) = T_0 + \left(\frac{1}{1 + \epsilon} - 1 \right) \frac{\ell_{\text{in}}}{\alpha} = T_0 - \frac{\epsilon \ell_{\text{in}}}{\alpha} + \frac{\epsilon^2 \ell_{\text{in}}}{\alpha} - \dots, \quad (21)$$

whose first-order part is exactly eq. (18). In the second term of eq. (20) the gradient is $\nabla_{\mathbf{y}} T(\epsilon; \mathbf{y}_0) = -\epsilon\mathbf{g}/(1 + \epsilon)$, which vanishes at $\epsilon = 0$ by Fermat's principle, while $\delta\mathbf{y} = O(\epsilon)$ because the ray moves continuously with the model. The ray-bending contribution to the travel time is therefore $O(\epsilon^2)$: the first-order delay time is obtained by integrating along the old ray, as claimed in the lecture. We can go one step further. Stationarity of the travel time at $\mathbf{y}_0 + \delta\mathbf{y}$ gives $\mathbf{H} \delta\mathbf{y} = \epsilon\mathbf{g} + O(\epsilon^2)$, and substituting into eq. (20),

$$T(\epsilon) - T(\epsilon; \mathbf{y}_0) = -\epsilon^2 \mathbf{g}^T \mathbf{H}^{-1} \mathbf{g} + \frac{1}{2} \epsilon^2 \mathbf{g}^T \mathbf{H}^{-1} \mathbf{g} + O(\epsilon^3) = -\frac{1}{2} \epsilon^2 \mathbf{g}^T \mathbf{H}^{-1} \mathbf{g} + O(\epsilon^3), \quad (22)$$

which is negative because $\mathbf{H}$ is positive definite at a minimum. Bending always shortens the travel time relative to the frozen path, whatever the sign of ϵ , since the true ray is the fastest path in the perturbed medium. For the present geometry the quantities are elementary. A straight segment of length L making angle θ with the x -axis has travel time $\sqrt{\Delta x^2 + \Delta y^2}/\alpha$ whose second derivative with respect to an end-point ordinate is $\cos^2\theta/\alpha L$, and so at $\mathbf{y}_0$

$$\mathbf{H} = \frac{\cos^3\theta}{\alpha} \begin{pmatrix} 1/a + 1/h & -1/h \\ -1/h & 1/a + 1/h \end{pmatrix}, \quad \mathbf{g} = \nabla T_{\text{in}} = \frac{\sin\theta}{\alpha} \begin{pmatrix} -1 \\ 1 \end{pmatrix}. \quad (23)$$

The vector $\mathbf{g}$ is an eigenvector of $\mathbf{H}$ with eigenvalue $(\cos^3 \theta / \alpha)(1/a + 2/h)$, and hence

$$-\frac{1}{2}\mathbf{g}^T \mathbf{H}^{-1} \mathbf{g} = -\frac{\sin^2 \theta ah}{\alpha \cos^3 \theta (h + 2a)} = -\frac{\ell_{\text{in}}}{\alpha} \frac{a \tan^2 \theta}{h + 2a}. \quad (24)$$

Collecting eqs. (21), (22) and (24), the exact delay time is

$$T(\epsilon) - T_0 = -\frac{\epsilon \ell_{\text{in}}}{\alpha} + \frac{\epsilon^2 \ell_{\text{in}}}{\alpha} \left(1 - \frac{a \tan^2 \theta}{h + 2a}\right) + O(\epsilon^3). \quad (25)$$

The two second-order pieces have distinct origins: the first is the non-linearity of the slowness $1/v$ in ϵ and would be absent had we perturbed the slowness rather than the velocity; the second is ray bending, and vanishes at normal incidence where the ray is not bent at all.

For the numerical values given, $\tan \theta = 500/500$, so $\theta = 45^\circ$, $\ell_{\text{in}} = h\sqrt{2} = 141.4$ km, $T_0 = 500\sqrt{2}/8 = 88.39$ s and $\ell_{\text{in}}/\alpha = 17.68$ s. The bending coefficient in eq. (24) is $-17.68 \times 0.4 = -7.07$ s, and the total second-order coefficient in eq. (25) is $17.68 \times 0.6 = 10.61$ s. Solving eq. (17) numerically we obtain the following.

ϵ	$-\epsilon \ell_{\text{in}}/\alpha$ (s)	$T(\epsilon) - T_0$ (s)	error (ms)	bending part (ms)
0.01	-0.17678	-0.17573	1.05	-0.70
0.03	-0.53033	-0.52114	9.19	-6.25
0.1	-1.76777	-1.67393	93.8	-66.9

Here “error” is the difference between the exact and the linearised delay times, and “bending part” is $T(\epsilon) - T(\epsilon; \mathbf{y}_0)$, the part of that difference due to the movement of the ray. Increasing ϵ by a factor of three multiplies the error by 8.8 and by a factor of ten multiplies it by 90, close to the factors of 9 and 100 expected for a quadratic, and the second-order formula $10.61 \epsilon^2$ s predicts 1.06, 9.5 and 106 ms, the discrepancies being the $O(\epsilon^3)$ terms. Even for a 10% velocity anomaly, far larger than anything in the mantle, the linearised delay time is in error by only 6%, and the ray-bending part of that error is smaller in magnitude than, and opposite in sign to, the part due to the non-linearity of the slowness in ϵ . This is why the linear relation $\mathbf{d} = \mathbf{A}\mathbf{m}$ is adequate for global tomography.

Example 4: Resolution and the Bayesian view. Return once more to case (a) of Example 1, $\mathbf{A} = (\ell \ \ell)$, and suppose the datum has variance σ_d^2 and that the prior on the model is Gaussian with mean $\mathbf{m}_0 = \mathbf{0}$ and covariance $\mathbf{C}_m = \sigma_m^2 \mathbf{I}$. Compute the posterior mean $\mathbf{m}_p$ and covariance $\mathbf{C}_p$ explicitly. Show that the posterior variance is reduced only along the direction $(1, 1)^T$ and is unchanged along the null-space direction $(1, -1)^T$, and relate the result to the regularised solution of Example 2. Evaluate everything for $d = 1.2$ s, $\sigma_d = 0.1$ s and $\sigma_m = 2 \times 10^{-3}$ s km⁻¹.

Solution. Let $\mathbf{P}_\parallel$ and $\mathbf{P}_\perp$ be the orthogonal projectors onto the unit vectors $(1, 1)^T / \sqrt{2}$ and $(1, -1)^T / \sqrt{2}$,

$$\mathbf{P}_\parallel = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{P}_\perp = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{P}_\parallel + \mathbf{P}_\perp = \mathbf{I}. \quad (26)$$

From Example 1, $\mathbf{A}^T \mathbf{A} = 2\ell^2 \mathbf{P}_{\parallel}$, and so the inverse posterior covariance of the lecture is

$$\mathbf{C}_p^{-1} = \frac{\mathbf{A}^T \mathbf{A}}{\sigma_d^2} + \frac{\mathbf{I}}{\sigma_m^2} = \left(\frac{2\ell^2}{\sigma_d^2} + \frac{1}{\sigma_m^2} \right) \mathbf{P}_{\parallel} + \frac{1}{\sigma_m^2} \mathbf{P}_{\perp}. \quad (27)$$

A matrix of the form $\mu_{\parallel} \mathbf{P}_{\parallel} + \mu_{\perp} \mathbf{P}_{\perp}$ is inverted by inverting its two eigenvalues, and hence

$$\mathbf{C}_p = c \mathbf{P}_{\parallel} + \sigma_m^2 \mathbf{P}_{\perp}, \quad c = \frac{\sigma_m^2 \sigma_d^2}{\sigma_d^2 + 2\ell^2 \sigma_m^2} < \sigma_m^2. \quad (28)$$

The posterior variance of the component of $\mathbf{m}$ along $(1, 1)^T / \sqrt{2}$ is c , which is smaller than the prior variance σ_m^2 by the factor $\sigma_d^2 / (\sigma_d^2 + 2\ell^2 \sigma_m^2)$, while that along $(1, -1)^T / \sqrt{2}$ is σ_m^2 , exactly as before the datum was collected. The data reduce our uncertainty only in the directions they are sensitive to; in the null space the posterior is the prior, and any apparent knowledge of the difference between the blocks in the final model comes entirely from the prior. Written out,

$$\mathbf{C}_p = \frac{1}{2} \begin{pmatrix} \sigma_m^2 + c & c - \sigma_m^2 \\ c - \sigma_m^2 & \sigma_m^2 + c \end{pmatrix}, \quad (29)$$

so the two blocks are negatively correlated a posteriori, with correlation coefficient $(c - \sigma_m^2) / (c + \sigma_m^2)$: the datum fixes the sum of the perturbations, and so if one block is faster than the posterior mean the other is probably slower. In the limit of a perfect datum, $\sigma_d \rightarrow 0$, we have $c \rightarrow 0$, $\mathbf{C}_p \rightarrow \sigma_m^2 \mathbf{P}_{\perp}$, and the correlation tends to -1 .

The posterior mean is $\mathbf{m}_p = \mathbf{C}_p \mathbf{A}^T d / \sigma_d^2$ since $\mathbf{m}_0 = \mathbf{0}$, and as $\mathbf{A}^T d = \ell d (1, 1)^T$ lies along $(1, 1)^T$,

$$\mathbf{m}_p = \frac{c \ell d}{\sigma_d^2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{\ell d}{2\ell^2 + \sigma_d^2 / \sigma_m^2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (30)$$

Comparing with eq. (14), this is the regularised least squares solution of Example 2 with

$$\lambda = \frac{\sigma_d^2}{\sigma_m^2}, \quad (31)$$

which is the lecture's identification $\lambda \mathbf{B} = \mathbf{C}_m^{-1}$ once the data variance, set to unity in Example 2, is restored: with $\mathbf{C} = \sigma_d^2 \mathbf{I}$ the regularised normal matrix is $\mathbf{A}^T \mathbf{A} / \sigma_d^2 + \lambda \mathbf{B}$, and $\lambda \mathbf{B} = \mathbf{I} / \sigma_m^2$ gives $\mathbf{A}^T \mathbf{A} + (\sigma_d^2 / \sigma_m^2) \mathbf{I}$. The damping parameter is thus the ratio of how noisy we think the data are to how large we think the model is: noisier data or a tighter prior both mean heavier damping.

It is also instructive to ask how the posterior mean is related to the true model. If the datum was generated as $d = \mathbf{A} \mathbf{m}_{\text{true}} + e$, then

$$\mathbf{m}_p = \mathbf{R} \mathbf{m}_{\text{true}} + \frac{\mathbf{C}_p \mathbf{A}^T e}{\sigma_d^2}, \quad \mathbf{R} = \frac{\mathbf{C}_p \mathbf{A}^T \mathbf{A}}{\sigma_d^2} = \frac{2\ell^2 c}{\sigma_d^2} \mathbf{P}_{\parallel} = \frac{2\ell^2 \sigma_m^2}{\sigma_d^2 + 2\ell^2 \sigma_m^2} \mathbf{P}_{\parallel}, \quad (32)$$

where $\mathbf{R}$ is called the resolution matrix. The recovered model is a filtered version of the truth: the mean of the two blocks is recovered with amplitude slightly less than one, and the difference between them is not recovered at all.

For the given numbers, $2\ell^2\sigma_m^2 = 0.08 \text{ s}^2$ and $\sigma_d^2 = 0.01 \text{ s}^2$, so $c = \sigma_m^2/9$: the posterior standard deviation along $(1, 1)^T/\sqrt{2}$ is $\sigma_m/3 = 0.67 \times 10^{-3} \text{ s km}^{-1}$, and along $(1, -1)^T/\sqrt{2}$ it remains $2 \times 10^{-3} \text{ s km}^{-1}$. Explicitly,

$$\mathbf{C}_p = \begin{pmatrix} 2.22 & -1.78 \\ -1.78 & 2.22 \end{pmatrix} \times 10^{-6} \text{ s}^2 \text{ km}^{-2}, \quad (33)$$

with marginal standard deviations $1.49 \times 10^{-3} \text{ s km}^{-1}$ for each block, a correlation coefficient of -0.8 , and $\mathbf{R} = (8/9)\mathbf{P}_{\parallel}$. The damping parameter is $\lambda = \sigma_d^2/\sigma_m^2 = 2500 \text{ km}^2$, i.e. $\lambda/\ell^2 = 0.25$, and the posterior mean is $\mathbf{m}_p = (5.33, 5.33) \times 10^{-3} \text{ s km}^{-1}$, corresponding to $\delta\alpha/\alpha = -4.3\%$ in each block. Note that each block's marginal uncertainty is reduced only modestly, from 2.0 to $1.5 \times 10^{-3} \text{ s km}^{-1}$, even though their sum is known three times better than a priori: marginal error bars on individual parameters can badly misrepresent what has been learned when the posterior is strongly correlated.

Example 5: Sensitivity along the ray. A ray passes through a homogeneous medium of velocity v . The velocity is perturbed by a small constant δ inside a sphere of radius R whose centre lies on the ray. Evaluate the linearised delay time, and comment on how it depends on R . What happens when the centre of the sphere is displaced a distance b perpendicular to the ray? Compare with the finite-frequency sensitivity kernels of Lecture 19.

Solution. The linearised delay time is the integral along the unperturbed ray,

$$\delta T = - \int_{\text{ray}} \frac{\delta v}{v^2} ds. \quad (34)$$

The ray is a straight line through the centre of the sphere, so it intersects the sphere along a diameter, of length $2R$, on which $\delta v = \delta$, and vanishes elsewhere. Hence

$$\delta T = -\frac{2R\delta}{v^2}, \quad (35)$$

a fast anomaly ($\delta > 0$) advancing the arrival and a slow one delaying it, as it should. For $v = 8 \text{ km s}^{-1}$, $R = 50 \text{ km}$ and a 1% slow anomaly, $\delta = -0.08 \text{ km s}^{-1}$, we obtain $\delta T = 0.125 \text{ s}$, a typical size for a teleseismic P-wave delay.

The delay time is proportional to R , the length of the ray inside the anomaly, and not to the volume $\frac{4}{3}\pi R^3$ of the anomaly: halving R halves δT but reduces the volume eightfold. This is the concrete meaning of the statement in the lecture that the functional derivative $DT(v)$ is a generalised function concentrated on the ray. If $DT(v)$ were an ordinary bounded function $K(\mathbf{x})$, then $\langle DT(v)|\delta v \rangle = \delta \int_{\text{sphere}} K d^3\mathbf{x}$ would be $O(R^3)$ as $R \rightarrow 0$, not $O(R)$. In the language of the block parameterisation, the sphere is a single basis function and $-2R/v^2$ is the corresponding element of $\mathbf{A}$.

If the centre is displaced a distance b from the ray, the ray cuts the sphere along a chord of length $2\sqrt{R^2 - b^2}$ for $b < R$, and misses it entirely for $b > R$, so

$$\delta T = \begin{cases} -\frac{2\sqrt{R^2 - b^2}\delta}{v^2}, & b < R, \\ 0, & b \geq R. \end{cases} \quad (36)$$

For the numbers above and $b = 30$ km the delay is 0.100 s, and for $b = 80$ km it is exactly zero. Ray theory therefore predicts that an anomaly is invisible to a ray that does not pass through it, however close it comes, and that the delay is largest when the ray passes through the centre. The finite-frequency kernels of Lecture 19 differ on both counts. They are non-zero throughout the first Fresnel zone, so an anomaly just off the ray still produces a delay, and they vanish on the ray itself, because a scatterer on the ray produces no phase delay relative to the direct wave. The two descriptions agree only when the anomaly is large compared with the width of the Fresnel zone, in which case the integral of the finite-frequency kernel over the anomaly reduces to the chord length and eq. (36) is recovered. For a 50 km anomaly and a 1 Hz teleseismic P-wave, whose Fresnel zone in the mantle is a few hundred kilometres wide, ray theory is already a rough guide.

Worked examples for Lecture 19: Waveform tomography

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These examples accompany Lecture 19. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. The first two apply the Lagrange multiplier method to systems small enough that every step can be written out, the third looks at the convergence of steepest descent, the fourth derives the adjoint traction for a cross-correlation delay time, and the last puts numbers to the cost of the adjoint method. Numerical results were checked with the scripts in the python directory.

Example 1: The adjoint method for a linear system. Let $\mathbf{A}(\mathbf{m})$ be an invertible $n \times n$ matrix depending smoothly on parameters $\mathbf{m} = (m_1, \dots, m_m)$, let $\mathbf{f}$ be a fixed vector, and define $\mathbf{u}(\mathbf{m})$ by $\mathbf{A}(\mathbf{m})\mathbf{u} = \mathbf{f}$. Data are predicted by $\mathbf{P}\mathbf{u}$, with $\mathbf{P}$ a fixed $p \times n$ matrix, and the misfit is $J = \frac{1}{2}\|\mathbf{P}\mathbf{u} - \mathbf{d}\|^2$ for observed data $\mathbf{d}$. Obtain $\partial J / \partial m_k$ (i) by direct differentiation and (ii) using the Lagrangian $L(\mathbf{u}, \mathbf{u}', \mathbf{m}) = J(\mathbf{u}) - \mathbf{u}'^T[\mathbf{A}(\mathbf{m})\mathbf{u} - \mathbf{f}]$. Count the number of linear solves each approach requires, and check the result on $\mathbf{A}(m) = \begin{pmatrix} 2 & -1 \\ -1 & m \end{pmatrix}$, $\mathbf{f} = (1, 0)^T$, $\mathbf{P} = (0 \ 1)$, $\mathbf{d} = \frac{1}{2}$ at $m = 1$.

Solution. This is the finite-dimensional skeleton of the calculation in the lecture: the matrix $\mathbf{A}(\mathbf{m})$ stands for the wave operator, $\mathbf{u}$ for the wavefield, $\mathbf{P}$ for the operation of recording the wavefield at the seismometer, and $\mathbf{u}'$ for the adjoint wavefield. Because the constraint is a single algebraic equation there is no need for a second multiplier playing the role of $\mathbf{w}'$.

Direct differentiation. Write $\mathbf{r} = \mathbf{P}\mathbf{u} - \mathbf{d}$ for the residual, so that $J = \frac{1}{2}\mathbf{r}^T\mathbf{r}$ and, by the chain rule,

$$\frac{\partial J}{\partial m_k} = \mathbf{r}^T \mathbf{P} \frac{\partial \mathbf{u}}{\partial m_k}. \quad (1)$$

Differentiating $\mathbf{A}\mathbf{u} = \mathbf{f}$ with respect to m_k gives $\frac{\partial \mathbf{A}}{\partial m_k}\mathbf{u} + \mathbf{A} \frac{\partial \mathbf{u}}{\partial m_k} = \mathbf{0}$, and hence

$$\frac{\partial \mathbf{u}}{\partial m_k} = -\mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial m_k} \mathbf{u}, \quad \frac{\partial J}{\partial m_k} = -\mathbf{r}^T \mathbf{P} \mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial m_k} \mathbf{u}. \quad (2)$$

To evaluate this we solve $\mathbf{A}\mathbf{u} = \mathbf{f}$ once, and then for each k solve $\mathbf{A} \partial_k \mathbf{u} = -(\partial_k \mathbf{A})\mathbf{u}$: a total of $m + 1$ linear solves with $\mathbf{A}$. This is precisely the cost of the finite-difference approach discussed in the lecture, of which the above is the exact limit.

Lagrangian. The Lagrangian is

$$L(\mathbf{u}, \mathbf{u}', \mathbf{m}) = \frac{1}{2}(\mathbf{P}\mathbf{u} - \mathbf{d})^T(\mathbf{P}\mathbf{u} - \mathbf{d}) - \mathbf{u}'^T[\mathbf{A}(\mathbf{m})\mathbf{u} - \mathbf{f}], \quad (3)$$

in which $\mathbf{u}$, $\mathbf{u}'$ and $\mathbf{m}$ are treated as independent. The Lagrange multiplier theorem states that $\partial J / \partial m_k = \partial L / \partial m_k$ provided the partial derivatives of L with respect to $\mathbf{u}$ and $\mathbf{u}'$ both vanish. Stationarity with respect to $\mathbf{u}'$ recovers the forward problem $\mathbf{A}\mathbf{u} = \mathbf{f}$. Stationarity with respect to $\mathbf{u}$ requires, for every $\delta\mathbf{u}$,

$$(\mathbf{P}\mathbf{u} - \mathbf{d})^T \mathbf{P} \delta\mathbf{u} - \mathbf{u}'^T \mathbf{A} \delta\mathbf{u} = \left[\mathbf{P}^T (\mathbf{P}\mathbf{u} - \mathbf{d}) - \mathbf{A}^T \mathbf{u}' \right]^T \delta\mathbf{u} = 0, \quad (4)$$

and hence $\mathbf{u}'$ satisfies the *adjoint equation*

$$\mathbf{A}^T \mathbf{u}' = \mathbf{P}^T (\mathbf{P}\mathbf{u} - \mathbf{d}). \quad (5)$$

The right-hand side is the residual mapped back into the state space by $\mathbf{P}^T$; it is the analogue of the adjoint traction h'_i in the lecture, which places the residual seismogram at the receiver location using a delta function. With both conditions satisfied, the model dependence of L is confined to the matrix $\mathbf{A}(\mathbf{m})$, and so

$$\frac{\partial J}{\partial m_k} = \frac{\partial L}{\partial m_k} = -\mathbf{u}'^T \frac{\partial \mathbf{A}}{\partial m_k} \mathbf{u}. \quad (6)$$

This agrees with eq. (2), since from eq. (5) $\mathbf{u}'^T = \mathbf{r}^T \mathbf{P} \mathbf{A}^{-1}$. The only work beyond the forward problem is the single solve in eq. (5); the gradient with respect to all m parameters then follows from eq. (6) by matrix-vector products, which cost no more than assembling $\mathbf{A}$ itself. The total is therefore two solves against $m + 1$. In the matrix setting the saving is smaller than it looks, because once $\mathbf{A}$ has been factorised each additional solve is cheap; the comparison is decisive when a “solve” means a full time-domain wavefield simulation, for which nothing analogous to a factorisation is available.

In the language of the appendix to the lecture, $\mathbf{a}(\mathbf{u}, \mathbf{m}) = \mathbf{A}(\mathbf{m})\mathbf{u} - \mathbf{f}$, so that $D_{\mathbf{u}}\mathbf{a} = \mathbf{A}$ and the adjoint operator appearing in the equation for the multiplier is $\mathbf{A}^T$. (The appendix writes the constraint term with a plus sign, which merely reverses the sign of $\mathbf{u}'$.)

The 2×2 example. Here $\det \mathbf{A} = 2m - 1$ and

$$\mathbf{u} = \mathbf{A}^{-1} \mathbf{f} = \frac{1}{2m-1} \begin{pmatrix} m \\ 1 \end{pmatrix}, \quad J = \frac{1}{2} \left(\frac{1}{2m-1} - d \right)^2, \quad (7)$$

and by direct differentiation

$$\frac{dJ}{dm} = -\frac{2}{(2m-1)^2} \left(\frac{1}{2m-1} - d \right). \quad (8)$$

For the adjoint route we note that $\mathbf{A}$ is symmetric, that $\mathbf{P}^T (\mathbf{P}\mathbf{u} - \mathbf{d}) = (0, u_2 - d)^T$, and that $\partial \mathbf{A} / \partial m$ has a single non-zero entry, equal to one, in the $(2, 2)$ position. Thus

$$\mathbf{u}' = \frac{u_2 - d}{2m-1} \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \frac{dJ}{dm} = -u'_2 u_2 = -\frac{2(u_2 - d)}{(2m-1)^2}, \quad (9)$$

in agreement with eq. (8). At $m = 1$, $d = \frac{1}{2}$ we have $\mathbf{u} = (1, 1)^T$, $J = \frac{1}{8}$, $\mathbf{u}' = (\frac{1}{2}, 1)^T$ and $dJ/dm = -1$; a central difference of J with step 10^{-6} gives -1.00000 . The lesson is that the adjoint variable $\mathbf{u}'$ is simply the residual propagated backwards through the transposed system, and that once it is known every component of the gradient is available without further solves.

Example 2: The adjoint method for a damped oscillator. A damped oscillator satisfies

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = f(t), \quad x(0) = \dot{x}(0) = 0,$$

for $0 \leq t \leq T$, with f a known forcing, and the misfit to an observed record $x^{\text{obs}}(t)$ is $J = \frac{1}{2} \int_0^T (x - x^{\text{obs}})^2 dt$. Using a Lagrangian of the same form as in the lecture, derive the adjoint equation, its terminal conditions and the derivatives $\partial J / \partial \omega_0^2$ and $\partial J / \partial \gamma$. Explain the sign of the damping term in the adjoint equation. Verify the result numerically for $f(t) = e^{-(t-2)^2}$, $T = 10$, $\omega_0 = 2$, $\gamma = 0.1$, with x^{obs} generated using $\omega_0 = 2.2$, $\gamma = 0.15$.

Solution. Following the lecture, we introduce a multiplier $x'(t)$ for the equation of motion and define

$$L(x, x', \omega_0^2, \gamma) = \frac{1}{2} \int_0^T (x - x^{\text{obs}})^2 dt - \int_0^T (\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x - f) x' dt. \quad (10)$$

The initial conditions are imposed directly on the class of admissible x , so that any variation satisfies $\delta x(0) = \delta \dot{x}(0) = 0$; this is exactly how the vanishing initial conditions were used in the lecture. Stationarity with respect to x' returns the equation of motion. Stationarity with respect to x requires

$$\int_0^T (x - x^{\text{obs}}) \delta x dt - \int_0^T (\delta \ddot{x} + 2\gamma \delta \dot{x} + \omega_0^2 \delta x) x' dt = 0 \quad (11)$$

for all admissible δx . We move the time derivatives onto x' by integrating by parts. Using $\delta x(0) = \delta \dot{x}(0) = 0$,

$$\int_0^T \delta \ddot{x} x' dt = [\delta \dot{x} x']_{t=0}^{t=T} - [\delta x \dot{x}']_{t=0}^{t=T} + \int_0^T \delta x \ddot{x}' dt, \quad (12)$$

$$\int_0^T 2\gamma \delta \dot{x} x' dt = 2\gamma [\delta x x']_{t=0}^{t=T} - \int_0^T 2\gamma \delta x \dot{x}' dt. \quad (13)$$

Substituting into eq. (11) and collecting terms,

$$\int_0^T [(x - x^{\text{obs}}) - \ddot{x}' + 2\gamma \dot{x}' - \omega_0^2 x'] \delta x dt - \delta \dot{x}(T) x'(T) + \delta x(T) [\dot{x}'(T) - 2\gamma x'(T)] = 0. \quad (14)$$

The values $\delta x(t)$ on the interior, $\delta x(T)$ and $\delta \dot{x}(T)$ can all be chosen independently. The interior term gives the adjoint equation

$$\ddot{x}' - 2\gamma \dot{x}' + \omega_0^2 x' = x - x^{\text{obs}}, \quad (15)$$

the coefficient of $\delta \dot{x}(T)$ gives $x'(T) = 0$, and the coefficient of $\delta x(T)$ then gives $\dot{x}'(T) = 0$. Thus the adjoint problem has *terminal* rather than initial conditions, exactly as in the lecture. Its source is the residual $x - x^{\text{obs}}$, the analogue of the adjoint traction h'_i .

The damping term has changed sign. Integrated forwards from $t = 0$ the adjoint equation would be unstable, since its homogeneous solutions grow like $e^{\gamma t}$. But it is to be integrated backwards from $t = T$, and if we set $t = T - t'$ and $y(t') = x'(T - t')$ then $\dot{y} = -\dot{x}'$, $\ddot{y} = \ddot{x}'$, and eq. (15) becomes

$$\ddot{y} + 2\gamma \dot{y} + \omega_0^2 y = (x - x^{\text{obs}})(T - t'), \quad y(0) = \dot{y}(0) = 0. \quad (16)$$

In reversed time the adjoint field satisfies the *same* damped oscillator equation as the forward field, driven by the time-reversed residual. This is the one-degree-of-freedom version of the footnote in the lecture: after time reversal the adjoint problem is identical in form to the forward problem, which is what allows the same numerical code to be used for both. Note also that the wave equation in the lecture had no first-derivative term, and so no sign changed there; a dissipative term is the simplest way to see that time reversal is genuinely involved.

With the two stationarity conditions satisfied, the gradient is obtained by differentiating L with respect to the parameters, which appear only in the constraint term:

$$\frac{\partial J}{\partial \omega_0^2} = \frac{\partial L}{\partial \omega_0^2} = - \int_0^T x x' dt, \quad \frac{\partial J}{\partial \gamma} = \frac{\partial L}{\partial \gamma} = -2 \int_0^T \dot{x} x' dt. \quad (17)$$

These are the analogues of the kernels K_A (the stiffness, which multiplies the field) and, in a loose sense, K_p (which involves time derivatives). If the forcing is also regarded as unknown, perturbing $f \rightarrow f + \delta f$ in eq. (10) gives $\delta J = \int_0^T x' \delta f dt$, so the adjoint field itself is the sensitivity kernel for the source, just as $K_{\bar{s}_{ij}} = \partial u'_i / \partial x_j$ in the lecture involves no product with the forward field.

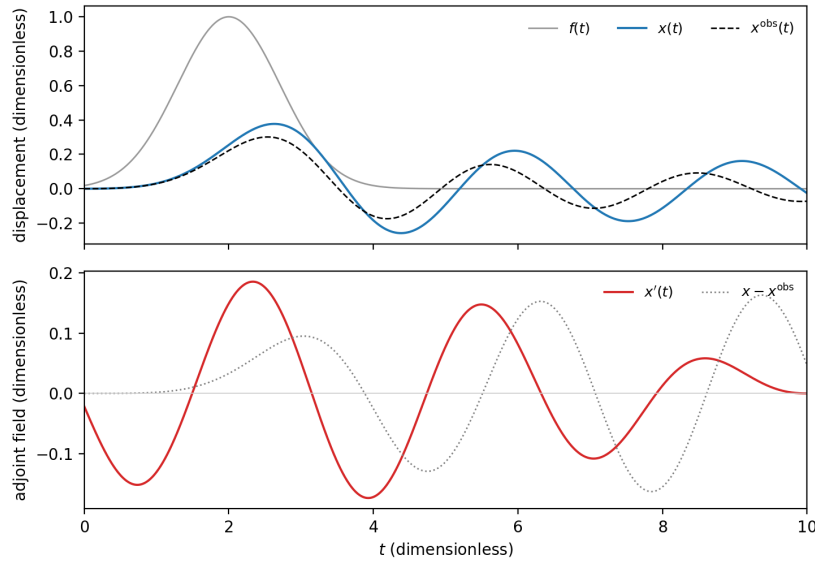


Fig. 1: Upper panel: the forcing f , the synthetic x for $\omega_0 = 2$, $\gamma = 0.1$, and the “observed” record x^{obs} generated with $\omega_0 = 2.2$, $\gamma = 0.15$. Lower panel: the adjoint field x' obtained by integrating eq. (15) backwards from $t = T = 10$, together with the residual that drives it. The adjoint field vanishes, with zero slope, at $t = T$ and is largest at early times.

Numerical check. The forward equation, the adjoint equation (backwards from $t = T$) and the two integrals in eq. (17) were computed with an eighth-order Runge–Kutta integrator at relative tolerance 10^{-12} ; the fields are shown in Fig. 1. The misfit is $J = 4.09742 \times 10^{-2}$ and the adjoint formulae give

$$\frac{\partial J}{\partial \omega_0^2} = -0.107897, \quad \frac{\partial J}{\partial \gamma} = -0.316192. \quad (18)$$

Central differences of J with step $h = 10^{-4}$ in each parameter give -0.107897 and -0.316192 respectively, the relative differences from eq. (18) being 9×10^{-11} and 3×10^{-7} ; with $h = 10^{-3}$ the finite-difference value for γ moves to -0.316201 , a change consistent with the $O(h^2)$ truncation error of the difference formula. Both derivatives are negative, as they should be: the observed record was made with larger ω_0 and γ , and so increasing either parameter reduces the misfit. For the source kernel, taking $\delta f = \epsilon \sin 3t$ gives $\int_0^T x' \sin 3t \, dt = -1.68008 \times 10^{-2}$ by the adjoint formula, against a finite-difference value of -1.68008×10^{-2} . Two integrations of a second-order equation have thus produced the derivatives with respect to every parameter, including the whole function f .

Example 3: Steepest descent on a quadratic misfit. Let $J(\mathbf{m}) = \frac{1}{2} \mathbf{m}^T \mathbf{H} \mathbf{m} - \mathbf{b}^T \mathbf{m}$ with $\mathbf{H}$ symmetric and positive definite, and let $\mathbf{g} = \mathbf{H} \mathbf{m} - \mathbf{b}$ be its gradient. Show that the step length minimising $f(\lambda) = J(\mathbf{m} - \lambda \mathbf{g})$ is $\lambda = \mathbf{g}^T \mathbf{g} / \mathbf{g}^T \mathbf{H} \mathbf{g}$. Writing $\mathbf{e} = \mathbf{m} - \bar{\mathbf{m}}$ for the error relative to the minimiser and $\|\mathbf{e}\|_{\mathbf{H}}^2 = \mathbf{e}^T \mathbf{H} \mathbf{e}$, show that one steepest-descent step satisfies $\|\mathbf{e}_1\|_{\mathbf{H}}^2 \leq \left(\frac{\kappa-1}{\kappa+1}\right)^2 \|\mathbf{e}_0\|_{\mathbf{H}}^2$, where κ is the condition number of $\mathbf{H}$. Illustrate with $\mathbf{H} = \text{diag}(1, 9)$, $\mathbf{b} = \mathbf{0}$, starting from $\mathbf{m}_0 = (9, 1)^T$.

Solution. Near a minimum any smooth misfit is approximately quadratic, with $\mathbf{H}$ the Hessian, so this example describes the asymptotic behaviour of the method of steepest descent described in the lecture. Since $\mathbf{H}$ is positive definite the unique minimiser is $\bar{\mathbf{m}} = \mathbf{H}^{-1} \mathbf{b}$, and expanding about it,

$$J(\mathbf{m}) = J(\bar{\mathbf{m}}) + \mathbf{e}^T (\mathbf{H} \bar{\mathbf{m}} - \mathbf{b}) + \frac{1}{2} \mathbf{e}^T \mathbf{H} \mathbf{e} = J(\bar{\mathbf{m}}) + \frac{1}{2} \|\mathbf{e}\|_{\mathbf{H}}^2, \quad (19)$$

so that the $\mathbf{H}$ -norm of the error measures the excess misfit, and $\mathbf{g} = \mathbf{H} \mathbf{e}$.

Step length. Along the search direction,

$$f(\lambda) = J(\mathbf{m} - \lambda \mathbf{g}) = J(\mathbf{m}) - \lambda \mathbf{g}^T \mathbf{g} + \frac{\lambda^2}{2} \mathbf{g}^T \mathbf{H} \mathbf{g}, \quad (20)$$

a convex quadratic in λ , and $f'(\lambda) = 0$ gives

$$\lambda = \frac{\mathbf{g}^T \mathbf{g}}{\mathbf{g}^T \mathbf{H} \mathbf{g}}. \quad (21)$$

This is the one-dimensional minimisation referred to in the lecture, here solvable in closed form. For a general misfit the same expression is obtained by fitting a parabola to $f(0)$, $f'(0) = -\|\mathbf{g}\|^2$ and one trial value $f(\lambda_t)$, which costs one extra forward calculation.

Contraction. With $\mathbf{e}_1 = \mathbf{e}_0 - \lambda \mathbf{g}$ and $\mathbf{g} = \mathbf{H} \mathbf{e}_0$,

$$\|\mathbf{e}_1\|_{\mathbf{H}}^2 = \mathbf{e}_0^T \mathbf{H} \mathbf{e}_0 - 2\lambda \mathbf{g}^T \mathbf{H} \mathbf{e}_0 + \lambda^2 \mathbf{g}^T \mathbf{H} \mathbf{g} = \|\mathbf{e}_0\|_{\mathbf{H}}^2 - \frac{(\mathbf{g}^T \mathbf{g})^2}{\mathbf{g}^T \mathbf{H} \mathbf{g}}, \quad (22)$$

where we used $\mathbf{g}^T \mathbf{H} \mathbf{e}_0 = \mathbf{g}^T \mathbf{g}$ and eq. (21). Since also $\|\mathbf{e}_0\|_{\mathbf{H}}^2 = \mathbf{g}^T \mathbf{H}^{-1} \mathbf{g}$,

$$\frac{\|\mathbf{e}_1\|_{\mathbf{H}}^2}{\|\mathbf{e}_0\|_{\mathbf{H}}^2} = 1 - \frac{(\mathbf{g}^T \mathbf{g})^2}{(\mathbf{g}^T \mathbf{H} \mathbf{g})(\mathbf{g}^T \mathbf{H}^{-1} \mathbf{g})}. \quad (23)$$

It remains to bound the second term from below, which is the content of the Kantorovich inequality. Let $0 < \mu_1 \leq \dots \leq \mu_n$ be the eigenvalues of $\mathbf{H}$, with orthonormal eigenvectors $\mathbf{v}_i$, and without loss of generality take $\|\mathbf{g}\| = 1$. Setting $c_i = (\mathbf{g} \cdot \mathbf{v}_i)^2$, so that $c_i \geq 0$ and $\sum_i c_i = 1$, we have $a \equiv \mathbf{g}^T \mathbf{H} \mathbf{g} = \sum_i c_i \mu_i$ and $b \equiv \mathbf{g}^T \mathbf{H}^{-1} \mathbf{g} = \sum_i c_i / \mu_i$. For any $\mu \in [\mu_1, \mu_n]$ we have $(\mu - \mu_1)(\mu_n - \mu) \geq 0$, which on dividing by μ reads $\mu + \mu_1 \mu_n / \mu \leq \mu_1 + \mu_n$. Applying this to each eigenvalue and summing with weights c_i gives $a + \mu_1 \mu_n b \leq \mu_1 + \mu_n$, and then by the arithmetic–geometric mean inequality

$$2\sqrt{\mu_1 \mu_n ab} \leq a + \mu_1 \mu_n b \leq \mu_1 + \mu_n \implies ab \leq \frac{(\mu_1 + \mu_n)^2}{4\mu_1 \mu_n}. \quad (24)$$

Inserting this into eq. (23), and writing $\kappa = \mu_n / \mu_1$,

$$\frac{\|\mathbf{e}_1\|_{\mathbf{H}}^2}{\|\mathbf{e}_0\|_{\mathbf{H}}^2} \leq 1 - \frac{4\mu_1 \mu_n}{(\mu_1 + \mu_n)^2} = \left(\frac{\mu_n - \mu_1}{\mu_n + \mu_1} \right)^2 = \left(\frac{\kappa - 1}{\kappa + 1} \right)^2. \quad (25)$$

Equivalently the $\mathbf{H}$ -norm of the error, and hence the square root of the excess misfit, contracts by at least the factor $(\kappa - 1)/(\kappa + 1)$ per iteration. The bound is attained when $\mathbf{g}$ has equal weight on the extreme eigenvectors and none elsewhere, and the iteration never escapes this situation, as the example below shows.

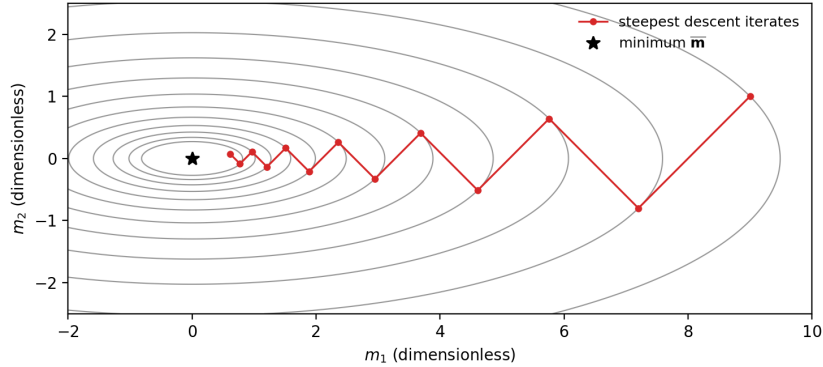


Fig. 2: Steepest descent with exact line search on $J = \frac{1}{2}(m_1^2 + 9m_2^2)$ from $\mathbf{m}_0 = (9, 1)^T$. Contours are drawn at the misfit values $J_k = 45 \times 0.64^k$ attained by the iterates, which zigzag between the two lines $m_2 = \pm m_1/9$.

The 2×2 example. For $\mathbf{H} = \text{diag}(1, 9)$ we have $\kappa = 9$ and $[(\kappa - 1)/(\kappa + 1)]^2 = 0.64$. With $\mathbf{b} = \mathbf{0}$ the minimiser is the origin and $\mathbf{e} = \mathbf{m}$. From $\mathbf{m}_0 = (9, 1)^T$ the gradient is $\mathbf{g}_0 = (9, 9)^T$, so $\lambda = 162/(81 + 729) = \frac{1}{5}$ and $\mathbf{m}_1 = (9, 1)^T - \frac{1}{5}(9, 9)^T = \frac{4}{5}(9, -1)^T$. The new gradient $\frac{4}{5}(9, -9)^T$ again has components of equal magnitude, so the step length is again $\frac{1}{5}$, and by induction

$$\mathbf{m}_k = \left(\frac{4}{5} \right)^k \begin{pmatrix} 9 \\ (-1)^k \end{pmatrix}, \quad J(\mathbf{m}_k) = 45 \times \left(\frac{16}{25} \right)^k. \quad (26)$$

The misfit falls by exactly the factor 0.64 predicted by eq. (25) at every step, and the iterates zigzag across the valley as shown in Fig. 2. To reduce the misfit by a factor of 10^6 requires $6 / \log_{10}(1/0.64) = 31.0$, i.e. 31 iterations ($J_{31}/J_0 = 0.64^{31} = 9.8 \times 10^{-7}$).

For $\kappa = 100$ the factor is $(99/101)^2 = 0.9608$ and the same reduction needs 346 iterations, while for $\kappa = 10^4$ it would need about 3.5×10^4 . By contrast, a start on an eigenvector, such as $\mathbf{m}_0 = (0, 3)^T$, reaches the minimum in one step. This slow, zigzagging convergence for ill-conditioned problems is why practical waveform inversions use conjugate-gradient or quasi-Newton directions built from the history of gradients, as remarked in the lecture; each iteration still costs the same adjoint calculation, but far fewer of them are needed.

Example 4: The adjoint traction for a cross-correlation delay time. Let $s(t) = w(t)\hat{v}_i u_i(\mathbf{x}_r, t)$ be a windowed component of the synthetic seismogram and $s^{\text{obs}}(t)$ the corresponding observed quantity, and let $\bar{\tau}$ be defined by $\dot{C}(\bar{\tau}) = 0$, where $C(\tau) = \int_{-\infty}^{\infty} s^{\text{obs}}(t - \tau)s(t) dt$ and the maximum of C is meant. By implicit differentiation find the first-order change $\delta\bar{\tau}$ produced by a perturbation $\delta s(t)$ of the synthetic, and hence the adjoint traction h'_i appropriate to the misfit $\bar{\tau}$. Simplify the result under the assumption that the observed and synthetic waveforms have the same shape, and verify the formulae numerically on a pair of pulses.

Solution. The lecture noted that the whole adjoint apparatus carries over to any misfit once the corresponding adjoint traction is known, since the form of J enters only through h'_i . Here the “misfit” is the delay time itself, and its functional derivative with respect to the wavefield is what we need.

Sign convention. It is worth being clear about what $\bar{\tau}$ measures. If the observed waveform is a delayed copy of the synthetic, $s^{\text{obs}}(t) = s(t - \Delta)$, then $C(\tau) = \int s(t - \Delta - \tau)s(t) dt$ is maximised when $\tau = -\Delta$. Thus with the definition used in the lecture, $\bar{\tau}$ is positive when the synthetic arrives *later* than the observation; it is the negative of the delay time as usually quoted. Some authors define the cross-correlation with $s^{\text{obs}}(t + \tau)$ or with $s(t - \tau)$ so as to reverse this, and all that changes below is the overall sign of h'_i .

Implicit differentiation. Differentiating under the integral sign, and using $\partial_\tau s^{\text{obs}}(t - \tau) = -\dot{s}^{\text{obs}}(t - \tau)$,

$$\dot{C}(\tau) = - \int_{-\infty}^{\infty} \dot{s}^{\text{obs}}(t - \tau)s(t) dt = \int_{-\infty}^{\infty} s^{\text{obs}}(t - \tau)\dot{s}(t) dt, \quad (27)$$

the second form following by an integration by parts in which the boundary terms vanish because the window w confines s to a finite interval. The defining condition $F(\bar{\tau}, s) \equiv \dot{C}(\bar{\tau}) = 0$ makes $\bar{\tau}$ an implicit function of s . Perturbing $s \rightarrow s + \delta s$ and $\bar{\tau} \rightarrow \bar{\tau} + \delta\bar{\tau}$ in the second form of eq. (27) and expanding to first order,

$$\underbrace{\int s^{\text{obs}}(t - \bar{\tau})\dot{s}(t) dt}_{=0} - \delta\bar{\tau} \int \dot{s}^{\text{obs}}(t - \bar{\tau})\dot{s}(t) dt + \int s^{\text{obs}}(t - \bar{\tau})\delta\dot{s}(t) dt = 0. \quad (28)$$

The coefficient of $\delta\bar{\tau}$ is $\ddot{C}(\bar{\tau})$, which is negative at a maximum, and the last integral can be integrated by parts. Hence

$$\delta\bar{\tau} = \frac{1}{\ddot{C}(\bar{\tau})} \int_{-\infty}^{\infty} \dot{s}^{\text{obs}}(t - \bar{\tau}) \delta s(t) dt, \quad (29)$$

where

$$\ddot{C}(\bar{\tau}) = - \int_{-\infty}^{\infty} \dot{s}^{\text{obs}}(t - \bar{\tau}) \dot{s}(t) dt = \int_{-\infty}^{\infty} s^{\text{obs}}(t - \bar{\tau}) \ddot{s}(t) dt. \quad (30)$$

This is exact, and involves only quantities available after the forward calculation: the shifted observed waveform and the synthetic. A useful check is a perturbation that simply delays the synthetic by ϵ , $\delta s = -\epsilon \dot{s}$; then eq. (29) gives $\delta \bar{\tau} = +\epsilon$ exactly, as the sign convention above requires, and this holds whatever the shape of s^{obs} .

The equal-shape approximation. The measured $\bar{\tau}$ is the shift that best aligns the two waveforms, and if they have the same shape, $s^{\text{obs}}(t - \bar{\tau}) \approx s(t)$, which in the lecture's convention is the statement $s^{\text{obs}}(t) \approx s(t + \bar{\tau})$. (The form $s^{\text{obs}}(t) \approx s(t - \bar{\tau})$ often written corresponds to the opposite sign convention for τ .) Under this assumption eq. (29) becomes

$$\delta \bar{\tau} \approx - \frac{\int_{-\infty}^{\infty} \dot{s}(t) \delta s(t) dt}{\int_{-\infty}^{\infty} \dot{s}(t)^2 dt}, \quad (31)$$

where the denominator, $\ddot{C}(\bar{\tau}) \approx \int s \ddot{s} dt = - \int \dot{s}^2 dt$, is manifestly negative. The observed data have disappeared entirely from the linearised relation: they enter only through the value of $\bar{\tau}$. An overall amplitude difference between s^{obs} and s also drops out, since it cancels between the numerator and denominator of eq. (29), so the approximation is really one of equal shape rather than equal waveform.

Adjoint traction. The window and the component $\hat{\nu}$ are fixed, so that a change in the wavefield produces $\delta s(t) = w(t) \hat{\nu}_i \delta u_i(\mathbf{x}_r, t)$, and the integrals may be restricted to $[0, T]$. Comparing with the form $\langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle = \int_0^T \int_{\partial M} h'_i \delta u_i dS dt$ used in the lecture, we read off

$$\begin{aligned} h'_i(\mathbf{x}, t) &= \frac{w(t) \dot{s}^{\text{obs}}(t - \bar{\tau})}{\ddot{C}(\bar{\tau})} \hat{\nu}_i \delta(\mathbf{x} - \mathbf{x}_r) \\ &\approx - \frac{w(t) \dot{s}(t)}{\int_0^T \dot{s}^2 dt} \hat{\nu}_i \delta(\mathbf{x} - \mathbf{x}_r). \end{aligned} \quad (32)$$

The adjoint traction is therefore, up to a normalising constant, the time derivative of the windowed synthetic, applied at the receiver in the direction of the measured component. Everything else in the lecture goes through unchanged: $\mathbf{u}'$ solves the adjoint wave equation with terminal conditions and this traction, and $\delta \bar{\tau}$ follows from the kernels K_ρ and $K_{A_{ijkl}}$. For an isotropic model, $\delta A_{ijkl} = (\delta \kappa - \frac{2}{3} \delta \mu) \delta_{ij} \delta_{kl} + \delta \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$, and with ρ held fixed $\delta \mu = 2\rho \beta \delta \beta$ and $\delta \kappa = 2\rho \alpha \delta \alpha - \frac{8}{3} \rho \beta \delta \beta$, which is how the kernels K_α and K_β of the lecture are assembled from $K_{A_{ijkl}}$. Because h'_i is proportional to $\dot{s}$ rather than to a residual, the kernel does not vanish when the data are fit perfectly; it is the sensitivity of a measurement, not of a misfit, and this is what gives the banana-doughnut kernels their universal form.

Numerical check. Take $s(t) = (t - t_0) e^{-(t-t_0)^2/\sigma^2}$ with $t_0 = 5$, $\sigma = 0.6$, and observed pulses $s^{\text{obs}}(t) = a(t - t_0 - \Delta) e^{-(t-t_0-\Delta)^2/\sigma_{\text{obs}}^2}$ with $\Delta = 0.4$, the record $[0, 12]$ containing both pulses entirely. In every case below the root of $\dot{C}$ is $\bar{\tau} = -0.4000$, confirming the sign convention; because both pulses are antisymmetric about their centres, C is symmetric about $\tau = -\Delta$ and the result is exact even when the shapes differ. We then perturb $s \rightarrow s + \epsilon \eta$ with $\eta(t) = \cos 2t e^{-(t-t_0-0.5)^2}$, recompute $\bar{\tau}(\pm \epsilon)$

by root finding with $\epsilon = 10^{-4}$, and compare the central difference with eqs. (29) and (31) evaluated with $\delta s = \eta$. For identical shapes ($a = 1$, $\sigma_{\text{obs}} = \sigma$) all three agree: $d\bar{\tau}/d\epsilon = 0.682928$. For $a = 1.2$ and $\sigma_{\text{obs}} = \sigma$ the numbers are unchanged, as predicted. For $a = 1.2$ and $\sigma_{\text{obs}} = 1.1\sigma$ the finite difference gives 0.747030, the exact formula eq. (29) gives 0.747030, and the approximate formula eq. (31) gives 0.682928, which is 8.6 per cent low; the discrepancy is entirely in the denominator, with $\ddot{C}(\bar{\tau}) = -0.7018$ against $-\int \dot{s}^2 dt = -0.5640$. For a 2 per cent width mismatch the error of eq. (31) is 1.8 per cent. The equal-shape approximation is thus accurate when it should be, and in practice it is adopted because it makes the adjoint source independent of the data and hence the kernels a property of the reference model alone.

Example 5: Counting the cost. A tomographic inversion uses n_{ev} earthquakes, performs n_{it} iterations of gradient descent, and each numerical solution of the wave equation costs a time c . Estimate the total computational cost of the inversion using (a) the adjoint method with two or three simulations per event and iteration, and (b) a finite-difference gradient with m model parameters. Put in the numbers from the lecture: $c \approx 10$ minutes, $n_{\text{ev}} = 10^2$ – 10^3 , $n_{\text{it}} \approx 15$, $m = 10^4$ – 10^6 . Explain why three simulations rather than two are usually performed.

Solution. The misfit is a sum over events, $J = \sum_e J_e$, and each J_e requires its own forward simulation because the source differs. The gradient is likewise a sum, $DJ = \sum_e DJ_e$, with each term requiring its own adjoint simulation, since the adjoint traction h'_i is built from that event's residuals.

Adjoint method. Per event and iteration we need one forward simulation, to obtain J_e and h'_i , and one adjoint simulation, to obtain $\mathbf{u}'$ and hence the kernels. The total cost is

$$\mathcal{C}_{\text{adj}} \approx 2 n_{\text{ev}} n_{\text{it}} c, \quad (33)$$

or $3 n_{\text{ev}} n_{\text{it}} c$ with re-simulation (below). The line search for the step length λ adds at least one further forward simulation per event, so that 3–4 simulations per event and iteration is a realistic count. Crucially, none of this depends on m : the kernels K_ρ , $K_{A_{ijkl}}$ and $K_{\bar{S}_{ij}}$ are fields, obtained on the numerical grid at the cost of a few multiplications per grid point and time step, and any parameterisation of the model can then be projected out of them.

Finite differences. The one-sided formula in the lecture requires $J(\mathbf{m})$ and $J(\mathbf{m} + \delta\mathbf{m}_k)$ for each of the m basis directions, each evaluation being a sum over events. Thus

$$\mathcal{C}_{\text{fd}} \approx (m + 1) n_{\text{ev}} n_{\text{it}} c, \quad (34)$$

and central differences would double this. The ratio $\mathcal{C}_{\text{adj}}/\mathcal{C}_{\text{fd}} = 3/(m + 1)$ is the factor quoted in the lecture.

Numbers. Take $n_{\text{ev}} = 10^3$, $n_{\text{it}} = 15$ and $c = 10$ min. The adjoint method with two simulations needs 3×10^4 simulations, or 3×10^5 min = 5000 h $\approx$ 208 days of cluster time; with three simulations this rises to 312 days, and with a trial step for the line search to about 420 days. With $n_{\text{ev}} = 10^2$ divide by ten. These are totals of machine time: different events are independent, so with a hundred simulations running concurrently the wall-clock time for the inversion is a few days, which is the scale on which the global models shown in the lecture were produced. Finite differences with

$m = 10^4$ need $1.5 \times 10^9 \text{ min} \approx 2.9 \times 10^3 \text{ years}$, and with $m = 10^6$ about $2.9 \times 10^5 \text{ years}$, regardless of how the work is distributed. There is no parameterisation coarse enough to rescue the finite-difference approach while retaining anything like the resolution of the grid on which the wave equation is solved.

Why three simulations. The kernels in the lecture are time integrals of products of the forward and adjoint fields, $K_\rho = \int_0^T \partial_t u_i \partial_t u'_i dt$ and so on. The adjoint field is obtained by integrating backwards from $t = T$, and so at the time step when $\mathbf{u}'(\mathbf{x}, t)$ is available, $\mathbf{u}(\mathbf{x}, t)$ was computed long before. Storing the forward field at every time step is impractical: for a global simulation with of order 10^9 grid points, 10^4 time steps and three components in single precision this is of order 10^{14} bytes per event. Instead one stores only the final state $\mathbf{u}(\mathbf{x}, T)$, $\partial_t \mathbf{u}(\mathbf{x}, T)$ (together with the wavefield on any absorbing boundaries), and during the adjoint calculation integrates the forward equations *backwards* from this final state alongside the adjoint equations; the elastic wave equation without attenuation is time-reversible, so this reconstructs $\mathbf{u}$ at each step as it is needed. The kernels are then accumulated on the fly. The price is a third simulation, and the factor of $3/(m + 1)$ in the lecture is the honest accounting for it.

Worked examples for Lecture 20: Self-gravitation and rotation

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Michaelmas Term, 2026

These examples accompany Lecture 20. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory. Throughout we take $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, and for the Earth a mass $M = 5.97 \times 10^{24} \text{ kg}$, a mean radius $b = 6371 \text{ km}$ and a rotation rate $\Omega = 7.29 \times 10^{-5} \text{ s}^{-1}$.

Example 1: The homogeneous sphere. A sphere of radius b has uniform density ρ and total mass $M = \frac{4}{3}\pi\rho b^3$. Solve Poisson's equation to find the gravitational potential ϕ and acceleration $\mathbf{g}$ inside and outside the sphere. Compute the gravitational binding energy $\mathcal{V}_g$ in two ways, using both expressions obtained in the lecture, and show that $\mathcal{V}_g = -\frac{3}{5}GM^2/b$. Evaluate the result for the Earth and compare it with the energy released by the largest earthquakes.

Solution. We take the reference body to be the sphere itself, so that the motion is the identity, the spatial and referential densities coincide, and $\zeta = \phi$, $\gamma = \mathbf{g}$. By symmetry ϕ depends only on $r = \|\mathbf{x}\|$, and Poisson's equation $\nabla^2\phi = 4\pi G\rho$ reads

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = \begin{cases} 4\pi G\rho, & r < b, \\ 0, & r > b. \end{cases} \quad (1)$$

Integrating twice, the solution that is regular at the origin and tends to zero at infinity is

$$\phi(r) = \begin{cases} \frac{2}{3}\pi G\rho r^2 + A, & r < b, \\ -\frac{C}{r}, & r > b, \end{cases} \quad (2)$$

with A and C constants. Since there are no surface masses, the lecture's continuity conditions require both ϕ and $g_r = -\frac{d\phi}{dr}$ to be continuous at $r = b$. Continuity of the radial derivative gives $\frac{4}{3}\pi G\rho b = C/b^2$, hence $C = \frac{4}{3}\pi G\rho b^3 = GM$, while continuity of ϕ then gives $A = -GM/b - \frac{2}{3}\pi G\rho b^2 = -GM/b - GM/(2b)$, where we have used $\rho = 3M/(4\pi b^3)$. Collecting terms,

$$\phi(r) = \begin{cases} -\frac{GM}{2b^3} (3b^2 - r^2), & r \leq b, \\ -\frac{GM}{r}, & r \geq b, \end{cases} \quad g_r(r) = \begin{cases} -\frac{GM}{b^3} r = -\frac{4}{3}\pi G\rho r, & r \leq b, \\ -\frac{GM}{r^2}, & r \geq b. \end{cases} \quad (3)$$

The exterior field is that of a point mass M at the origin, while inside the sphere the acceleration grows linearly with radius, as only the mass interior to r contributes. The potential at the centre is $-\frac{3}{2}GM/b$, one and a half times its surface value.

For the binding energy we first use the volume form $\mathcal{V}_g = \frac{1}{2} \int_{M_t} \varrho \phi \, d^3\mathbf{y}$ from the lecture, where the integral extends only over the body. With $d^3\mathbf{y} = 4\pi r^2 \, dr$,

$$\begin{aligned} \mathcal{V}_g &= \frac{1}{2} \rho \int_0^b \left[-\frac{GM}{2b^3} (3b^2 - r^2) \right] 4\pi r^2 \, dr \\ &= -\frac{\pi \rho GM}{b^3} \left(b^5 - \frac{b^5}{5} \right) = -\frac{4\pi \rho GM b^2}{5} = -\frac{3}{5} \frac{GM^2}{b}, \end{aligned} \quad (4)$$

where the last step again uses $\rho = 3M/(4\pi b^3)$. Next we use the field-energy form $\mathcal{V}_g = -\frac{1}{8\pi G} \int_{\mathbb{R}^3} \|\nabla \phi\|^2 \, d^3\mathbf{y}$, for which the integral extends over all space. Here $\|\nabla \phi\|^2 = g_r^2$, and

$$\begin{aligned} \int_{\mathbb{R}^3} \|\nabla \phi\|^2 \, d^3\mathbf{y} &= 4\pi \left(\frac{G^2 M^2}{b^6} \int_0^b r^4 \, dr + G^2 M^2 \int_b^\infty \frac{dr}{r^2} \right) \\ &= 4\pi G^2 M^2 \left(\frac{1}{5b} + \frac{1}{b} \right) = \frac{24\pi}{5} \frac{G^2 M^2}{b}, \end{aligned} \quad (5)$$

so that

$$\mathcal{V}_g = -\frac{1}{8\pi G} \cdot \frac{24\pi}{5} \frac{G^2 M^2}{b} = -\frac{3}{5} \frac{GM^2}{b}, \quad (6)$$

in agreement with eq. (4). It is worth noticing how differently the two formulae distribute the same total: in eq. (5) five sixths of the field energy resides *outside* the sphere, whereas eq. (4) has no exterior contribution at all. The two expressions differ by a divergence, and only their integrals agree.

For the Earth, with the values quoted in the preamble, $GM^2/b = 3.73 \times 10^{32} \text{ J}$ and hence

$$\mathcal{V}_g = -\frac{3}{5} \frac{GM^2}{b} \approx -2.24 \times 10^{32} \text{ J}. \quad (7)$$

For comparison, the same data give a mean density $3M/(4\pi b^3) \approx 5510 \text{ kg m}^{-3}$ and a surface gravity $GM/b^2 \approx 9.82 \text{ m s}^{-2}$. The energy radiated by the largest recorded earthquakes, such as the 1960 Chile event, is of order 10^{18} – 10^{19} J , so the gravitational binding energy exceeds the energy of the largest earthquake by a factor of 10^{13} – 10^{14} . Because the real Earth's mass is concentrated towards its centre, the true binding energy is somewhat larger in magnitude than the homogeneous estimate. The comparison makes plain why gravity cannot be ignored in the long-period dynamics: even a minute fractional change in the gravitational energy is enormous on the scale of seismic energies.

Example 2: Referential gravity under a rigid motion. Let φ be a motion of the body, and let $\tilde{\varphi} = \mathbf{a} + \mathbf{Q}\varphi$ be the motion obtained by composing it with a rigid translation $\mathbf{a}$ and rotation $\mathbf{Q}$ (so that $Q_{ki}Q_{kj} = \delta_{ij}$). Show directly from the integral expressions of the lecture that the referential gravitational potential is unchanged, $\tilde{\zeta} = \zeta$, while the referential gravitational acceleration transforms as $\tilde{\gamma}_i = Q_{ij}\gamma_j$.

Comment on the significance of these results for the Lagrangian and for the equations of motion in a rotating frame.

Solution. The referential potential is given in the lecture by

$$\zeta(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')}{\|\varphi(\mathbf{x}, t) - \varphi(\mathbf{x}', t)\|} d^3\mathbf{x}', \quad (8)$$

and depends on the motion only through the separation $\varphi(\mathbf{x}, t) - \varphi(\mathbf{x}', t)$ of pairs of material points. Under the rigid motion this separation becomes

$$\tilde{\varphi}(\mathbf{x}, t) - \tilde{\varphi}(\mathbf{x}', t) = \mathbf{a} + \mathbf{Q}\varphi(\mathbf{x}, t) - \mathbf{a} - \mathbf{Q}\varphi(\mathbf{x}', t) = \mathbf{Q} [\varphi(\mathbf{x}, t) - \varphi(\mathbf{x}', t)], \quad (9)$$

so the translation cancels identically. Writing $\mathbf{d} = \varphi(\mathbf{x}, t) - \varphi(\mathbf{x}', t)$, orthogonality of $\mathbf{Q}$ gives

$$\|\mathbf{Q}\mathbf{d}\|^2 = Q_{ki}d_i Q_{kj}d_j = \delta_{ij}d_i d_j = \|\mathbf{d}\|^2, \quad (10)$$

and hence every denominator in eq. (8) is unchanged. The referential density $\rho(\mathbf{x}')$ is a property of the reference body and does not depend on the motion at all. It follows at once that $\tilde{\zeta}(\mathbf{x}, t) = \zeta(\mathbf{x}, t)$.

The referential acceleration is

$$\gamma_i(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')d_i}{\|\mathbf{d}\|^3} d^3\mathbf{x}', \quad (11)$$

with $\mathbf{d}$ as above. Under the rigid motion $d_i \rightarrow Q_{ij}d_j$ in the numerator while, by eq. (10), the denominator is unchanged. Since $\mathbf{Q}$ does not depend on $\mathbf{x}'$ it can be taken outside the integral, and so

$$\tilde{\gamma}_i(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')Q_{ij}d_j}{\|\mathbf{d}\|^3} d^3\mathbf{x}' = Q_{ij}\gamma_j(\mathbf{x}, t), \quad (12)$$

as required.

The same conclusions can be reached from the spatial description. If the spatial density of the original motion is $\varrho(\mathbf{y}, t)$, that of the rigidly moved body is $\tilde{\varrho}(\mathbf{y}, t) = \varrho[\mathbf{Q}^T(\mathbf{y} - \mathbf{a}), t]$, and because the Laplacian is invariant under rigid motions the solution of Poisson's equation is simply carried along, $\tilde{\phi}(\mathbf{y}, t) = \phi[\mathbf{Q}^T(\mathbf{y} - \mathbf{a}), t]$. Evaluating at $\mathbf{y} = \tilde{\varphi}(\mathbf{x}, t)$ gives $\tilde{\zeta}(\mathbf{x}, t) = \phi[\varphi(\mathbf{x}, t), t] = \zeta(\mathbf{x}, t)$, while the chain rule applied to $\tilde{g}_i = -\partial\tilde{\phi}/\partial y_i$ produces the factor Q_{ij} .

Two consequences are worth drawing out. First, the gravitational binding energy $\mathcal{V}_g = \frac{1}{2} \int_M \rho \zeta d^3\mathbf{x}$ is unchanged by a rigid motion. The Lagrangian of the lecture is therefore frame indifferent in the same sense as the elastic strain energy, which is essential if the equations of motion are to be physically meaningful. Second, taking $\mathbf{a} = \mathbf{\Phi}(t)$ and $\mathbf{Q} = \mathbf{R}(t)$ to be the translation and rotation introduced in the lecture for the co-rotating frame, eq. (12) is precisely the identity $\gamma_i = R_{ij}\bar{\gamma}_j$ used there, with $\bar{\gamma}$ computed from the relative motion $\bar{\varphi}$. Gravity, unlike the inertial terms, is oblivious to the rotation of the frame; it is only the acceleration that acquires Coriolis and centrifugal contributions.

Example 3: A steadily rotating frame. Let $\mathbf{R}(t)$ be the rotation through angle Ωt about the $\hat{\mathbf{e}}_3$ axis. Compute the matrix $R_{kj} \frac{dR_{kl}}{dt}$, verify that it is anti-symmetric, and use the lecture's relation $R_{kj} \frac{dR_{kl}}{dt} = \epsilon_{jml} \Omega_m$ to identify the angular velocity vector $\boldsymbol{\Omega}$. Write out the Coriolis and centrifugal accelerations in components. Show that the centrifugal acceleration $\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m$ is the gradient of $\psi = -\frac{1}{2} \Omega^2 (x_1^2 + x_2^2)$, and check this against the general expression $\psi = \frac{1}{2} (\Omega_i \Omega_j - \Omega_k \Omega_k \delta_{ij}) x_i x_j$ valid for an arbitrary rotation vector. Evaluate the centrifugal acceleration at the Earth's equator.

Solution. Writing $c = \cos \Omega t$ and $s = \sin \Omega t$, the rotation matrix and its time derivative are

$$\mathbf{R}(t) = \begin{pmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \frac{d\mathbf{R}}{dt} = \Omega \begin{pmatrix} -s & -c & 0 \\ c & -s & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (13)$$

The quantity $R_{kj} \frac{dR_{kl}}{dt}$ is the (j, l) element of $\mathbf{R}^T \frac{d\mathbf{R}}{dt}$, and multiplying out,

$$\begin{aligned} \mathbf{R}^T \frac{d\mathbf{R}}{dt} &= \Omega \begin{pmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -s & -c & 0 \\ c & -s & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= \Omega \begin{pmatrix} -cs + sc & -c^2 - s^2 & 0 \\ s^2 + c^2 & sc - cs & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\Omega & 0 \\ \Omega & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \end{aligned} \quad (14)$$

which is manifestly anti-symmetric and, notably, independent of time. To identify $\boldsymbol{\Omega}$ we compare with $\epsilon_{jml} \Omega_m$. Taking $(j, l) = (1, 2)$ gives $\epsilon_{1m2} \Omega_m = \epsilon_{132} \Omega_3 = -\Omega_3$, which must equal $-\Omega$, while $(j, l) = (2, 3)$ gives $\epsilon_{2m3} \Omega_m = \Omega_1 = 0$ and $(j, l) = (3, 1)$ gives $\Omega_2 = 0$. Hence $\boldsymbol{\Omega} = \Omega \hat{\mathbf{e}}_3$, as expected for a rotation about the third axis. More generally, contracting the lecture's relation with ϵ_{jnl} and using $\epsilon_{jml} \epsilon_{jnl} = 2\delta_{mn}$ gives the explicit formula

$$\Omega_n = \frac{1}{2} \epsilon_{jnl} R_{kj} \frac{dR_{kl}}{dt}, \quad (15)$$

which recovers the angular velocity from any rotation history $\mathbf{R}(t)$.

With $\boldsymbol{\Omega} = \Omega \hat{\mathbf{e}}_3$ the Coriolis acceleration $2\epsilon_{ijk} \Omega_j \bar{v}_k$ is $2\boldsymbol{\Omega} \times \bar{\mathbf{v}}$, with components

$$2\epsilon_{ijk} \Omega_j \bar{v}_k = 2\Omega (-\bar{v}_2, \bar{v}_1, 0), \quad (16)$$

so it is perpendicular to both the rotation axis and the relative velocity, and vanishes for motion parallel to the axis. For the centrifugal term we use the identity $\epsilon_{ijk} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$ to write

$$\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m = \Omega_i \Omega_j x_j - \Omega_j \Omega_j x_i = [\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x})]_i, \quad (17)$$

and for $\boldsymbol{\Omega} = \Omega \hat{\mathbf{e}}_3$ this is

$$\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m = \Omega^2 (0, 0, x_3) - \Omega^2 (x_1, x_2, x_3) = -\Omega^2 (x_1, x_2, 0), \quad (18)$$

which points towards the rotation axis with magnitude Ω^2 times the distance from it.

Now let $\psi = -\frac{1}{2}\Omega^2(x_1^2 + x_2^2)$. Its gradient is $\nabla\psi = -\Omega^2(x_1, x_2, 0)$, which by eq. (18) is exactly the centrifugal acceleration. Thus the term appearing on the left-hand side of the lecture's equations of motion is $\rho \partial\psi/\partial x_i$, and when it is carried to the right-hand side it becomes the centrifugal body force $-\rho \partial\psi/\partial x_i$ per unit volume, directed away from the axis; ψ is the **centrifugal potential**. To check the general formula, differentiate $\psi = \frac{1}{2}(\Omega_i\Omega_j - \Omega_k\Omega_k\delta_{ij})x_ix_j$, noting that the coefficient matrix is symmetric, to find

$$\frac{\partial\psi}{\partial x_i} = (\Omega_i\Omega_j - \Omega_k\Omega_k\delta_{ij})x_j = \Omega_i\Omega_jx_j - \Omega_k\Omega_kx_i, \quad (19)$$

which is eq. (17). Substituting $\Omega = \Omega\hat{\mathbf{e}}_3$ into the general formula gives

$$\psi = \frac{1}{2} \left[\Omega^2x_3^2 - \Omega^2(x_1^2 + x_2^2 + x_3^2) \right] = -\frac{1}{2}\Omega^2(x_1^2 + x_2^2), \quad (20)$$

in agreement with the special case. Note that $\nabla^2\psi = -2\Omega^2 \neq 0$, so unlike the gravitational potential in free space the centrifugal potential is not harmonic.

At the equator the distance from the axis is b , and the centrifugal acceleration has magnitude

$$\Omega^2b = (7.29 \times 10^{-5})^2 \times 6.371 \times 10^6 \text{ m s}^{-2} \approx 3.39 \times 10^{-2} \text{ m s}^{-2}, \quad (21)$$

which is $\Omega^2b^3/GM \approx 3.4 \times 10^{-3}$, or about one part in 290, of the surface gravity found in Example 1. This small ratio is what permits the Earth's equilibrium figure to be treated as a slightly flattened sphere in the following lecture, where the centrifugal and gravitational potentials are combined into a single geopotential.

Example 4: Scaling analysis with numbers. Using the order-of-magnitude estimates of the lecture, with $\bar{\rho} = 5000 \text{ kg m}^{-3}$ and $\bar{v} = 8000 \text{ m s}^{-1}$ as typical density and wave speed, evaluate the ratio of gravitational to elastic forces, the ratio ΩT of Coriolis to inertial terms, and the ratio of centrifugal to inertial terms, for (a) a 1 Hz body wave with $L \sim 10 \text{ km}$; (b) a 100 s surface wave with $L \sim 400 \text{ km}$; and (c) the gravest spheroidal mode ${}_0S_2$, with period 54 min and $L \sim b$. Tabulate the results and state where gravity and rotation matter.

Solution. From the lecture, the ratios of the various terms in the linearised equations of motion to the elastic and inertial terms are

$$\frac{\text{gravitational}}{\text{elastic}} = \frac{4\pi G\bar{\rho}L^2}{\bar{v}^2}, \quad \frac{\text{Coriolis}}{\text{inertial}} = \Omega T, \quad \frac{\text{centrifugal}}{\text{inertial}} = \frac{2\bar{\rho}\Omega^2U}{\bar{\rho}U/T^2} = 2(\Omega T)^2, \quad (22)$$

where the final ratio follows from the lecture's estimate $2\bar{\rho}\Omega^2U$ for the centrifugal term; the factor of two is an upper bound, attained when $\mathbf{u}$ is perpendicular to Ω , and is of no consequence in an order-of-magnitude argument. It is convenient to note the combination

$$4\pi G\bar{\rho} = 4\pi \times 6.674 \times 10^{-11} \times 5000 \text{ s}^{-2} = 4.19 \times 10^{-6} \text{ s}^{-2}, \quad (23)$$

so that the gravitational ratio is $(L/L_g)^2$ with $L_g = \bar{v}/\sqrt{4\pi G\bar{\rho}} \approx 3900 \text{ km}$, while the rotational ratios involve T compared with $\Omega^{-1} \approx 1.37 \times 10^4 \text{ s} \approx 3.8 \text{ h}$. Substituting the three sets of values gives the following table.

Case	T (s)	L (km)	grav./elastic	ΩT	$2(\Omega T)^2$
(a) body wave	1	10	6.55×10^{-6}	7.29×10^{-5}	1.06×10^{-8}
(b) surface wave	100	400	1.05×10^{-2}	7.29×10^{-3}	1.06×10^{-4}
(c) ${}_0S_2$	3240	6371	2.66	0.236	0.112

For the body wave every correction is utterly negligible: gravity enters at the level of a few parts per million and rotation at the level of one part in 10^4 or below, so the elastodynamics of the earlier lectures is exact for all practical purposes. For the 100 s surface wave the gravitational correction has grown to about one per cent, as expected from the L^2 scaling, and although still small it is comparable to the fractional variations in wave speed that surface-wave tomography seeks to resolve, so gravity is routinely included in surface-wave calculations at these periods. Rotation, by contrast, remains below one per cent. For ${}_0S_2$ the gravitational ratio is of order unity – this is the lecture’s estimate of about 3 – and gravity is as important as elasticity; indeed, the restoring force for this mode is largely gravitational. The Coriolis ratio $\Omega T \approx 0.24$ is no longer small, and it is responsible for the observable splitting of the gravest modes into multiplets. The centrifugal ratio $2(\Omega T)^2 \approx 0.1$ is smaller but not negligible at this period.

The table illustrates the general conclusion of the lecture. Gravity matters for deformations whose length scale approaches L_g of a few thousand kilometres, which for seismic waves means periods of several hundred seconds and longer, while rotation matters for time scales that approach Ω^{-1} , i.e. for the longest-period normal modes, and is dominant only for phenomena such as tides with periods comparable to a day.

Example 5: Linearised gravitational acceleration for rigid displacements. The lecture gives the first-order perturbation to the referential gravitational acceleration as

$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{\|\mathbf{x} - \mathbf{x}'\|^3} - \frac{3(x_i - x'_i)(x_j - x'_j)}{\|\mathbf{x} - \mathbf{x}'\|^5} \right) (u_j - u'_j) d^3 \mathbf{x}'. \quad (24)$$

Show that a rigid translation $\mathbf{u} = \mathbf{u}_0$ gives $\gamma^1 = \mathbf{0}$, and that an infinitesimal rigid rotation $\mathbf{u} = \boldsymbol{\theta} \times \mathbf{x}$, with $\boldsymbol{\theta}$ a constant vector, gives $\gamma^1 = \boldsymbol{\theta} \times \gamma^0$, where γ^0 is the equilibrium referential acceleration. Explain why both results are to be expected.

Solution. For the translation, $u_j - u'_j = u_{0j} - u_{0j} = 0$ for every pair of points, so the integrand in eq. (24) vanishes identically and $\gamma^1 = \mathbf{0}$. Nothing needs to be assumed about the density or the shape of the body. Physically, $\gamma_i(\mathbf{x}, t) = g_i[\boldsymbol{\varphi}(\mathbf{x}, t), t]$ is the acceleration felt by the material particle labelled $\mathbf{x}$. Translating the whole body translates its gravitational field along with it, and each particle continues to feel exactly the field it felt before; only in the spatial description, at fixed $\mathbf{y}$, does the field change. This is the linearised form of the translational invariance established in Example 2.

For the rotation write $\mathbf{r} = \mathbf{x} - \mathbf{x}'$ and $r = \|\mathbf{r}\|$. Then

$$u_j - u'_j = \epsilon_{jkl} \theta_k (x_l - x'_l) = \epsilon_{jkl} \theta_k r_l = (\boldsymbol{\theta} \times \mathbf{r})_j, \quad (25)$$

which is perpendicular to $\mathbf{r}$. Contracting with the kernel in eq. (24),

$$\left(\frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5} \right) (\boldsymbol{\theta} \times \mathbf{r})_j = \frac{(\boldsymbol{\theta} \times \mathbf{r})_i}{r^3} - \frac{3r_i \mathbf{r} \cdot (\boldsymbol{\theta} \times \mathbf{r})}{r^5} = \frac{(\boldsymbol{\theta} \times \mathbf{r})_i}{r^3}, \quad (26)$$

because $\mathbf{r} \cdot (\boldsymbol{\theta} \times \mathbf{r}) = 0$. The second, traceless part of the kernel therefore contributes nothing, and

$$\gamma_i^1 = -G \int_M \rho' \frac{\epsilon_{ikl} \theta_k r_l}{r^3} d^3 \mathbf{x}' = \epsilon_{ikl} \theta_k \left(-G \int_M \rho' \frac{x_l - x'_l}{\|\mathbf{x} - \mathbf{x}'\|^3} d^3 \mathbf{x}' \right) = \epsilon_{ikl} \theta_k \gamma_l^0, \quad (27)$$

where the bracketed integral is the equilibrium referential acceleration obtained by setting $\varphi(\mathbf{x}) = \mathbf{x}$ in the lecture's integral formula. That is, $\gamma^1 = \boldsymbol{\theta} \times \gamma^0$.

This too is what Example 2 predicts. A finite rotation $\mathbf{Q}$ transforms the referential acceleration as $\gamma \rightarrow \mathbf{Q}\gamma$, and for an infinitesimal rotation $Q_{ij} = \delta_{ij} + s \epsilon_{ikj} \theta_k + O(s^2)$, corresponding to the motion $\varphi_i = x_i + s \epsilon_{ikj} \theta_k x_j + O(s^2)$, so that $\gamma_i = \gamma_i^0 + s \epsilon_{ikj} \theta_k \gamma_j^0 + O(s^2)$; the first-order term is exactly eq. (27). The gravitational field is rigidly rotated with the body, and each particle feels the rotated field. In the non-rotating case, Example 2 shows that a rigid motion of an equilibrium configuration is again an equilibrium configuration, so rigid translations and rotations must be zero-frequency solutions of the linearised equations of motion; the two calculations here are the gravitational part of that statement, and the remaining terms must conspire in the same way. When the frame rotates the argument no longer applies as it stands, because the centrifugal term $\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m$ does not vanish for a rigid translation transverse to the axis, and the rigid-body modes of a rotating planet require separate treatment.

Worked examples for Lecture 21: Equilibrium figures

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These examples accompany Lecture 21. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory. Throughout, $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, the mean radius of the Earth is $b = 6371 \text{ km}$, its mass is $M = 5.972 \times 10^{24} \text{ kg}$, its mean density is $\bar{\rho} = 5515 \text{ kg m}^{-3}$, and its angular velocity is $\Omega = 7.292 \times 10^{-5} \text{ s}^{-1}$. Values quoted for the Preliminary Reference Earth Model (PREM) are from Dziewonski & Anderson (1981).

Example 1: Pressure inside a homogeneous non-rotating planet. A non-rotating planet of radius b has uniform density ρ . Show that the hydrostatic pressure is

$$p(r) = \frac{2}{3}\pi G\rho^2(b^2 - r^2),$$

and express the central pressure in terms of the mass M and the surface gravity $g(b)$. Evaluate the central pressure for a planet with the mean radius and mean density of the Earth, compare with the value $3.64 \times 10^{11} \text{ Pa}$ at the centre of PREM, and explain the discrepancy.

Solution. For a spherically symmetric non-rotating planet the hydrostatic equilibrium equation reduces to

$$\frac{dp}{dr} + \rho g_0 = 0, \quad g_0(r) = \frac{d\phi_0}{dr} = \frac{GM(r)}{r^2}, \quad (1)$$

where $M(r)$ is the mass within radius r , and $g_0 > 0$ is the magnitude of the gravitational acceleration. The expression for g_0 follows from Gauss's law applied to Poisson's equation, or equivalently from Newton's shell theorem. For uniform density $M(r) = \frac{4}{3}\pi\rho r^3$, and so

$$g_0(r) = \frac{4}{3}\pi G\rho r, \quad (2)$$

which increases linearly from the centre to the surface. Integrating eq. (1) inwards from the surface, where $p(b) = 0$,

$$p(r) = \int_r^b \rho g_0(r') dr' = \frac{4}{3}\pi G\rho^2 \int_r^b r' dr' = \frac{2}{3}\pi G\rho^2(b^2 - r^2), \quad (3)$$

as required. The pressure is a parabola in r , and its central value is

$$p(0) = \frac{2}{3}\pi G\rho^2 b^2. \quad (4)$$

Substituting $\rho = 3M/(4\pi b^3)$ we can write this as

$$p(0) = \frac{3GM^2}{8\pi b^4} = \frac{3g(b)^2}{8\pi G}, \quad (5)$$

where $g(b) = GM/b^2$ is the surface gravity. The second form is worth remembering: it says that the central pressure of a homogeneous planet is fixed by its surface gravity alone, and does not otherwise depend on its size.

For the Earth's mean density and radius, eq. (4) gives

$$p(0) = \frac{2}{3}\pi \times 6.674 \times 10^{-11} \times (5515)^2 \times (6.371 \times 10^6)^2 \text{ Pa} = 1.73 \times 10^{11} \text{ Pa}, \quad (6)$$

which is less than half the PREM value of 3.64×10^{11} Pa. The discrepancy arises because the Earth's mass is concentrated towards its centre: the density in the core is roughly twice that in the mantle, and within each region density increases with depth through self-compression. Two effects then act in the same direction. First, concentrating mass towards the centre increases $M(r)/r^2$ at depth, so that g_0 within the Earth is everywhere larger than the linear profile of eq. (2); in PREM, g_0 is roughly constant at about 10 m s^{-2} through the whole mantle, and only falls towards zero within the core. Second, the weight per unit volume ρg_0 at depth is further increased by the larger density there. Both effects steepen the pressure gradient at depth, and hence raise the central pressure. The homogeneous model gets the mass and surface gravity right, but the pressure within a planet depends on how that mass is distributed with radius, and a homogeneous model places too much of it near the surface. The next example quantifies this with the simplest model that has a dense core.

Example 2: Two-layer planet. A non-rotating planet consists of a core of uniform density ρ_c and radius c , surrounded by a mantle of uniform density ρ_m extending to the surface $r = b$. Obtain the gravitational acceleration $g_0(r)$ and the hydrostatic pressure $p(r)$ in both layers. Taking $\rho_c = 11 \times 10^3 \text{ kg m}^{-3}$, $\rho_m = 4.5 \times 10^3 \text{ kg m}^{-3}$, $c = 3480 \text{ km}$, and $b = 6371 \text{ km}$, compute the mass, the moment of inertia factor $I/(Mb^2)$, the pressure at the core–mantle boundary (CMB), and the central pressure. Compare with the Earth's values $I/(Mb^2) = 0.3307$, and with PREM's $p(\text{CMB}) = 1.36 \times 10^{11} \text{ Pa}$ and $p(0) = 3.64 \times 10^{11} \text{ Pa}$.

Solution. The mass within radius r is

$$M(r) = \begin{cases} \frac{4}{3}\pi\rho_c r^3 & 0 \leq r \leq c, \\ \frac{4}{3}\pi [\rho_c c^3 + \rho_m(r^3 - c^3)] & c \leq r \leq b, \end{cases} \quad (7)$$

and so, with $g_0 = GM(r)/r^2$ as in Example 1,

$$g_0(r) = \begin{cases} \frac{4}{3}\pi G\rho_c r & 0 \leq r \leq c, \\ \frac{4}{3}\pi G \left[\rho_m r + (\rho_c - \rho_m)\frac{c^3}{r^2} \right] & c \leq r \leq b. \end{cases} \quad (8)$$

In the mantle, g_0 is the sum of the linearly increasing field of a homogeneous planet of density ρ_m and the inverse-square field of the excess core mass $\frac{4}{3}\pi(\rho_c - \rho_m)c^3$. These

two terms vary in opposite senses with r , which is why g_0 is so nearly uniform through the Earth's mantle. Note that g_0 is continuous at $r = c$, as it must be since $M(r)$ is continuous.

The pressure follows from $dp/dr = -\rho g_0$ with $p(b) = 0$. In the mantle,

$$p(r) = \rho_m \int_r^b g_0(r') dr' = \frac{2}{3}\pi G \rho_m^2 (b^2 - r^2) + \frac{4}{3}\pi G \rho_m (\rho_c - \rho_m) c^3 \left(\frac{1}{r} - \frac{1}{b} \right), \quad c \leq r \leq b. \quad (9)$$

The first term is the pressure in a homogeneous planet of density ρ_m , while the second is the additional pressure needed to support the mantle against the attraction of the excess core mass. Setting $r = c$ gives the pressure at the CMB,

$$p(c) = \frac{2}{3}\pi G \rho_m^2 (b^2 - c^2) + \frac{4}{3}\pi G \rho_m (\rho_c - \rho_m) c^2 \left(1 - \frac{c}{b} \right). \quad (10)$$

Within the core the density is uniform and g_0 is linear in r , so the calculation of Example 1 applies with the constant of integration fixed by continuity of p at the CMB,

$$p(r) = p(c) + \frac{2}{3}\pi G \rho_c^2 (c^2 - r^2), \quad 0 \leq r \leq c. \quad (11)$$

The pressure is continuous at $r = c$, but its gradient $-\rho g_0$ jumps by the factor ρ_c/ρ_m , since g_0 is continuous while ρ is not.

We now put in the numbers. From eq. (7) with $r = b$, the mass is

$$M = \frac{4}{3}\pi \left[\rho_c c^3 + \rho_m (b^3 - c^3) \right] = 6.02 \times 10^{24} \text{ kg}, \quad (12)$$

within one per cent of the Earth's mass. The moment of inertia about any axis through the centre of a spherically symmetric body is $I = \frac{8}{3}\pi \int_0^b \rho r^4 dr$, and hence

$$I = \frac{8}{15}\pi \left[\rho_c c^5 + \rho_m (b^5 - c^5) \right] = 8.47 \times 10^{37} \text{ kg m}^2, \quad \frac{I}{Mb^2} = 0.347. \quad (13)$$

For a homogeneous sphere $I/(Mb^2) = 0.4$, and the observed value of 0.3307 for the Earth is one of the classical indications that its density increases with depth. The two-layer model accounts for most of the reduction, but not all of it, because within each layer the real density increases with depth under compression, which the model neglects. The gravitational acceleration from eq. (8) is $g_0(b) = 9.90 \text{ m s}^{-2}$ at the surface and $g_0(c) = 10.70 \text{ m s}^{-2}$ at the CMB, the values in PREM being 9.82 and 10.68 m s^{-2} . For the pressure, the two terms in eq. (10) evaluate to $8.06 \times 10^{10} \text{ Pa}$ and $4.49 \times 10^{10} \text{ Pa}$, giving

$$p(c) = 1.26 \times 10^{11} \text{ Pa}, \quad (14)$$

and the core then adds $\frac{2}{3}\pi G \rho_c^2 c^2 = 2.05 \times 10^{11} \text{ Pa}$, so that from eq. (11)

$$p(0) = 3.30 \times 10^{11} \text{ Pa}. \quad (15)$$

These are within 8 and 9 per cent of the PREM values of $1.36 \times 10^{11} \text{ Pa}$ and $3.64 \times 10^{11} \text{ Pa}$, and almost twice the central pressure of the homogeneous model of Example 1. The model errs on the low side for the same reason that $I/(Mb^2)$ errs on the high side: neglecting self-compression places too little mass at depth. A model with only two free densities nevertheless reproduces the mass, moment of inertia, and pressures of the Earth to within about ten per cent, which illustrates how strongly the gross structure of a planet is controlled by the existence of a dense core.

Example 3: Centrifugal potential and Y_{20} . For $\mathbf{\Omega} = \Omega \hat{\mathbf{e}}_3$ show that the centrifugal potential $\psi = \frac{1}{2}(\Omega_i \Omega_j - \Omega_k \Omega_k \delta_{ij}) x_i x_j$ defined in the lecture can be written

$$\psi = -\frac{1}{3}\Omega^2 r^2 + \frac{1}{3}\Omega^2 r^2 P_2(\cos \theta),$$

with θ the colatitude and P_2 the Legendre polynomial of degree two, and hence that its aspherical part is proportional to $r^2 Y_{20}$. Explain why the spherical part can be absorbed into the reference model, and deduce that all first-order perturbations of a slowly rotating hydrostatic planet have Y_{20} angular dependence. Using the lecture's definition of the ellipticity, show that the surface of the planet is $r(\theta) = b [1 - \frac{2}{3}\epsilon(b) P_2(\cos \theta)]$ and verify that $\epsilon(b)$ is the difference between the equatorial and polar radii divided by the mean radius.

Solution. With $\Omega_i = \Omega \delta_{i3}$ the definition gives

$$\psi = \frac{1}{2} (\Omega^2 x_3^2 - \Omega^2 x_k x_k) = -\frac{1}{2}\Omega^2 (x_1^2 + x_2^2) = -\frac{1}{2}\Omega^2 r^2 \sin^2 \theta, \quad (16)$$

which is minus one half of the square of the distance from the rotation axis times Ω^2 . Its gradient is $-\Omega^2(x_1, x_2, 0)$, the outward centrifugal acceleration $-\mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{x})$, as it should be. Recalling that $P_2(\cos \theta) = \frac{1}{2}(3 \cos^2 \theta - 1)$, we have

$$\sin^2 \theta = 1 - \cos^2 \theta = 1 - \frac{1}{3} [2P_2(\cos \theta) + 1] = \frac{2}{3} [1 - P_2(\cos \theta)], \quad (17)$$

and substituting into eq. (16) gives

$$\psi = -\frac{1}{3}\Omega^2 r^2 + \frac{1}{3}\Omega^2 r^2 P_2(\cos \theta), \quad (18)$$

as required. The zonal spherical harmonic of degree two is $Y_{20} = \sqrt{5/(4\pi)} P_2(\cos \theta)$, with the normalisation $\int |Y_{20}|^2 dS = 1$ over the unit sphere, and so the aspherical part of ψ is

$$\psi^{\text{asph}} = \sqrt{\frac{4\pi}{5}} \frac{1}{3} \Omega^2 r^2 Y_{20}. \quad (19)$$

As a check, eq. (16) gives $\nabla^2 \psi = -2\Omega^2$, and since $\nabla^2 r^2 = 6$ the spherical part of eq. (18) supplies the whole of this, while $r^2 Y_{20}$ is a solid harmonic and hence contributes nothing.

The spherical part $-\frac{1}{3}\Omega^2 r^2$ depends on r alone. Adding it to the gravitational potential of the reference model simply changes the zeroth-order equilibrium equation to $dp_0/dr + \rho_0(g_0 - \frac{2}{3}\Omega^2 r) = 0$, that is, the spherical model feels a slightly reduced effective gravity. The relative change, $\frac{2}{3}\Omega^2 r/g_0$, is at most $\frac{2}{3}\Omega^2 b^3/(GM) \approx 2 \times 10^{-3}$ for the Earth, so this is a first-order correction to a spherically symmetric model. It changes the reference values of p_0 and g_0 by a fraction of order the small parameter, but does not alter the spherical symmetry of the reference state, and it is this spherical symmetry alone that the first-order theory relies upon. We may therefore regard the spherical part as already included in ρ_0 , p_0 , and ϕ_0 , and take $\psi_1 = \psi^{\text{asph}}$. This is precisely the freedom used in the lecture when the perturbations were assumed to average to zero over spherical surfaces.

Now consider the first-order problem. The boundary value problem for ϕ_1 is linear in ϕ_1 and ψ_1 , its coefficients $4\pi G\rho'_0/g_0$ and $4\pi G\rho_0/g_0$ depend on r alone, and the Laplacian commutes with rotations. Substituting the separated form $\phi_1 = f(r)Y_{20}$, the angular dependence factors out of every term, leaving an ordinary differential equation in r for f whose forcing is proportional to r^2 . Any other spherical harmonic component of ϕ_1 would satisfy the same problem with no forcing, and must vanish for the perturbation to be unique.¹ Hence ϕ_1 , and therefore $\gamma_1 = \phi_1 + \psi_1$, is proportional to Y_{20} . The remaining first-order fields are given in terms of γ_1 by the algebraic relations $p_1 = -\rho_0\gamma_1$, $h_1 = -\gamma_1/g_0$, and $\rho_1 = (\rho'_0/g_0)\gamma_1$, with coefficients depending on r alone, and so all have the same Y_{20} dependence. Since Y_{20} is independent of longitude and even in $\cos\theta$, the perturbed planet is axisymmetric and symmetric about its equatorial plane.

Finally, the lecture defines the ellipticity by $\gamma_1 = \sqrt{4\pi/5} \frac{2}{3} r g_0 \epsilon Y_{20} = \frac{2}{3} r g_0 \epsilon P_2(\cos\theta)$, and therefore the surface displacement is

$$h_1 = -\frac{\gamma_1}{g_0} \Big|_{r=b} = -\frac{2}{3} b \epsilon(b) P_2(\cos\theta), \quad r(\theta) = b \left[1 - \frac{2}{3} \epsilon(b) P_2(\cos\theta) \right]. \quad (20)$$

At the equator $P_2(0) = -\frac{1}{2}$ and at the poles $P_2(\pm 1) = 1$, so the equatorial and polar radii are

$$r_{\text{eq}} = b \left[1 + \frac{1}{3} \epsilon(b) \right], \quad r_{\text{pol}} = b \left[1 - \frac{2}{3} \epsilon(b) \right], \quad (21)$$

and their difference is $r_{\text{eq}} - r_{\text{pol}} = b \epsilon(b)$. For $\epsilon > 0$ the planet is oblate, with the equatorial bulge twice as far from the mean sphere as the polar depression, as required by the vanishing of the spherical average of P_2 . That average also shows that b is the mean radius of the perturbed surface, and that its volume is unchanged to first order. Hence $\epsilon(b) = (r_{\text{eq}} - r_{\text{pol}})/b$, which is the geometrical interpretation quoted in the lecture; for the Earth $r_{\text{eq}} - r_{\text{pol}} \approx 21.4 \text{ km}$ and $\epsilon(b) \approx 1/298$. The lesson is that the entire aspherical structure of a slowly rotating hydrostatic planet is described by one radial function, because the forcing has a single angular component and the problem is linear.

Example 4: Clairaut's equation for a homogeneous planet. Show that for a planet of uniform density ρ_0 one has $8\pi G\rho_0/g_0 = 6/r$, so that Clairaut's equation reduces to $\epsilon'' + 6\epsilon'/r = 0$, whose regular solution is $\epsilon = \text{const}$. Apply the surface boundary condition to obtain Newton's result

$$\epsilon = \frac{5}{4} \frac{\Omega^2 b^3}{GM} = \frac{5}{4} m,$$

where $m = \Omega^2 b^3/(GM)$. Evaluate $1/\epsilon$ for a homogeneous Earth, compare with the observed value of 298, and explain the sign of the discrepancy.

Solution. For constant ρ_0 we found in Example 1 that $g_0 = \frac{4}{3}\pi G\rho_0 r$, and so

$$\frac{8\pi G\rho_0}{g_0} = \frac{8\pi G\rho_0}{\frac{4}{3}\pi G\rho_0 r} = \frac{6}{r}. \quad (22)$$

¹ The one exception is the degree-one part, which corresponds to a rigid translation of the whole planet and is removed by keeping the centre of mass at the origin.

More generally, $g_0 = \frac{4}{3}\pi G \bar{\rho}(r) r$ with $\bar{\rho}(r)$ the mean density inside radius r , and the coefficient in Clairaut's equation is $6\rho_0(r)/[r\bar{\rho}(r)]$; it is the ratio of the local density to the mean density within r that controls the solution. Substituting eq. (22) into Clairaut's equation,

$$\frac{d^2\epsilon}{dr^2} + \frac{6}{r} \left(\frac{d\epsilon}{dr} + \frac{\epsilon}{r} \right) - \frac{6\epsilon}{r^2} = \frac{d^2\epsilon}{dr^2} + \frac{6}{r} \frac{d\epsilon}{dr} = 0, \quad (23)$$

the terms in ϵ/r^2 cancelling exactly. This is an equation of Euler type, and trying $\epsilon = r^n$ gives $n(n-1) + 6n = n(n+5) = 0$, so the general solution is

$$\epsilon = A + Br^{-5}. \quad (24)$$

The second solution is singular at the centre and violates the condition $\epsilon'(0) = 0$, so we must take $B = 0$: within a homogeneous planet the ellipticity is the same on every level surface, and all the level surfaces are similar spheroids.

The constant A is fixed by the surface condition. With $\epsilon'(b) = 0$ the boundary condition from the lecture,

$$\left. \frac{d\epsilon}{dr} \right|_{r=b} = \frac{1}{b} \left[\frac{5\Omega^2 b^3}{2GM} - 2\epsilon(b) \right], \quad (25)$$

becomes $2\epsilon = 5\Omega^2 b^3 / (2GM)$, and so

$$\epsilon = \frac{5}{4} \frac{\Omega^2 b^3}{GM} = \frac{5}{4} m, \quad m = \frac{\Omega^2 b^3}{GM} = \frac{\Omega^2 b}{g(b)} = \frac{3\Omega^2}{4\pi G \rho_0}, \quad (26)$$

which is Newton's result. The parameter m is the ratio of the centrifugal to the gravitational acceleration at the equator; the last form shows it to be, up to a numerical factor, the ratio of centrifugal to gravitational potentials used in the lecture to define slow rotation. For the Earth, $m = (7.292 \times 10^{-5})^2 (6.371 \times 10^6)^3 / (3.986 \times 10^{14}) = 3.45 \times 10^{-3}$, so that

$$\epsilon = \frac{5}{4} m = 4.31 \times 10^{-3}, \quad \frac{1}{\epsilon} = 232, \quad (27)$$

compared with the observed $1/\epsilon = 298$. The homogeneous planet is thus too flattened by about 30 per cent.²

To understand the sign of the discrepancy it helps to see where the factor $5/4$ comes from. Write $\gamma_1 = \frac{2}{3} r g_0 \epsilon P_2(\cos \theta)$ as in Example 3. Displacing the surface of a homogeneous planet by $h_1 = -\frac{2}{3} b \epsilon P_2$ is equivalent, to first order, to placing a surface mass density $\sigma = \rho_0 h_1$ on the sphere $r = b$. The interior potential of a surface density $\sigma_2 P_2(\cos \theta)$ on a sphere of radius b is $-\frac{4}{5} \pi G b \sigma_2 (r/b)^2 P_2(\cos \theta)$, a standard consequence of the Legendre expansion of $1/\|\mathbf{x} - \mathbf{x}'\|$, and so

$$\phi_1 = \frac{8}{15} \pi G \rho_0 \epsilon r^2 P_2(\cos \theta), \quad r < b. \quad (28)$$

² Newton obtained $1/230$ in the "Principia" using a slightly different value of m . The French geodetic expeditions to Lapland and Peru in the 1730s confirmed that the Earth is oblate, against the prolate Earth favoured by the Cassinis, but accurate values of the flattening came only later from geodesy and from the analysis of the Moon's motion. Clairaut's theory of 1743 explained why the Earth is less flattened than a homogeneous planet.

This is positive at the poles and negative at the equator: the extra mass in the equatorial bulge deepens the potential well there. Adding the aspherical centrifugal potential of eq. (18) and equating to $\gamma_1 = \frac{2}{3}rg_0\epsilon P_2 = \frac{8}{9}\pi G\rho_0\epsilon r^2 P_2$ gives

$$\frac{8}{9}\pi G\rho_0\epsilon = \frac{8}{15}\pi G\rho_0\epsilon + \frac{1}{3}\Omega^2 \implies \epsilon = \frac{15\Omega^2}{16\pi G\rho_0} = \frac{5}{4}m, \quad (29)$$

in agreement with eq. (26), and providing an independent check on the signs in Clairaut's equation and its surface condition. The proportionality to r^2 of every term is the reason ϵ is constant. Had we ignored the self-gravity of the bulge, dropping the first term on the right of eq. (29), we would have found $\epsilon = 3\Omega^2/(8\pi G\rho_0) = \frac{1}{2}m$, the value for a massless fluid envelope surrounding a point mass, for which $1/\epsilon = 580$ for the Earth. The self-attraction of the equatorial bulge therefore amplifies the flattening of a homogeneous planet by a factor of 5/2.

The real Earth lies between these two limits. In a centrally condensed planet the equatorial bulge contains a smaller fraction of the total mass, its self-attraction is weaker, and the flattening is correspondingly smaller than $\frac{5}{4}m$, though larger than $\frac{1}{2}m$. The observed $1/\epsilon = 298 > 232$ is therefore a further indication, independent of the moment of inertia, that the Earth's density increases with depth. Solving Clairaut's equation by shooting for the two-layer model of Example 2 gives $1/\epsilon(b) = 285$ at the surface and $1/\epsilon = 374$ at the CMB, so that ϵ increases outwards from the centre. The same trend is seen in the lower panel of the figure in Lecture 21, where $1/\epsilon$ is about 300 at the surface and increases inwards, and it is a general property of solutions of Clairaut's equation for a density that does not increase outwards. The level surfaces deep within a planet are less flattened than its outer surface because they feel the centrifugal potential, which grows as r^2 , less strongly relative to the gravitational attraction of the mass within them.

Example 5: Hydrostatic level surfaces. In the lecture it was shown that in a hydrostatic planet the level surfaces of p , ρ , and γ coincide. Verify this directly for the first-order solution of the slowly rotating problem, $p_1 = -\rho_0\gamma_1$ and $\rho_1 = (\rho'_0/g_0)\gamma_1$, by showing that the level surfaces of all three fields through the reference radius r are displaced radially by the same amount $-\gamma_1/g_0$. Show also that the radial component of the first-order equilibrium equation, $\nabla p_1 + \rho_0\nabla\gamma_1 + \rho_1g_0\hat{\mathbf{r}} = \mathbf{0}$, is then satisfied identically, and interpret the function $\epsilon(r)$ in the light of this result.

Solution. Consider a field $F = F_0(r) + sF_1(r, \theta, \varphi) + \dots$ whose zeroth-order part depends on r alone. The level surface of F that passes through radius r in the reference model is, to first order, $r + s\delta r_F(\theta, \varphi)$, where

$$F_0(r + s\delta r_F) + sF_1(r, \theta, \varphi) = F_0(r) \implies \delta r_F = -\frac{F_1}{F'_0}, \quad (30)$$

provided $F'_0 \neq 0$. This is the same expansion that produced the boundary condition $p_1 + p'_0h_1 = 0$ on the perturbed surface in the lecture, the outer surface being the level surface $p = 0$. Applying eq. (30) in turn to the three fields, and using the zeroth-order

hydrostatic equation $p'_0 = -\rho_0 g_0$ together with $\gamma'_0 = \phi'_0 = g_0$, we find

$$\delta r_p = -\frac{p_1}{p'_0} = -\frac{-\rho_0 \gamma_1}{-\rho_0 g_0} = -\frac{\gamma_1}{g_0}, \quad (31)$$

$$\delta r_\rho = -\frac{\rho_1}{\rho'_0} = -\frac{1}{\rho'_0} \frac{\rho'_0}{g_0} \gamma_1 = -\frac{\gamma_1}{g_0}, \quad (32)$$

$$\delta r_\gamma = -\frac{\gamma_1}{\gamma'_0} = -\frac{\gamma_1}{g_0}. \quad (33)$$

The three displacements are identical, and equal to the surface displacement h_1 of the lecture evaluated at radius r . Thus at every depth the perturbed level surfaces of pressure, density, and gravity potential coincide, as the exact theory requires. Where $\rho'_0 = 0$ the density has no level surfaces to speak of, but there $\rho_1 = 0$ as well, and there is nothing to check.

The relations for p_1 and ρ_1 can be read in the same way. Writing $h(r, \theta, \varphi) = -\gamma_1/g_0$ for the displacement of the level surface through r , we have

$$p_1 = -\rho_0 \gamma_1 = -p'_0 h, \quad \rho_1 = \frac{\rho'_0}{g_0} \gamma_1 = -\rho'_0 h, \quad (34)$$

which are exactly the changes at a fixed point produced by displacing the spherical profiles $p_0(r)$ and $\rho_0(r)$ radially outwards through h . The first-order solution is, in other words, nothing more than the reference model with each of its level surfaces carried onto the corresponding level surface of the perturbed gravity potential.

Now take the radial component of the first-order equilibrium equation. Substituting $p_1 = -\rho_0 \gamma_1$,

$$\frac{\partial p_1}{\partial r} + \rho_0 \frac{\partial \gamma_1}{\partial r} + \rho_1 g_0 = -\rho'_0 \gamma_1 - \rho_0 \frac{\partial \gamma_1}{\partial r} + \rho_0 \frac{\partial \gamma_1}{\partial r} + \rho_1 g_0 = g_0 \left(\rho_1 - \frac{\rho'_0}{g_0} \gamma_1 \right), \quad (35)$$

which vanishes by the expression for ρ_1 . The tangential components are satisfied because $p_1 + \rho_0 \gamma_1 = 0$ identically. Note that the argument can be run in the other direction: once the tangential components have given $p_1 = -\rho_0 \gamma_1$, the radial component yields $\rho_1 = (\rho'_0/g_0) \gamma_1$ directly, without taking the curl of the equation as was done in the lecture. The two routes must agree, because the curl of the equilibrium equation is a consequence of the equation itself.

Finally, with $\gamma_1 = \frac{2}{3} r g_0 \epsilon(r) P_2(\cos \theta)$ the common displacement is

$$h = -\frac{\gamma_1}{g_0} = -\frac{2}{3} r \epsilon(r) P_2(\cos \theta), \quad (36)$$

which is eq. (20) of Example 3 with b replaced by r . Every level surface of the perturbed planet is therefore an oblate spheroid of mean radius r whose equatorial and polar radii differ by $r \epsilon(r)$, and $\epsilon(r)$ is the ellipticity of the level surface of mean radius r . The lecture's geometrical interpretation of $\epsilon(b)$ is the special case of the outer surface, and the same interpretation applies to the CMB and to any other internal boundary, each of which must be a level surface in a hydrostatic planet. This is why the profile of $1/\epsilon$ plotted in the lecture, and computed for the two-layer model in Example 4, can be read directly as the flattening of the Earth's internal structure: a two-layer Earth in hydrostatic equilibrium would have a CMB with equatorial and polar radii differing by $c/374 \approx 9$ km.

Worked examples for Lecture 22: Free oscillations

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These examples accompany Lecture 22. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Example 1: A two-degree-of-freedom analogue. Two particles of mass m and $2m$ move along a line. The first is attached to a fixed wall by a spring of stiffness k , the two particles are joined by a spring of stiffness $2k$, and the second is attached to a second fixed wall by a spring of stiffness $2k$. Let $u_1(t)$ and $u_2(t)$ be the displacements from equilibrium, and let external forces $f_1(t)$ and $f_2(t)$ act on the particles. (a) Write the equations of motion in the form $\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f}$. (b) Find the eigenfrequencies and eigenvectors, show that the eigenvectors are orthogonal with respect to $\mathbf{M}$, and normalise them so that $\mathbf{s}_k^T \mathbf{M} \mathbf{s}_k = 1$. (c) Show that in the frequency domain $\tilde{\mathbf{u}} = \sum_k \frac{\mathbf{s}_k (\mathbf{s}_k^T \tilde{\mathbf{f}})}{\omega_k^2 - \omega^2}$. (d) For the step force $\mathbf{f}(t) = (0, F_0)^T H(t)$, with H the Heaviside function, obtain the time-domain solution and check its long-time average against the static solution.

Solution. (a) The spring forces on the first particle are $-ku_1$ from the wall spring and $-2k(u_1 - u_2)$ from the coupling spring, while the second particle feels $-2k(u_2 - u_1)$ and $-2ku_2$. Newton's second law therefore gives

$$\begin{pmatrix} m & 0 \\ 0 & 2m \end{pmatrix} \begin{pmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{pmatrix} + k \begin{pmatrix} 3 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}, \quad (1)$$

which defines $\mathbf{M}$ and $\mathbf{K}$. Both matrices are symmetric, and both are positive definite. This is the finite-dimensional counterpart of the weak form $\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle$ in the lecture, with the correspondences $\langle \mathbf{w} | P | \mathbf{u} \rangle \leftrightarrow \mathbf{w}^T \mathbf{M} \mathbf{u}$, $\langle \mathbf{w} | H | \mathbf{u} \rangle \leftrightarrow \mathbf{w}^T \mathbf{K} \mathbf{u}$ and $\langle \mathbf{w} | \mathbf{f} \rangle \leftrightarrow \mathbf{w}^T \mathbf{f}$. Since everything is real we can work with real vectors, and the symmetry of $\mathbf{M}$ and $\mathbf{K}$ is the analogue of the self-adjointness of the kinetic and potential energy forms.

(b) Substituting $\mathbf{u} = \mathbf{s} e^{i\omega t}$ with $\mathbf{f} = \mathbf{0}$ gives the generalised eigenvalue problem $\mathbf{K} \mathbf{s} = \omega^2 \mathbf{M} \mathbf{s}$. Non-trivial solutions require

$$\det(\mathbf{K} - \omega^2 \mathbf{M}) = (3k - m\omega^2)(4k - 2m\omega^2) - 4k^2 = 2(m\omega^2 - k)(m\omega^2 - 4k) = 0, \quad (2)$$

so that the squared eigenfrequencies are $\omega_1^2 = k/m$ and $\omega_2^2 = 4k/m$. For ω_1^2 the matrix $\mathbf{K} - \omega_1^2 \mathbf{M}$ has rows proportional to $(1, -1)$, giving $\mathbf{s}_1 \propto (1, 1)^T$; for ω_2^2 the rows are

proportional to $(1, 2)$, giving $\mathbf{s}_2 \propto (2, -1)^T$. The mass-weighted inner product of these vectors is $m(1 \times 2) + 2m(1 \times (-1)) = 0$, so they are $\mathbf{M}$ -orthogonal even though they are not orthogonal in the usual sense. This is a general fact: from $\mathbf{s}_2^T \mathbf{K} \mathbf{s}_1 = \omega_1^2 \mathbf{s}_2^T \mathbf{M} \mathbf{s}_1$ and $\mathbf{s}_1^T \mathbf{K} \mathbf{s}_2 = \omega_2^2 \mathbf{s}_1^T \mathbf{M} \mathbf{s}_2$, the symmetry of $\mathbf{K}$ and $\mathbf{M}$ shows that the left-hand sides are equal, and hence $(\omega_1^2 - \omega_2^2) \mathbf{s}_1^T \mathbf{M} \mathbf{s}_2 = 0$, exactly as in the lecture. Normalising, we take

$$\mathbf{s}_1 = \frac{1}{\sqrt{3m}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{s}_2 = \frac{1}{\sqrt{6m}} \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{s}_j^T \mathbf{M} \mathbf{s}_k = \delta_{jk}. \quad (3)$$

Because the two eigenvectors are linearly independent, any vector can be expanded as $\mathbf{u} = \sum_k (\mathbf{s}_k^T \mathbf{M} \mathbf{u}) \mathbf{s}_k$, which is the two-dimensional version of the completeness relation in the lecture.

(c) With the transform convention of the lecture, $\tilde{\mathbf{u}}(\omega) = \int_0^\infty \mathbf{u}(t) e^{-i\omega t} dt$, and zero initial conditions, two integrations by parts give $\tilde{\ddot{\mathbf{u}}} = -\omega^2 \tilde{\mathbf{u}}$, and so $-\omega^2 \mathbf{M} \tilde{\mathbf{u}} + \mathbf{K} \tilde{\mathbf{u}} = \tilde{\mathbf{f}}$. Writing $\tilde{\mathbf{u}} = \sum_k c_k \mathbf{s}_k$, using $\mathbf{K} \mathbf{s}_k = \omega_k^2 \mathbf{M} \mathbf{s}_k$, and premultiplying by $\mathbf{s}_j^T$, we find that $(\omega_j^2 - \omega^2) c_j = \mathbf{s}_j^T \tilde{\mathbf{f}}$, and so

$$\tilde{\mathbf{u}}(\omega) = \sum_{k=1}^2 \frac{\mathbf{s}_k (\mathbf{s}_k^T \tilde{\mathbf{f}})}{\omega_k^2 - \omega^2}, \quad (4)$$

which is the eigenfunction expansion solution of the lecture with the bra-kets replaced by matrix products. The resonances at $\omega = \pm\omega_k$ are evident.

(d) The time-domain solution is easiest to obtain by projecting the equations of motion rather than by inverting eq. (4). Let $q_k(t) = \mathbf{s}_k^T \mathbf{M} \mathbf{u}(t)$ be the modal coordinates, so that $\mathbf{u} = \sum_k q_k \mathbf{s}_k$. Premultiplying eq. (1) by $\mathbf{s}_k^T$, and using the transposed eigenvalue equation $\mathbf{s}_k^T \mathbf{K} = \omega_k^2 \mathbf{s}_k^T \mathbf{M}$, we obtain the uncoupled oscillator equations

$$\ddot{q}_k + \omega_k^2 q_k = \mathbf{s}_k^T \mathbf{f}(t), \quad q_k(0) = \dot{q}_k(0) = 0. \quad (5)$$

For the step force $\mathbf{s}_k^T \mathbf{f} = F_k H(t)$ with $F_k = \mathbf{s}_k^T(0, F_0)^T$ constant, and one checks by differentiation that the solution for $t > 0$ is

$$q_k(t) = F_k \frac{1 - \cos \omega_k t}{\omega_k^2}, \quad (6)$$

since $\ddot{q}_k = F_k \cos \omega_k t$ and hence $\ddot{q}_k + \omega_k^2 q_k = F_k$, while q_k and $\dot{q}_k = F_k \sin(\omega_k t)/\omega_k$ both vanish at $t = 0$. Here $F_1 = F_0/\sqrt{3m}$ and $F_2 = -F_0/\sqrt{6m}$, so with $\omega_2 = 2\omega_1$ we find

$$\begin{aligned} u_1(t) &= \frac{F_0}{3k} (1 - \cos \omega_1 t) - \frac{F_0}{12k} (1 - \cos 2\omega_1 t), \\ u_2(t) &= \frac{F_0}{3k} (1 - \cos \omega_1 t) + \frac{F_0}{24k} (1 - \cos 2\omega_1 t). \end{aligned} \quad (7)$$

Averaging over the common period $2\pi/\omega_1$ replaces each factor $1 - \cos \omega_k t$ by unity, and gives $\bar{u}_1 = F_0/4k$ and $\bar{u}_2 = 3F_0/8k$; these are exactly the components of the static solution $\mathbf{K}^{-1}(0, F_0)^T$, as they should be. The response is the static deflection plus an undamped oscillation about it, and since $\omega_2 = 2\omega_1$ the motion here is periodic. As a consistency check on eq. (4), note that for $\text{Im } \omega < 0$ the transform of $H(t)$ is $1/(i\omega)$, while

$$\int_0^\infty \frac{1 - \cos \omega_k t}{\omega_k^2} e^{-i\omega t} dt = \frac{1}{\omega_k^2} \left(\frac{1}{i\omega} - \frac{i\omega}{\omega_k^2 - \omega^2} \right) = \frac{1}{i\omega(\omega_k^2 - \omega^2)}, \quad (8)$$

so the transform of eq. (6) is precisely $F_k \tilde{H}(\omega) / (\omega_k^2 - \omega^2)$, the k th coefficient in eq. (4). Everything in the lecture's derivation, including the $\frac{1 - \cos \omega_k t}{\omega_k^2}$ time dependence used in the second problem set, is already present in this two-dimensional model; the continuum case differs only in having infinitely many modes.

Example 2: A free-free rod. A uniform elastic rod of density ρ and Young's modulus E occupies $0 \leq x \leq L$ and has traction-free ends. Its longitudinal displacement $u(x, t)$ satisfies $\rho \ddot{u} - Eu'' = -\partial_x \bar{S}$ with $Eu' = \bar{S}$ at $x = 0, L$, where $\bar{S}(x, t)$ is a stress glut. (a) Write down the weak form and the normalised eigenfunctions $|k\rangle$ and eigenfrequencies ω_k . (b) Show that the rigid-translation mode $k = 0$ is not excited by any stress glut. (c) For the point source $\bar{S} = M_0 \delta(x - x_s) H(t)$ with $0 < x_s < L$, write down the response using the time-domain mode sum of the second problem set, check the initial conditions and, by direct substitution, the equation of motion, and interpret the long-time average of the solution.

Solution. (a) Multiplying the equation of motion by a test function w^* , integrating over the rod, and integrating by parts using the boundary condition $Eu' - \bar{S} = 0$ at the ends gives

$$\int_0^L \rho w^* \ddot{u} \, dx + \int_0^L E \frac{\partial w^*}{\partial x} \frac{\partial u}{\partial x} \, dx = \int_0^L \bar{S} \frac{\partial w^*}{\partial x} \, dx, \quad (9)$$

which is $\langle w|P|\ddot{u}\rangle + \langle w|H|u\rangle = \langle w|f\rangle$ in one dimension. Time-harmonic free solutions $s(x)e^{i\omega t}$ satisfy $Es'' = -\omega^2 \rho s$ with $s'(0) = s'(L) = 0$, so $s \propto \cos(k\pi x/L)$ with $k = 0, 1, 2, \dots$ and

$$\omega_k = \frac{k\pi c}{L}, \quad c = \sqrt{\frac{E}{\rho}}. \quad (10)$$

Since $\int_0^L \cos^2(k\pi x/L) \, dx = L/2$ for $k \geq 1$ and L for $k = 0$, the eigenfunctions normalised by $\langle k|P|k'\rangle = \int_0^L \rho |k\rangle |k'\rangle \, dx = \delta_{kk'}$ are

$$|0\rangle = \frac{1}{\sqrt{\rho L}}, \quad |k\rangle = \sqrt{\frac{2}{\rho L}} \cos \frac{k\pi x}{L} \quad (k \geq 1). \quad (11)$$

Each eigenspace is one-dimensional, so the second label m of the lecture is not needed.

(b) The force term for the k th mode is $\langle k|f\rangle = \int_0^L \bar{S} \partial_x |k\rangle \, dx$. For $k = 0$ the eigenfunction is constant, its derivative vanishes, and so $\langle 0|f\rangle = 0$ whatever $\bar{S}$ may be. This is the one-dimensional version of the lecture's statement that the trivial modes cannot be excited by a seismic source: an internal source exerts no net force, and the centre of mass, $\int_0^L \rho u \, dx \propto \langle 0|P|u\rangle$, remains at rest.

(c) For the point source, $\langle k|f\rangle(t) = M_0 H(t) \partial_x |k\rangle(x_s) = F_k H(t)$ with

$$F_k = -M_0 \sqrt{\frac{2}{\rho L}} \frac{k\pi}{L} \sin \frac{k\pi x_s}{L}. \quad (12)$$

The mode sum from the second problem set requires the time integrals $\int_0^t \sin[\omega_k(t -$

$t')]$ $H(t') dt' / \omega_k = (1 - \cos \omega_k t) / \omega_k^2$, and so

$$u(x, t) = \sum_{k=1}^{\infty} F_k \frac{1 - \cos \omega_k t}{\omega_k^2} |k\rangle(x) = -\frac{2M_0}{\pi E} \sum_{k=1}^{\infty} \frac{1}{k} \sin \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L} \left(1 - \cos \frac{k\pi ct}{L}\right), \quad (13)$$

where we used $\omega_k^2 = Ek^2\pi^2/(\rho L^2)$ to simplify the coefficients. At $t = 0$ every term vanishes, and the time derivative brings down a factor $\omega_k \sin \omega_k t$ that also vanishes, so $u(x, 0) = \dot{u}(x, 0) = 0$ as required. The boundary conditions hold term by term, since $\partial_x \cos(k\pi x/L)$ vanishes at both ends, in agreement with $Eu' = \bar{S} = 0$ there.

For the direct substitution we apply $\rho \partial_t^2 - E \partial_x^2$ to each term of eq. (13). Since $E(k\pi/L)^2 = \rho \omega_k^2$, the k th term is multiplied by $\rho \omega_k^2 [\cos \omega_k t + (1 - \cos \omega_k t)] = \rho \omega_k^2$ for $t > 0$, so

$$\rho \ddot{u} - Eu'' = -\frac{2M_0}{\pi E} \sum_{k=1}^{\infty} \frac{\rho \omega_k^2}{k} \sin \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L} = -\frac{2\pi M_0}{L^2} \sum_{k=1}^{\infty} k \sin \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L}. \quad (14)$$

To identify the right-hand side we use the completeness relation, $u(x) = \sum_k \langle k|P|u\rangle |k\rangle$, which on writing out the inner product becomes $\rho \sum_k |k\rangle(x) |k\rangle(x') = \delta(x - x')$, and which, with eq. (11), is the cosine series

$$\delta(x - x_s) = \frac{1}{L} + \frac{2}{L} \sum_{k=1}^{\infty} \cos \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L}. \quad (15)$$

Differentiating with respect to x_s , and using $\partial_{x_s} \delta(x - x_s) = -\delta'(x - x_s)$, gives

$$\delta'(x - x_s) = \frac{2\pi}{L^2} \sum_{k=1}^{\infty} k \sin \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L}, \quad (16)$$

and comparison with eq. (14) shows that $\rho \ddot{u} - Eu'' = -M_0 \delta'(x - x_s) H(t) = -\partial_x \bar{S}$, as required. The series in eqs. (15) and (16) converge only in the sense of distributions, which is the sense in which a point source satisfies the equation of motion; the series in eq. (13) for the displacement itself has coefficients decaying like $1/k$ and converges, though slowly, to a function with a jump.

That jump is the physical content of the long-time average. Averaging over time replaces each $1 - \cos \omega_k t$ by 1, and the resulting series is the cosine series of the static solution. Indeed, the static problem $Eu' = \bar{S}$ integrates to $u = (M_0/E)H(x - x_s) + \text{const}$, and the constant is fixed by $\langle 0|P|u\rangle = 0$, i.e. zero mean displacement. The function $g(x) = (M_0/E)[H(x - x_s) - (L - x_s)/L]$ has $\int_0^L g dx = 0$ and cosine coefficients

$$\frac{2}{L} \int_0^L g(x) \cos \frac{k\pi x}{L} dx = \frac{2M_0}{EL} \int_{x_s}^L \cos \frac{k\pi x}{L} dx = -\frac{2M_0}{\pi Ek} \sin \frac{k\pi x_s}{L}, \quad (17)$$

which are exactly the time-averaged coefficients in eq. (13). The point stress glut is therefore a dislocation: the material on the two sides of x_s is permanently offset by M_0/E , the mean displacement stays zero, and the dynamic solution consists of the waves radiated by the sudden appearance of this offset, reflecting for ever between the free ends because we have neglected attenuation. Numerically, a partial sum with 20 000 terms reproduces the step to four decimal places away from x_s , and the mode sum for a Gaussian-smoothed glut agrees with a finite-difference solution of the rod to the discretisation error of the latter.

Example 3: Verifying the time-domain mode sum. The second problem set quotes the time-domain solution of the forced problem in a non-rotating earth model,

$$\mathbf{u}(\mathbf{x}, t) = \sum_{km} \left\{ \int_0^t \frac{\sin[\omega_k(t - t')]}{\omega_k} \int_M \bar{S}_{ij}(\mathbf{x}', t') \frac{\partial |km\rangle^*}{\partial x'_j}(\mathbf{x}') d^3\mathbf{x}' dt' \right\} |km\rangle(\mathbf{x}). \quad (18)$$

Show by direct substitution that this satisfies the weak form $\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle$ for every test function $\mathbf{w}$, together with the initial conditions $\mathbf{u}(\mathbf{x}, 0) = \dot{\mathbf{u}}(\mathbf{x}, 0) = \mathbf{0}$. Then show that its Fourier transform is the frequency-domain eigenfunction expansion of the lecture, and specialise to a source that is a step in time, $\bar{S}_{ij}(\mathbf{x}, t) = \bar{S}_{ij}(\mathbf{x})H(t)$.

Solution. The inner spatial integral in eq. (18) is the force term of the lecture evaluated on the (k, m) th eigenfunction,

$$F_{km}(t) = \langle km | \mathbf{f} \rangle(t) = \int_M \bar{S}_{ij}(\mathbf{x}, t) \frac{\partial |km\rangle_i^*}{\partial x_j} d^3\mathbf{x}, \quad (19)$$

so that eq. (18) reads $\mathbf{u} = \sum_{km} q_{km}(t) |km\rangle$ with the modal coordinates

$$q_{km}(t) = \int_0^t \frac{\sin[\omega_k(t - t')]}{\omega_k} F_{km}(t') dt'. \quad (20)$$

A word about the trivial modes. If $\omega_k = 0$ then $\sin[\omega_k(t - t')]/\omega_k$ is to be read as its limit $t - t'$, but for these modes $|km\rangle_i = a_i + B_{ij}x_j$ with B_{ij} anti-symmetric, so $\partial_j |km\rangle_i = B_{ij}$, and its contraction with the symmetric $\bar{S}_{ij}$ in eq. (19) vanishes. The trivial modes therefore contribute nothing, and in what follows we may take $\omega_k > 0$.

Differentiating eq. (20) with respect to its upper limit and under the integral sign,

$$\dot{q}_{km}(t) = \frac{\sin 0}{\omega_k} F_{km}(t) + \int_0^t \cos[\omega_k(t - t')] F_{km}(t') dt' = \int_0^t \cos[\omega_k(t - t')] F_{km}(t') dt', \quad (21)$$

and differentiating once more,

$$\ddot{q}_{km}(t) = F_{km}(t) - \omega_k \int_0^t \sin[\omega_k(t - t')] F_{km}(t') dt' = F_{km}(t) - \omega_k^2 q_{km}(t). \quad (22)$$

Thus each modal coordinate satisfies the forced oscillator equation $\ddot{q}_{km} + \omega_k^2 q_{km} = F_{km}$, and from eqs. (20) and (21) both $q_{km}(0)$ and $\dot{q}_{km}(0)$ vanish, since the integrals are over an interval of zero length. Hence $\mathbf{u}(\mathbf{x}, 0) = \dot{\mathbf{u}}(\mathbf{x}, 0) = \mathbf{0}$. Eq. (20) is just Duhamel's formula for this oscillator.

Now substitute into the weak form, differentiating the sum term by term. Using the sesquilinearity of the forms and the eigenvalue equation $\langle \mathbf{w} | H | km \rangle = \omega_k^2 \langle \mathbf{w} | P | km \rangle$, which holds for every test function $\mathbf{w}$,

$$\begin{aligned} \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle &= \sum_{km} [\ddot{q}_{km} \langle \mathbf{w} | P | km \rangle + q_{km} \langle \mathbf{w} | H | km \rangle] \\ &= \sum_{km} (\ddot{q}_{km} + \omega_k^2 q_{km}) \langle \mathbf{w} | P | km \rangle = \sum_{km} F_{km} \langle \mathbf{w} | P | km \rangle. \end{aligned} \quad (23)$$

It remains to identify the last sum with $\langle \mathbf{w} | \mathbf{f} \rangle$. Expanding the test function using completeness, $\mathbf{w} = \sum_{km} \langle km | P | \mathbf{w} \rangle | km \rangle$, and recalling that $\langle \mathbf{w} | \mathbf{f} \rangle$ is conjugate-linear in $\mathbf{w}$, we have

$$\langle \mathbf{w} | \mathbf{f} \rangle = \sum_{km} \langle km | P | \mathbf{w} \rangle^* \langle km | \mathbf{f} \rangle = \sum_{km} \langle \mathbf{w} | P | km \rangle F_{km}, \quad (24)$$

where the self-adjointness of the kinetic energy form, $\langle km | P | \mathbf{w} \rangle^* = \langle \mathbf{w} | P | km \rangle$, was used in the last step. Eqs. (23) and (24) agree for every $\mathbf{w}$, so eq. (18) solves the weak form, and because the weak form incorporates the boundary conditions, the traction condition on ∂M is satisfied automatically by the eigenfunctions. Only three ingredients were needed: the eigenvalue equation, the orthonormality and completeness of the eigenfunctions, and the self-adjointness of P .

For the connection with the frequency domain, we take the transform of eq. (20), namely $\tilde{q}_{km}(\omega) = \int_0^\infty q_{km} e^{-i\omega t} dt$. Writing the exponential as $e^{-i\omega(t-t')} e^{-i\omega t'}$ and exchanging the order of integration over the region $0 \leq t' \leq t < \infty$ gives the causal convolution theorem $\tilde{q}_{km} = \tilde{g}_k \tilde{F}_{km}$ with

$$\tilde{g}_k(\omega) = \int_0^\infty \frac{\sin \omega_k t}{\omega_k} e^{-i\omega t} dt = \frac{1}{2i\omega_k} \left[\frac{1}{i(\omega - \omega_k)} - \frac{1}{i(\omega + \omega_k)} \right] = \frac{1}{\omega_k^2 - \omega^2}, \quad (25)$$

where the integral converges for $\text{Im } \omega < 0$, the standard device that corresponds physically to a slight attenuation moving the poles off the real axis. Hence

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} | km \rangle, \quad (26)$$

which is the eigenfunction expansion solution derived in the lecture; eq. (18) is its inverse transform, and the contour integration mentioned in the lecture simply evaluates the residues at $\omega = \pm \omega_k$ to produce $\sin(\omega_k t)/\omega_k$. Finally, for a step-in-time source $F_{km}(t) = F_{km} H(t)$ with $F_{km} = \int_M \bar{S}_{ij}(\mathbf{x}) \partial_j | km \rangle_i^* d^3 \mathbf{x}$ constant, the time integral in eq. (20) is elementary and

$$\mathbf{u}(\mathbf{x}, t) = \sum_{km} F_{km} \frac{1 - \cos \omega_k t}{\omega_k^2} | km \rangle(\mathbf{x}), \quad (27)$$

the form used in Examples 1 and 2. Each mode is then displaced to a new static equilibrium F_{km}/ω_k^2 about which it oscillates for ever with amplitude equal to that offset, which is why a step source excites the gravest modes, with the smallest ω_k^2 , so strongly.

Example 4: Spectral resolution and record length. A seismogram is recorded for a time T , which we model by the boxcar window $h(t) = 1$ for $0 \leq t \leq T$ and $h(t) = 0$ otherwise. (a) Compute $\tilde{h}(\omega)$, locate the first zero of $|\tilde{h}|$ and find the full width at half maximum (FWHM) of the peak, both for the power spectrum $|\tilde{h}|^2$ and the amplitude spectrum $|\tilde{h}|$. (b) Show that the observed spectrum of an undamped mode $u(t) = A \cos \omega_0 t$ is a copy of $\tilde{h}$ centred on ω_0 , and state the criterion for two modes with nearby frequencies to be resolved. (c) The five singlets of the multiplet ${}_0S_2$ are centred on $f_0 \approx 0.3094$ mHz and adjacent singlets are separated by about $4.6 \mu\text{Hz}$. Estimate the record length needed to resolve them, and compare it with the decay time of the mode, for which $Q \approx 510$.

Solution. (a) With the lecture's convention,

$$\tilde{h}(\omega) = \int_0^T e^{-i\omega t} dt = \frac{1 - e^{-i\omega T}}{i\omega} = T e^{-i\omega T/2} \frac{\sin(\omega T/2)}{\omega T/2}, \quad (28)$$

so $|\tilde{h}(\omega)| = T |\text{sinc}(\omega T/2)|$ with $\text{sinc } x = \sin x/x$. In terms of the frequency $f = \omega/2\pi$ the argument is $x = \pi f T$, and the zeros of $|\tilde{h}|$ lie at $f = n/T$ for $n = \pm 1, \pm 2, \dots$; the first zero is at $f = 1/T$. The power spectrum $|\tilde{h}|^2 = T^2 \text{sinc}^2 x$ falls to half its peak value where $\sin x/x = 1/\sqrt{2}$, which has the root $x_{1/2} = 1.3916$, and the full width of the peak is therefore $2x_{1/2}/\pi$ in units of $1/T$:

$$\Delta f_{\text{FWHM}}(|\tilde{h}|^2) = \frac{2x_{1/2}}{\pi T} = \frac{0.886}{T}. \quad (29)$$

For the amplitude spectrum the half-maximum condition is $\sin x/x = 1/2$, with root $x = 1.8955$, giving $\Delta f_{\text{FWHM}}(|\tilde{h}|) = 1.207/T$. Whichever definition is used, the spectral line has a width of order $1/T$, and the main lobe between the first zeros has total width $2/T$.

(b) A single undamped mode observed through the window has transform

$$\tilde{u}_{\text{obs}}(\omega) = \int_0^T A \cos(\omega_0 t) e^{-i\omega t} dt = \frac{A}{2} [\tilde{h}(\omega - \omega_0) + \tilde{h}(\omega + \omega_0)], \quad (30)$$

which is the convolution result of the lecture made explicit: the transform of the infinite record, with its poles at $\pm\omega_0$, is replaced by copies of $\tilde{h}$ centred on $\pm\omega_0$. Near $\omega \approx \omega_0 > 0$ the second term is smaller by a factor of order $1/(\omega_0 T)$ and can be neglected whenever the record contains many cycles (for $0S_2$ and $T = 24 \text{ h}$, $\omega_0 T \approx 170$). Two modes at f_1 and f_2 each contribute such a line, and by the usual Rayleigh-type criterion they are resolved when the peak of one lies no closer than the first zero of the other, i.e. when $|f_1 - f_2| \gtrsim 1/T$. Fig. 1 shows the amplitude spectrum of two equal-amplitude cosines separated by $4.6 \mu\text{Hz}$ for two record lengths.

(c) With adjacent singlets $4.6 \mu\text{Hz}$ apart the criterion of part (b) requires

$$T \gtrsim \frac{1}{4.6 \times 10^{-6} \text{ Hz}} = 2.2 \times 10^5 \text{ s} \approx 60 \text{ h}, \quad (31)$$

that is, two and a half days. A one-day record has $1/T = 11.6 \mu\text{Hz}$, larger than the width of the entire multiplet ($4 \times 4.6 \approx 18 \mu\text{Hz}$), and shows a single broad peak, while a five-day record has $1/T = 2.3 \mu\text{Hz}$ and separates the singlets clearly, as in Fig. 1.

Whether such records are available is set by attenuation. A mode with quality factor Q decays as $e^{-\omega_0 t/2Q} = e^{-\pi f_0 t/Q}$, so the amplitude e-folding time is

$$\tau = \frac{Q}{\pi f_0} = \frac{510}{\pi \times 3.094 \times 10^{-4} \text{ s}^{-1}} = 5.2 \times 10^5 \text{ s} \approx 6 \text{ days}. \quad (32)$$

This is comfortably longer than the 60 h required: after five days the amplitude has fallen only to $e^{-120/146} \approx 0.44$ of its initial value, so for a sufficiently large earthquake the signal remains above the noise for long enough. Attenuation also sets a floor on the achievable resolution, because the transform of $e^{-\pi f_0 t/Q} \cos \omega_0 t$ is a Lorentzian rather

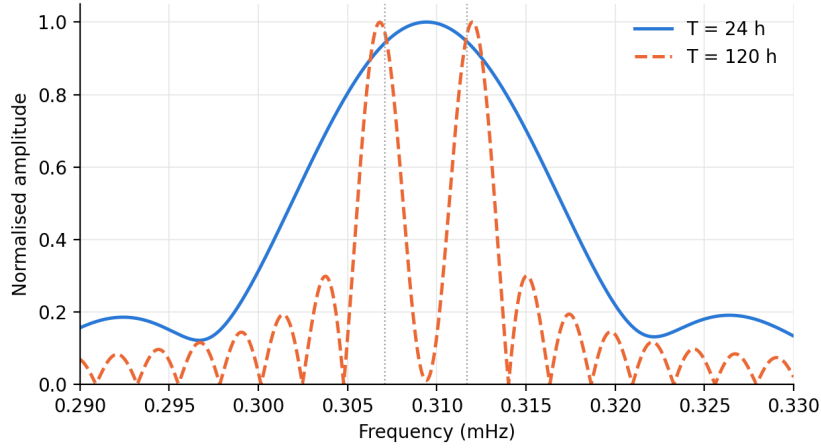


Fig. 1: Amplitude spectrum, normalised to its maximum, of the sum of two equal-amplitude cosines at the frequencies of two adjacent singlets of ${}_0S_2$ (dotted lines, separated by $4.6 \mu\text{Hz}$) observed through a boxcar window of length 24 h and 120 h. Attenuation is neglected.

than a delta function, with FWHM $f_0/Q \approx 0.6 \mu\text{Hz}$ in the power spectrum. Since this is well below the singlet spacing, the splitting of ${}_0S_2$ is resolvable, and it is indeed observed after the very largest earthquakes. Extending the record much beyond a few τ gains nothing, as the window then contains mostly noise. The situation deteriorates at higher frequencies, where f/Q grows to values comparable with the widths of the multiplets, and this is why individual peaks become hard to identify in the higher-frequency part of the spectrum shown in the lecture.

Example 5: Pointwise stability of an isotropic solid. In an isotropic solid without pre-stress the elastic tensor is $A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$. (a) For a complex symmetric tensor e_{ij} , decompose e_{ij} into isotropic and deviatoric parts and show that $A_{ijkl} e_{ij}^* e_{kl} = \kappa |e_{kk}|^2 + 2\mu d_{ij}^* d_{ij}$, where d_{ij} is the deviatoric part and $\kappa = \lambda + \frac{2}{3}\mu$. (b) Show that the pointwise stability condition $A_{ijkl} e_{ij}^* e_{kl} \geq c_0 e_{ij}^* e_{ij}$ holds for some $c_0 > 0$ if and only if $\kappa > 0$ and $\mu > 0$, and find the largest admissible c_0 . (c) Show that these conditions imply $\alpha > \beta > 0$ for the P- and S-wave speeds, but that the converse fails, using $\lambda = -0.9\mu$ as a counter-example. Explain physically why the two conditions differ.

Solution. (a) Write e_{ij} as the sum of an isotropic part proportional to the identity and a traceless remainder,

$$e_{ij} = \frac{1}{3} \theta \delta_{ij} + d_{ij}, \quad \theta = e_{kk}, \quad d_{kk} = 0. \quad (33)$$

The two parts are orthogonal: $\delta_{ij} d_{ij} = d_{kk} = 0$, so that

$$e_{ij}^* e_{ij} = \frac{1}{9} |\theta|^2 \delta_{ij} \delta_{ij} + d_{ij}^* d_{ij} + \frac{1}{3} (\theta^* d_{kk} + \theta d_{kk}^*) = \frac{1}{3} |\theta|^2 + d_{ij}^* d_{ij}. \quad (34)$$

Contracting the elastic tensor with $e_{ij}^* e_{kl}$, the λ term gives $\lambda e_{ii}^* e_{kk} = \lambda |\theta|^2$, and the two μ terms give $\mu (e_{ij}^* e_{ij} + e_{ji}^* e_{ji}) = 2\mu e_{ij}^* e_{ij}$ by the symmetry of e_{ij} . Using eq. (34),

$$A_{ijkl} e_{ij}^* e_{kl} = \lambda |\theta|^2 + 2\mu \left(\frac{1}{3} |\theta|^2 + d_{ij}^* d_{ij} \right) = \kappa |\theta|^2 + 2\mu d_{ij}^* d_{ij}, \quad (35)$$

with $\kappa = \lambda + \frac{2}{3}\mu$ the bulk modulus. The strain energy density $\frac{1}{2}A_{ijkl}e_{ij}e_{kl} = \frac{1}{2}\kappa\theta^2 + \mu d_{ij}d_{ij}$ thus splits into a part that resists change of volume, governed by κ , and a part that resists change of shape, governed by μ .

(b) Combining eqs. (34) and (35),

$$A_{ijkl}e_{ij}^*e_{kl} - c_0e_{ij}^*e_{ij} = (3\kappa - c_0)\frac{1}{3}|\theta|^2 + (2\mu - c_0)d_{ij}^*d_{ij}. \quad (36)$$

If $c_0 \leq 3\kappa$ and $c_0 \leq 2\mu$ the right-hand side is non-negative for every symmetric e_{ij} . Conversely, the choice $e_{ij} = \delta_{ij}$ (a pure dilatation, $\theta = 3$, $d_{ij} = 0$) makes the right-hand side $3(3\kappa - c_0)$, and a traceless choice such as $e_{12} = e_{21} = 1$ with all other components zero makes it $2(2\mu - c_0)$, so both inequalities are also necessary. Pointwise stability with some $c_0 > 0$ therefore holds if and only if $\kappa > 0$ and $\mu > 0$, and the largest admissible constant is

$$c_0 = \min(3\kappa, 2\mu). \quad (37)$$

In other words, regarded as a symmetric linear map on the six-dimensional space of symmetric tensors, A_{ijkl} has the eigenvalue 3κ on the one-dimensional space of isotropic tensors and the eigenvalue 2μ on the five-dimensional space of deviatoric tensors, and c_0 is its smallest eigenvalue.

(c) From the Christoffel equation in an isotropic material, $\rho\alpha^2 = \lambda + 2\mu = \kappa + \frac{4}{3}\mu$ and $\rho\beta^2 = \mu$. If $\kappa > 0$ and $\mu > 0$ then $\beta^2 > 0$ and $\rho(\alpha^2 - \beta^2) = \kappa + \frac{1}{3}\mu > 0$, so $\alpha > \beta > 0$. The converse fails: real wave speeds with $\alpha > \beta$ require only $\mu > 0$ and $\lambda > -2\mu$, whereas $\kappa > 0$ requires the stronger $\lambda > -\frac{2}{3}\mu$. For $\lambda = -0.9\mu$ with $\mu > 0$ we have

$$\kappa = -0.9\mu + \frac{2}{3}\mu = -\frac{7}{30}\mu < 0, \quad \rho\alpha^2 = 1.1\mu > 0, \quad \alpha = \sqrt{1.1}\beta \approx 1.05\beta, \quad (38)$$

so P- and S-waves propagate with real speeds and $\alpha > \beta > 0$, yet the uniform dilatation $e_{ij} = \delta_{ij}$ has $A_{ijkl}e_{ij}e_{kl} = 9\kappa = -2.1\mu < 0$ and the material is not pointwise stable.

The reason is that plane waves test the elastic tensor only on strains of a very special kind. A plane wave with polarisation $\mathbf{a}$ and unit propagation direction $\hat{\mathbf{p}}$ produces the rank-one strain $e_{ij} = \frac{1}{2}(a_i\hat{p}_j + a_j\hat{p}_i)$, for which, using the symmetries of A_{ijkl} , $A_{ijkl}e_{ij}e_{kl} = A_{ijkl}a_i\hat{p}_ja_k\hat{p}_l = \rho a_i\Gamma_{ik}(\hat{\mathbf{p}})a_k$ with Γ_{ik} the Christoffel matrix. Such a strain has $\theta = \mathbf{a} \cdot \hat{\mathbf{p}}$ and $e_{ij}e_{ij} = \frac{1}{2}[\|\mathbf{a}\|^2 + (\mathbf{a} \cdot \hat{\mathbf{p}})^2]$, so by eq. (34) its deviatoric part satisfies $d_{ij}d_{ij} = \frac{1}{2}\|\mathbf{a}\|^2 + \frac{1}{6}(\mathbf{a} \cdot \hat{\mathbf{p}})^2$, and eq. (35) gives

$$A_{ijkl}e_{ij}e_{kl} = \mu\|\mathbf{a}\|^2 + \left(\kappa + \frac{1}{3}\mu\right)(\mathbf{a} \cdot \hat{\mathbf{p}})^2 = \rho \left[\beta^2\|\mathbf{a}\|^2 + (\alpha^2 - \beta^2)(\mathbf{a} \cdot \hat{\mathbf{p}})^2\right], \quad (39)$$

in agreement with the Christoffel matrix $\Gamma_{ik} = \alpha^2\hat{p}_i\hat{p}_k + \beta^2(\delta_{ik} - \hat{p}_i\hat{p}_k)$ of Lecture 15. Even the longitudinal case $\mathbf{a} = \hat{\mathbf{p}}$ has $d_{ij}d_{ij} = \frac{2}{3}$ against $\frac{1}{3}\theta^2 = \frac{1}{3}$: a rank-one strain is never a pure dilatation, and its deviatoric part always outweighs its isotropic part. A sufficiently positive shear modulus can therefore mask a negative bulk modulus in eq. (39), which stays positive so long as $\kappa > -\frac{4}{3}\mu$. A homogeneous compression, by contrast, has no deviatoric part at all; it is not a plane wave of any finite wavelength, and it is precisely the strain that a negative κ makes energetically favourable.

The consequence for a finite body is decisive. Take $u_i = x_i$ on any body M made of the material in eq. (38), so that $e_{ij} = \delta_{ij}$ everywhere and, in the non-gravitating case,

$\langle \mathbf{u} | H | \mathbf{u} \rangle = \int_M 9\kappa d^3\mathbf{x} < 0$. Expanding $\mathbf{u} = \sum_{km} c_{km} |km\rangle$ with $c_{km} = \langle km | P | \mathbf{u} \rangle$, the eigenvalue equation with test function $\mathbf{u}$ and the self-adjointness of P give

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \sum_{km} c_{km} \langle \mathbf{u} | H | km \rangle = \sum_{km} \omega_k^2 c_{km} \langle \mathbf{u} | P | km \rangle = \sum_{km} \omega_k^2 |c_{km}|^2, \quad (40)$$

which can only be negative if at least one squared eigenfrequency is negative. The body therefore has a mode that grows exponentially in time and is unstable, even though plane waves travel through the same material perfectly well. Positive wave speeds are a condition on the propagation of disturbances of finite wavelength; point-wise stability is the stronger condition that every strain, including a uniform change of volume, costs energy, and it is the latter that the lecture needs to conclude that all the squared eigenfrequencies of a non-gravitating earth model are non-negative.

Worked examples for Lecture 23: Mode splitting and coupling

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These examples accompany Lecture 23. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Example 1 works through the toroidal mode problem for a homogeneous earth model in the detail referred to in the lecture summary. Numerical results were checked with the scripts in the python directory.

Example 1: Toroidal modes of a homogeneous sphere. In the lecture it was stated that, within an entirely solid and spherically symmetric earth model of radius b , the eigenvalue problem for a toroidal mode of degree l reduces to

$$-\omega^2 \int_0^b \rho W'^* W r^2 dr + \int_0^b \mu \left(r \frac{dW'^*}{dr} - W'^* \right) \left(r \frac{dW}{dr} - W \right) dr + (\zeta^2 - 2) \int_0^b \mu W'^* W dr = 0, \quad (1)$$

which is to hold for all test functions W' , where $\zeta^2 = l(l+1)$ and W is the radial eigenfunction. (i) By integrating by parts, show that eq. (1) is equivalent to an ordinary differential equation of Sturm–Liouville type for W on $0 < r < b$ together with a boundary condition at $r = b$, and interpret this boundary condition physically. (ii) Show that when ρ and μ are constant the solution regular at the centre is $W(r) = j_l(kr)$, with $k = \omega/\beta$, $\beta = \sqrt{\mu/\rho}$, and j_l the spherical Bessel function of the first kind, and obtain the equation that determines the eigenfrequencies. (iii) Solve this equation numerically for $l = 2$ and $l = 3$, tabulating the periods of the first three overtones for a homogeneous sphere with $b = 6371$ km and $\beta = 6.3$ km s⁻¹, and compare the period of ${}_0T_2$ with the observed value of 44.2 min.

Solution. Throughout this example there is no time dependence, and so we use a dot to denote d/dr ; as in the lecture, W' is the test function and not a derivative.

(i) *The radial equation and its boundary condition.* Expanding the product in the second integral of eq. (1) gives

$$\mu (r \dot{W}'^* - W'^*) (r \dot{W} - W) = \mu r^2 \dot{W}'^* \dot{W} - \mu r \dot{W}'^* W - \mu r W'^* \dot{W} + \mu W'^* W, \quad (2)$$

and we integrate by parts the two terms that contain the derivative of the test function,

$$\int_0^b \mu r^2 \dot{W}'^* \dot{W} \, dr = \left[\mu r^2 W'^* \dot{W} \right]_0^b - \int_0^b W'^* \frac{d}{dr} (\mu r^2 \dot{W}) \, dr, \quad (3)$$

$$- \int_0^b \mu r \dot{W}'^* W \, dr = - \left[\mu r W'^* W \right]_0^b + \int_0^b W'^* \frac{d}{dr} (\mu r W) \, dr. \quad (4)$$

The boundary terms at $r = 0$ vanish so long as W , $\dot{W}$ and W' remain bounded at the centre, which we assume. Since $\frac{d}{dr}(\mu r W) = r \dot{\mu} W + \mu W + \mu r \dot{W}$, the term $\mu r \dot{W}$ cancels against the third term on the right of eq. (2), and collecting everything eq. (1) becomes

$$\int_0^b W'^* \left[-\frac{d}{dr} (\mu r^2 \dot{W}) + (\zeta^2 \mu + r \dot{\mu}) W - \omega^2 \rho r^2 W \right] \, dr + b^2 W'^*(b) T(b) = 0, \quad (5)$$

where we have introduced the function

$$T(r) = \mu \left(\dot{W} - \frac{W}{r} \right) = \mu r \frac{d}{dr} \left(\frac{W}{r} \right). \quad (6)$$

Eq. (5) must hold for every test function. Considering first those that vanish at $r = b$ but are otherwise arbitrary, the square bracket must vanish for $0 < r < b$, and so W satisfies

$$\frac{d}{dr} \left(\mu r^2 \frac{dW}{dr} \right) - \left(\zeta^2 \mu + r \frac{d\mu}{dr} \right) W + \omega^2 \rho r^2 W = 0. \quad (7)$$

Given eq. (7), the integral in eq. (5) vanishes identically, and what remains must vanish for arbitrary values of $W'(b)$. Hence we also require

$$T(b) = \mu \left(\frac{dW}{dr} - \frac{W}{r} \right) \Big|_{r=b} = 0. \quad (8)$$

Conversely, if W satisfies eqs. (7) and (8) then eq. (5), and hence eq. (1), holds for every test function; the two formulations are equivalent.

Eq. (7) has the Sturm–Liouville form $\frac{d}{dr}(p\dot{W}) - qW + \lambda wW = 0$ with $p = \mu r^2$, $q = \zeta^2 \mu + r \dot{\mu}$, weight $w = \rho r^2$, and eigenvalue $\lambda = \omega^2$. Standard theory for such problems then gives directly what the lecture stated in general: the squared eigenfrequencies are real, eigenfunctions with distinct eigenfrequencies are orthogonal with respect to $\int_0^b \rho W_n W_{n'} r^2 \, dr$, which is just the kinetic energy form restricted to toroidal modes of degree l , and there is a countable infinity of eigenvalues accumulating at infinity, labelled by the overtone number $n = 0, 1, 2, \dots$. The boundary condition in eq. (8) was not imposed but emerged from the weak form; it is a *natural* boundary condition. Its physical meaning follows from computing the traction on a spherical surface. For a toroidal displacement $\mathbf{s} = W \mathbf{C}_{lm}$ the radial component vanishes, and the shear strains in an isotropic model give $\sigma_{r\theta} = \mu(\dot{W} - W/r)C_\theta$ and $\sigma_{r\phi} = \mu(\dot{W} - W/r)C_\phi$, so that the traction acting on the surface $r = \text{const}$ is $T(r) \mathbf{C}_{lm}$. Eq. (8) is therefore precisely the traction-free condition at the free surface, as was to be expected from the discussion in Lecture 22 of how boundary conditions are built into the weak form. It is also worth noting the identity $r\dot{W} - W = r^2 \frac{d}{dr}(W/r)$: the first potential energy term in eq. (1) vanishes for $W \propto r$, which for $l = 1$ is a rigid rotation, and we return to this point below.

(ii) *The homogeneous sphere.* If ρ and μ are constant then $\dot{\mu} = 0$, and dividing eq. (7) by μr^2 we obtain

$$\frac{d^2 W}{dr^2} + \frac{2}{r} \frac{dW}{dr} + \left[k^2 - \frac{l(l+1)}{r^2} \right] W = 0, \quad k = \frac{\omega}{\beta}, \quad \beta = \sqrt{\frac{\mu}{\rho}}. \quad (9)$$

In terms of $x = kr$ this is the spherical Bessel equation of order l , with general solution $W = A j_l(kr) + B y_l(kr)$. Near the centre, trying $W \propto r^\alpha$ in eq. (9) gives at leading order $\alpha(\alpha+1) = l(l+1)$, so that $\alpha = l$ or $\alpha = -(l+1)$; these are the behaviours $j_l(x) \sim x^l / (2l+1)!!$ and $y_l(x) \sim -(2l-1)!! / x^{l+1}$ of the two solutions. The second is unbounded, and would give an infinite kinetic energy since $\int_0 \rho W^2 r^2 dr$ then behaves like $\int_0 r^{-2l} dr$ for $l \geq 1$. Hence $B = 0$, and up to an arbitrary amplitude

$$W(r) = j_l(kr). \quad (10)$$

The eigenfrequencies follow from the traction-free condition in eq. (8), which reads $\mu[kj'_l(kb) - j_l(kb)/b] = 0$. Writing $x = kb = \omega b / \beta$, this is

$$x j'_l(x) = j_l(x), \quad \text{or equivalently} \quad \frac{d}{dx} \left[\frac{j_l(x)}{x} \right] = 0. \quad (11)$$

Using the recurrence relation $j'_l(x) = \frac{l}{x} j_l(x) - j_{l+1}(x)$ we can also write the frequency equation as

$$(l-1)j_l(x) = x j_{l+1}(x). \quad (12)$$

For each degree this transcendental equation has a countable infinity of positive roots $x_0 < x_1 < x_2 < \dots$, and the mode ${}_n T_l$ has eigenfrequency and period

$$\omega = \frac{\beta x_n}{b}, \quad \mathcal{T} = \frac{2\pi b}{\beta x_n}. \quad (13)$$

For $l = 2$, inserting $j_1(x) = \sin x / x^2 - \cos x / x$ and $j_2(x) = (3/x^3 - 1/x) \sin x - 3 \cos x / x^2$ into eq. (11) and clearing denominators gives the explicit form

$$\tan x = \frac{x(x^2 - 12)}{5x^2 - 12}, \quad (14)$$

which shows clearly that the roots must be found numerically. For large x we have $j_l(x) \approx \sin(x - l\pi/2)/x$, so eq. (11) becomes $\tan(x - l\pi/2) \approx x/2$ and successive roots are separated by approximately π : the overtones of a given degree become equally spaced in frequency, with spacing $\Delta f = \beta/2b$, which is the inverse of the two-way travel time of an S-wave across a diameter of the sphere. Finally, we note the special case $l = 1$, for which eq. (12) reduces to $j_2(kb) = 0$. This has the root $k = 0$, corresponding to $\omega = 0$ and, from the small-argument form of j_1 , to $W \propto r$. Since $r\mathbf{C}_{1m}$ is a rigid rotation (see Example 5), this is one of the trivial zero-frequency modes discussed in Lecture 22, and there is no fundamental toroidal mode of degree one.

(iii) *Numerical results.* The roots of eq. (11) were found by locating sign changes of $x j'_l(x) - j_l(x)$ on a fine grid and refining each by bisection, with j_l and j'_l evaluated using `scipy.special.spherical_jn`. With $b = 6371$ km and $\beta = 6.3$ km s⁻¹ we obtain the values in the table below. The shear wave speed chosen is a crude average over the mantle, whose values in PREM range from about 4.5 km s⁻¹ beneath the crust to about 7.3 km s⁻¹ at its base.

mode	$x_n = kb$	$f = \omega/2\pi$ (mHz)	period (min)
${}_0T_2$	2.5011	0.3936	42.3
${}_1T_2$	7.1360	1.1231	14.8
${}_2T_2$	10.5146	1.6548	10.1
${}_0T_3$	3.8647	0.6082	27.4
${}_1T_3$	8.4449	1.3291	12.5
${}_2T_3$	11.8817	1.8700	8.9

The predicted period of ${}_0T_2$ is 42.3 min, about 4% shorter than the observed 44.2 min ($f = 0.377$ mHz); a shear wave speed of 6.03 km s^{-1} would reproduce the observation exactly. Given how crude the model is, this agreement is better than we have any right to expect, and it is partly fortuitous. The Earth has a fluid outer core in which toroidal motion cannot be sustained, so the real mode is confined to the mantle, a shell occupying only the outer 45% of the radius; this stiffening of the problem raises the frequency. Against this, the low shear wave speeds of the upper mantle, where the fundamental mode has its largest displacements (Fig. 1), lower the frequency, and the two effects roughly compensate. For ${}_0T_3$ the model gives 27.4 min against an observed period of about 28.4 min. Fig. 1 shows the radial eigenfunctions of the three lowest $l = 2$ modes. The number of interior zeros of W equals the overtone number n , as the oscillation theorem for Sturm–Liouville problems guarantees, and the higher overtones oscillate increasingly rapidly with radius while the fundamental mode is smooth and largest at the surface.

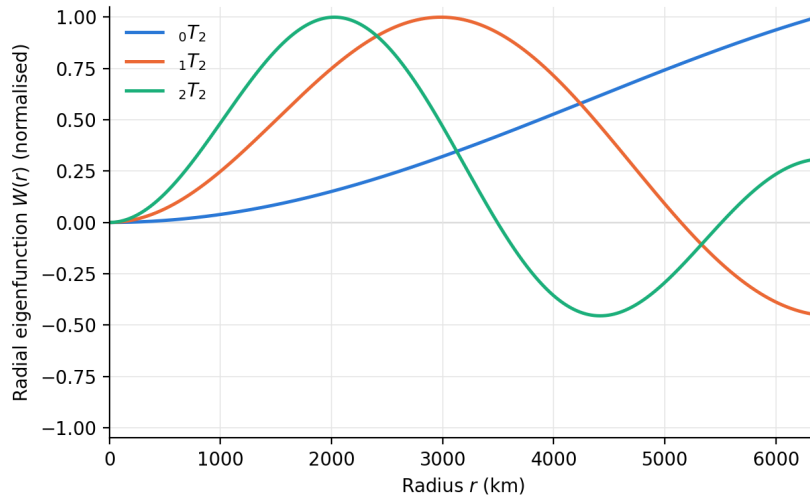


Fig. 1: Radial eigenfunctions $W(r) = j_2(x_n r/b)$ of the toroidal modes ${}_0T_2$, ${}_1T_2$ and ${}_2T_2$ of a homogeneous sphere with $b = 6371$ km, each normalised to unit maximum absolute value. The mode ${}_nT_2$ has n zeros in the interior, and all three satisfy the traction-free condition $\dot{W} = W/b$ at the surface.

This example illustrates how the abstract eigenvalue problem of the lecture becomes a concrete boundary value problem once spherical symmetry is used, and that even the simplest conceivable earth model predicts the gravest toroidal period to within a few per cent.

Example 2: The diagonal sum rule for a degree-one multiplet. For a multiplet with $l = 1$ the first-order splitting problem of the lecture is the 3×3 Hermitian eigenvalue problem $\sum_m M_{m'm} a_m = \delta\omega a_{m'}$, with $m, m' \in \{-1, 0, 1\}$ and

$$M_{m'm} = \frac{1}{2\omega_k} \left[-\omega_k^2 \langle km' | P_1 | km \rangle + i\omega_k \langle km' | W | km \rangle + \langle km' | H_1 | km \rangle \right].$$

Suppose that, ordering the indices as $m = -1, 0, 1$, this splitting matrix is

$$M = \begin{pmatrix} -\Delta & \epsilon & 0 \\ \epsilon & 0 & \epsilon \\ 0 & \epsilon & \Delta \end{pmatrix},$$

with $\Delta > 0$ and ϵ real, the diagonal part representing splitting by rotation (see Example 3) and the off-diagonal part a small contribution from lateral heterogeneity.

- (i) Find the three singlet eigenfrequency perturbations and verify that the arithmetic mean of the split eigenfrequencies equals the degenerate eigenfrequency ω_k .
- (ii) Find the corresponding coefficient vectors a_m , verify that they are orthonormal, and explain what this implies for the zeroth-order eigenfunctions $\mathbf{s} = \sum_m a_m |km\rangle$.
- (iii) Explain why the conclusion of part (i) holds for any Hermitian splitting matrix with zero trace.

Solution. (i) The eigenvalues $\delta\omega$ are the roots of $\det(M - \delta\omega I) = 0$. Expanding along the first row,

$$\begin{aligned} \det(M - \delta\omega I) &= (-\Delta - \delta\omega) \left[-\delta\omega(\Delta - \delta\omega) - \epsilon^2 \right] - \epsilon [\epsilon(\Delta - \delta\omega)] \\ &= -\delta\omega (\delta\omega^2 - \Delta^2 - 2\epsilon^2), \end{aligned} \quad (15)$$

where the terms in $\Delta\epsilon^2$ cancel. The singlet perturbations are therefore

$$\delta\omega = -\Lambda, \quad 0, \quad +\Lambda, \quad \Lambda = \sqrt{\Delta^2 + 2\epsilon^2}, \quad (16)$$

and the split eigenfrequencies are $\omega_k - \Lambda$, ω_k and $\omega_k + \Lambda$, whose arithmetic mean is ω_k . Rotation alone ($\epsilon = 0$) gives three equally spaced lines with spacing Δ ; the heterogeneity leaves the central singlet where it is but pushes the outer two further apart, since $\Lambda > \Delta$. As an illustration, if $\Delta/2\pi = 3 \mu\text{Hz}$ and $\epsilon/2\pi = 2 \mu\text{Hz}$ then $\Lambda/2\pi = \sqrt{17} \mu\text{Hz} = 4.12 \mu\text{Hz}$.

(ii) For $\delta\omega = 0$ the equations $M\mathbf{a} = 0$ give $a_0 = (\Delta/\epsilon)a_{-1}$ and $a_1 = -a_{-1}$, while for $\delta\omega = \pm\Lambda$ the first and third rows give $a_0 = (\Delta \pm \Lambda)a_{-1}/\epsilon$ and $a_0 = (\Delta \mp \Lambda)a_1/\epsilon$. Using $\Lambda^2 - \Delta^2 = 2\epsilon^2$ to simplify, and normalising, we find

$$\mathbf{a}^{(0)} = \frac{1}{\Lambda} \begin{pmatrix} \epsilon \\ \Delta \\ -\epsilon \end{pmatrix}, \quad \mathbf{a}^{(\pm)} = \frac{1}{2\Lambda} \begin{pmatrix} \Lambda \mp \Delta \\ \pm 2\epsilon \\ \Lambda \pm \Delta \end{pmatrix}. \quad (17)$$

It is readily checked that $M\mathbf{a}^{(\pm)} = \pm\Lambda\mathbf{a}^{(\pm)}$; for example the middle row of $M\mathbf{a}^{(+)}$ is $\epsilon(\Lambda - \Delta) + \epsilon(\Lambda + \Delta) = 2\epsilon\Lambda$, which is Λ times the middle entry of $\mathbf{a}^{(+)}$. For the norms, $\|\mathbf{a}^{(0)}\|^2 = (\Delta^2 + 2\epsilon^2)/\Lambda^2 = 1$ and

$$\|\mathbf{a}^{(\pm)}\|^2 = \frac{(\Lambda \mp \Delta)^2 + 4\epsilon^2 + (\Lambda \pm \Delta)^2}{4\Lambda^2} = \frac{2\Lambda^2 + 2\Delta^2 + 4\epsilon^2}{4\Lambda^2} = 1, \quad (18)$$

while for the inner products

$$\begin{aligned}\mathbf{a}^{(+)} \cdot \mathbf{a}^{(-)} &= \frac{(\Lambda^2 - \Delta^2) - 4\epsilon^2 + (\Lambda^2 - \Delta^2)}{4\Lambda^2} = 0, \\ \mathbf{a}^{(0)} \cdot \mathbf{a}^{(\pm)} &= \frac{\epsilon(\Lambda \mp \Delta) \pm 2\epsilon\Delta - \epsilon(\Lambda \pm \Delta)}{2\Lambda^2} = 0.\end{aligned}\quad (19)$$

The three coefficient vectors are thus orthonormal, as the eigenvectors of a Hermitian matrix with distinct eigenvalues must be. Because the singlets satisfy $\langle km|P_0|km'\rangle = \delta_{mm'}$, the zeroth-order eigenfunctions $\mathbf{s}^{(j)} = \sum_m a_m^{(j)} |km\rangle$ obey

$$\langle \mathbf{s}^{(j)} | P_0 | \mathbf{s}^{(j')} \rangle = \sum_m a_m^{(j)*} a_m^{(j')} = \delta_{jj'}, \quad (20)$$

so they are orthonormal with respect to the kinetic energy form, exactly as Lecture 22 requires of eigenfunctions with distinct eigenfrequencies. Note that for $\epsilon \neq 0$ each of these eigenfunctions is a mixture of all three singlets; the heterogeneity has selected a particular orthonormal basis of the degenerate eigenspace, and it is this basis, not the original singlets, that the perturbed eigenfunctions approach as the perturbation is switched off. As $\epsilon \rightarrow 0$ we recover $\mathbf{a}^{(\pm)} \rightarrow (0, 0, 1)$ and $(1, 0, 0)$, the singlets themselves.

(iii) For any Hermitian matrix we can write $M = U \text{diag}(\delta\omega_1, \dots, \delta\omega_{2l+1}) U^\dagger$ with U unitary, and the cyclic property of the trace gives $\text{tr } M = \text{tr}(\text{diag}(\delta\omega_j) U^\dagger U) = \sum_j \delta\omega_j$. Equivalently, the coefficient of $\delta\omega^{2l}$ in the characteristic polynomial is $-\text{tr } M$ up to sign, as eq. (15) illustrates. Hence if $\text{tr } M = 0$ the singlet perturbations sum to zero, and the mean of the $2l + 1$ split eigenfrequencies is ω_k . The content of the diagonal sum rule of the lecture is the statement, following from the selection rules for the matrix elements, that the trace does vanish for splitting by rotation and by lateral heterogeneity, so that averaging the observed singlet frequencies recovers the eigenfrequency of the spherically averaged Earth.

Example 3: Rotational splitting is linear in m . Within a multiplet $|km\rangle$, $-l \leq m \leq l$, of a spherically symmetric earth model it can be shown, using the conventions of the lecture, that the matrix elements of the Coriolis form are

$$\langle km' | W | km \rangle = -2i\Omega\beta_k m \delta_{mm'},$$

where Ω is the angular rate of rotation and β_k is a dimensionless constant, the **splitting parameter**, that depends on the radial eigenfunctions of the multiplet. Take this result as given. (i) Check that it is consistent with the anti-self-adjointness of the Coriolis form. (ii) Neglecting the perturbations P_1 and H_1 (which for a rotating earth model describe the ellipticity of figure and the centrifugal potential), solve the first-order Hermitian eigenvalue problem of the lecture, and show that $\delta\omega_m = m\Omega\beta_k$, so that the multiplet splits into $2l + 1$ equally spaced lines. (iii) For ${}_0S_2$, with degenerate frequency $f_k = 0.3094$ mHz and $\beta_k \approx 0.40$, give the spacing of adjacent singlets and the total width of the multiplet in μHz , and comment on the length of record required to resolve the singlets.

Solution. (i) In Lecture 22 we showed that $\langle \mathbf{w} | W | \mathbf{u} \rangle = -\langle \mathbf{u} | W | \mathbf{w} \rangle^*$. Applied to the singlets this requires $\langle km' | W | km \rangle = -\langle km | W | km' \rangle^*$. With the stated matrix elements the right-hand side is $-(-2i\Omega\beta_k m' \delta_{m'm})^* = -2i\Omega\beta_k m' \delta_{mm'}$, which equals the

left-hand side because the Kronecker delta forces $m' = m$. In particular the diagonal elements $\langle km|W|km\rangle$ are purely imaginary, as anti-self-adjointness demands. They are also odd in m : the singlets $\pm m$ correspond to patterns $e^{i(\pm m\varphi + \omega t)}$ travelling in opposite directions around the rotation axis, and the Coriolis force acts on them with opposite sign.

(ii) Dropping P_1 and H_1 , the first-order eigenvalue problem of the lecture reduces to

$$\frac{1}{2\omega_k} \sum_{m=-l}^l i\omega_k \langle km'|W|km\rangle a_m = \delta\omega a_{m'}. \quad (21)$$

Inserting the matrix elements, the sum collapses to the single term $m = m'$, and

$$\frac{i}{2} (-2i\Omega\beta_k m') a_{m'} = \Omega\beta_k m' a_{m'} = \delta\omega a_{m'}, \quad -l \leq m' \leq l. \quad (22)$$

The splitting matrix is thus already diagonal, $M = \Omega\beta_k \text{diag}(-l, \dots, l)$, with real entries, as the diagonal of a Hermitian matrix must be. Its eigenvalues and eigenvectors can be read off: for each m ,

$$\delta\omega_m = m\Omega\beta_k, \quad a_{m'} = \delta_{m'm}, \quad (23)$$

so the zeroth-order eigenfunctions are the singlets $|km\rangle$ themselves, and the perturbed eigenfrequencies are

$$\omega_m = \omega_k + m\Omega\beta_k, \quad m = -l, \dots, l. \quad (24)$$

The multiplet splits into $2l + 1$ lines, equally spaced by $\Omega\beta_k$ and placed symmetrically about ω_k , which is the direct analogue of the normal Zeeman effect, $\delta E = m\mu_B B$, for an atom in a magnetic field. The shifts sum to $\Omega\beta_k \sum_m m = 0$, in accordance with the diagonal sum rule. The result holds to first order in Ω ; the neglected terms P_1 and H_1 add a contribution that is even in m and so slightly breaks the equal spacing, but for the gravest modes it is smaller than the Coriolis term.

For toroidal modes the splitting parameter can be found explicitly. With $|km\rangle = W\mathbf{C}_{lm}$ and $\mathbf{\Omega} = \Omega\hat{\mathbf{z}}$ the definition of the Coriolis form gives

$$\langle km|W|km\rangle = 2\Omega \int_0^b \rho W^2 r^2 dr \int_{S^2} \mathbf{C}_{lm}^* \cdot (\hat{\mathbf{z}} \times \mathbf{C}_{lm}) dS, \quad (25)$$

and a short calculation with the explicit components of $\mathbf{C}_{lm}$, which we omit, shows that the surface integral equals $-im$. With the normalisation $\zeta^2 \int_0^b \rho W^2 r^2 dr = 1$ from the lecture this gives $\langle km|W|km\rangle = -2i\Omega m/\zeta^2$, so that $\beta_k = 1/l(l+1)$ exactly for every toroidal multiplet. For ${}_0T_2$ the singlets are therefore spaced by $\Omega/(12\pi) = 1.93 \mu\text{Hz}$.

(iii) Using the sidereal day, $\Omega = 2\pi/86164 \text{ s} = 7.292 \times 10^{-5} \text{ s}^{-1}$, or $\Omega/2\pi = 11.61 \mu\text{Hz}$. For ${}_0S_2$ with $\beta_k = 0.40$ the spacing between adjacent singlets is

$$\delta f = \frac{\beta_k \Omega}{2\pi} = 4.64 \mu\text{Hz}, \quad (26)$$

and the five singlets span a total width of $2l\delta f = 18.6 \mu\text{Hz}$, which is 1.5% of f_k per singlet spacing. This agrees well with the observed singlet frequencies of ${}_0S_2$, which are spaced by about $4.6 \mu\text{Hz}$. To resolve two spectral lines separated by δf we saw in

the worked examples for Lecture 22 that the record length T must satisfy $T \gtrsim 1/\delta f$, which here is 2.2×10^5 s, or about 60 hours. The intrinsic width of each line due to attenuation is $f_k/Q \approx 0.6 \mu\text{Hz}$ for $Q \approx 510$, an order of magnitude smaller than the splitting, and the corresponding decay time $Q/\pi f_k \approx 6$ days comfortably exceeds the required record length. The rotational splitting of ${}_0S_2$ can therefore be resolved from a single seismogram several days long after a great earthquake, and indeed it was such a resolved splitting that motivated the first theoretical treatments of the problem.

Example 4: Temperature or composition? Write $\delta \ln \beta = \delta \beta / \beta$, $\delta \ln \gamma = \delta \gamma / \gamma$ and $\delta \ln \rho = \delta \rho / \rho$ for the fractional perturbations of shear wave speed, bulk sound speed and density relative to the spherical average at some depth in the lower mantle. Take the following illustrative values of the partial derivatives, of the magnitudes suggested by mineral physics. With respect to temperature, $\partial \ln \beta / \partial T = -1.0 \times 10^{-4} \text{ K}^{-1}$, $\partial \ln \gamma / \partial T = -3.0 \times 10^{-5} \text{ K}^{-1}$ and $\partial \ln \rho / \partial T = -1.5 \times 10^{-5} \text{ K}^{-1}$. With respect to a compositional variable X , taken to be the molar iron fraction $\text{Fe}/(\text{Mg} + \text{Fe})$, $\partial \ln \beta / \partial X = -0.30$, $\partial \ln \gamma / \partial X = +0.10$ and $\partial \ln \rho / \partial X = +0.30$. (i) Show that for purely thermal anomalies $\delta \ln \gamma / \delta \ln \beta > 0$, and evaluate the ratio. (ii) Tomography suggests that within an LLSVP $\delta \ln \beta \approx -2\%$ and $\delta \ln \gamma \approx +0.5\%$. Show that these values cannot be explained by temperature alone, and solve for the temperature and compositional anomalies δT and δX that explain both. (iii) Compute the implied density anomaly for (a) a purely thermal interpretation of the shear wave anomaly alone and (b) the joint solution of part (ii), and comment.

Solution. (i) For a purely thermal anomaly the lecture's linearised relations give $\delta \ln \beta = (\partial \ln \beta / \partial T) \delta T$ and $\delta \ln \gamma = (\partial \ln \gamma / \partial T) \delta T$, so that

$$\frac{\delta \ln \gamma}{\delta \ln \beta} = \frac{\partial \ln \gamma / \partial T}{\partial \ln \beta / \partial T} = \frac{-3.0 \times 10^{-5}}{-1.0 \times 10^{-4}} = 0.3. \quad (27)$$

The ratio is positive because both derivatives are negative: heating a crystalline solid reduces its elastic moduli more rapidly than thermal expansion reduces its density, so all seismic wave speeds fall. Thermal anomalies therefore produce positively correlated shear and bulk sound speed anomalies, with the bulk sound speed the less sensitive of the two.

(ii) If the shear wave anomaly were thermal we would need $\delta T = \delta \ln \beta / (\partial \ln \beta / \partial T)$, which is $(-0.02) / (-1.0 \times 10^{-4}) = 200 \text{ K}$, and eq. (27) would then predict $\delta \ln \gamma = 0.3 \times (-2\%) = -0.6\%$. The observed value of $+0.5\%$ has the opposite sign, so no choice of δT alone can fit both observations. Allowing for composition, the lecture's relations become the linear system

$$\begin{pmatrix} -1.0 \times 10^{-4} & -0.30 \\ -3.0 \times 10^{-5} & 0.10 \end{pmatrix} \begin{pmatrix} \delta T \\ \delta X \end{pmatrix} = \begin{pmatrix} -0.020 \\ 0.005 \end{pmatrix}, \quad (28)$$

in which δT is in kelvin. The determinant is $(-1.0 \times 10^{-4})(0.10) - (-0.30)(-3.0 \times 10^{-5}) = -1.9 \times 10^{-5}$, and Cramer's rule gives

$$\begin{aligned} \delta T &= \frac{(0.10)(-0.020) - (-0.30)(0.005)}{-1.9 \times 10^{-5}} = 26 \text{ K}, \\ \delta X &= \frac{(-1.0 \times 10^{-4})(0.005) - (-3.0 \times 10^{-5})(-0.020)}{-1.9 \times 10^{-5}} = 0.058. \end{aligned} \quad (29)$$

With these numbers the LLSVP is barely warmer than its surroundings and is enriched in iron by nearly six mole per cent; almost the whole of the shear wave anomaly is compositional.

(iii) For interpretation (a), $\delta T = 200$ K and

$$\delta \ln \rho = \frac{\partial \ln \rho}{\partial T} \delta T = -1.5 \times 10^{-5} \times 200 = -0.3\%, \quad (30)$$

so the LLSVP would be lighter than the ambient mantle and buoyant: the “super plume” picture. For interpretation (b),

$$\delta \ln \rho = \frac{\partial \ln \rho}{\partial T} \delta T + \frac{\partial \ln \rho}{\partial X} \delta X = -1.5 \times 10^{-5} \times 26 + 0.30 \times 0.058 = +1.7\%, \quad (31)$$

so the same region would be denser than its surroundings and would tend to remain at the base of the mantle. The two interpretations thus differ not merely in the size but in the sign of the density anomaly, and hence in the dynamical role of the LLSVPs. The joint solution is, moreover, sensitive to derivatives that are poorly known: repeating the calculation with $\partial \ln \gamma / \partial X = +0.05$ instead of $+0.10$ gives $\delta T = -36$ K, $\delta X = 0.079$ and $\delta \ln \rho = +2.4\%$, so even the sign of the temperature anomaly is not robust, although the density excess is. What this example shows is that wave speeds alone, however well resolved, cannot determine whether LLSVPs are light or dense once compositional variations are admitted, and this is why direct constraints on lateral density variations from free oscillations and tides are of such interest.

Example 5: Vector spherical harmonics of degree one. The spherical harmonic $Y_{10} = \sqrt{3/4\pi} \cos \theta$ is real and normalised. (i) Compute $\mathbf{A}_{10}$, $\mathbf{B}_{10}$ and $\mathbf{C}_{10}$ explicitly, and verify that $\int_{S^2} \mathbf{B}_{10}^* \cdot \mathbf{C}_{10} dS = 0$ and $\int_{S^2} \|\mathbf{B}_{10}\|^2 dS = \int_{S^2} \|\mathbf{C}_{10}\|^2 dS = l(l+1) = 2$. (ii) Show that $\mathbf{A}_{10} + \mathbf{B}_{10}$ is a constant vector and that $r\mathbf{C}_{10}$ is a rigid rotation, and interpret these results in terms of the trivial modes of Lecture 22.

Solution. (i) Since Y_{10} does not depend on φ , the tangential gradient is $\nabla_1 Y_{10} = \hat{\theta} \partial Y_{10} / \partial \theta$, and from the definitions in the lecture

$$\begin{aligned} \mathbf{A}_{10} &= \sqrt{\frac{3}{4\pi}} \cos \theta \hat{\mathbf{r}}, & \mathbf{B}_{10} &= \nabla_1 Y_{10} = -\sqrt{\frac{3}{4\pi}} \sin \theta \hat{\theta}, \\ \mathbf{C}_{10} &= -\hat{\mathbf{r}} \times \nabla_1 Y_{10} = \sqrt{\frac{3}{4\pi}} \sin \theta \hat{\varphi}, \end{aligned} \quad (32)$$

where we have used $\hat{\mathbf{r}} \times \hat{\theta} = \hat{\varphi}$. Because $\hat{\theta}$ and $\hat{\varphi}$ are orthogonal at every point, $\mathbf{B}_{10}^* \cdot \mathbf{C}_{10} = 0$ pointwise and its integral vanishes trivially. With $dS = \sin \theta d\theta d\varphi$,

$$\int_{S^2} \|\mathbf{B}_{10}\|^2 dS = \frac{3}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi \sin^3 \theta d\theta = \frac{3}{4\pi} \times 2\pi \times \frac{4}{3} = 2, \quad (33)$$

using $\int_0^\pi \sin^3 \theta d\theta = \int_{-1}^1 (1-c^2) dc = 4/3$, and the same integrand gives $\int \|\mathbf{C}_{10}\|^2 dS = 2$. Both agree with $l(l+1) = 2$. That the answer is $l(l+1)$ in general follows by integrating by parts on the sphere, $\int \nabla_1 Y_{lm}^* \cdot \nabla_1 Y_{lm} dS = -\int Y_{lm}^* \nabla_1^2 Y_{lm} dS = l(l+1)$, using the defining eigenvalue relation for Y_{lm} ; the present calculation is just this identity made explicit for one harmonic.

(ii) Recalling that $\hat{\mathbf{z}} = \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}$, eq. (32) gives

$$\mathbf{A}_{10} + \mathbf{B}_{10} = \sqrt{\frac{3}{4\pi}} \hat{\mathbf{z}}, \quad (34)$$

a constant vector. Similarly $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \sin \theta \hat{\boldsymbol{\phi}}$, so that

$$r\mathbf{C}_{10} = \sqrt{\frac{3}{4\pi}} \hat{\mathbf{z}} \times \mathbf{x}, \quad (35)$$

which is the displacement produced by an infinitesimal rigid rotation about the z -axis. In the language of the lecture, the spheroidal displacement $U\mathbf{A}_{10} + V\mathbf{B}_{10}$ with $U = V = \text{const}$ is a rigid translation along $\hat{\mathbf{z}}$, and the toroidal displacement $W\mathbf{C}_{10}$ with $W \propto r$ is a rigid rotation about $\hat{\mathbf{z}}$; the harmonics with $m = \pm 1$ supply the translations and rotations about the x - and y -axes. These are the six zero-frequency trivial modes of Lecture 22, and we see that they are all of degree one. This is consistent with Example 1, where the $l = 1$ frequency equation was found to have the root $\omega = 0$ with $W \propto r$, and it explains why the mode catalogues of the Earth contain no ${}_0T_1$ and no ${}_0S_1$. A gravitating earth model does have a non-trivial degree-one spheroidal mode, ${}_1S_1$, in which the inner core translates against the rest of the Earth, but it is not a rigid motion.